

Read this before using the recipes.

This is a patient-created culinary working edition, not medical advice, a treatment, or a substitute for your nephrologist or renal dietitian. CKD stage, dialysis modality, residual kidney function, medications, labs, diabetes, fluid limits, and individual targets vary. Nutrition figures in this book are recipe estimates, not personal limits; brands, substitutions, preparation, and serving size can change them. Check labels and review ingredients, portions, supplements, herbs, and unfamiliar products with your care team. Feedback a dietitian gave the creator on his personal food-rule sheet helped inform this caution-first revision; it was not a formal product review, endorsement, clinical certification, or validation of every recipe.

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RENALWISE

Digital MasterCook Book

CKD Kitchen Freeware Edition v09

200+ Recipes · 37 Chapters · 3 Appendices

FREE SEARCHABLE BROWSER EDITION + PRINTABLE PDF

“Eat like you love food and life whether your kidneys work or not.”

NorthStar Prime · 2026

37 Chapters · 3 Appendices

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Pull up a chair.

Stay awhile.

PREFACE

Introduction & Dedication

Dedication

For everyone who was told "you can't eat that anymore."
And made it anyway.

"One cannot think well, love well, sleep well, if one has not dined well."
— Virginia Woolf

Introduction

This is a cookbook for people whose kidneys don't work.
That sentence should make you sad, and it does, for about three seconds,
and then you turn the page and there's a recipe for Taco Bell's \$7 Luxe Box
that has 65% less sodium than the drive-thru version and tastes 95%
identical, and sadness becomes a luxury you can't afford because you're
too busy making chalupa shells from scratch.

"The only time to eat diet food is while waiting for the steak to cook." —
Julia Child

You are handed a laminated sheet at the clinic. The sheet says NO. No
potassium. No phosphorus. No sodium. No fun. The sheet is medically
accurate and spiritually catastrophic. It tells you what you can't have. It
never tells you what you *can*.

This book tells you what you can.

"Cooking is like love. It should be entered into with abandon or not at all."
— Harriet Van Horne

200+ recipes. Thai street food. Taco Bell dupes. Mango sticky rice. Country
fried steak with white gravy. Hot sauces with names that would make your
doctor blush. Smoothies that taste like a Caribbean vacation. Gummies that
glow in the dark. A Persian ketchup from a food cart in Cornelius, Oregon,
that changed everything.

All of it CKD-aware, with portions and labs doing the real safety work. All of
it actually good. All of it made by someone who decided the kitchen doesn't
quit.

"Tell me what you eat, and I will tell you what you are." — Jean Anthelme
Brillat-Savarin

I am steak tacos with fresh pico at midnight. I am chocolate Rice Krispie
treats at 2am. I am Tom Kha Gai with too much lime. I am my mother's
radiatore pasta salad with white wine vinegar, and I will be that salad until I
die, and then I will be the memory of that salad, which is the same thing,
because food is memory and memory is food and both are love.

"People who love to eat are always the best people." — Julia Child

A Note on How to Use This Book

"Perfection is achieved, not when there is nothing more to add, but when
there is nothing left to take away." — Antoine de Saint-Exupery

This is the working edition. Recipes only. No essays. No equipment lists.
No CKD nutrition lectures. Just ingredients, methods, and results.

The full version — with nutrition data per serving, equipment
recommendations, ingredient sourcing, clinical notes, and philosophical
tangents about consciousness — lives online at northstarprime.com/ckd-kitchen.

This version is for the kitchen counter. It gets flour on it. That's what it's for.

The Emoji Soup Restaurant Menu

(translated from the original trilingual symbolic
notation)

The menu predates the restaurant. The restaurant predates the building.
The building predates the city. All of this is somehow the fault of a filing
error in the Multnomah County Department of Records.

"Item one: a soup. Price: nothing. Description: everything." — The
Archaeological Record, Southeast Hawthorne Boulevard, 1987

The Six Languages of the Menu

The original menu is not written in one language. It is written in six: glyph,
name, kitchen action, hunger, price, and law. The glyph tells the eye what
the dish is. The name gives the mouth something to say. The action tells
the cook what must happen. The hunger tells the soul why it came. The
price tells the customer what will be exchanged. The law tells everyone
what the dish refuses to lie about.

Glyph	Name	Action	Need	Price	Law
	The Bread	tear/rest	first gift	notice	arrives before permission
	The Soup	simmer/wait	origin	nothing	contains everything but explains nothing
	Mango Rice Dream Pop	cook/blend/freeze	soft sweetness	patience	rice must become porridge before it becomes memory
	Mango Habanero Afterfire	ferment/finish	courage with fruit	gloves	fresh fruit comes after the fire learns manners
	The Check	total/return	consequence	exact change	the bill is always itemized, never unfair

Second Menu Board

This menu now reads as a working grammar for the novels, the cookbook,
and the sauce lab. A glyph can foreshadow a chapter. A sauce label can
become a waiver. A receipt can become the kindest possible accusation.

Glyph	Name	Action	Need	Price	
	Mercy Butter	soften/spread	ease	one lowered defense	me me arg witt
	Permit Pickles	brine/file	permission	one honest form	sol bec offi whi witr
	Bob's Mushroom Theorem	saute/prove	generosity	one shared plate	the is n opt
	The Courier's Spoon	carry/return	direction	fuel and humility	eve deli deli the
	Black Orchard Adobo	char/blend/log	courage in shadow	gloves and honesty	sm mu hid frui
	Gold Pheasant Peach	gleam/blend/warn	good fortune with teeth	respect	aus doe me
	Pineapple Thai Basil Lightning	chop/flash/cool	sudden courage	today	ligh not left
	Floccinaucinihilipilification Pudding	set/reflect	worth	one admitted value	the ran pric

Appetizers for the Apocalypse

- **The Bread** — Arrives before you order. Warm. Endless. Do not question the bread. The bread is correct.
- **The Bisque** — What it's a bisque OF depends on who's asking and whether they're ready for the answer.
- **Breadsticks of Diminished Fear** — The largest things are always drawn smallest by people who are afraid of them.

Entrees That Remember You

- **The Soup** — There is only one soup. It contains everything. It has always contained everything. You've been eating it your whole life. You just didn't know it had a name.
- **Bob's Mushroom Theorem** — A dish that proves, mathematically, that generosity is the only rational economic strategy. Served with rice. The rice is optional. The proof is not.
- **The Thing You Came Here For** — You don't order this. It orders you. It arrives when you're ready. You are never ready. It arrives anyway.

Desserts That Eat You

- **The Tea of Enlightenment** — Tastes like whatever you need it to taste like. Served at exactly the temperature that makes you tell the truth.
- **Floccinaucinihilipilification Pudding** — The act of estimating something as worthless, served in a ramekin. The pudding is worthless. The ramekin is priceless. The joke is on you. The joke is also the pudding.

The Check

The check never arrives. The meal is free. You pay by remembering what it tasted like and telling someone. This is the only currency the restaurant accepts. This is also how love works, and language, and music, and every other thing that matters.

"There is no sincerer love than the love of food." — George Bernard Shaw

A Few Words Before We Cook

"The secret of success in life is to eat what you like and let the food fight it out inside." — Mark Twain

Medical safety note: This cookbook is a planning aid, not medical advice. Use your prescribed sodium, potassium, phosphorus, protein, and fluid limits; confirm recipes with your nephrologist or renal dietitian, especially if you are on dialysis, using binders, managing diabetes, immunocompromised, pregnant, or taking transplant medications.

CHAPTER 1

Breakfast Classics

"The waffle iron was patented by Cornelius Swartwout on August 24, 1869."

French Toast, Waffles, Pancakes & More — Morning Done Right

"The waffle iron was patented by Cornelius Swartwout on August 24, 1869."

"All happiness depends on a leisurely breakfast." — John Gunther

Breakfast Intelligence

Before you cook a single thing, internalize these:

Principle	Why It Matters
Eggs are king	High biological value protein, moderate P (~95mg/egg), low Na (~70mg/egg). Two eggs is one example serving; whether it fits depends on the rest of your day and your individualized targets.
White bread > whole wheat	Whole wheat = higher P (phosphorus bound to phytates) and higher K. Brioche, white flour, sourdough — all better choices.
Phosphorus-free baking powder	Rumford aluminum-free. Regular baking powder is loaded with sodium aluminum phosphate. This applies to ALL baked items in this chapter.
K-leach every potato	Hash browns MUST soak 4+ hours in cold water, drain, rinse, squeeze. Removes 40-50% of potassium. Non-negotiable.

Label check: Nutrition estimates are working estimates. Packaged foods, salt substitutes, potassium chloride, phosphate additives, sauces, broths, and restaurant dupes can change the renal math quickly. Check labels and use the limits from your care team.

Evidence language: In this edition, "CKD-aware" means a recipe is built around lower sodium/potassium/phosphorus assumptions as portioned; it does not mean universally safe. National Kidney Foundation guidance emphasizes that sodium, potassium, phosphorus, protein, and fluid targets vary by labs, dialysis status, medications, and care-team instructions. Source: <https://www.kidney.org/kidney-topics/nutrition-and-kidney-disease-stages-1-5-not-dialysis>

But also: make the gumbo. Fry the chicken. Bake the carrot cake. Pour the white gravy over the country fried steak. Squeeze the lime into the Tom Kha. Coat the gummies in nuclear sour dust and eat three before bed.

"Life is uncertain. Eat dessert first." — Ernestine Ulmer

The restrictions are in the *ingredients*, not the *recipes*. Margarine instead of butter. Almond milk instead of whole. Leach the potatoes. Rinse the beans three times. Use the no-salt seasoning. Skip the olives.

Same plate. Same flavor. Same Saturday morning breakfast. Different margarine. Your kidneys can't tell the difference. Your taste buds can't either.

"After a good dinner, one can forgive anybody, even one's own relatives." — Oscar Wilde

Now wash your hands. Put on some music. Open this book to whatever page you land on.

Cook like you love food and life whether your kidneys work or not. Because you do.

"All sorrows are less with bread." — Miguel de Cervantes

Andrew D White Jr. Portland, Oregon 2026

988 — You are not alone. 988 + 11 = 999; 999 is symbolic hope, not a crisis number.

Principle	Why It Matters
Turkey sausage > pork sausage	Lower phosphorus, lower sodium per serving. Read labels — some turkey sausage is still injected with sodium phosphate. Buy CLEAN.
Maple syrup is safe	Basically sugar + water + trace minerals. Low K (~40mg/tbsp), low P (~3mg/tbsp), low Na (~2mg/tbsp). Drizzle without guilt.
Cream cheese > butter for frosting	Lower P per tablespoon, better mouthfeel, and it's what cinnamon rolls demand anyway.
Almond milk replaces dairy	Unsweetened almond milk: ~5mg P per cup vs whole milk: ~230mg P per cup. Massive difference.

1. Golden Sunday French Toast

"French toast first appears in the Apicius, a Roman cookbook from the 4th or 5th century AD, where it was called 'Pan Dulcis' — bread soaked in milk and egg, then fried."

Ingredient	Amount	CKD Notes
Brioche or thick-cut white bread	6 slices (3/4" thick)	NOT whole wheat — brioche is lower P and makes better French toast anyway
Eggs	2 large	~190mg P total for the whole batch, split across servings
Unsweetened almond milk	1/3 cup	Replaces whole milk — ~2mg P vs 75mg
Vanilla extract	1 tsp	
Ground cinnamon	1 tsp	
Nutmeg	1/4 tsp (fresh grated ideal)	
Sugar	1 tbsp	In the custard for caramelization
Salt	Pinch	
Margarine	2 tbsp (for pan)	Lower P than butter

Ingredient	Amount	CKD Notes
Pure maple syrup	For serving	CKD-aware — low K/P/Na
Powdered sugar	For dusting (optional)	
Fresh berries	For topping	Strawberries and blueberries are lower-K fruits

Method: 1. Whisk eggs, almond milk, vanilla, cinnamon, nutmeg, sugar, and salt in a wide shallow bowl until uniform. 2. Heat a large skillet or griddle over medium heat. Melt 1 tbsp margarine and swirl to coat. 3. Dip each bread slice into the custard — 15 seconds per side. Let excess drip off. You want saturated, not dripping. 4. Cook 2-3 min per side until deep golden brown and slightly crispy on the edges. The sugars in brioche caramelize beautifully. 5. Add more margarine between batches. Keep finished slices warm on a sheet pan in a 200F oven. 6. Serve stacked, dusted with powdered sugar, drizzled with maple syrup, berries alongside.

Per serving (2 slices + 2 tbsp syrup): ~120mg Na | ~105mg P | ~130mg K

2. Iron-Crisp Waffles

"The first known waffle recipe printed in English appeared in a 1725 London cookbook by Robert Smith, calling for a pint of cream, seven eggs, and a pound of butter."

Ingredient	Amount	CKD Notes
All-purpose flour	2 cups	White flour only — lower P than whole wheat
Phosphorus-free baking powder (Rumford)	1 tbsp	CRITICAL — regular baking powder adds ~350mg P per batch
Baking soda	1/2 tsp	
Sugar	2 tbsp	
Salt	1/4 tsp	
Eggs	2 large, separated	Whites whipped = the secret to crispy exterior/fluffy interior
Unsweetened almond milk	1.5 cups	
Avocado oil	1/3 cup	Neutral flavor, high smoke point
Vanilla extract	1 tsp	
White vinegar	1 tbsp	Added to almond milk = "buttermilk" tang, activates baking soda

Method: 1. Add vinegar to almond milk, stir, let sit 5 min (it curdles slightly — that's correct). 2. Whisk flour, baking powder, baking soda, sugar, salt in a large bowl. 3. In a separate bowl, whisk egg yolks, almond milk mixture, oil, vanilla. 4. In a third bowl (or stand mixer), beat egg whites to stiff peaks. This is the move that separates diner waffles from restaurant waffles. 5. Pour wet into dry, stir until JUST combined — lumps are fine, overworking = tough waffles. 6. Fold in whipped egg whites gently. The batter should look like a cloud with flour in it. 7. Preheat waffle iron to medium-high. Lightly oil or spray. Pour batter to fill 2/3 of the iron surface. 8. Cook until steam stops and waffle is deep golden (3-5 min depending on iron). DO NOT open early — you'll tear it. 9. Serve immediately or keep crispy on a wire rack in a 200F oven. Never stack fresh waffles (steam = soggy).

Makes: 4-6 waffles depending on iron size

Per waffle: ~110mg Na | ~95mg P | ~90mg K

3. Cloud-Stack Pancakes

"Otzi the Iceman, the 5,300-year-old mummy found in the Alps in 1991, had traces of ground einkorn wheat in his stomach — possibly the oldest evidence of something resembling a pancake."

Ingredient	Amount	CKD Notes
All-purpose flour	1.5 cups	White flour — lower P
Phosphorus-free baking powder (Rumford)	2 tsp	
Baking soda	1/2 tsp	
Sugar	2 tbsp	
Salt	1/4 tsp	

Ingredient	Amount	CKD Notes
Egg	1 large	
Unsweetened almond milk	1.25 cups	
White vinegar	1 tbsp	"Buttermilk" conversion
Avocado oil	2 tbsp	In the batter for tender crumb
Vanilla extract	1 tsp	

Berry Compote:

Ingredient	Amount	CKD Notes
Mixed berries (strawberries + blueberries)	2 cups (fresh or frozen)	Lower-K fruit choices
Sugar	2 tbsp	
Lemon juice	1 tbsp	
Water	2 tbsp	

Method — Pancakes: 1. Add vinegar to almond milk, stir, let sit 5 min. 2. Whisk flour, baking powder, baking soda, sugar, salt in a large bowl. 3. Whisk egg, almond milk mixture, oil, vanilla in a separate bowl. 4. Pour wet into dry. Stir until JUST combined — 10-15 strokes max. The batter should be lumpy. Walk away from it. Overmixed pancakes are rubber discs. 5. Let batter rest 5 min (baking powder activates, bubbles form = fluffier pancakes). 6. Heat griddle or skillet to medium (325F). Lightly oil. 7. Pour 1/4 cup batter per pancake. Wait until bubbles form on the surface AND the edges look set (2-3 min), then flip once. Cook 1-2 min more. 8. One flip only. Every extra flip compresses the pancake. Patience.

Method — Berry Compote: 1. Combine berries, sugar, lemon juice, water in a small saucepan over medium heat. 2. Bring to a simmer, stirring occasionally. Mash some berries with a fork for texture. 3. Cook 8-10 min until syrupy but still chunky. Serve warm over pancakes.

Per serving (3 pancakes + 3 tbsp compote): ~140mg Na | ~90mg P | ~150mg K

4. Soft Landing Eggs Benedict

"Eggs Benedict was allegedly invented in 1894 at the Waldorf Hotel in New York City when a Wall Street broker named Lemuel Benedict ordered 'buttered toast, poached eggs, crisp bacon, and a hooker of hollandaise' to settle his hangover."

Ingredient	Amount	CKD Notes
English muffins	2, split and toasted	White flour English muffins — check label, some brands add calcium phosphate
Eggs	4 large (2 for poaching, 2 yolks for hollandaise)	~380mg P total across 4 eggs, split into 2 servings
Turkey bacon or Canadian bacon	4 slices	Turkey bacon preferred — lower Na/P than traditional Canadian bacon
White vinegar	1 tbsp (for poaching water)	
Fresh chives	For garnish	
Paprika	Pinch, for garnish	

CKD Hollandaise:

Ingredient	Amount	CKD Notes
Egg yolks	2 large	
Margarine	1/2 cup (1 stick), melted and hot	Lower P than butter
Lemon juice	1 tbsp	Fresh — bottled has additives
Cayenne pepper	Pinch	
Salt	1/8 tsp	Traditional hollandaise uses 1/2 tsp+ — this is plenty with the lemon
Water	1 tbsp (warm)	To adjust consistency

Method — Hollandaise (make first, keep warm): 1. Whisk egg yolks + lemon juice in a heatproof bowl over a pot of barely simmering water

(double boiler). Whisk constantly until yolks thicken and lighten in color (2-3 min). The bowl should NEVER touch the water. 2. Remove from heat. Slowly drizzle in hot melted margarine while whisking vigorously — a thin steady stream. It will emulsify into a creamy sauce. 3. Add cayenne, salt, warm water to adjust consistency. It should coat the back of a spoon. 4. Keep warm by leaving the bowl over the pot (heat OFF). Stir occasionally.

Method — Assembly: 1. Heat turkey bacon in a skillet until crispy. Set aside. 2. Bring a wide pot of water to a gentle simmer (NOT a rolling boil — turbulence shreds the whites). Add vinegar. 3. Crack each egg into a small cup. Create a gentle whirlpool with a spoon, then slide the egg into the center. Poach 3-4 min for runny yolk. 4. Remove with a slotted spoon, drain on a paper towel. 5. Toast English muffin halves. Layer: muffin → bacon → poached egg → hollandaise → chives + paprika.

Per serving (2 halves): ~280mg Na | ~230mg P | ~200mg K

5. Sunrise Wrap Burrito

"The breakfast burrito as we know it was popularized in the 1970s in New Mexico, with Tia Sophia's restaurant in Santa Fe often credited as the first to put a complete breakfast inside a flour tortilla."

Ingredient	Amount	CKD Notes
Low-sodium flour tortillas (10")	2 large	Look for <300mg Na per tortilla — Mission has a low-sodium line
Eggs	4 large, scrambled	
Turkey breakfast sausage	4 oz (2 patties), crumbled	Read label — no sodium phosphate
Bell pepper (red or green)	1/2 cup, diced	Red peppers slightly higher K than green, but both fine in this quantity
White onion	1/4 cup, diced	
Sharp cheddar cheese	1/4 cup, shredded	
Avocado oil	1 tbsp	
Garlic powder	1/4 tsp	
Cumin	1/4 tsp	
Paprika	1/4 tsp	
Black pepper	1/4 tsp	
Salt	Pinch	The sausage and cheese carry enough salt
Hot sauce	To taste	

Optional toppings:

Topping	CKD Notes
Sour cream (1 tbsp)	Lower P than extra cheese
Pico de gallo (2 tbsp)	Moderate K from tomato — small portion OK
Salsa verde	Often lower Na than red salsa — check label

Method: 1. Brown crumbled turkey sausage in a skillet over medium heat until cooked through. Remove, set aside. 2. In the same skillet, add oil. Saute bell pepper and onion until soft (3-4 min). Add garlic powder, cumin, paprika, pepper. 3. Push veggies to one side. Add beaten eggs to the other side. Scramble on medium-low, stirring gently — large soft curds, not rubber. Pull off heat when eggs are STILL slightly wet (carryover cooking finishes them). 4. Mix eggs, sausage, and veggies together. Add cheese while hot so it melts in. 5. Warm tortillas in a dry skillet 15 seconds per side (or 10 seconds in microwave under a damp paper towel). 6. Load filling down the center, fold sides in, then roll from the bottom. Tight but not bursting. 7. Optional: sear the seam side down in a dry skillet for 30 seconds for a crispy seal.

Per burrito: ~350mg Na | ~220mg P | ~280mg K

6. Sunday Gravy Biscuits

"Biscuits and gravy became a staple of the American South during the Revolutionary War era, when wheat flour biscuits served with gravy made from pan drippings provided a cheap, calorie-dense meal for laborers and soldiers."

Flaky CKD Biscuits:

Ingredient	Amount	CKD Notes
All-purpose flour	2 cups	White flour — lower P
Phosphorus-free baking powder (Rumford)	1 tbsp	The single most important swap in CKD baking
Sugar	1 tsp	
Salt	1/2 tsp	
Cold margarine, cubed	1/3 cup	Keep COLD — visible margarine chunks = flaky layers
Unsweetened almond milk	3/4 cup, cold	

Turkey Sausage Gravy:

Ingredient	Amount	CKD Notes
Turkey breakfast sausage	8 oz	No sodium phosphate on the label
Margarine	2 tbsp	
All-purpose flour	3 tbsp	The roux — equal parts fat and flour
Unsweetened almond milk	2 cups	Whole milk would add ~460mg P. Almond milk adds ~10mg.
Black pepper	1 tsp (generous)	Pepper IS the seasoning in gravy — don't be timid
Garlic powder	1/4 tsp	
Onion powder	1/4 tsp	
Salt	1/4 tsp	The sausage has salt already
Dried sage	1/4 tsp	Optional but traditional

Method — Biscuits: 1. Whisk flour, baking powder, sugar, salt in a large bowl. 2. Add cold cubed margarine. Cut in with a pastry cutter (or two knives, or your hands — fast) until mixture resembles coarse crumbs with pea-sized margarine pieces. Those chunks become flaky layers. 3. Add cold almond milk. Stir with a fork until JUST combined. The dough should be shaggy, not smooth. 4. Turn onto a floured surface. Pat (don't roll) to 1" thick. Fold in half, pat again. Repeat 3 times. This lamination creates layers. 5. Pat to final 1" thickness. Cut with a 2.5" biscuit cutter (or a glass). Press straight down — don't twist (twisting seals the edges and kills the rise). 6. Place on a parchment-lined baking sheet with biscuits touching (they help each other rise). Brush tops with almond milk. 7. Bake at 425F for 12-15 min until golden on top. Let rest 5 min.

Method — Gravy: 1. Brown turkey sausage in a skillet over medium heat, breaking into small crumbles (5-6 min). If the sausage is very lean, add the margarine now to compensate for missing fat. 2. Add margarine to the pan (if not already added). Once melted, sprinkle flour over the sausage. Stir and cook 1-2 min — this is the roux, it should smell nutty, not raw. 3. Slowly pour in almond milk, whisking constantly. The gravy will look thin — keep whisking. It thickens as it simmers (3-5 min). 4. Add pepper (generous), garlic powder, onion powder, sage, salt. Taste and adjust. Good sausage gravy should be peppery. 5. Split biscuits, ladle gravy over both halves. Eat immediately.

Per serving (2 biscuits + 1/2 cup gravy): ~320mg Na | ~170mg P | ~190mg K

7. Kanel Morning Rolls

"The kanelbulle, Sweden's cinnamon roll, is so culturally important that October 4th is officially 'Kanelbullens Dag' (Cinnamon Roll Day) — a national holiday since 1999."

Dough:

Ingredient	Amount	CKD Notes
All-purpose flour	3.5 cups	White flour — lower P
Unsweetened almond milk	1 cup, warmed to 110F	
Active dry yeast	2.25 tsp (1 packet)	
Sugar	1/4 cup	Feeds the yeast and sweetens the dough
Egg	1 large	
Margarine	1/4 cup, melted	

Ingredient	Amount	CKD Notes
Salt	1/2 tsp	

Filling:

Ingredient	Amount	CKD Notes
Margarine	1/4 cup, softened	Spread on rolled dough
Brown sugar	1/2 cup, packed	
Ground cinnamon	1.5 tbsp	

Cream Cheese Frosting:

Ingredient	Amount	CKD Notes
Cream cheese	4 oz, softened	Lower P than butter-based frosting
Powdered sugar	1 cup	
Vanilla extract	1 tsp	
Unsweetened almond milk	1-2 tbsp	To adjust drizzle consistency

Method — Dough: 1. Warm almond milk to 110F (not hotter — you'll kill the yeast). Stir in yeast + 1 tsp of the sugar. Let sit 5-10 min until foamy. If it doesn't foam, your yeast is dead — get new yeast. 2. In a large bowl (or stand mixer with dough hook), combine flour, remaining sugar, salt. 3. Add yeast mixture, egg, melted margarine. Mix until a soft dough forms. 4. Knead 8-10 min (by hand) or 5 min (dough hook on medium) until smooth and elastic. The dough should spring back when poked. 5. Place in an oiled bowl, cover with plastic wrap or damp towel. Let rise in a warm spot 1-1.5 hours until doubled.

Method — Assembly: 1. Punch down dough. Roll out on a floured surface to a 16x12" rectangle. 2. Spread softened margarine evenly over the entire surface. 3. Mix brown sugar + cinnamon. Sprinkle evenly over the margarine. Press gently so it sticks. 4. Roll up tightly from the long side (you want a 16" log). Pinch the seam. 5. Cut into 12 equal rolls. Dental floss or unflavored thread works better than a knife (slide under the log, cross the ends, pull to cut — clean slices). 6. Place in a greased 9x13 baking pan, cut side up. Cover and let rise 30-45 min until puffy and touching. 7. Bake at 350F for 22-28 min until golden on top. Center rolls should be set but still soft. 8. Let cool 10 min (not longer — they need to be warm for the frosting).

Method — Frosting: 1. Beat cream cheese until smooth. Add powdered sugar and vanilla. Beat until fluffy. 2. Add almond milk 1 tbsp at a time until you reach a thick but pourable consistency. 3. Spread or drizzle over warm cinnamon rolls. The warmth melts it slightly — that's the point.

Makes: 12 rolls

Per roll: ~190mg Na | ~80mg P | ~85mg K

8. Dawn Patrol Breakfast Sandwich

"The Egg McMuffin was invented by Herb Peterson, a McDonald's franchise owner in Santa Barbara, California, in 1972 — he was trying to create a fast-food version of Eggs Benedict."

Ingredient	Amount	CKD Notes
White English muffin	1, split and toasted	Check label — avoid brands with calcium phosphate
Egg	1 large	
Turkey breakfast sausage	1 patty (2 oz)	Or turkey bacon (2 slices)
American cheese	1 slice	Yes, processed — but 1 slice is ~260mg Na, which fits the budget. The meltability matters here.
Margarine	1 tsp (for egg ring)	
Salt	Tiny pinch	
Black pepper	To taste	

Method: 1. Form turkey sausage into a thin patty roughly the diameter of the English muffin. Season with pepper, a tiny bit of garlic powder. Cook in a skillet over medium heat — 3 min per side until cooked through and slightly crispy. Set aside. 2. **The egg ring trick:** Use a metal ring mold (or a clean tuna can with both ends removed) in the same skillet. Grease with margarine. Crack egg into the ring. Pierce the yolk gently with a fork (this is

a fast food dupe — flat egg, not a fried egg). Season with salt and pepper. Cook 2 min, flip the ring with the egg in it, cook 1 more min. 3. Toast English muffin halves in the toaster or face-down in a dry skillet. 4. Build: bottom muffin → sausage patty → cheese slice (place on hot sausage so it melts) → egg → top muffin. 5. Wrap in foil or parchment for 1 min. The steam inside melds everything together. This is the step that separates homemade from sad.

Per sandwich: ~380mg Na | ~200mg P | ~170mg K

Compared to McDonald's Egg McMuffin: ~820mg Na, ~300mg P. You just cut sodium in half.

9. Crispy Skillet Hash Browns

"The term 'hash brown' first appeared in print in 1888 in a menu from Dayton, Ohio, described as 'hashed brown potatoes' — referring to the technique of hashing (chopping) and browning."

Ingredient	Amount	CKD Notes
Russet potatoes	2 large (~1 lb)	MUST BE K-LEACHED — see step 1
Avocado oil	3 tbsp	For crispy pan-fry
Salt	1/4 tsp	After leaching, these need some seasoning back
Black pepper	1/4 tsp	
Onion powder	1/4 tsp	
Garlic powder	1/4 tsp	Optional

Method: 1. **K-LEACH (NON-NEGOTIABLE):** Peel and shred potatoes using a box grater or food processor shredding disc. Place shredded potatoes in a large bowl of cold water. Soak for **minimum 4 hours** — overnight is better. Change the water at least once halfway through. This removes 40-50% of the potassium. Drain, rinse under cold water, then squeeze dry in a clean kitchen towel or cheesecloth. Squeeze **HARD** — the drier they are, the crispier they get. Moisture is the enemy of crunch. 2. Toss the squeezed-dry shreds with salt, pepper, onion powder, garlic powder. 3. Heat avocado oil in a large cast iron or non-stick skillet over medium-high heat. The oil should shimmer. 4. Spread the potatoes in an even thin layer. Press down with a spatula. **DO NOT TOUCH THEM** for 5-6 min. Let the bottom crust form undisturbed. 5. When the bottom is deep golden (lift a corner to check), flip in sections. Press down again. Cook another 4-5 min. 6. The edges should be lacy and shatteringly crispy. The center should be tender. If your hash browns are pale and floppy, your oil wasn't hot enough or your potatoes were too wet.

Per serving (1/2 the batch): ~120mg Na | ~55mg P | ~250mg K (*without leaching this would be ~450mg K — leaching is critical*)

10. Garden Fold Omelette Bar

"The word 'omelette' derives from the French 'alumelle,' meaning thin blade — a reference to the flat shape of the egg dish, which first appeared in French cookbooks in the 1600s."

Base (same for all three):

Ingredient	Amount	CKD Notes
Eggs	3 large	~285mg P, ~210mg Na for 3 eggs
Unsweetened almond milk	1 tbsp	Makes eggs silkier
Salt	Pinch	
Black pepper	To taste	
Margarine	1 tbsp (for pan)	

Method (The French Technique — works for all three): 1. Whisk eggs, almond milk, salt, pepper until uniform — no streaks of white. 2. Heat an 8-10" non-stick skillet over medium heat. Add margarine, swirl to coat. 3. Pour in egg mixture. Let it set for 30 seconds without touching. 4. Using a silicone spatula, gently push the edges toward the center while tilting the pan to let raw egg flow to the edges. Repeat around the pan. 5. When the top is **STILL** slightly wet (it finishes cooking from residual heat), add filling to one half. 6. Fold the other half over. Slide onto plate. The inside should be barely set, not dry.

Filling Combo 1: The Western

Ingredient	Amount	CKD Notes
Diced ham (low-sodium)	2 tbsp	Use low-sodium deli ham — regular ham is a Na bomb
Bell pepper, diced	2 tbsp	
White onion, diced	2 tbsp	
Sharp cheddar, shredded	2 tbsp	

Saute ham, pepper, onion in a separate pan first (2 min). Add to omelette.

Per omelette: ~310mg Na | ~290mg P | ~250mg K

Filling Combo 2: The Mediterranean

Ingredient	Amount	CKD Notes
Fresh spinach	Small handful (1/4 cup)	Wilted — raw spinach is HIGH K, but a small wilted amount is acceptable
Sun-dried tomatoes (not in oil)	2 pieces, chopped	Small portion — concentrated K, keep it minimal
Feta cheese, crumbled	1 tbsp	Salty — 1 tbsp keeps Na in check
Fresh basil	3-4 leaves, torn	

Wilt spinach in the pan before adding eggs, or add raw and let the omelette's heat wilt it.

Per omelette: ~290mg Na | ~280mg P | ~270mg K

Filling Combo 3: The Garden

Ingredient	Amount	CKD Notes
White mushrooms, sliced	3-4 mushrooms	Lower K than portobello
Zucchini, diced small	2 tbsp	Low K vegetable
Red bell pepper, diced	2 tbsp	
Fresh herbs (chives, parsley)	1 tbsp, chopped	
Swiss cheese	1 slice, torn	

Saute veggies 3-4 min until tender before adding to omelette.

Per omelette: ~270mg Na | ~275mg P | ~280mg K

11. Candlemas Crepes

"The tradition of eating crepes on Candlemas (February 2) in France dates back to the 5th century, when Pope Gelasius I distributed crepes to pilgrims arriving in Rome — the round golden shape symbolized the sun and the approaching spring."

Crepe Batter:

Ingredient	Amount	CKD Notes
All-purpose flour	1 cup	White flour
Eggs	2 large	
Unsweetened almond milk	1.25 cups	
Water	1/4 cup	Thins the batter — crepes need to be thin
Avocado oil	2 tbsp	In the batter for non-stick and tender crepes
Sugar	1 tbsp	For sweet crepes. Omit for savory.
Salt	Pinch	
Vanilla extract	1 tsp	For sweet crepes. Omit for savory.

Method: 1. Blend ALL batter ingredients in a blender for 15 seconds. Scrape sides, blend 5 more seconds. (You can whisk by hand, but blender = perfectly smooth batter.) 2. Refrigerate batter 30 min minimum (1 hour ideal). This lets the gluten relax — no rubbery crepes. 3. Heat an 8-10" non-stick skillet over medium heat. Lightly oil with a paper towel dipped in avocado oil. 4. Pour ~3 tbsp batter into the center, then IMMEDIATELY tilt and swirl the pan in a circle to spread the batter into a thin even layer. Speed matters — you have about 3 seconds before it sets. 5. Cook 1-2 min until the edges lift and the bottom is light golden with brown lace spots. 6.

Flip with a thin spatula (or your fingers if you're brave). Cook 30 seconds more. 7. Stack finished crepes with parchment between them to prevent sticking. They'll stay pliable.

Makes: 10-12 crepes

Sweet Fillings:

Strawberry Cream Cheese:

Filling	Amount
Cream cheese, softened	2 oz
Powdered sugar	2 tbsp
Fresh strawberries, sliced	1/2 cup
Maple syrup drizzle	1 tbsp

Spread cream cheese mixture on crepe, layer strawberries, fold into quarters, drizzle syrup.

Cinnamon Sugar: Margarine + cinnamon + sugar. The Parisian street-cart classic.

Savory Fillings:

Ham & Cheese:

Filling	Amount
Low-sodium deli ham	2 slices
Swiss cheese	1 slice
Dijon mustard	1/2 tsp

Layer ham and cheese on crepe, fold into quarters, heat in pan until cheese melts.

Egg & Veggie: Scrambled egg + sauteed bell pepper + onion + herbs. Breakfast crepe.

Per crepe (unfilled): ~55mg Na | ~45mg P | ~40mg K **Per crepe (strawberry cream cheese filling):** ~90mg Na | ~65mg P | ~80mg K

12. Maple Crunch Granola

"Granola was invented in 1863 by Dr. James Caleb Jackson at a health spa in Dansville, New York — he called it 'granula' and it was so hard it had to be soaked overnight before eating."

Ingredient	Amount	CKD Notes
Old-fashioned rolled oats	3 cups	Oats are moderate P (~150mg/cup) — portion control matters, but homemade avoids phosphate additives
Rice cereal (puffed rice or Rice Krispies)	1 cup	Bulks up the granola with a low-P filler — plus the crunch is excellent
Avocado oil	1/4 cup	
Honey	1/3 cup	CKD-aware sweetener
Brown sugar	2 tbsp	
Vanilla extract	1 tsp	
Ground cinnamon	1 tsp	
Salt	1/4 tsp	
Unsweetened coconut flakes	1/3 cup	Lower P than almonds/walnuts
Dried cranberries	1/3 cup	Lower K than raisins or dried apricots

What's NOT in here (and why):

Common Granola Ingredient	Why It's Excluded
Almonds/walnuts/cashews	High P (130-160mg per oz) — if you MUST, limit to 1 tbsp sliced almonds
Dried banana chips	High K
Raisins	High K (~300mg per 1/4 cup)
Chocolate chips	Often contain phosphate additives — check label if adding
Store-bought granola	Almost always contains sodium phosphate, calcium phosphate, or other P additives in the ingredient list

Method: 1. Preheat oven to 300F. Line a large sheet pan with parchment. 2. Mix oats and rice cereal in a large bowl. 3. In a small saucepan, warm oil + honey + brown sugar over low heat until sugar dissolves. Stir in vanilla, cinnamon, salt. 4. Pour wet mixture over dry. Toss until every flake is coated. 5. Spread in an even layer on the sheet pan. Press down firmly with a spatula (this creates clusters). 6. Bake 40-45 min, stirring ONCE at the 20-minute mark. The granola should be golden and fragrant. It will still feel soft — it crisps as it cools. 7. Let cool COMPLETELY on the pan without touching it. Seriously. This is how clusters form. 8. Once cool, break into chunks. Fold in coconut flakes and dried cranberries. 9. Store in an airtight container. Lasts 2 weeks at room temperature.

Per 1/2 cup serving: ~60mg Na | ~95mg P | ~110mg K

13. Morning Jar Oats

"Bircher muesli, the direct ancestor of overnight oats, was created around 1900 by Swiss physician Maximilian Bircher-Benner for patients at his sanatorium in Zurich."

Base Recipe:

Ingredient	Amount	CKD Notes
Old-fashioned rolled oats	1/2 cup	~150mg P per cup, so 1/2 cup = ~75mg P
Unsweetened almond milk	1/2 cup	
Plain yogurt (non-Greek)	2 tbsp	Regular yogurt, NOT Greek — Greek yogurt has nearly double the P
Maple syrup or honey	1 tbsp	
Vanilla extract	1/2 tsp	
Chia seeds	1 tsp	Small amount — chia is moderate P, 1 tsp is fine

Method: 1. Combine all ingredients in a jar or container with a lid. 2. Stir well (or shake with lid on). 3. Refrigerate overnight (minimum 6 hours). 4. In the morning, stir. Add a splash more almond milk if too thick. 5. Eat cold or microwave 1-2 min if you prefer warm.

Flavor variations (mix in before refrigerating or top in the morning):

Variation	Add-ins	Extra CKD Notes
Blueberry Vanilla	1/4 cup blueberries + extra vanilla	Blueberries are one of the lowest-K fruits
Cinnamon Apple	1/4 cup diced apple + 1/2 tsp cinnamon + 1 tsp brown sugar	Apples are renal-aware, ~150mg K per medium apple
Strawberry	1/4 cup sliced strawberries + 1 tsp honey drizzle	
PB&J	1 tbsp peanut butter + 1 tbsp strawberry jam	PB adds ~60mg P per tbsp — occasional treat, not daily
Cranberry Coconut	2 tbsp dried cranberries + 1 tbsp coconut flakes	Low K combo

Per serving (base recipe): ~50mg Na | ~100mg P | ~140mg K

14. North African Skillet Eggs

"Shakshuka likely originated in Ottoman North Africa — the word comes from the Berber or Arabic for 'a mixture,' and the dish traveled across the Middle East before becoming Israel's national breakfast by the mid-20th century."

Ingredient	Amount	CKD Notes
Eggs	4 large	Poached directly in the sauce
Canned diced tomatoes (no salt added)	1 can (14.5 oz)	NO SALT ADDED is critical — regular canned tomatoes = 400mg+ Na per can. Tomatoes are moderate K, but the portion per serving is controlled.
Red bell pepper	1 medium, diced	
White onion	1 medium, diced	
Garlic	3 cloves, minced	
Avocado oil	2 tbsp	

Ingredient	Amount	CKD Notes
Cumin	1 tsp	
Paprika	1 tsp	
Smoked paprika	1/2 tsp	The smoky depth that separates OK shakshuka from great shakshuka
Cayenne	1/4 tsp (or to taste)	
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Sugar	1/2 tsp	Cuts tomato acidity
Fresh cilantro or parsley	For garnish	
Crusty white bread	For serving	For dipping — sourdough or white baguette
Feta cheese	1 tbsp crumbled (optional)	Small portion — adds salt and tang

Method: 1. Heat oil in a deep 10-12" skillet (cast iron ideal) over medium heat. 2. Saute onion and bell pepper until soft (5-6 min). 3. Add garlic, cook 30 seconds until fragrant. Add cumin, paprika, smoked paprika, cayenne, black pepper. Stir and toast spices 1 min — they should bloom in the oil and smell incredible. 4. Add diced tomatoes (with liquid), salt, sugar. Stir and bring to a simmer. Cook 10-12 min, stirring occasionally, until the sauce thickens and reduces slightly. You want a thick sauce, not soup. 5. Taste and adjust seasoning. The sauce should be bold and slightly smoky. 6. Make 4 wells in the sauce with the back of a spoon. Crack one egg into each well. 7. Cover the skillet with a lid. Reduce heat to medium-low. Cook 5-8 min: 5 min for runny yolks, 8 min for set yolks. Check frequently. 8. Remove from heat. Garnish with cilantro/parsley and optional feta crumbles. 9. Serve in the skillet with crusty white bread for scooping. Shakshuka is communal — everyone eats from the pan.

Per serving (2 eggs + 1/2 the sauce): ~250mg Na | ~200mg P | ~380mg K

The K is on the higher side from the tomatoes. If you're on a strict K limit, use 3/4 of the tomato can instead of the full can, and add 2 tbsp water to compensate.

15. Stale-Tortilla Magic Chilaquiles

"Chilaquiles comes from the Nahuatl word 'chilaquiltil,' meaning 'chilis and greens' — the dish was originally an ancient Aztec method for using stale tortillas, predating European contact in the Americas."

Ingredient	Amount	CKD Notes
Corn tortillas	6-8 (stale or oven-dried)	Cut into triangles and fried/baked — corn tortillas are lower Na than flour
Avocado oil	3 tbsp (for frying chips)	Or bake for a lighter version
Salsa verde (homemade or low-sodium store)	1 cup	Tomatillo-based — LOWER K than red tomato salsa. Check Na on store brands (<300mg per 1/2 cup)
Eggs	2-4, fried or scrambled	Served on top
Queso fresco or feta	2 tbsp, crumbled	Lower Na per oz than cheddar
Sour cream or Mexican crema	2 tbsp	
White onion	2 tbsp, thinly sliced	
Fresh cilantro	For garnish	
Lime	1 wedge per serving	
Avocado	2-3 slices per serving (optional)	Moderate K (~150mg per 1/4 avocado) — small portion OK

Quick CKD Salsa Verde (if making from scratch):

Ingredient	Amount
Tomatillos	6 medium, husked and rinsed
Jalapeno	1, halved and seeded
Garlic	2 cloves
White onion	1/4, chunked
Cilantro	Small bunch
Lime juice	1 tbsp
Salt	1/4 tsp
Water	2-3 tbsp (to thin)

Broil tomatillos, jalapeno, garlic, onion on a sheet pan 5-6 min until charred. Blend with cilantro, lime, salt, water. This salsa is MUCH lower Na than anything in a jar.

Method: 1. Make chips: Cut corn tortillas into triangles (6 per tortilla). Either fry in oil until golden and crispy (2-3 min per batch) OR brush with oil and bake at 375F for 12-15 min. 2. Warm salsa verde in a large skillet over medium heat. 3. Add the chips to the salsa. Toss gently to coat. Cook 2-3 min — you want the chips to soften slightly on the outside but stay crispy in the center. This is the soul of chilaquiles: the tension between crisp and sauced. 4. While chips cook in salsa, fry eggs in a separate skillet (sunny side up or over easy — runny yolk is mandatory). 5. Plate the sauced chips. Top with fried eggs. Garnish with crumbled cheese, sour cream, sliced onion, cilantro, lime wedge, avocado slices. 6. Eat immediately. Chilaquiles wait for no one — they go from perfect to mush in 5 minutes.

Per serving (half the batch + 2 eggs): ~290mg Na | ~210mg P | ~320mg K

CKD Breakfast Safety Reference

CKD Principle	Application in This Chapter
Phosphorus-free baking powder	Used in ALL baked items: waffles, pancakes, biscuits, cinnamon rolls. Rumford aluminum-free is the standard. Regular baking powder contains sodium aluminum phosphate — a phosphorus additive with near-100% absorption rate.
White flour > whole wheat	Every recipe specifies all-purpose or brioche. Whole wheat has ~120mg more P per cup and higher K. Refined flour is one of the rare cases where the "less healthy" option is the correct CKD choice.
K-leaching potatoes	Hash browns require 4+ hour cold water soak. Non-negotiable. Removes 40-50% of potassium. Without leaching, a serving of hash browns can hit 500mg+ K.
Turkey sausage > pork sausage	Lower P and Na per ounce. Always check the label for "sodium phosphate" in the ingredients — some turkey sausage brands add it back. If it says phosphate anywhere, put it back on the shelf.

CHAPTER 2

American Soul Food

"Comfort has no restrictions."

Country Fried Steak, Fried Chicken, Mac & Cheese, Gumbo, BBQ, Breakfast

"Comfort has no restrictions."

"One cannot think well, love well, sleep well, if one has not dined well."
— Virginia Woolf

FAMILY FAVORITES

Bob, Susan, and Stephanie recipes gathered into the Comfort chapter.

BOB'S RECIPES

Bob's Split Pea Soup

"No one who cooks, cooks alone." — Laurie Colwin

CKD Principle	Application in This Chapter
Almond milk replaces dairy	~5mg P per cup vs ~230mg P per cup (whole milk). Used in every recipe that would traditionally call for milk. Unsweetened only — flavored varieties add unnecessary sugar and sometimes phosphate.
Eggs: 2-3 per meal	Eggs are CKD's best breakfast protein. High biological value, ~95mg P per egg, ~70mg Na per egg. Two eggs is one example serving. A larger serving changes phosphorus and protein totals, so compare it with your individualized targets and the rest of the day.
Maple syrup: check the portion and glucose plan	~40mg K, ~3mg P, ~2mg Na per tablespoon. Basically sugar and water with trace minerals. Use it.
Cream cheese for frosting	28mg P per oz (cream cheese) vs 40mg P per oz (butter). Also lower Na. And it tastes better on cinnamon rolls anyway.

Quick Renal Number Reference — All 15 Recipes

#	Recipe	Na (mg)	P (mg)	K (mg)
1	French Toast (2 slices + syrup)	120	105	130
2	Waffles (1 waffle)	110	95	90
3	Pancakes (3 + compote)	140	90	150
4	Eggs Benedict (2 halves)	280	230	200
5	Breakfast Burrito (1)	350	220	280
6	Biscuits & Gravy (2 + gravy)	320	170	190
7	Cinnamon Roll (1)	190	80	85
8	Breakfast Sandwich (1)	380	200	170
9	Hash Browns (1/2 batch, leached)	120	55	250
10a	Omelette — Western	310	290	250
10b	Omelette — Mediterranean	290	280	270
10c	Omelette — Garden	270	275	280
11	Crepe (strawberry cream cheese)	90	65	80
12	Granola (1/2 cup)	60	95	110
13	Overnight Oats (base)	50	100	140
14	Shakshuka (2 eggs + sauce)	250	200	380
15	Chilaquiles (half batch + 2 eggs)	290	210	320

General CKD daily targets for reference: Na <2,000mg | P <800-1,000mg | K varies by labs (typically 2,000-3,000mg). These are breakfast numbers — plan your other meals accordingly.

Ingredient	Amount	CKD Notes
Green split peas	1 cup, rinsed	Moderate K — rinse well, and the serving size controls it
Low-sodium chicken broth	4 cups	NOT regular broth
Turkey bacon	4 strips, chopped	Bob's version would have ham. Turkey bacon keeps the smoke.
Onion	1 medium, diced	
Carrots	2, diced	Moderate K — small amount across 6 servings
Celery	2 stalks, diced	
Garlic	3 cloves, minced	
Dried thyme	1 tsp	
Bay leaf	1	
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Apple cider vinegar	1 tsp (finish)	Brightens everything at the end

Method: 1. Cook turkey bacon in Dutch oven until crispy. Remove, leave drippings. 2. Sauté onion + carrots + celery in drippings 5 min. 3. Add garlic 1 min. 4. Add split peas + broth + thyme + bay leaf + pepper. 5. Bring to boil. Reduce to low simmer. Cover. 6. Cook 45-60 min, stirring occasionally, until peas dissolve into thick soup. 7. Remove bay leaf. Stir in vinegar + salt. 8. Serve topped with crumbled turkey bacon. 9. Make enough for six. Give five bowls away.

Per serving (6 servings): ~180mg Na | ~130mg P | ~350mg K

Bob's Box - "The Costco Run & Wine Pairing"

Costco runs. Good wine. Enough for everyone.

Compartment	What Goes In	Why
1 — Cheese	Aged cheddar, Brie wedge, or Kirkland Manchego — wine demands real cheese	Bob knows wine. Wine demands a cheese board. Costco's cheese section is elite.
2 — Crackers	Costco water crackers or fig & olive crisps	The cracker that doesn't compete with the wine or the cheese.
3 — Nuts	Marcona almonds or Costco roasted cashews (1/4 cup)	The wine pairing nut. Marcona > regular almonds. Bob knows.
4 — Fruit	Red grapes + dried apricots + fresh figs (when in season)	Classic wine accompaniments. Grapes with wine is redundant and perfect.
5 — Meat	Costco Columbus salami or prosciutto (thin-sliced, rolled)	The charcuterie lane. Wine + cured meat = civilization.
6 — Sweet	Dark chocolate (70%+) squares + Kirkland chocolate covered almonds	The wine closer. Dark chocolate + a good red = the end of the evening.

Wine note: Bob pairs. Red wine is lower-K than white (Andrew's CKD note if sharing). A 4oz glass is manageable. Bob would know the vintage, the region, and the story behind the bottle before the cork is pulled.

Bob's Theorem applied: This box costs ~\$5 to fill from Costco bulk — but it pairs with a \$40 bottle he already has. Bob would fill 10 boxes and hand them out with a wine recommendation card tucked in. That's who he is.

Box style: No-frills clear Gladware. Function over form. Bob doesn't need a sticker. Bob IS the sticker.

BOB'S SPECIALS

"Generosity is the only rational economic strategy." — Bob's Theorem

Bob's Ultra-Fresh Sweet & Spicy Ceviche

"Raw fish cured in citrus. No heat required. The acid does the cooking. The chili does the rest."

Base (works with any seafood):

Ingredient	Amount	CKD Notes
Fresh seafood (pick one or mix):		
Wild shrimp, peeled + deveined	1 lb	Andrew's #1 — lowest P seafood
Wild salmon, sushi-grade, 1/2" dice	1 lb	Wild only — never farmed
Wild cod or tilapia, 1/2" dice	1 lb	Mild, clean, takes the marinade well
Scallops, quartered	1 lb	Sweet + luxurious

The Cure (sweet + spicy + ultra fresh):

Ingredient	Amount	CKD Notes
Lime juice	1 cup (8-10 limes)	The house fix — best fresh squeezed, not bottled
Lemon juice	1/4 cup	Brightens the lime
Fresh mango, small dice	1/2 cup	Sweet counterpoint
Fresh pineapple, small dice	1/4 cup	More sweet acid
Red onion, finely diced	1/4 cup	Rinse in cold water first to tame raw bite
Jalapeño	1, seeded, minced	Seeds out = sweet heat, seeds in = war

Ingredient	Amount	CKD Notes
Serrano	1, minced	The real spice — add half, taste, then decide
Habanero	1/4, minced (optional — for the brave)	Bob would want this in there
Fresh cilantro	1/2 cup, rough chopped	
Cucumber, peeled, seeded, diced	1/2 cup	Cool + crunch
Avocado	1, diced (add at the end)	Moderate K — worth it for texture
Olive oil	1 tbsp	Silky finish
Honey	1 tbsp	The sweet in sweet & spicy
Salt	1/4 tsp	

Method: 1. Cut seafood into 1/2" pieces. Place in glass bowl (never metal — acid reacts). 2. Pour lime + lemon juice over seafood. Every piece must be submerged. Cover. 3. Refrigerate **minimum 30 min** (shrimp) to **2 hours** (firm fish). The acid "cooks" the protein — flesh turns opaque. 4. Drain about half the citrus juice (too much = rubbery). 5. Fold in mango, pineapple, cucumber, onion, jalapeño, serrano, cilantro, honey, olive oil. 6. Add habanero if Bob's ghost is watching. 7. Add avocado last — fold gently so it doesn't mush. 8. Serve immediately in small bowls or on tostadas. Or eat it with a spoon standing at the counter.

Per serving (4 servings): ~130mg Na | ~180mg P | ~350mg K

Makes 4 servings.

Bob's Drumstick Ice Cream Cone Dupe

"The ice cream truck isn't coming. But this is better anyway."

Chocolate-Dipped Waffle Cone:

Ingredient	Amount	CKD Notes
Sugar cones (store-bought)	6	Low P, low Na — read label for phosphate additives
Semi-sweet chocolate chips	1/2 cup	
Coconut oil	1 tsp	Thins the chocolate for dipping
Chopped peanuts	3 tbsp	Moderate P — small amount for the crunch coating

CKD Vanilla Ice Cream Filling:

Ingredient	Amount	CKD Notes
Heavy cream	1 cup	Moderate P — portion controlled across 6 cones
Unsweetened almond milk	1 cup	Cuts the dairy P in half
Sugar	2/3 cup	
Vanilla extract	2 tsp	Real vanilla
Egg yolks	3	
Pinch of salt		

The Magic Shell Top:

Ingredient	Amount
Semi-sweet chocolate chips	1/2 cup
Coconut oil	2 tbsp

Method: 1. **Ice cream:** Heat cream + almond milk + sugar until steaming. Temper into yolks. Return to heat, stir to 170F. Strain, add vanilla. Chill overnight. Churn in ice cream maker. 2. **Prep cones:** Melt 1/2 cup chocolate + 1 tsp coconut oil. Dip inside rim of each cone in chocolate (seals the bottom so it doesn't leak). Roll outer rim in chopped peanuts. Set upside down on parchment. Freeze 10 min. 3. **Fill:** Scoop ice cream into prepared cones. Pack it tight. Round the top into a dome. Freeze 1 hour until firm. 4. **Magic shell:** Melt 1/2 cup chocolate + 2 tbsp coconut oil. Let cool 2 min (still liquid but not screaming hot). 5. **Dip:** Dip the ice cream dome into magic shell. It hardens on contact (the coconut oil is the trick). Immediately press a few more chopped peanuts into the wet chocolate. 6. Eat immediately or wrap individually in plastic and freeze.

Per cone: ~110mg Na | ~95mg P | ~140mg K — That's a real Drumstick at half the phosphorus.
Makes 6 cones. \$0.80 each.

Bob's Mushroom Theorem

A dish that proves, mathematically, that generosity is the only rational economic strategy.

Ingredient	Amount
Mushrooms	As many as you can carry
Rice	1 cup
Margarine	Some
Garlic	4 cloves
Thyme	Fresh
The knowledge that you're making this for someone else	Required

Method: Cook the mushrooms. Cook the rice. Combine them. Put them on two plates. Give one away. The plate you gave away tastes better than the plate you kept. This is not a metaphor. This is thermodynamics.
Proof: Let G = generosity, R = return. For all $G > 0$, R approaches ∞ as the number of plates given away increases. The limit does not exist because there is no upper bound on what comes back when you feed someone without asking for anything. ■

SUSAN'S KITCHEN

"After a good dinner, one can forgive anybody, even one's own relatives."
— Oscar Wilde

Susan's Cornbread with Hot Dogs

It's exactly what it sounds like. It's perfect.
The Original (Susan's Way): Cut hot dogs into coins. Pour cornbread batter over them in a cast iron skillet. Bake until golden. Eat with mustard. Don't explain it to anyone. Don't apologize.

CKD Version:

Ingredient	Amount	CKD Notes
Turkey hot dogs	4, sliced into coins	Lower Na than beef franks
Cornbread batter (GF cornbread recipe, Ch 11)	1 batch	Rumford BP, almond milk, margarine
Yellow mustard	For serving	Lowest Na condiment

Method: Grease cast iron skillet. Scatter hot dog coins across the bottom. Pour cornbread batter over them. Bake 400F 20-25 min until golden. Flip onto cutting board. The hot dogs are embedded in the bottom like fossils. Cut into wedges. Dip in mustard. Remember.

Susan's Cream of Mushroom Everything

She put cream of mushroom soup in things that had no business containing cream of mushroom soup. It worked every time.
The Original: 1 can Campbell's Cream of Mushroom + literally any protein + rice or noodles = dinner.

CKD Version — The Susan Casserole Formula:

Ingredient	Amount	CKD Notes
Homemade cream of mushroom soup (recipe #2 above)	2 cups	Not the can. The can has 870mg Na per cup.
Chicken thighs, cooked and shredded	1 lb	
Jasmine rice, cooked	2 cups	
Frozen green beans	1 cup	Frozen is fine here — Susan would insist
Sharp cheddar, shredded	1/2 cup (on top)	
Black pepper	1/4 tsp	

Method: Mix mushroom soup + chicken + rice + green beans in a baking dish. Top with cheddar. Bake 375F 20 min until bubbly. This is Susan's dinner. The mushroom soup is the glue. The recipe is the love.
Per serving (4 servings): ~190mg Na | ~180mg P | ~360mg K (vs canned version: ~900mg Na)

Susan's Sprite with Splenda

A beverage that exists because Susan said it did.
The Original: Sprite. Splenda. Stir. Serve. That's it. That's the recipe.

CKD Note: Sprite is actually one of the more renal-aware sodas — no phosphoric acid (that's Coke/Pepsi), moderate K. Adding Splenda cuts sugar if you're watching glucose. Susan was accidentally ahead of her time.
The CKD upgrade (if you want one): Sparkling water + fresh lime juice + Splenda or stevia. Same vibe. Zero sodium. Susan would say the original was fine and she'd be right.

Susan's Tuna Noodle Casserole

The Midwest's answer to everything.
The Original: Egg noodles + canned tuna + cream of mushroom soup + frozen peas + crushed potato chips on top.
CKD Version:

Ingredient	Amount	CKD Notes
Egg noodles (or GF rice noodles)	8 oz, cooked	
Canned tuna in water	1 can (5oz), drained and rinsed	Andrew doesn't like canned fish — sub shredded chicken for him. Lauren gets the tuna.
Homemade cream of mushroom soup (recipe #2)	1.5 cups	
Frozen peas	1/2 cup	Moderate K — small amount is fine
Sharp cheddar	1/4 cup, shredded	
Panko breadcrumbs	1/3 cup	Replaces potato chips — still crunchy
Margarine	1 tbsp (melted, tossed with panko)	
Black pepper	1/4 tsp	

Method: Mix noodles + tuna (or chicken) + mushroom soup + peas in baking dish. Top with cheddar + buttered panko. Bake 375F 20 min. The panko gets golden. The cheese bubbles. Susan nods.

Susan's Hamburger Helper (From Scratch)

Because Susan's version came from a box and it was still good and that's okay.

Ingredient	Amount	CKD Notes
Ground turkey or chicken	1 lb	No pork. Lower P than beef.
Elbow pasta	8 oz, cooked	
No-salt-added tomato paste	2 tbsp	
Low-sodium beef broth	1 cup	
Unsweetened almond milk	1/2 cup	
Sharp cheddar	1/2 cup, shredded (stir in at end)	
Onion powder	1 tsp	
Garlic powder	1 tsp	
Paprika	1/2 tsp	
Costco no-salt seasoning	1 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp	

Method: Brown meat, drain fat. Add tomato paste, stir 1 min. Add broth + almond milk + all seasonings. Simmer 5 min. Add cooked pasta + cheddar. Stir until melted and creamy. Serve in bowls. Tell no one it's homemade. Let them think it came from a box. Susan would.

Per serving (4 servings): ~200mg Na | ~190mg P | ~320mg K

Susan's Jell-O "Salad"

It's not a salad. It never was. Midwest calls it a salad anyway.

The Original: Lime Jell-O + cottage cheese + crushed pineapple + Cool Whip. Called "salad" at potlucks. Brought in a Pyrex dish with tin foil. Nobody questions it.

CKD Version:

Ingredient	Amount	CKD Notes
Lime Jell-O	1 small box	Low K/P/Na — one of the safest CKD desserts
Boiling water	1 cup	
Cold water	1/2 cup	
Crushed pineapple, drained	1/2 cup (canned, drained well)	Draining removes K
Whipped topping (Cool Whip or coconut whipped cream)	1 cup	

Method: Dissolve Jell-O in boiling water. Add cold water. Chill until partially set (~45 min). Fold in drained pineapple + whipped topping. Refrigerate until firm. Cut into squares. Call it salad. Bring it in a Pyrex dish. Cover with tin foil. Susan's spirit is in the tin foil.

Susan's Meatloaf

Every Midwest mom's meatloaf is different. Every Midwest mom's meatloaf is the same.

Ingredient	Amount	CKD Notes
Ground turkey	1.5 lbs	Susan used beef. We use turkey. She'd understand.
Breadcrumbs	1/2 cup	
Egg	1	
Onion, finely diced	1/2 cup	
Garlic powder	1 tsp	
Costco no-salt seasoning	1 tsp	
Worcestershire sauce	1 tbsp	Small amount = controlled Na
Black pepper	1/2 tsp	
Salt	1/4 tsp	

Glaze:

Ingredient	Amount
No-salt-added tomato paste	2 tbsp
Brown sugar	1 tbsp
Yellow mustard	1 tsp
Apple cider vinegar	1 tsp

Method: Mix everything (except glaze) with your hands. Don't overmix — tough meatloaf is a crime. Shape into loaf on sheet pan or in loaf pan. Mix glaze, brush on top. Bake 375F 45-50 min until 165F internal. Rest 10 min before slicing. Serve with mashed potatoes (K-leached, recipe in Ch 1). The glaze caramelizes and gets sticky and sweet and Susan would say "it's ready" without checking the temperature because she always knew.

Per serving (6 servings): ~190mg Na | ~170mg P | ~280mg K

Susan's Totino's Pizza Rolls (From Scratch)

Invented 1968 by Jeno Paulucci in Duluth, Minnesota. 290+ billion sold since. Wonton wrapper version.

Filling:

Ingredient	Amount	CKD Notes
Turkey pepperoni, diced small	1/4 cup	Lower Na than regular pepperoni

Ingredient	Amount	CKD Notes
No-salt-added tomato paste	2 tbsp	
Water	2 tbsp	
Italian seasoning	1/2 tsp	
Garlic powder	1/4 tsp	
Mozzarella, shredded	1/2 cup	Fresh mozz diced small works too
Black pepper	Pinch	

Wrappers:

Option	Notes
Wonton wrappers	Easiest — buy at Fred Meyer/Asian aisle
Egg roll wrappers, cut into 3" squares	Thicker, crunchier
Homemade: flour + water + salt dough, rolled thin	Control everything

Method: 1. Mix tomato paste + water + Italian seasoning + garlic powder into a thick sauce. 2. Lay wonton wrappers flat. Drop 1 tsp sauce + a few pepperoni pieces + pinch of mozzarella in center. 3. Wet edges with finger dipped in water. Fold corner to corner (triangle) or bring all corners to center (bundle). Press to seal. 4. **Bake:** 400F on parchment, brush with olive oil, 12-15 min until golden and blistered. 5. **Or fry:** 350F oil, 2 min per side until golden and puffy. 6. Let cool 2 min. The inside is molten. Susan knew this and ate them immediately anyway.

Makes ~30 rolls. Per 5 rolls: ~140mg Na | ~65mg P | ~80mg K (vs Totino's: ~400mg Na per 6 rolls)

Lauren's Ham & Cheese Hot Pockets / Toaster Strudels

"The after-school food that Lauren elevated to a lifestyle."

Dough (makes 8 pockets):

Ingredient	Amount
All-purpose flour	2 cups
Cold margarine, cubed	1/2 cup
Cream cheese	2 oz, cold
Cold water	3-4 tbsp
Salt	1/2 tsp

Ham & Cheese Filling:

Ingredient	Amount	CKD Notes
Deli turkey (sub for ham — or use real ham for Lauren, she's not CKD)	4 oz, chopped	Turkey for Andrew, ham for Lauren
Havarti, shredded	1/2 cup	Lauren's cheese
Mozzarella, shredded	1/4 cup	
Dijon mustard	1 tsp	Inside the pocket
Black pepper	Pinch	

Method: 1. **Dough:** Pulse flour + salt + margarine + cream cheese in food processor until pea-sized. Add cold water until dough holds. Wrap, chill 30 min. 2. Roll dough thin. Cut into rectangles (~4x6"). 3. Spoon filling onto one half. Fold over. Crimp edges with fork. Poke top twice with fork (steam vents). 4. **Egg wash:** Brush with beaten egg. 5. **Bake:** 400F on parchment, 18-22 min until deep golden and puffed.

Toaster Strudel version: Same recipe but thinner dough, pipe cream cheese icing on top after baking (2 oz cream cheese + 1/4 cup powdered sugar + splash of almond milk, whisked). The icing is the move.

Per pocket: ~180mg Na | ~95mg P | ~100mg K

Girl Scout Samoa Cookie Dupe

"Caramel. Coconut. Chocolate. The cookie that funds an empire."

Ingredient	Amount	CKD Notes
Shortbread base:		
All-purpose flour	1.5 cups	
Margarine, softened	1/2 cup	

Ingredient	Amount	CKD Notes
Powdered sugar	1/3 cup	
Vanilla	1/2 tsp	
Salt	1/4 tsp	
Caramel-coconut topping:		
Caramels (unwrapped)	25 pieces (~8oz)	Or homemade: sugar + margarine + cream
Unsweetened almond milk	2 tbsp	Thin the caramel
Shredded coconut, toasted	2 cups	Toast in dry skillet until golden
Salt	Pinch	
Chocolate drizzle:		
Semi-sweet chocolate chips	1/2 cup	
Coconut oil	1 tsp	

Method: 1. **Shortbread:** Cream margarine + sugar + vanilla. Mix in flour + salt. Roll to 1/4" thick. Cut 2" rounds. Use a small cutter or bottle cap to punch center hole (the Samoa ring). Bake 325F 12-14 min until just barely golden. Cool completely. 2. **Caramel-coconut:** Melt caramels with almond milk in saucepan over low heat, stirring constantly. Fold in toasted coconut. Work fast — it sets up. 3. **Assemble:** Spoon caramel-coconut onto each cookie ring, pressing to coat the top and sides. Set on parchment. 4. **Chocolate:** Melt chocolate + coconut oil. Dip bottoms of cookies in chocolate. Then drizzle stripes across the top. Refrigerate 15 min to set. 5. Try not to eat them all while they're setting. You will fail. That's okay.
Per cookie (makes ~20): ~55mg Na | ~30mg P | ~45mg K

"The shared meal elevates eating from a mechanical process to a ceremony." — Wendell Berry

CKD Considerations

RENAL CHECK – Mushroom Farm & Susan's Kitchen
K (Potassium): LOW-MED (mushrooms are moderate K, but per-serving amounts are controlled)
P (Phosphorus): LOW-MED (homemade cream of mushroom = no phosphate preservatives from cans)
Na (Sodium): LOW (homemade everything, no canned soups, low-sodium broth)
NOTE: The cream of mushroom soup recipe (#2) replaces canned in ALL Susan recipes.
One batch of homemade = 140mg Na/serving. One can of Campbell's = 870mg Na/serving.
Susan's recipes live on. The sodium doesn't.

"There is no sincerer love than the love of food." — George Bernard Shaw

STEPHANIE'S CLASSICS

"Tell me what you eat, and I will tell you what you are. — Brillat-Savarin"

S.O.S., Beijing Beef, Empanadas & Stroganoff

"Tell me what you eat, and I will tell you what you are." — Brillat-Savarin

S.O.S. — Mashed Potatoes with Hamburger Gravy
Originally "Creamed Chipped Beef on Toast" — U.S. military mess halls, WWII era. Soldiers called it S.O.S. The name stuck longer than the war.
K-Leached Mashed Potatoes:

Ingredient	Amount	CKD Notes
Russet potatoes	2 lbs, peeled, 1" cubes	LEACH: soak 2-4 hours in cold water, drain, rinse, boil in fresh water, drain again
Unsweetened almond milk	1/2 cup, warmed	
Margarine	3 tbsp	

Ingredient	Amount	CKD Notes
Garlic powder	1/2 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp	

Ingredient	Amount	CKD Notes
Ground beef (90/10)	1 lb	Mom used beef. We use beef.
All-purpose flour	3 tbsp	
Low-sodium beef broth	2 cups	
Onion powder	1/2 tsp	
Garlic powder	1/2 tsp	
Worcestershire sauce	1 tsp	
Black pepper	1/2 tsp (generous)	The pepper is the whole point
Salt	1/4 tsp	

Method: 1. Brown ground beef in skillet, breaking into small crumbles. Drain most fat, keep 2 tbsp. 2. Sprinkle flour over meat + fat. Stir 1 min. 3. Slowly add broth, whisking. Add all seasonings. 4. Simmer until thick and bubbling, 4-5 min. 5. Mash the leached potatoes with almond milk + margarine + seasonings. 6. Pile potatoes on plate. Ladle hamburger gravy over the top. The gravy goes on the potatoes. The potatoes go in your mouth. Nothing else matters.
Per serving (4 servings): ~220mg Na | ~210mg P | ~350mg K

Mom's Beijing Beef (Panda Express Dupe)
Panda Express dupe. Crispy flank steak in sweet ginger-tomato sauce.

Ingredient	Amount	CKD Notes
Flank steak	1 lb, sliced thin against the grain	
Cornstarch	1/4 cup (for coating)	
Avocado oil	For frying (1" deep)	
Bell peppers (red + green)	1 each, cut into 1" pieces	
Onion	1/2, cut into wedges	
Dried red chili peppers	4-6 (for wok toss, not eating)	

Ingredient	Amount	CKD Notes
No-salt-added tomato paste	2 tbsp	
Rice vinegar	2 tbsp	
Low-sodium soy sauce	1 tbsp	Small amount — the sauce is sweet-forward
Brown sugar	3 tbsp	The sweetness IS the dish
Garlic	3 cloves, minced	
Ginger	1" piece, minced	
Sesame oil	1 tsp	
Cornstarch slurry	1 tbsp cornstarch + 2 tbsp water	

Method: 1. Slice steak thin. Toss in cornstarch until fully coated. 2. Fry in 1" of oil at 375F, 2-3 min per batch until crispy. Drain on rack. 3. Mix all sauce ingredients (except slurry) in a bowl. 4. Hot wok: stir-fry peppers + onion + dried chilies 2 min. 5. Add sauce. Bring to bubble. Add cornstarch slurry. Stir until thick and glossy. 6. Toss in crispy beef. Coat everything. Serve immediately over jasmine rice. 7. The beef stays crispy for about 4 minutes. Eat fast. Mom did.
Per serving (4 servings): ~250mg Na | ~200mg P | ~380mg K

Ingredient	Amount	CKD Notes
Ground beef (90/10)	1 lb	

Ingredient	Amount	CKD Notes
Canned green chiles (diced, mild or hot)	1 can (4oz)	Low K, low Na
Onion	1/2, finely diced	
Garlic	3 cloves, minced	
Cumin	1 tsp	
Chili powder	1/2 tsp	
Cream cheese	2 oz	Binds the filling, adds richness
Salt	1/4 tsp	
Black pepper	1/4 tsp	

Dough (or shortcut):

Ingredient	Amount
All-purpose flour	2.5 cups
Cold margarine, cubed	1/2 cup
Cold water	6-8 tbsp
Salt	1/2 tsp
Egg wash	1 egg beaten with 1 tbsp water

Shortcut: Use Goya discos (frozen empanada dough) — Mom would understand.

Method: 1. **Filling:** Brown beef + onion. Add garlic 1 min. Add chiles + spices. Stir in cream cheese until melted and combined. Cool 15 min. 2. **Dough:** Pulse flour + salt + margarine in food processor until pea-sized. Add cold water 1 tbsp at a time until dough holds together. Wrap, chill 30 min. Roll thin, cut 5" circles. (Or: use Goya discos.) 3. **Assemble:** Spoon 2 tbsp filling onto each circle. Fold in half. Crimp edges with fork. 4. **Bake:** Brush with egg wash. Bake 400F on parchment-lined sheet, 18-22 min until golden. 5. **Or fry:** 350F oil, 3 min per side until golden and blistered.

Per empanada (makes ~14): ~130mg Na | ~75mg P | ~110mg K
The green chiles + cream cheese inside the crispy shell is the move. Mom knew.

Mom's Beef Stroganoff

"Egg noodles. Sour cream. Beef. Mushrooms. Nothing else needed to happen."

Ingredient	Amount	CKD Notes
Beef sirloin or top round	1 lb, sliced into thin strips	
Egg noodles (wide)	8 oz	Or GF rice noodles
White mushrooms	8 oz, sliced	
Onion	1 medium, sliced thin	
Garlic	3 cloves, minced	
Margarine	2 tbsp	
All-purpose flour	2 tbsp	
Low-sodium beef broth	1.5 cups	
Sour cream	1/2 cup	Stir in OFF HEAT — boiling curdles it
Worcestershire sauce	1 tbsp	
Dijon mustard	1 tsp	Mom's secret — most people skip this
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Fresh parsley	For garnish	

Method: 1. Season beef strips with pepper + salt. Sear in hot skillet with 1 tbsp margarine, 2 min per side. Remove. Don't crowd — do batches. 2. Same pan: add remaining margarine. Sauté mushrooms 5 min until golden. Add onion 3 min. Add garlic 1 min. 3. Sprinkle flour over vegetables. Stir 1 min. 4. Add broth + Worcestershire + mustard. Simmer until thickened, 4-5 min. 5. Return beef to pan. Simmer 2 min. 6. **REMOVE FROM HEAT.** Stir in

sour cream. (If you add sour cream to boiling liquid it breaks. Mom learned this once and never forgot.) 7. Serve over egg noodles. Garnish with parsley. The sauce should coat everything like velvet.

Per serving (4 servings): ~240mg Na | ~220mg P | ~400mg K

"Cooking is at once child's play and adult joy. And cooking done with care is an act of love." — Craig Claiborne

CKD Considerations

RENAL CHECK — Mom's Classics
K (Potassium): MED (leached potatoes, controlled portions)
P (Phosphorus): MED (beef is moderate P — portion to 4oz per serving)
Na (Sodium): LOW (homemade everything — no canned soups, no packets)
NOTE: Mom used regular salt, regular butter, regular everything.
We use margarine, almond milk, low-sodium broth, leached potatoes.
Same food. Same taste. Different math.

"People who love to eat are always the best people." — Julia Child

BREAKFAST

1. Country Fried Steak & Gravy

Ingredient	Amount	CKD Notes
Cube steak	4 pieces (4-5oz each)	Tenderized beef — Andrew's #1 protein
All-purpose flour	1 cup	
Egg	1, beaten	
Unsweetened almond milk	1/4 cup	Mixed with egg for wash
Garlic powder	1 tsp	
Onion powder	1 tsp	
Paprika	1 tsp	
Black pepper	1/2 tsp	
Cayenne	1/4 tsp	
Salt	1/4 tsp	Minimal — the gravy adds flavor
Avocado oil	For frying (1/2" deep)	

White Country Gravy:

Ingredient	Amount	CKD Notes
Pan drippings	3 tbsp	From frying the steak
All-purpose flour	3 tbsp	
Unsweetened almond milk	1.5 cups	Instead of whole milk — lower P
Black pepper	1/2 tsp (generous)	The pepper IS the gravy
Salt	1/4 tsp	
Garlic powder	1/4 tsp	

Method: 1. Season flour with garlic powder, onion powder, paprika, pepper, cayenne, salt 2. Dip steaks in egg/milk wash, then dredge in seasoned flour. Double-dredge for extra crispy (egg wash again, flour again) 3. Fry in avocado oil (350F) 3-4 min per side until deep golden. Drain on wire rack. 4. **Gravy:** Keep 3 tbsp drippings in pan. Whisk in flour, cook 1 min. Slowly add almond milk, whisking constantly. Simmer until thick (3-4 min). Season with pepper + salt + garlic powder. 5. Serve steak with gravy poured over. Pair with CKD mashed potatoes (below).

2. Brown Gravy (Mushroom)

Ingredient	Amount
Margarine	2 tbsp
White mushrooms, sliced	1 cup
All-purpose flour	2 tbsp
Low-sodium beef broth	1.5 cups
Worcestershire sauce	1 tsp
Black pepper	1/2 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp

Method: Saut mushrooms in margarine 5 min. Add flour, stir 1 min. Gradually add broth, whisking. Add Worcestershire + spices. Simmer until thick. Serve over mashed potatoes, steak, or chicken.

3. Hearty Breakfast Plate

Component	CKD Version
Eggs	Scrambled (2 eggs, cooked in avocado oil) — moderate P, 2 is fine
Bacon	Turkey bacon (lower Na/P than pork bacon) — 2 strips max
Hash browns	LEACHED potatoes: shred, soak in cold water 2+ hours, rinse, squeeze dry, then fry. Leaching removes ~50% of potassium.
Toast	Sourdough, lightly toasted, with margarine
Grits	CKD-aware as portioned — low K/P/Na. Cook with almond milk + margarine. Add sharp cheddar.
Biscuits	Homemade with phosphorus-free baking powder (see below)
Gravy	White country gravy (recipe above)

4. Cloud Biscuits

Ingredient	Amount
All-purpose flour	2 cups
Phosphorus-free baking powder (Rumford)	1 tbsp
Sugar	1 tsp
Salt	1/2 tsp
Cold margarine, cubed	1/3 cup
Unsweetened almond milk	3/4 cup

Method: Mix flour + BP + sugar + salt. Cut in cold margarine until pea-sized. Add milk, stir until just combined (DON'T overmix). Pat to 1" thick, cut rounds. Bake 425F 12-15 min until golden top.

MAINS

5. Golden Skillet Fried Chicken

The big one. Fried chicken is usually a CKD disaster (phosphate-injected meat, high-Na brine). Homemade fixes everything.

Ingredient	Amount	CKD Notes
Chicken thighs or drumsticks	8 pieces	FRESH, not injected/marinated — check label for "enhanced"
Unsweetened almond milk	1 cup	Buttermilk sub (add 1 tbsp vinegar, let sit 10 min)
All-purpose flour	1.5 cups	
Cornstarch	1/4 cup	Extra crunch
Paprika	1 tbsp	
Garlic powder	1 tsp	
Onion powder	1 tsp	
Cayenne	1/2 tsp	
Black pepper	1 tsp	
Dried thyme	1/2 tsp	
Salt	1/2 tsp	Way less than KFC
Avocado oil	For deep frying (3-4" deep)	

Method: 1. Soak chicken in almond milk "buttermilk" for 1 hour minimum (overnight better) 2. Mix flour + cornstarch + ALL spices in a bowl 3. Remove chicken from milk, dredge in flour mix. Press firmly. Let sit on rack 15 min (this sets the coating) 4. Heat oil to 325F (NOT 350 — lower temp = juicier interior, crispier exterior, less oil absorption) 5. Fry in batches, don't crowd. 14-16 min total, turning once. Internal temp 165F. 6. Rest on wire rack 5 min before serving. DO NOT rest on paper towels (traps steam, gets soggy).

Per 2 pieces: ~180mg Na, ~220mg P, ~350mg K (vs KFC: ~1,200mg Na per 2 pieces)

6. Velvet Mac & Cheese

Ingredient	Amount	CKD Notes
Elbow pasta (or GF)	8 oz	
Cream cheese	4 oz	Lower P than making roux with milk
Sharp cheddar, shredded	1 cup	Andrew's preferred cheese
Smoked gouda, shredded	1/2 cup	Andrew's preferred — adds smoky depth
Unsweetened almond milk	1/2 cup	
Margarine	2 tbsp	
Garlic powder	1/2 tsp	
Paprika	1/4 tsp	
Dry mustard	1/4 tsp	Secret ingredient — lifts the cheese flavor
Black pepper	1/4 tsp	
Salt	1/4 tsp	
Panko breadcrumbs	1/3 cup (optional topping)	

Method: 1. Cook pasta 1 min SHORT of al dente. Drain. 2. In same pot: melt margarine, add cream cheese + almond milk, stir until smooth 3. Add shredded cheddar + gouda, stir until melted 4. Add garlic powder, paprika, mustard, pepper, salt 5. Add pasta, fold gently 6. **Optional baked:** Transfer to baking dish, top with panko + paprika, bake 375F 15 min until golden bubbling top

Why cream cheese base: Traditional mac uses a flour roux + lots of milk = higher phosphorus. Cream cheese gives the SAME creamy texture with lower P. The smoked gouda adds the depth that the missing parmesan would have provided.

7. Bayou Sunday Gumbo

Ingredient	Amount	CKD Notes
Avocado oil	1/4 cup	For the roux
All-purpose flour	1/4 cup	For the roux
Onion	1, diced	The trinity
Celery	2 stalks, diced	The trinity — moderate K, small amount
Bell pepper	1 green, diced	The trinity
Garlic	4 cloves	
Chicken thighs	1 lb, boneless, cubed	
Andouille sausage	6 oz, sliced	Watch Na — use sparingly or sub chicken sausage
Low-sodium chicken broth	4 cups	NOT regular broth
No-salt-added diced tomatoes	1 can (14oz)	
Okra	1 cup, sliced	Thickener + vegetable — low K
Bay leaves	2	
Dried thyme	1 tsp	
Cayenne	1/2 tsp	
Paprika	1 tsp	
File powder	1 tsp (optional — sassafras leaf, traditional)	
Hot sauce (homemade)	To taste	

Method: 1. **THE ROUX:** Oil + flour in heavy pot over medium heat. Stir CONSTANTLY 20-30 min until dark chocolate brown. This is the soul of gumbo. Do not walk away. Do not answer the phone. Stir. 2. Add trinity (onion, celery, bell pepper). Stir 5 min. 3. Add garlic, stir 1 min. 4. Add broth SLOWLY (it will bubble violently). Stir smooth. 5. Add tomatoes, bay leaves, thyme, cayenne, paprika. 6. Add chicken + sausage. Simmer 30 min. 7.

Add okra. Simmer 15 min more. 8. Remove bay leaves. Serve over jasmine rice with hot sauce + file powder.

Per serving: ~350mg Na (vs restaurant gumbo: 1,500mg+), ~250mg P, ~450mg K

8. Big Easy Jambalaya

Ingredient	Amount
Jasmine rice (dry)	1.5 cups
Chicken thighs, cubed	1 lb
Andouille or chicken sausage	6 oz, sliced
Shrimp	8 oz, peeled (optional — Andrew likes shrimp)
Onion	1, diced
Bell pepper	1, diced
Celery	2 stalks, diced
Garlic	4 cloves
No-salt-added diced tomatoes	1 can (14oz)
Low-sodium chicken broth	2.5 cups
Cajun seasoning (homemade — see below)	2 tbsp
Bay leaf	1
Hot sauce	To taste

Homemade CKD Cajun Seasoning (NO SALT):

Spice	Amount
Paprika	2 tbsp
Garlic powder	1 tbsp
Onion powder	1 tbsp
Cayenne	1 tsp
Black pepper	1 tsp
Dried oregano	1 tsp
Dried thyme	1 tsp
White pepper	1/2 tsp

Method: 1. Season chicken with Cajun seasoning. Brown in oil 4 min. Remove. 2. Brown sausage 3 min. Remove. 3. Saut trinity 5 min. Add garlic 1 min. 4. Add tomatoes + broth + remaining Cajun seasoning + bay leaf. Bring to boil. 5. Add rice + chicken + sausage. Cover tightly. Reduce to LOW. 6. Cook 20-25 min without lifting lid. 7. Add shrimp in last 5 min (they cook fast — don't overcook). 8. Remove bay leaf. Fluff with fork. Serve with hot sauce.

BBQ / COOKOUT

9. Backyard Smokehouse Ribs

Ingredient	Amount	CKD Notes
Beef back ribs or beef short ribs	1 rack (~2.5 lbs)	Beef over pork — Andrew's preference. Lamb ribs also work.
Dry Rub (NO SALT):		
Brown sugar	2 tbsp	
Paprika	1 tbsp	
Garlic powder	1 tsp	
Onion powder	1 tsp	
Cumin	1 tsp	
Chili powder	1 tsp	
Black pepper	1 tsp	
Cayenne	1/2 tsp	
Mustard powder	1/2 tsp	

CKD BBQ Sauce:

Ingredient	Amount
No-salt-added tomato paste	3 tbsp

Ingredient	Amount
Water	3 tbsp
Brown sugar	2 tbsp
Apple cider vinegar	2 tbsp
Garlic powder	1/2 tsp
Onion powder	1/2 tsp
Smoked paprika	1 tsp
Cayenne	1/4 tsp
Worcestershire	1 tsp
Liquid smoke	1/2 tsp

Method (Oven — no smoker needed): 1. Remove membrane from back of ribs 2. Rub both sides generously with dry rub. Wrap in foil. 3. Bake 300F for 2.5-3 hours until tender 4. Unwrap. Brush with BBQ sauce. 5. Broil 3-4 min or grill over high heat 2-3 min per side until caramelized 6. Rest 10 min. Cut between bones. Serve with extra sauce.

Store-bought BBQ sauce: 300-600mg Na per 2 tbsp. This homemade version: ~40mg Na per 2 tbsp.

10. Double-Rinse Mashed Potatoes

THE trick that makes mashed potatoes CKD-possible

Ingredient	Amount	CKD Notes
Russet potatoes	2 lbs, peeled and cubed 1"	
Unsweetened almond milk	1/2 cup, warmed	
Margarine	3 tbsp	
Garlic powder	1/2 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp	

THE LEACHING METHOD (critical): 1. Peel potatoes. Cut into small 1" cubes (more surface area = more K removed) 2. Soak in a large bowl of cold water for **MINIMUM 2 hours** (4 hours better, overnight best) 3. **DRAIN. RINSE. REFILL** with fresh cold water at least once during soaking 4. Drain AGAIN. Place in pot with fresh cold water. 5. Boil 15-20 min until fork-tender. 6. DRAIN. (You're throwing away the K-loaded water) 7. Mash with almond milk + margarine + seasonings

The science: Raw potatoes have ~900mg K per cup. Leaching (soaking + double-boiling) removes 40-50% of potassium. Final mashed potatoes: ~450-500mg K per cup. A reasonable serving (1/2 cup) = ~225-250mg K. That's manageable.

For both gravies: Pour white or brown gravy over the leached mashed potatoes. The whole plate together is a CKD-legal country dinner.

11. Backyard Baked Beans Dupe

Baked beans are **EXTREMELY** high-K (600mg+ per cup). This version uses a trick.

Ingredient	Amount	CKD Notes
Canned white beans (cannellini)	1 can (15oz), DRAINED AND RINSED x3	Rinsing removes 30-40% of K + Na
No-salt-added tomato paste	2 tbsp	
Water	1/4 cup	
Brown sugar	2 tbsp	
Apple cider vinegar	1 tbsp	
Yellow mustard	1 tsp	
Garlic powder	1/2 tsp	
Onion powder	1/2 tsp	
Smoked paprika	1/2 tsp	
Liquid smoke	1/4 tsp	
Turkey bacon	2 strips, chopped and cooked	
Cayenne	Pinch	

Method: 1. DRAIN + RINSE beans 3 times (critical for K reduction) 2. Mix tomato paste + water + brown sugar + vinegar + mustard + spices 3. Combine beans + sauce + bacon in oven-safe dish 4. Bake 350F 25-30 min until thick and bubbly

Per 1/2 cup: ~280mg K (vs regular baked beans: 400mg+). The triple-rinse is the difference.

12. Pulled Beef or Chicken (Slow Cooker)

Ingredient	Amount
Beef chuck roast or boneless chicken thighs	3-4 lbs
Homemade Cajun seasoning (above)	2 tbsp
Garlic powder	1 tsp
Onion, quartered	1
Apple cider vinegar	2 tbsp
Low-sodium beef or chicken broth	1/2 cup
CKD BBQ sauce (above)	For serving

Method: Rub meat with seasoning. Place in slow cooker with onion + vinegar + broth. Cook LOW 8-10 hours (beef) or 5-6 hours (chicken). Shred with two forks. Toss with BBQ sauce. Serve on sourdough buns with coleslaw.

DATE NIGHT

13. Steak Dinner (Andrew's Ultimate Meal)

Component	CKD Version
Steak	Ribeye or NY strip (6-8oz), seasoned with salt + pepper + garlic powder ONLY. Cast iron sear: avocado oil screaming hot, 3-4 min per side for medium-rare. Rest 5 min.
Compound butter	Margarine softened + minced garlic + fresh herbs (thyme, rosemary) + squeeze of lemon. Form into log, chill, slice medallion on top of resting steak.
Mashed potatoes	K-leached recipe above, piped through star tip for fancy presentation
Asparagus	Moderate K — 6 spears max. Roast 400F 12 min with olive oil + garlic.
Wine	Small glass (4oz). Red wine is lower K than white.

14. Shrimp Scampi

Ingredient	Amount
Large shrimp, peeled	1 lb
Olive oil	2 tbsp
Margarine	2 tbsp
Garlic	6 cloves, minced
Dry white wine	1/3 cup (or sub low-Na chicken broth)
Lemon juice	2 tbsp
Red pepper flakes	1/4 tsp
Fresh parsley	2 tbsp, chopped
GF or regular pasta (or rice)	8 oz

Method: Cook pasta. Heat oil + margarine. Saut garlic 1 min. Add shrimp, cook 2 min per side. Add wine + lemon + pepper flakes. Simmer 2 min. Toss with pasta. Top with parsley.

Shrimp is actually one of the LOWEST phosphorus proteins. Lower P than chicken. Andrew's #3 protein choice and perfect for date night.

15. Lobster Tail (Special Occasion)

Ingredient	Amount
Lobster tails	2 (6-8oz each)
Margarine, melted	3 tbsp
Garlic	2 cloves, minced
Lemon juice	1 tbsp
Paprika	1/4 tsp
Fresh parsley	Garnish

Method: Split lobster tails down the top. Gently pull meat up and rest on top of shell (butterfly style). Mix margarine + garlic + lemon + paprika. Brush over meat. Broil 8-10 min until meat is opaque and slightly charred on edges. Serve with remaining garlic margarine for dipping + lemon wedges.

STATE FAIR / SPECIAL OCCASION

16. Rosemary-Crusted Lamb Chops with Mint Jelly

Andrew requested. Lauren's favorite.

Ingredient	Amount	CKD Notes
Rack of lamb (or 8 individual loin chops)	~2 lbs	Fresh, not pre-marinated — check for sodium phosphate
Panko breadcrumbs	1/2 cup	
Fresh rosemary, minced	1 tbsp	
Fresh thyme, minced	1 tbsp	
Garlic, minced	3 cloves	
Olive oil	2 tbsp	Binds the crust
Dijon mustard	1 tbsp	Thin coat — adhesive for crumbs
Black pepper	1/2 tsp	
Salt	1/4 tsp	Lamb is naturally flavorful — needs less

Homemade Mint Jelly (not that neon abomination):

Ingredient	Amount	CKD Notes
Fresh mint leaves, packed	1 cup	
Apple juice (unsweetened)	1 cup	Lower K than grape juice
Sugar	1/4 cup	
Apple cider vinegar	2 tbsp	
Unflavored gelatin	1 packet (1/4 oz)	

Method — Lamb: 1. Season chops with salt + pepper. Let sit 20 min at room temp. 2. Mix breadcrumbs + rosemary + thyme + garlic + olive oil in a bowl. 3. Sear chops in screaming hot cast iron with avocado oil — 2 min per side. Remove. 4. Brush thin layer of Dijon on the meat side (not the bone). 5. Press herb-crumb mixture firmly onto the mustard. 6. Roast 400F for 12-15 min (medium-rare 130F internal). Rest 5 min.

Method — Mint Jelly: 1. Bring apple juice + vinegar + sugar to a boil. Stir until sugar dissolves. 2. Remove from heat. Add mint leaves. Steep 15 min (like tea). 3. Strain out mint. Return liquid to low heat. 4. Bloom gelatin in 2 tbsp cold water (1 min), then whisk into warm liquid. 5. Pour into small jar. Refrigerate 2+ hours until set. 6. Serve a spoonful alongside each chop. The color is natural pale green — real mint, not dye.

Per serving (2 chops + 1 tbsp jelly): ~160mg Na | ~190mg P | ~280mg K

17. Green Spark Chimichurri

Two versions. Both slap. Put it on anything that used to be alive.

Classic Green Chimichurri:

Ingredient	Amount	CKD Notes
Fresh flat-leaf parsley, finely chopped	1 cup	Low K herb — the base
Fresh cilantro, finely chopped	1/2 cup	Optional — some people have the gene
Garlic, minced	4 cloves	
Red wine vinegar	3 tbsp	
Extra virgin olive oil	1/2 cup	The carrier
Dried oregano	1 tsp	
Red pepper flakes	1/2 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp	Barely any — the vinegar does the work

Red Chimichurri (variation):

Additional/Swap	Amount	CKD Notes
Roasted red pepper (jarred, drained + rinsed)	1/2 cup, finely diced	Rinse to cut Na from brine
Smoked paprika	1 tsp	Replaces some red pepper flakes
Fresh parsley	Reduce to 1/2 cup	Red pepper takes center stage

Method (both versions): 1. Combine all chopped herbs + garlic in a bowl. 2. Add vinegar + oil + spices. Stir. 3. That's it. No blender. No food processor. Knife and a cutting board. This is Argentine, not French. 4. Let sit 30 min minimum before serving — the flavors need to meet each other. 5. Keeps 5-7 days in the fridge. Gets better on day 2.

Pairs with: Steak (recipe #13), lamb chops (#16), grilled chicken, shrimp, roasted vegetables, eggs. Honestly — a spoon.

Per 1 tbsp (green): ~15mg Na | ~5mg P | ~20mg K **Per 1 tbsp (red):** ~20mg Na | ~6mg P | ~25mg K

18. County Fair Corn Dogs

Midway classic. Turkey dog, homemade batter, stick included. Dignity optional.

Ingredient	Amount	CKD Notes
Turkey hot dogs	8	Lower Na/P than beef franks — still check label
All-purpose flour	1 cup	
Yellow cornmeal	3/4 cup	
Sugar	2 tbsp	
Phosphorus-free baking powder (Rumford)	1 tbsp	
Salt	1/4 tsp	
Egg	1	
Unsweetened almond milk	3/4 cup	
Honey	1 tbsp	Helps browning
Avocado oil	For deep frying (3-4" deep)	
Wooden skewers or chopsticks	8	

Method: 1. Pat hot dogs VERY dry with paper towels. Insert sticks. Dust lightly with flour (batter won't stick to wet dogs). 2. Whisk flour + cornmeal + sugar + baking powder + salt. 3. Whisk egg + almond milk + honey. Combine wet into dry. Batter should be thick enough to coat — add flour if too thin. 4. Heat oil to 350F. Pour batter into a tall glass for easy dipping. 5. Dip each dog, twirl to coat evenly. Straight into oil. 6. Fry 3-4 min, turning once, until deep golden brown. 7. Drain on wire rack. Serve with yellow mustard (lowest Na condiment at the fair).

Per corn dog: ~280mg Na | ~95mg P | ~120mg K

19. Midway Funnel Cake

Fried dough with powdered sugar at a carnival. What exactly was the kidney problem here? Right — there wasn't one.

Ingredient	Amount	CKD Notes
All-purpose flour	1.5 cups	
Egg	1	
Sugar	2 tbsp	
Unsweetened almond milk	1 cup	Lower P than whole milk
Vanilla extract	1 tsp	
Phosphorus-free baking powder (Rumford)	1 tsp	
Salt	Pinch	
Avocado oil	For deep frying (1.5" deep)	
Powdered sugar	For dusting	

Method: 1. Whisk flour + sugar + baking powder + salt. 2. Beat egg + almond milk + vanilla. Pour into dry. Whisk until smooth — should be pourable like thin pancake batter. 3. Heat oil to 375F in a wide skillet or Dutch oven. 4. Pour batter into a squeeze bottle or funnel (or a zip-lock bag with the corner snipped). 5. Drizzle batter in a circular/crisscross pattern into hot oil. Work fast. 6. Fry 1-2 min per side until golden. Flip carefully with tongs + spatula. 7. Drain on wire rack. Blizzard of powdered sugar on top. Serve immediately.

Per funnel cake (1/4 batch): ~85mg Na | ~55mg P | ~70mg K — A lower-estimated-mineral dessert option; confirm the portion and ingredients against your plan.

20. Midway Fried Cookie Bites

The state fair food that shouldn't work but does. Surprisingly low in everything that matters.

Ingredient	Amount	CKD Notes
Oreos	12	Regular, not double-stuf — P is in the filling
Pancake mix (just-add-water kind)	1 cup	Check label — avoid brands with sodium aluminum phosphate
Water	2/3 cup	Or per pancake mix instructions
Vanilla extract	1/2 tsp	
Avocado oil	For deep frying (2" deep)	
Powdered sugar	For dusting	

Method: 1. Heat oil to 350F. 2. Mix pancake batter with water + vanilla. Should coat a spoon thickly. 3. Freeze Oreos 15 min first (they hold up better in the fryer). 4. Dip each Oreo in batter, fully coating. Let excess drip off. 5. Fry 2-3 min, flipping once, until golden and puffed. 6. Drain on wire rack. Dust with powdered sugar while hot. 7. Eat within 5 minutes. These do not age well. They age like milk.

Per 2 deep fried Oreos: ~180mg Na | ~40mg P | ~55mg K

21. State Fair Turkey Legs

The food that makes you feel like Henry VIII at a Renaissance fair.

Ingredient	Amount	CKD Notes
Turkey legs (drumsticks)	4 (~1 lb each)	Fresh, not pre-brined — this is critical
CKD Carnival Dry Rub:		
Smoked paprika	1 tbsp	
Brown sugar	1 tbsp	
Garlic powder	1 tsp	
Onion powder	1 tsp	
Black pepper	1 tsp	
Cumin	1/2 tsp	
Cayenne	1/4 tsp	
Chili powder	1/2 tsp	
Salt	1/2 tsp	Split across 4 legs — minimal per serving

Method (Oven — no smoker needed): 1. Pat turkey legs dry. Coat generously with dry rub on all sides. Press it in. 2. Wrap each leg individually in foil. 3. Bake 300F for 2.5 hours (low and slow — collagen breaks down, meat falls off the bone). 4. Unwrap. Brush with a thin coat of CKD BBQ sauce (recipe #9) if desired. 5. Broil 4-5 min until skin is dark and crispy, turning once. 6. Rest 5 min. Grab by the bone. Walk around your yard like you own the kingdom.

Smoker version: 250F for 3-4 hours over apple or cherry wood. Skip the foil. Spritz with apple juice every hour.

Per turkey leg: ~220mg Na | ~250mg P | ~380mg K — Turkey is LOWER in P than chicken breast, ounce for ounce.

22. Sweet-Salty Kettle Corn

Sweet, salty, crunchy, and one of the most kidney-friendly snacks on earth. Not a typo.

Ingredient	Amount	CKD Notes
Popcorn kernels	1/3 cup	Plain popcorn: 1mg Na, 30mg P, 30mg K per cup
Avocado oil	3 tbsp	
Sugar	2 tbsp	Added right in the pot
Salt	1/4 tsp	Total. For the whole batch.

Method: 1. Heat oil in a large pot with a lid over medium-high. Toss in 3 test kernels. 2. When test kernels pop, add ALL remaining kernels + sugar + salt at once. 3. Shake the pot continuously (lid cracked for steam). The sugar caramelizes ON the kernels as they pop. 4. When popping slows to 2 seconds between pops, dump immediately into a large bowl. 5. Toss. Let cool 2 min (the sugar coating hardens). Eat by the fistful.

Per 3 cups: ~75mg Na | ~30mg P | ~35mg K — This is basically free food on a renal diet. Snack without guilt.

23. Street-Corner Elote

Grilled corn, creamy sauce, spice, lime. The cotija sub is the only detour — everything else is authentic.

Ingredient	Amount	CKD Notes
Ears of corn, husked	6	Corn is moderate K — 1 ear is fine per serving
Avocado oil mayonnaise	3 tbsp	Lower Na than regular mayo
Cream cheese, softened	2 tbsp	Cotija substitute — similar tang, way less Na/P
Lime juice	2 tbsp	
Chili powder	1 tsp	
Smoked paprika	1/2 tsp	
Garlic powder	1/4 tsp	
Cayenne	1/4 tsp	
Fresh cilantro, chopped	2 tbsp	
Lime wedges	For serving	

Why cream cheese instead of cotija: Cotija has ~400mg Na per ounce. Cream cheese has ~85mg Na per ounce. Same creamy crumble effect when you dollop it on. Your tongue won't know. Your labs will.

Method: 1. Grill corn directly over high heat (or broil), turning every 2-3 min until charred in spots all around. ~8-10 min total. 2. Mix mayo + cream cheese + lime juice + chili powder + paprika + garlic powder + cayenne in a bowl. 3. While corn is hot, brush or smear the cream mixture all over each ear. 4. Sprinkle with extra chili powder + cilantro. Squeeze lime over the top. 5. Insert a stick or skewer in the bottom if you want the full street vendor experience.

Per ear: ~95mg Na | ~80mg P | ~240mg K

24. Lauren's Prosciutto & Brie Board (Date Night Fondue Prep)

"Twenty-four cheeses in the fridge at once. This is not a phase."

Component	What	Notes
Prosciutto	4 oz, thin-sliced, draped in rosettes	Lauren's favorite. The fancy fold matters.
Brie	4 oz wedge, room temp 30 min	Let it breathe. Cold Brie is a crime.
Havarti	3 oz, sliced	Creamy, mild, Lauren-approved
Fresh mozzarella	4 oz, torn by hand	Burrata if New Seasons has it
Accompaniments	Marcona almonds, fig jam, honeycomb, Raincoast Crisps, champagne grapes, cornichons	The supporting cast

Assembly: Arrange on a slate or wooden board. Brie center stage. Prosciutto draped like fabric. Mozzarella torn, not sliced. Almonds scattered. Fig jam in a tiny bowl. Honeycomb broken, not cut. The board should look like it belongs in a magazine Lauren would actually buy.

CKD note for Andrew: Cheese is moderate-P. If sharing this board, Andrew sticks to 2-3 oz total cheese + prosciutto and loads up on crackers + fruit.

PIZZA

Saturday Supreme Pizza

The supreme was invented by Pizza Hut in 1958. "Supreme" just means "everything on it." The only pizza where olives are acceptable. — Andrew

Component	CKD Version
Dough	GF pizza dough (recipe in Ch 11) OR store-bought (check Na — Trader Joe's is lowest)
Sauce	Fresh blender marinara (recipe in Ch 8) — NOT jarred
Cheese	Shredded mozzarella (2/3 cup — controlled P)
Turkey pepperoni	15-20 slices — lower Na than pork pepperoni
Turkey sausage	1/4 cup crumbled, browned — homemade breakfast sausage (Ch 3)
Bell peppers	1/4 cup mixed colors, diced
Red onion	2 tbsp, thinly sliced
Mushrooms	1/4 cup, sliced
Black olives	2 tbsp sliced — the ONE place olives are allowed

Method: Stretch dough on oiled sheet pan or pizza stone. Spread sauce thin (1/3 cup — too much = soggy). Layer cheese, then ALL toppings. Bake 450F directly on stone/lowest rack, 12-15 min until crust is golden and cheese bubbles. The supreme doesn't hold back and neither should you.

Per 2 slices: ~280mg Na | ~180mg P | ~300mg K

Experimental Pizza Lab

The Breakfast Pizza Scrambled eggs + turkey sausage + sharp cheddar + hot sauce drizzle on pizza dough. Bake. Eat at noon. Call it breakfast.

The Mac & Cheese Pizza CKD mac & cheese (Ch 1) spread on pizza dough as the "sauce." Extra cheddar on top. Bake. The carb-on-carb crime that tastes like a warm hug.

The Taco Pizza Seasoned ground turkey + nacho cheese sauce (Ch 8) instead of marinara. Top with shredded lettuce + pico de gallo + sour cream AFTER baking. Taco Bell on a crust.

The Thai Peanut Pizza Peanut sauce (from PB&J chapter) as base instead of marinara. Chicken thighs + bean sprouts + cilantro + lime + sriracha drizzle. Southeast Asian decoder pizza.

The Dessert Pizza Sugar cookie dough pressed into pizza pan. Bake 350F 12 min. Spread cream cheese frosting. Top with fresh fruit (strawberries, blueberries, kiwi slices). Drizzle honey. The Vorathic Rainbow Dessert Pizza if you use all the colors.

The Hot Sauce Gauntlet Pizza 8 slices. Each slice gets a DIFFERENT hot sauce from the NorthStar lineup (Ch 9). Slice 1: Baby's Bathwater (mild). Slice 8: The Final Session (500K SHU). Eat in order. The pizza that's also a dare.

The Nothing Pizza Dough. Olive oil. Salt. Garlic. That's it. No sauce. No cheese. No toppings. Focaccia energy. The pizza equivalent of Recipe #42 — it orders you.

CKD Considerations for Soul Food

The reason these foods are "banned" is that RESTAURANT/PROCESSED versions are sodium and phosphorus bombs. Homemade versions with CKD substitutions are dramatically safer:

Swap	Why
Homemade seasoning → store packets	Eliminates 500-1000mg Na per serving
Margarine → butter	Lower P

Swap	Why
Almond milk → whole milk	Lower P
Phosphorus-free baking powder → regular	Eliminates phosphorus additive
Leached potatoes → regular	Cuts K by 40-50%
Triple-rinsed canned beans → straight from can	Cuts K by 30-40%
Homemade BBQ sauce → store-bought	Cuts Na by 80%

CHAPTER 3

Chicken & Protein Mains

“The KitchenAid grinder attachment earns its keep tonight.”

Chicken Sausage Spaghetti, Chicken Tinga, Chicken Cacciatore, Steak Tacos w/ Pico

Plus: From-Scratch Sausage Grinding & Sauce Canning

“The KitchenAid grinder attachment earns its keep tonight.”

1. Grinder-Night Chicken Sausage Spaghetti

From-Scratch Chicken Sausage (KitchenAid Grinder)

Making your own sausage = total sodium/phosphorus control. No preservatives, no mystery meat.

Ingredient	Amount	CKD Notes
Chicken thighs, boneless skinless	2 lbs	Cut into 1" cubes, partially freeze 30 min (grinds better cold)
Garlic	4 cloves, minced	Mix into meat
Fresh basil	2 tbsp, chopped	
Fresh parsley	2 tbsp, chopped	
Dried oregano	1 tsp	
Fennel seeds	1 tsp, lightly crushed	THE Italian sausage flavor
Red pepper flakes	1/2 tsp	
Black pepper	1/2 tsp	
Paprika	1 tsp	
Salt	1/2 tsp (total — vs 800mg+ per link in store-bought)	
Olive oil	1 tbsp	

KitchenAid Grinder Method: 1. Cut chicken into 1" cubes. Spread on sheet pan, freeze 20-30 min (firm but not solid) 2. Set up KitchenAid with grinder attachment, medium plate 3. Mix all seasonings in a bowl 4. Feed chicken through grinder into a large bowl 5. Add seasoning mix to ground chicken. Mix with hands until evenly distributed (don't overmix — gets tough) 6. Form into patties, links (using casings if you have them), or just crumble directly into the pan

Cooking: Brown sausage in skillet over medium-high, breaking into chunks, 6-8 min.

Per 4oz serving: ~120mg Na (vs store-bought ~600mg), ~180mg P, ~250mg K

Sunday Red Sauce (Canning-Ready)

Makes ~4 pints — can in mason jars for shelf storage

Ingredient	Amount	CKD Notes
No-salt-added crushed tomatoes	1 can (28oz)	Lowest sodium option
No-salt-added tomato paste	2 tbsp	Concentrate flavor without more liquid K
Olive oil	2 tbsp	
Onion	1 medium, diced	

Swap	Why
Fresh chicken → "enhanced"/injected chicken	Eliminates sodium phosphate injection

The thesis: You don't have to eat like you're dying. You eat like you're LIVING. The restrictions are in the **INGREDIENTS**, not the **RECIPES**. Same plate, same flavors, same Saturday morning breakfast — different margarine, different milk, leached potatoes. Your kidneys can't tell the difference. Your taste buds can't either.

Ingredient	Amount	CKD Notes
Garlic	6 cloves, minced	
Fresh basil	1/4 cup, torn	Add at end
Dried oregano	1 tsp	
Dried thyme	1/2 tsp	
Sugar	1 tsp	Balances acidity
Red pepper flakes	1/4 tsp	
Black pepper	1/2 tsp	
Salt	1/4 tsp only	
Bay leaf	1	Remove before canning

Method: 1. Saut?? onion in olive oil 5 min until soft 2. Add garlic, cook 1 min 3. Add tomato paste, stir 1 min (caramelizes) 4. Add crushed tomatoes, all dried spices, sugar, bay leaf 5. Simmer LOW 30-45 min, stirring occasionally 6. Remove bay leaf. Add fresh basil. 7. Taste — adjust salt/sugar/pepper

Canning (water bath method): 1. Sterilize mason jars + lids in boiling water 10 min 2. Add 1 tbsp lemon juice per pint jar (REQUIRED for acidity/safety) 3. Ladle hot sauce into hot jars, leave 1/2" headspace 4. Wipe rims. Apply lids + rings finger-tight 5. Process in boiling water bath: pints 35 min, quarts 40 min 6. Remove, cool undisturbed 12 hours. Check seal (lid shouldn't flex) 7. Sealed jars store 12-18 months at room temp

Per 1/2 cup sauce: ~45mg Na, ~30mg P, ~280mg K

Assembly

Cook spaghetti per package (no salt in the water — you don't need it). Toss with red sauce + crumbled chicken sausage. Top with a little shredded mozzarella if desired (small amount).

2. Smoky Chicken Tinga

Mexican shredded chicken in smoky chipotle tomato sauce — served in tacos, tostadas, or over rice

Ingredient	Amount	CKD Notes
Chicken thighs, boneless	1.5 lbs	Thighs stay moist when braised
No-salt-added diced tomatoes	1 can (14oz)	
Chipotle pepper in adobo	2 peppers + 1 tbsp sauce	From the can you added to cart
Onion	1 medium, sliced	
Garlic	4 cloves	
Oregano	1 tsp	Mexican oregano if available
Cumin	1/2 tsp	
Bay leaf	1	
Oil	1 tbsp	
Salt	1/4 tsp	
Lime juice	1 tbsp (finish)	

Method: 1. Sear chicken thighs in oil 3 min per side (browning = flavor). Remove. 2. Same pan: cook onion 5 min. Add garlic 1 min. 3. Add diced tomatoes, chipotles + adobo sauce, oregano, cumin, bay leaf 4. Nestle chicken back in. Bring to simmer. 5. Cover. Low heat 25-30 min until chicken shreds easily 6. Remove chicken. Shred with two forks. 7. Return

shredded chicken to sauce. Simmer uncovered 5 min to thicken 8. Remove bay leaf. Finish with lime juice.

Serve in: Guerrero corn tortillas, over rice, on tostadas, or in a bowl with sour cream + cilantro + pickled onion

Per serving (4 servings): ~200mg Na, ~190mg P, ~380mg K

3. Hunter's Chicken Cacciatore

Italian hunter's chicken — braised with tomatoes, olives, capers, peppers

Ingredient	Amount	CKD Notes
Chicken thighs, bone-in skin-on	4 pieces (~2 lbs)	Bone-in = more flavor. Remove skin after browning to cut fat if desired
No-salt-added crushed tomatoes	1 can (14oz)	
Bell pepper	1, sliced	Any color
Onion	1, sliced	
Garlic	4 cloves, sliced	
Black olives	1/4 cup, sliced	OPTIONAL — Andrew: skip (dislikes olive flavor)
Capers	1 tbsp, rinsed	RINSE to reduce sodium
Dry white wine	1/4 cup (or sub chicken broth)	Alcohol cooks off
Dried oregano	1 tsp	
Dried basil	1 tsp	
Red pepper flakes	1/4 tsp	
Fresh parsley	Garnish	
Oil	2 tbsp	
Salt	1/4 tsp	

Method: 1. Season chicken lightly with salt + pepper. Brown skin-side down in oil 5 min. Flip 3 min. Remove. 2. Same pan: cook onion + bell pepper 5 min 3. Add garlic 1 min 4. Deglaze with wine, scrape up browned bits 5. Add crushed tomatoes, oregano, basil, red pepper flakes 6. Nestle chicken back in, skin-side up 7. Add capers (skip olives per Andrew preference) around chicken 8. Cover. Simmer 30-35 min until chicken is 165F and falling off bone 9. Garnish with parsley

Serve over: Spaghetti, rice, or with crusty bread

Per serving (4 servings): ~250mg Na, ~220mg P, ~420mg K

4. Midnight Steak Tacos with Pico de Gallo

The Steak

Ingredient	Amount
Flank steak	1 lb
Lime juice	2 tbsp
Garlic	3 cloves, minced
Cumin	1 tsp
Chili powder	1 tsp
Paprika	1/2 tsp
Oil	1 tbsp
Salt	1/4 tsp
Black pepper	1/2 tsp

Marinate steak minimum 30 min (overnight better). Grill or cast-iron sear over HIGH heat 4-5 min per side for medium. Rest 5 min. Slice THIN against the grain.

Fresh Pico de Gallo

Ingredient	Amount	CKD Notes
Fresh tomato	1 medium, diced small	Moderate K — one tomato across 4 servings = controlled
White onion	1/4 cup, finely diced	Low K

Ingredient	Amount	CKD Notes
Cilantro	1/4 cup, chopped	
jalapeño	1, seeded and minced	Seeds out = less heat, the pepper amount is nutritionally tiny
Lime juice	2 tbsp	
Salt	Pinch	

Mix everything. Let sit 15 min for flavors to meld. Best fresh, same day.

Per 1/4 cup pico: ~80mg K, ~8mg P, ~25mg Na

Assembly

Guerrero corn tortillas (warmed on dry skillet 30 sec per side) → sliced steak → pico → squeeze of lime → optional sour cream drizzle

Hot sauce recipes are in the Hot Sauces chapter — 30+ sauces across three lineups.

6. FROM-SCRATCH SAUSAGE GRINDING & STORAGE

KitchenAid Grinder Setup

- 1. Attach grinder to KitchenAid stand mixer
- 2. Use **medium grinding plate** for Italian-style sausage (coarse for breakfast, fine for hot dogs)
- 3. Keep meat COLD — partially freeze 20-30 min before grinding
- 4. Cut into 1" cubes that fit the feed tube
- 5. Run through once for standard texture, twice for smoother

Sausage Flavors to Grind

Italian Sweet (chicken or turkey)

Fennel seeds + garlic + basil + parsley + paprika + black pepper

Italian Hot

Same as sweet + red pepper flakes + cayenne

Breakfast Style (Andrew's Favorite — Costco/Inesto's Clone)

"The link that makes the whole plate make sense."

Ingredient	Amount
Chicken thighs, boneless	2 lbs, cubed + partially frozen
Sage (dried, rubbed)	1 tbsp
Thyme (dried)	1 tsp
Black pepper	1 tsp
Brown sugar	1 tbsp
Nutmeg	1/4 tsp
Costco no-salt seasoning blend	1 tsp (or sub garlic powder 1/2 tsp + onion powder 1/2 tsp)
Cayenne	1/4 tsp
Salt	1/2 tsp
Maple syrup	1 tsp (optional — maple breakfast sausage variant)

Grind chicken through medium plate. Mix in all seasonings by hand. Form into 2oz patties or links. Pan-fry in avocado oil 3-4 min per side until 165F.

Per 2oz patty: ~80mg Na | ~90mg P | ~130mg K (vs Costco/store: ~400mg Na)

Sausage Gravy (Biscuits & Gravy)

Ingredient	Amount
Chicken breakfast sausage (above)	4 patties, crumbled
All-purpose flour	3 tbsp
Unsweetened almond milk	2 cups
Black pepper	1/2 tsp (generous)
Garlic powder	1/4 tsp
Salt	1/4 tsp

Method: Brown crumbled sausage in skillet. Add flour, stir 1 min. Slowly add almond milk, whisking. Simmer until thick. Season. Serve over CKD biscuits (recipe in soul food chapter).

Thanksgiving Turkey-Sausage Stuffing

"The stuffing that earns the bird's respect."

Ingredient	Amount
Chicken or turkey breakfast sausage (above)	1 lb, crumbled and browned
Sourdough bread, cubed and dried	6 cups
Onion, diced	1 medium
Celery, diced	2 stalks
Margarine	3 tbsp
Low-sodium chicken broth	1.5 cups
Fresh sage	1 tbsp, chopped
Fresh thyme	1 tsp
Black pepper	1/2 tsp
Salt	1/4 tsp
Egg	1, beaten

Method: Sauté onion + celery in margarine 5 min. Combine with bread cubes, browned sausage, herbs, pepper. Toss with broth + egg until evenly moist. Transfer to greased baking dish. Bake 375F 30-35 min until golden top. The sausage makes this — every bite has sage and thyme and meat.

Per serving (8 servings): ~180mg Na | ~140mg P | ~200mg K

Mexican Chorizo Style

Chili powder + paprika + cumin + oregano + garlic + vinegar

Thai Basil (fusion)

Thai basil + lemongrass (minced fine) + garlic + ginger + lime zest + chili flakes

CHAPTER 4

Tex-Mex & Fast Food Dupes

"Cravings don't negotiate."

Renal-Aware Fast Food at Home

"Cravings don't negotiate."

Why Taco Bell Dupes?

Taco Bell is one of the most CKD-hostile fast food chains: high sodium (1000-2000mg per item), high phosphorus (cheese, beans, processed meat), high potassium (tomatoes, beans, guac). A single Crunchwrap Supreme has more sodium than most people on a renal diet should eat in an ENTIRE DAY.

These dupes recreate the flavor and experience with renal-aware substitutions.

The Recipes

1. Bellbreaker Crunchwrap

The original: ~1,300mg sodium, ~400mg phosphorus **The dupe:** ~450mg sodium, ~180mg phosphorus

Component	Original	CKD Dupe
Tortilla	12" flour tortilla	12" low-sodium flour tortilla (Mission low-sodium)
Meat	Seasoned ground beef	Ground turkey + homemade taco seasoning (cumin, paprika, garlic, onion powder, oregano — NO packet mix, packets are sodium bombs)
Nacho cheese	Nacho cheese sauce	Homemade cheese sauce: cream cheese + a little cheddar + milk + paprika, melted (cream cheese is lower P than processed cheese)
Sour cream	Regular sour cream	Regular sour cream (actually CKD-okay in small amounts)

Storage

- Fresh (fridge):** Use within 3 days in sealed container
- Frozen patties:** Form into 3oz patties, separate with parchment, freeze in ziplock. Good 3 months.
- Vacuum sealed:** If you get a vacuum sealer, frozen sausage keeps 6+ months
- Casings (optional):** Natural hog casings from the butcher counter. Soak 30 min in water, feed onto grinder stuffing tube, fill while grinding. Twist into links every 5-6".

CKD Advantage of Homemade Sausage

	Store-Bought (per link)	Homemade (per link)
Sodium	500-800mg	80-120mg
Phosphorus	200-350mg	150-180mg
Preservatives	Nitrites, phosphates, MSG	NONE
Cost	\$1.50-2.50/link	~\$0.80/link

You control EVERYTHING. No hidden sodium phosphate (a common meat preservative that's a kidney killer). No nitrites. No mystery.

CKD Considerations

RENAL CHECK — Chicken Mains
K (Potassium): MED (no-salt-added tomato products, controlled portions)
P (Phosphorus): LOW-MED (homemade sausage = no phosphate preservatives)
Na (Sodium): LOW (homemade everything, no-salt-added canned goods)
Protein: GOOD (chicken thighs = quality renal protein)

"The KitchenAid earned its spot on the counter tonight. Every link is a ceremony."

Component	Original	CKD Dupe
Tostada shell	Flat tostada	Baked corn tortilla round (lower sodium than store tostada)
Lettuce	Iceberg	Iceberg (fine — low K, low P)
Tomato	Diced tomato	SKIP or use 1 tbsp max (tomatoes are high-K) — sub with diced bell pepper for crunch+color

Assembly: Tortilla flat → meat → cheese sauce → tostada shell → sour cream → lettuce → pepper. Fold edges in, flip seam-down, toast in dry skillet 2 min per side.

Per serving: ~450mg Na, ~180mg P, ~280mg K

2. Soft-Crunch Gordita

Component	CKD Dupe
Outer flatbread	Low-sodium flour tortilla, warmed
Spicy ranch	Avocado oil mayo + a pinch of cayenne + garlic powder + dried dill
Inner taco shell	Regular hard taco shell (moderate sodium, 1 is okay)
Meat	Turkey taco meat (same seasoning as above)
Three cheese blend	Small amount of cheddar + cream cheese blend (portion control)
Lettuce	Iceberg shredded

3. Borderline Mexican Pizza

Component	CKD Dupe
Pizza "crust"	Two flour tortillas, brushed with oil, baked crispy at 400F for 5 min
Beef layer	Turkey taco meat
Bean layer	SKIP refried beans (very high K+P) — use a thin layer of cream cheese as the "glue" instead
Pizza sauce	1 tbsp low-sodium marinara (not more — tomato = K)
Cheese	Small amount of shredded cheddar

Component	CKD Dupe
Toppings	Diced green onion (low K), diced bell pepper, a few black olive slices

Bake assembled pizza at 400F for 8-10 min until cheese melts and tortilla is crispy.

4. Griddle-Melt Chicken Quesadilla

Component	CKD Dupe
Tortilla	Low-sodium flour tortilla
Chicken	Plain grilled chicken breast, shredded (no pre-marinated — marinades are sodium bombs)
Cheese	Cream cheese + small amount of mozzarella (lower P than cheddar)
Creamy jalapeño sauce	Avocado oil mayo + a few drops of jalapeño hot sauce + garlic powder

Cook in skillet with light oil spray, 3 min per side until golden and cheese melts.

5. Five-Layer Beef Burrito

Component	CKD Dupe
Tortilla	Large low-sodium flour tortilla
Beef	Turkey taco meat
Beans	SKIP (too high K+P) — add extra rice instead
Rice	White rice (renal-aware, low K when cooked)
Sour cream	Regular sour cream, 2 tbsp
Nacho cheese	Homemade cream cheese sauce

6. Big Bell Nachos

Component	CKD Dupe
Chips	Unsalted tortilla chips (or low-sodium)
Beef	Turkey taco meat
Beans	SKIP
Cheese	Homemade cream cheese sauce drizzled
Sour cream	Dollop
Tomato	1 tbsp max diced, or sub diced bell pepper
Green onion	Small amount (low K)

7. Cinnamon Cloud Bites

Component	CKD Dupe
Dough	Bisquick (phosphorus-free baking powder version) + milk + sugar, rolled into balls
Filling	Cream cheese + powdered sugar + vanilla
Coating	Cinnamon sugar (Ceylon cinnamon — Cinnamon Cloud crossover!)

Scoop dough, press flat, add cream cheese filling, seal into ball, deep fry at 350F for 2-3 min until golden. Roll in cinnamon sugar while warm.

8. Baja Storm Freeze

The original: Mountain Dew Baja Blast + ice **The CKD issue:** Phosphoric acid in Mountain Dew is a phosphorus source + high sugar + fluid counts

Component	CKD Dupe
Base	Sprite Zero or 7UP Zero (no phosphoric acid, unlike Dew)
Color	Blue food coloring + green food coloring (2 drops each for the teal)
Flavor	Lime juice (1 tbsp) + a few drops of pineapple extract
Freeze	Blend with ice, serve in clear cup

Fluid note: This counts toward daily fluid restriction. Make it a treat, not a daily thing. ~8oz serving.

The Master Taco Seasoning (make a jar, use for everything)

Spice	Amount	Notes
Cumin	3 tbsp	The backbone
Paprika	2 tbsp	Color + mild heat
Garlic powder	1 tbsp	Not garlic salt!
Onion powder	1 tbsp	Not onion salt!
Oregano	1 tsp	Mexican oregano if you can find it
Cayenne	1/2 tsp	Adjust to heat preference
Black pepper	1 tsp	
Coriander	1/2 tsp	Optional — adds authenticity

NO SALT in this mix. The whole point is controlling sodium yourself. Store in a jar. Use 2 tbsp per pound of meat + 1/4 cup water, simmer until absorbed.

Masa Press Lab: Tortilla Press Technique + Four Masa Harina Colors

White, yellow, blue, and red masa harina behave like siblings, not clones. The press is simple; the hydration is the craft.

Tortilla Press Setup

Tool	Best Setup	Why It Matters
Tortilla press	Cast iron or aluminum press on a stable counter	Even pressure. Do not crank the handle like a clamp; press in stages.
Liners	Cut a zip-top freezer bag into two thick plastic sheets	Better than flimsy wrap. Peels cleanly without tearing the masa.
Scale	35g balls for street tacos, 45-50g for 6 inch tortillas	Consistent balls make consistent cook times.
Comal or skillet	Dry cast iron, carbon steel, or heavy nonstick	No oil for tortillas. Heat and steam create the puff.
Towel warmer	Clean towel in a covered bowl or tortilla keeper	Steam finishes the tortilla and keeps it flexible.

Master Press Technique

- Hydrate by feel, not just by the cup.** Start with the recipe amount below, mix with your hand, then add warm water 1 tsp at a time until the dough feels like soft Play-Doh. It should not crumble at the edges or smear wet onto your palm.
- Rest the dough.** Cover 20-30 minutes. Masa harina keeps drinking water after mixing. Blue and red masa usually need the full 30 minutes.
- Re-check hydration after resting.** Pinch a golf-ball-size piece and flatten it between your palms. Cracked edge = add water. Sticky surface = dust in 1 tsp masa harina and wait 5 minutes.
- Roll smooth balls.** No cracks before pressing. A cracked ball becomes a cracked tortilla.
- Press in two light passes.** Place the ball slightly behind center, close press gently, open, rotate the plastic 180 degrees, press again. This keeps the tortilla round and avoids one thick edge.
- Peel plastic from tortilla, not tortilla from plastic.** Hold the tortilla flat on your palm and pull the plastic away slowly. If it tears, the dough is too wet, too dry, or unrested.
- Cook hot and fast.** Preheat until a drop of water skates. Cook side 1 for 20-30 seconds, flip, cook side 2 for 45-60 seconds, flip again and press gently with a spatula for 15-30 seconds. Puff is the win.
- Steam stack immediately.** Move each tortilla into a towel-lined bowl and cover. Ten minutes in its own steam makes it bend instead of crack.

Heat Target

Medium-high is usually right: about 425-475F surface heat. Too cool gives pale, dry tortillas. Too hot burns spots before the center sets. The first tortilla is the test tortilla; adjust before making the whole batch.

White Masa Harina Tortillas

Ingredient	Amount	Notes
White masa harina	2 cups	Clean baseline flavor; best for tacos and enchiladas.
Warm water	1 1/2 cups, plus 1-3 tbsp as needed	Warm tap water hydrates faster than cold.
Fine salt	1/4 tsp optional	Skip or reduce for stricter sodium days.
Avocado oil	1 tsp optional	Not traditional; helps flexibility for beginners.

Method: Mix, rest 20 minutes, portion into 12 balls, press, cook hot, steam stack. Best use: everyday taco, quesadilla, enchilada dip-and-roll.

Yellow Masa Harina Tortillas

Ingredient	Amount	Notes
Yellow masa harina	2 cups	More roasted-corn aroma and deeper color.
Warm water	1 2/3 cups, plus 1-2 tbsp as needed	Yellow often drinks slightly more water than white.
Fine salt	1/4 tsp optional	Optional.
Avocado oil	1 tsp optional	Useful if making tostadas or reheating later.

Method: Same as white, but rest 25 minutes. Best use: crunchy tacos, tostadas, tortilla chips, breakfast chilaquiles.

Blue Masa Harina Tortillas

Ingredient	Amount	Notes
Blue masa harina	2 cups	Earthier, nuttier, more dramatic on camera.
Warm water	1 3/4 cups, plus 1-4 tbsp as needed	Blue masa usually needs more hydration and longer rest.
Fine salt	1/4 tsp optional	Optional.
Lime juice	1/2 tsp optional	Brightens flavor; keep tiny so the dough does not tighten.

Method: Mix thoroughly, rest 30 minutes, press slightly thicker than white masa, cook 5-10 seconds longer per side. Best use: steak tacos, photo plates, tostadas, three-color tortilla flights.

Red Masa Harina Tortillas

Ingredient	Amount	Notes
Red corn masa harina	2 cups	If you have true red masa harina, use it straight.
Warm water	1 2/3 to 1 3/4 cups	Hydrate like blue masa: patient and warm.
Fine salt	1/4 tsp optional	Optional.
Avocado oil	1 tsp optional	Helps flexibility.

If you only have white or yellow masa: make red adobo masa with 2 cups masa harina, 1 2/3 cups warm water, 1 tbsp mild chile powder or paprika, 1 tsp annatto or achiote powder, 1/4 tsp cumin, and 1 tsp vinegar. Rest 30 minutes. This is not true red corn, but it gives the red taco-shop look and a gentle chile aroma.

Tostadas, Chips, and Taco Shells

Cook tortillas first, then crisp them. Raw pressed masa does not become a good chip. For tostadas, dry cooked tortillas uncovered 20-30 minutes, then fry at 350F until rigid and lightly browned, or brush with oil and bake at 375F for 10-14 minutes, flipping once. For crunchy taco shells, fry a cooked tortilla 10 seconds flat, fold with tongs, then hold open in the oil until crisp.

Troubleshooting

Problem	Cause	Fix
Cracked edge after pressing	Dough too dry or unrested	Knead in water 1 tsp at a time; rest 10 more minutes.
Sticks to plastic	Dough too wet	Dust in masa harina 1 tsp at a time; wait 5 minutes.

Problem	Cause	Fix
No puff	Heat too low, tortilla too dry, or first side cooked too long	Raise heat; cook first side only until dry spots appear.
Hard after cooking	Overcooked or left uncovered	Shorten cook time and steam stack immediately.
One side thick	Ball centered wrong or one heavy press	Place ball slightly behind center; press, rotate, press again.

RENAL CHECK - Masa Press Lab
K (Potassium): LOW-MED by portion; corn is usually manageable in small tortillas.
P (Phosphorus): LOW-MED; avoid phosphate-added packaged tortillas when possible.
Na (Sodium): CONTROLLED; salt is optional and can be skipped.
Fluid: Water in dough mostly cooks off, but count beverages served with the meal.

MEXICAN RESTAURANT DUPES

9. Carne Asada Fry Plate

Origin: San Diego, CA — emerged from taco shops in the 1990s as border-town street food. Now a stadium staple from Petco Park to backyard parties.

Component	CKD Dupe
Fries	Russet potatoes, peeled, cut into fries, soaked in water 4+ hours (K-leaching — pulls out ~50% potassium), drained, patted dry, fried in avocado oil at 375F until crispy
Carne asada	Flank steak marinated in Master Carne Asada Marinade (Recipe 14 below), grilled hot and fast, sliced thin against the grain
Nacho cheese	Homemade cheese sauce (Ch 8 recipe — cream cheese + cheddar + milk + paprika)
Sour cream	Regular sour cream, 2 tbsp
Pico de gallo	2 tbsp max — diced tomato + white onion + cilantro + lime juice + pinch of salt (portion-controlled for K)
Guacamole	1/4 avocado mashed with lime juice + cilantro + pinch of garlic powder (small portion — avocado is high-K, but 1/4 is manageable)

Assembly: Fries on plate → sliced carne asada → drizzle cheese sauce → dollops of sour cream + guac → scatter pico. Eat immediately.

Per serving: ~480mg Na | ~200mg P | ~520mg K

10. Sizzle Plate Carne Asada

Origin: The standard plate at every taqueria from Tijuana to East LA — meat, rice, beans, tortillas. Simple. Complete.

Component	CKD Dupe
Carne asada	Flank steak, Master Marinade (Recipe 14), grilled, sliced against the grain
Spanish rice	White rice cooked with 1 tbsp no-salt-added tomato sauce + cumin + garlic powder + onion powder + avocado oil. Low-sodium chicken broth or water as liquid
Refried beans	Canned pinto beans — drain, rinse 3x thoroughly (removes ~40% sodium + potassium), mash in skillet with avocado oil + cumin + garlic powder. No lard
Tortillas	Guerrero corn tortillas (corn, water, lime — 3 ingredients, low sodium)
Pico de gallo	2 tbsp, same as Recipe 9

Plate it: Steak sliced fanned on one side, rice + beans side by side, two warm corn tortillas, pico on top of the meat. Lime wedge on the side.

Per serving: ~440mg Na | ~220mg P | ~580mg K

11. Green-Chile Chicken Enchiladas

Origin: Enchiladas predate the Spanish conquest — Aztec markets in Tenochtitlan served tortillas dipped in chili sauce. The cream-cheese filling twist is Tex-Mex, circa 1960s-70s.

Component	CKD Dupe
Tortillas	Guerrero corn tortillas, lightly fried in avocado oil to soften (5 seconds per side)

Component	CKD Dupe
Filling	Shredded chicken breast (poached or baked, no marinade packets) + 2 oz cream cheese per 4 enchiladas, mixed together
Enchilada sauce	1 can (8 oz) no-salt-added tomato sauce + 1 tbsp chili powder + 1 tsp cumin + 1/2 tsp garlic powder + 1/2 tsp oregano + 1/4 tsp cayenne + 1/4 cup water. Simmer 10 min
Cheese topping	Shredded cheddar, light layer (~1 oz per 2 enchiladas — portion control keeps P in check)

Assembly: Dip each softened tortilla in warm enchilada sauce. Fill with chicken-cream cheese mixture, roll seam-down in baking dish. Pour remaining sauce over top, sprinkle cheddar. Bake 375F for 20 min until bubbly.

Makes 4 enchiladas (2 per serving).

Per serving (2 enchiladas): ~390mg Na | ~210mg P | ~420mg K

12. Mazatlan Fajita Burrito

Origin: Named for Mazatlan Mexican Restaurant — the kind of neighborhood spot where the burrito is the size of your forearm and one order is two meals. Sinaloa port-city energy, Pacific coast heat.

Component	CKD Dupe
Tortilla	Large (12") flour tortilla (regular — one tortilla's sodium is acceptable in context)
Chicken	Boneless skinless chicken thighs (more flavor than breast, similar renal profile), sliced thin, seasoned with homemade fajita seasoning: 1 tsp cumin + 1 tsp chili powder + 1 tsp paprika + 1/2 tsp garlic powder + 1/2 tsp onion powder + 1/2 tsp oregano. Grilled or pan-seared in avocado oil, high heat
Peppers + onions	Sliced bell peppers (green + red) + white onion, sauteed in avocado oil until charred edges, still crisp
Rice	White rice, plain or Spanish-style (same as Recipe 10)
Cheese	Shredded cheddar or Monterey Jack, ~1 oz
Sour cream	2 tbsp inside the wrap

Assembly: Warm tortilla on dry skillet. Layer: rice down the center → chicken → peppers and onions → cheese → sour cream. Fold bottom up, sides in, roll tight. Cut in half on the bias. Serve with a lime wedge.

Per serving (1 burrito): ~520mg Na | ~240mg P | ~490mg K

13. Lauren's Warm Campechana

Origin: Campechana is a Gulf Coast Mexican seafood cocktail from Campeche/Veracruz — traditionally served cold in a copa. Lauren's version is served warm, which changes the whole character. The heat opens up the citrus and softens the seafood without making it rubbery.

Component	CKD Dupe
Shrimp	1/2 lb medium shrimp, peeled and deveined, poached in lightly salted water until just pink (~3 min), chopped
Crab	4 oz imitation crab (surimi) — lower phosphorus than real crab, or use real lump crab if budget allows (portion-control for P)
Cocktail sauce base	1 cup no-salt-added tomato sauce + 1 tbsp lime juice + 1 tsp hot sauce (or to taste) + 1/2 tsp cumin. Warm in saucepan over medium heat
Avocado	1/4 avocado, diced (portion-controlled for K)
Vegetables	2 tbsp diced white onion + 2 tbsp diced cucumber + 1 tbsp diced jalapeno (seeded) + 2 tbsp chopped cilantro
Citrus	Juice of 1 lime, plus wedges for serving

Assembly: Warm the tomato sauce base in a saucepan. Add chopped shrimp and crab, stir gently, heat through 2 min (do not boil — keeps the seafood tender). Remove from heat, fold in onion, cucumber, jalapeno, cilantro, avocado, lime juice. Serve warm in a bowl or copa glass with tostadas or low-sodium crackers on the side.

The "warm not cold" move: Heating the cocktail base first and adding the seafood to it — instead of chilling everything — blooms the cumin and lime, gives it a soup-adjacent comfort without losing the bright acidity. Lauren's call.

Per serving: ~380mg Na | ~190mg P | ~450mg K

14. Citrus Fire Carne Asada Marinade

Origin: Every taqueria has a version. The lime-soy-cilantro trinity is northern Mexico meets Pacific Rim — the soy sauce arrived via Chinese immigrants in Baja California in the 1800s and never left.

Ingredient	Amount	Notes
Lime juice	1/4 cup (fresh)	The acid — tenderizes + brightens. Andrew's sour/citrus preference lives here
Low-sodium soy sauce	2 tbsp	Umami depth without the sodium bomb of regular soy (~50% less Na)
Garlic	4 cloves, minced	Fresh, not powder — marinade wants the moisture
Cilantro	1/4 cup, chopped	Stems included — that's where the flavor hides
Cumin	1 tsp	Warmth
Chili powder	1 tsp	Heat + color
Avocado oil	2 tbsp	Carries the fat-soluble flavors into the meat
Black pepper	1/2 tsp	No salt needed — the soy and lime do the work

Method: Whisk everything together. Place flank steak in a zip-lock bag or glass dish, pour marinade over, coat all surfaces. Refrigerate minimum 2 hours, overnight is better. Remove from fridge 30 min before grilling.

Grilling: Hottest part of the grill (or cast iron screaming hot with avocado oil). 4-5 min per side for medium-rare on a 1-inch flank steak. Rest 5 min. Slice thin, against the grain, on the bias.

Yield: Enough for 1.5-2 lbs flank steak (4 servings).

Per serving (marinade contribution only): ~180mg Na | ~30mg P | ~90mg K

CKD Considerations (every recipe)

RENAL CHECK — Taco Bell CKD Dupes (Recipes 1-8)
K (Potassium): LOW-MED (beans removed, tomato minimized)
P (Phosphorus): LOW-MED (cream cheese over cheddar, no processed cheese)
Na (Sodium): LOW (no-salt seasoning, low-sodium tortillas)
Fluid: Count beverages toward daily limit
Protein: MODERATE (turkey is good renal protein)

RENAL CHECK — Mexican Restaurant Dupes (Recipes 9-14)
K (Potassium): MED (K-leached potatoes, portion-controlled avocado/tomato, triple-rinsed beans)
P (Phosphorus): LOW-MED (cream cheese sauces, cheddar portion-controlled, fresh proteins over processed)
Na (Sodium): LOW-MED (low-sodium soy in marinade, no-salt-added tomato sauce, homemade seasoning throughout)
Fluid: Campechana broth counts toward daily limit
Protein: MODERATE-HIGH (flank steak, chicken thighs, shrimp — all quality renal proteins)

"The Drive-Thru closed. Your kitchen opened."

REESER'S BURRITO DUPES

Reeser's Fine Foods has been making frozen burritos in Topeka, Kansas since 1969 — the same year as the moon landing.

15. Reeser's Red Hot Beef Burrito

"Reeser's first plant opened at 31st and Topeka Blvd — a building that had been a dairy. By the mid-1970s they were shipping a million burritos a week across the Great Plains."

The original: Gas station legend. Red chile sauce, seasoned beef, processed cheese, flour tortilla. ~900mg sodium, ~350mg phosphorus per burrito. **The dupe:** ~380mg sodium, ~160mg phosphorus

Component	Original	CKD Dupe
Tortilla	10" flour tortilla	10" low-sodium flour tortilla
Meat	Seasoned ground beef	Ground turkey, browned and drained, seasoned with Master Taco Seasoning (2 tbsp per lb)
Red chile sauce	Processed red chile sauce (sodium bomb + phosphate additives)	Homemade CKD red chile sauce (see below)

Component	Original	CKD Dupe
Cheese	Processed cheese food	Cream cheese (2 tbsp per burrito) — lower P than processed cheese, melts smooth

Homemade CKD Red Chile Sauce:

Ingredient	Amount	Notes
No-salt-added tomato sauce	1 cup (8 oz can)	Controls sodium — regular tomato sauce has 600mg+ Na per cup
Chili powder	1 tbsp	The heat backbone
Cumin	1 tsp	Warmth + earthiness
Garlic powder	1/2 tsp	Not garlic salt
Onion powder	1/2 tsp	Depth
Cayenne	1/4 tsp	Adjust up for more heat — the original is SPICY
Water	2 tbsp	Thin to saucy consistency

Whisk all together in a small saucepan, simmer 5 min. Makes enough for 4-5 burritos.

Method: Spoon ~3 tbsp turkey down the center of the tortilla. Drizzle 2 tbsp red chile sauce over the meat. Add 2 tbsp cream cheese in small dollops. Fold bottom up, sides in, roll tight. Wrap individually in foil. **Bake at 375F for 20 min OR microwave unwrapped on a plate for 2 min**, flipping halfway.

Per serving (1 burrito): ~380mg Na | ~160mg P | ~290mg K

16. Reeser's BBQ Beef Burrito

"Reeser's BBQ variant showed up in the early 1980s — Kansas is barbecue country, and someone in Topeka figured if you could put red chile in a tortilla, you could put KC-style sweet smoke in one too."

The original: Sweet-tangy BBQ sauce, seasoned beef, flour tortilla. ~850mg sodium, ~330mg phosphorus. **The dupe:** ~360mg sodium, ~155mg phosphorus

Component	Original	CKD Dupe
Tortilla	10" flour tortilla	10" low-sodium flour tortilla
Meat	Seasoned ground beef	Ground turkey, browned and drained, seasoned with Master Taco Seasoning (light hand — 1 tbsp per lb, the BBQ sauce carries the flavor)
BBQ sauce	Processed BBQ sauce (high-fructose corn syrup + sodium + phosphates)	Homemade CKD-aware BBQ sauce (see below)
Cheese	Processed cheese food	Cream cheese (2 tbsp per burrito)

Homemade CKD-Aware BBQ Sauce:

Ingredient	Amount	Notes
No-salt-added ketchup OR no-salt-added tomato paste	1/2 cup ketchup or 3 tbsp paste + 1/4 cup water	Tomato paste is more concentrated — either works
Brown sugar	2 tbsp (packed)	The KC-style sweetness
Apple cider vinegar	1 tbsp	Tang — the soul of BBQ sauce
Smoked paprika	1 tsp	Smoke without the sodium of liquid smoke
Garlic powder	1/2 tsp	
Onion powder	1/2 tsp	
Worcestershire sauce	1/2 tsp	Just a whisper — Worcestershire has sodium but 1/2 tsp across the whole batch is negligible
Black pepper	1/4 tsp	
Cayenne	Pinch	Optional heat

Whisk all together in a small saucepan, simmer 5-8 min until thickened slightly. Makes enough for 4-5 burritos.

Method: Same as Recipe 15. Spoon ~3 tbsp turkey down the center, drizzle 2 tbsp BBQ sauce, add 2 tbsp cream cheese. Fold, roll, wrap in foil. **Bake at 375F for 20 min OR microwave unwrapped 2 min**, flipping halfway.

Per serving (1 burrito): ~360mg Na | ~155mg P | ~280mg K

17. Reeser's Bean & Cheese Burrito

"The bean and cheese is the oldest burrito format in the Americas — pre-Columbian Mesoamerican people wrapped beans in corn flatbreads centuries before flour tortillas existed. Reeser's just figured out how to freeze it and sell it at a Shell station."

The original: Refried beans, processed cheese, flour tortilla. ~780mg sodium, ~300mg phosphorus, ~450mg potassium. **The dupe:** ~340mg Na, ~150mg P, ~310mg K

Component	Original	CKD Dupe
Tortilla	10" flour tortilla	10" low-sodium flour tortilla
Beans	Refried beans (high Na, high K, high P)	Triple-rinsed pinto beans, mashed (see K-management note below)
Cheese	Processed cheese food	Cream cheese (2 tbsp) + small amount of shredded cheddar (~1 tbsp) for flavor contrast

Bean Prep — K-Management Protocol:

1. Start with canned pinto beans (low-sodium if available, but regular works because you're rinsing)
2. Drain ALL liquid — the canning liquid is where most of the sodium hides
3. Rinse under cold running water in a colander for 30 seconds
4. Soak in fresh cold water for 5 minutes
5. Drain and rinse again — two more times (three total rinses)
6. **This triple-rinse removes approximately 40% of potassium and 40-50% of sodium**
7. Mash beans in a skillet with 1 tsp avocado oil + 1/2 tsp cumin + 1/4 tsp garlic powder. Add water 1 tbsp at a time to reach refried consistency

CKD Note on Beans: Beans are the most misunderstood renal food. The National Kidney Foundation updated guidance — properly prepared beans CAN fit a renal diet. The triple-rinse + portion control (1/4 cup per burrito) keeps K and P in the safe zone. Don't skip the rinse steps.

Method: Spoon ~1/4 cup mashed beans down the center. Add 2 tbsp cream cheese in dollops + sprinkle of cheddar. Fold, roll, wrap in foil. **Bake at 375F for 20 min OR microwave unwrapped 2 min**, flipping halfway.

Per serving (1 burrito): ~340mg Na | ~150mg P | ~310mg K

MEAL PREP: Reeser's Freezer Stash

"Reeser's whole business model was: make it once, freeze it, eat it later. Your kitchen can do the same thing — minus the phosphate preservatives."

Batch Size: 20 burritos (mix and match any of Recipes 15-17)

What You Need: - 20 low-sodium flour tortillas (10") - 2 lbs ground turkey (for ~12-14 meat burritos) - 2 cans pinto beans (for ~6-8 bean burritos) - 1 block cream cheese (8 oz = 16 tbsp, covers all 20) - 1 batch red chile sauce + 1 batch BBQ sauce (or double one if you have a favorite) - 20 sheets of aluminum foil (~12" squares) - Gallon freezer bags or a large freezer bin - A Sharpie

Assembly Line Method:

1. **Cook all protein first.** Brown 2 lbs turkey in a large skillet, drain, season with Master Taco Seasoning. Set aside. Prep beans (triple-rinse, mash, season) in a second skillet. Set aside.
2. **Make both sauces.** Red chile and BBQ — 10 min total.
3. **Set up the line:** Tortillas laid out flat (work in batches of 5) → protein/beans → sauce → cream cheese → fold → wrap.
4. **Fill each burrito:** ~3 tbsp filling + ~2 tbsp sauce + ~2 tbsp cream cheese. Don't overfill — they need room to fold without tearing.
5. **Roll tight:** Bottom up, sides in, roll away from you. Tight = no air pockets = no freezer burn.
6. **Wrap in foil:** Place burrito seam-down on foil square, wrap snugly. Fold foil ends under.
7. **Label with Sharpie:** Write the flavor + date directly on the foil. ("RH 4/12" = Red Hot, April 12th. "BBQ 4/12". "B&C 4/12".)
8. **Flash freeze:** Lay flat on a sheet pan in a single layer, freeze 2 hours until solid.
9. **Store:** Transfer to gallon freezer bags or stack in a freezer bin. Remove as much air as possible from bags.

Reheat Instructions:

Method	From Frozen	From Fridge
Oven (best texture)	Remove foil. Place on baking sheet. 375F for 25-30 min, flipping once at 15 min. Tortilla gets slightly crispy.	375F for 15-18 min
Microwave (fastest)	Remove foil (!). Wrap in damp paper towel. Microwave 2.5-3 min, flipping halfway. Let stand 1 min.	1.5-2 min
Air Fryer	Remove foil. 350F for 12-15 min, flipping once.	350F for 8-10 min

Freezer Life: 3 months at 0F or below. After 3 months they're still safe but quality drops — the tortilla gets tough and the filling dries out. Label your date and rotate stock.

Thaw Option: Move from freezer to fridge the night before — thaw overnight, then reheat from fridge temps for faster/more even heating.

Yield per batch: - ~60 min total prep + assembly - 20 burritos = roughly 2-3 weeks of grab-and-go meals - Total cost: ~\$15-20 for 20 burritos (\$0.75-1.00 each vs. \$1.50-2.00 at the gas station, and yours won't destroy your kidneys)

RENAL CHECK — Reeser's Burrito Dupes (Recipes 15-17)
K (Potassium): LOW-MED (triple-rinsed beans, no raw tomato, cream cheese over processed cheese)
P (Phosphorus): LOW (cream cheese base, no phosphate additives — the additives in gas station burritos are the real P danger)
Na (Sodium): LOW (low-sodium tortillas, no-salt-added tomato sauce, homemade seasoning throughout)
Fluid: Minimal fluid content — these are grab-and-go solid meals
Protein: MODERATE (turkey is quality renal protein, beans add plant protein without the P load of processed meat)

"The Drive-Thru closed. Your kitchen opened."

Baja Blast Zero + Chicken Chalupa + Grilled 5-Layer Burrito + Crunchy Taco

Every sauce. Every component. Every ingredient. Built for the renal kitchen.

"\$7 at Taco Bell could put you in the ER. \$7 at home keeps you alive AND full."

THE ORIGINAL \$7 LUXE BOX PROBLEM

Item	Sodium	Phosphorus	Potassium
Chicken Chalupa Supreme	~800mg Na	~350mg P	~400mg K
Beefy 5-Layer Burrito	~1,100mg Na	~400mg P	~500mg K
Crunchy Taco	~310mg Na	~200mg P	~200mg K
Baja Blast (medium)	~65mg Na	~50mg P (phosphoric acid)	~10mg K
BOX TOTAL	~2,275mg Na	~1,000mg P	~1,110mg K

That's MORE than the entire daily sodium limit in ONE MEAL.

THE HOMEMADE DUPE: EVERY COMPONENT

ITEM 1: Chicken Chalupa Supreme

The Chalupa Shell

Ingredient	Amount	CKD Notes
All-purpose flour	1 cup	Fine
Baking powder	1 tsp	Use PHOSPHORUS-FREE (Rumford aluminum-free)
Salt	1/4 tsp	Minimal — the original shell has ~400mg Na alone
Warm water	1/3 cup	Counts toward fluid

Ingredient	Amount	CKD Notes
Oil for frying	Enough to shallow-fry	Canola or vegetable

Method: Mix flour + baking powder + salt. Add warm water, knead 2 min until smooth. Rest 15 min. Divide into 4 balls. Roll each into oval, ~6". Fry in 1" oil at 350F for ~45 seconds per side until puffed and golden. Drain on paper towels. Shape over a rolling pin while warm to form the taco-boat shape.

The Chicken Filling

Ingredient	Amount	CKD Notes
Chicken breast	1 large, diced small	Good renal protein
Homemade taco seasoning	1.5 tbsp (see master recipe)	NO PACKET — packets are 500mg+ Na each
Water	2 tbsp	To simmer seasoning into chicken
Oil	1 tsp	For cooking

Method: Dice chicken into 1/2" pieces. Cook in oil over medium-high 4-5 min. Add seasoning + water, stir until absorbed and chicken is coated.

The Toppings

Topping	Amount	CKD Notes
Lettuce	1/4 cup shredded iceberg	Low everything — safe
Sour cream	1 tbsp	Fine in small amounts
Cheese	1 tbsp shredded cheddar	Small portion keeps P low
Tomato	1 tbsp diced	OR skip and use diced bell pepper — tomato = K

Chalupa Assembly

Shell → chicken → sour cream → cheese → lettuce → tomato/pepper

Per chalupa: ~180mg Na, ~150mg P, ~200mg K (vs original's 800/350/400)

ITEM 2: Grilled Five-Layer Burrito

The Components

Layer	Ingredient	Amount	CKD Notes
Tortilla	Low-sodium flour tortilla (10")	1	Mission low-sodium has ~300mg Na vs standard ~600mg
Layer 1: Beef	Ground turkey + taco seasoning	2 oz	Turkey is lower P than beef
Layer 2: Beans	SKIP	—	Beans are very high K + P. Replace with:
Layer 2 ALT: Rice	White rice, cooked	2 tbsp	Low-K, acts as the "filler" layer
Layer 3: Sour cream	Regular sour cream	1.5 tbsp	The creamy layer
Layer 4: Cheese	Nacho cheese sauce (homemade)	1 tbsp	See sauce recipes below
Layer 5: Cheese	Shredded cheddar	1 tbsp	Small amount

Assembly & Grilling

1. Lay tortilla flat
2. Spread sour cream across center third
3. Layer: rice → meat → nacho cheese → shredded cheese
4. Fold bottom up, sides in, top down (burrito fold)
5. Place seam-down in dry skillet or panini press
6. Grill 2 min per side until golden brown and cheese melts
7. Cut in half diagonally

Per burrito: ~380mg Na, ~180mg P, ~250mg K (vs original's 1,100/400/500)

ITEM 3: Crunchy Taco

The Shell

Option	CKD Notes
Store-bought hard taco shell	~120mg Na per shell — acceptable
OR: Fry your own corn tortilla	Lower sodium, crispier

The Filling

Component	Amount	CKD Notes
Turkey taco meat	2 oz	Same seasoning as above
Lettuce	2 tbsp shredded iceberg	
Cheese	1 tbsp shredded cheddar	Small portion
Sour cream	1 tsp	Optional
Hot sauce	Dash	See CKD hot sauce below

Per taco: ~200mg Na, ~120mg P, ~150mg K (vs original's 310/200/200 — close but still lower)

ITEM 4: Baja Blast Zero

Ingredient	Amount	CKD Notes
Sprite Zero	12 oz	No phosphoric acid (Mountain Dew HAS phosphoric acid — that's why we sub)
Blue food coloring	2 drops	
Green food coloring	1 drop	
Lime juice	1 tsp	That citrus tang
Pineapple extract	2-3 drops	OR 1 tbsp pineapple juice (adds ~15mg K)
Ice	Fill cup	

Method: Mix Sprite Zero + food coloring + lime + pineapple in glass with ice. Stir. That's it. Looks and tastes like Baja Blast, zero phosphoric acid.

Per serving: ~40mg Na, ~0mg P (vs Dew's ~50mg phosphoric acid P), ~15mg K

Fluid note: This is 12+ oz of fluid. Counts toward daily limit.

ITEM 5: Cinnamon Twists Dupe

The important trick: use Mexican *duros de harina* or wheat twist pellets. Do not use dry rotini pasta; pasta stays hard and unpleasant. Duros puff in hot oil in seconds.

Ingredient	Amount	CKD Notes
Duros de harina / wheat twist pellets	1 cup dry pellets	Check label; choose low-sodium pellets with no phosphate additives.
Neutral oil	2 inches in a small pot	Avocado, canola, or vegetable oil.
Granulated sugar	1/3 cup	Coating.
Ceylon cinnamon	1 1/2 tsp	Cinnamon Cloud crossover; use Ceylon for the house standard.
Fine salt	Pinch optional	Optional; skip for tighter sodium days.

1. Mix sugar, cinnamon, and optional tiny pinch of salt in a large bowl.
2. Heat oil to 350-365F. Test one pellet: it should puff within 3-5 seconds.
3. Fry a small handful at a time. They expand hard and fast, so do not crowd the pot.
4. Remove as soon as fully puffed, usually 10-20 seconds. Drain briefly on a rack or paper towel.
5. While still hot, toss in cinnamon sugar until every ridge is coated.
6. Cool completely before bagging. Store airtight up to 3 days.

Air fryer note: raw duros are unreliable in an air fryer. If avoiding deep frying, buy plain puffed duros/chicharrones de harina and warm them 1-2 minutes before tossing with cinnamon sugar.

Per small handful: sodium depends on pellet label; homemade coating keeps phosphorus additives out and avoids the heavy restaurant sodium load.

ALL THE SAUCES (Lower-Sodium Dupes)

Taco Bell Creamy jalapeño Sauce

Ingredient	Amount
Avocado oil mayo (Primal Kitchen)	3 tbsp
Pickled jalapeño juice	1 tsp (from a jar of jalapeños)
Minced pickled jalapeño	1 tsp
Garlic powder	1/4 tsp
Cumin	Pinch
Cayenne	Pinch
Lime juice	1/2 tsp

Mix and refrigerate 30 min for flavors to meld. Keeps 1 week.

Taco Bell Nacho Cheese Sauce

Ingredient	Amount
Cream cheese	2 oz (1/4 block)
Unsweetened almond milk	3 tbsp
Cheddar cheese, shredded	2 tbsp
Paprika	1/4 tsp
Garlic powder	1/8 tsp
Cayenne	Pinch
Turmeric	Pinch (for yellow color)

Microwave cream cheese 20 sec. Add milk, stir smooth. Add cheddar, microwave 15 sec, stir. Add spices. Adjust thickness with milk.

Why cream cheese base: Velveeta/processed cheese = very high phosphorus + sodium. Cream cheese is lower P and you control the sodium.

Taco Bell Red Sauce

Ingredient	Amount
Tomato paste	1 tbsp (small amount = controlled K)
Water	3 tbsp
Cumin	1/4 tsp
Chili powder	1/4 tsp
Garlic powder	1/8 tsp
Onion powder	1/8 tsp
Cayenne	Pinch
Vinegar	1/2 tsp

Whisk everything together. Simmer 2 min. Thin, pourable, tastes exactly like the packets.

Why small tomato paste: Tomato is high-K, but 1 tbsp of paste across 4 servings = ~30mg K per serving. Controlled.**

Taco Bell Hot Sauce (packet dupe)

Ingredient	Amount
Cayenne pepper	1/2 tsp
Cumin	1/4 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp
Vinegar	2 tbsp
Water	1 tbsp
Tomato paste	1/2 tsp
Sugar	1/4 tsp

Blend or whisk. Store in small squeeze bottle. Dime-sized squirts on tacos.

Taco Bell Chipotle Sauce

Ingredient	Amount
Avocado oil mayo (Primal Kitchen)	2 tbsp
Chipotle pepper in adobo	1 pepper, minced (canned — use one pepper, not the whole can)
Adobo sauce	1/2 tsp
Lime juice	1/2 tsp
Garlic powder	Pinch

Mix. Smoky, creamy, a little spicy. One canned chipotle pepper goes a long way.

Taco Bell Avocado Ranch

Ingredient	Amount
Avocado oil mayo (Primal Kitchen)	2 tbsp
1/4 avocado	Mashed smooth
Lime juice	1 tsp
Garlic powder	1/8 tsp
Dried dill	1/8 tsp
Dried chives	1/8 tsp
Onion powder	Pinch

Mix smooth. Use within 2 days (avocado browns).

K note: 1/4 avocado = ~180mg K. Use sparingly or skip for stricter K limits.

CKD LUXE BOX TOTAL vs ORIGINAL

Nutrient	Original \$7 Luxe Box	CKD Dupe Box	Reduction
Sodium	~2,275mg	~800mg	65% less
Phosphorus	~1,000mg	~450mg	55% less
Potassium	~1,110mg	~615mg	45% less
Cost	\$7.00 + tax + gas	~\$5-6 in ingredients (feeds 2)	Cheaper
Flavor	Taco Bell	95% identical	Worth it

MEAL PREP VERSION

Make a batch on Sunday, eat Taco Bell all week: 1. Cook 2 lbs turkey taco meat with seasoning. Portion into 4 containers. 2. Make all sauces. Store in squeeze bottles. 3. Fry 8 chalupa shells. Store in airtight container with paper towels. 4. Shred lettuce, cheese. Store separately. 5. Make Baja Blast mix (minus the Sprite) as a concentrate. Add Sprite when serving.

Assembly time per meal: 5 minutes. Faster than the drive-thru.

CKD Considerations

RENAL CHECK — \$7 Luxe Box CKD Dupe
K (Potassium): MODERATE (bell pepper over tomato, no beans, controlled avocado)
P (Phosphorus): LOW-MED (cream cheese sauces, no processed cheese, turkey over beef)
Na (Sodium): LOW (no-salt seasoning, low-sodium tortillas, homemade sauces)
Fluid: Baja Blast = 12+ oz fluid — TRACK IT
Protein: GOOD (chicken + turkey = quality renal protein)

"The whole box. Every sauce. Every crunch. Your kidneys don't even know."

10 Wings. 10 Sauces. Zero Compromise.

"The best questions come after the fourth wing." -- Sean Evans, sort of

THE WINGS (Base Recipe)

Baked Crispy Wings -- Universal Base for All 10 Rounds

In 1964, Teresa Bellissimo at the Anchor Bar in Buffalo, NY deep-fried leftover wings and tossed them in cayenne hot sauce, accidentally inventing the "Buffalo wing" -- previously, wings were considered throwaway parts only fit for stock. (Source: The New Yorker, 1980)

Ingredient	Amount	CKD Note
Chicken wing sections (flats + drums)	3 lbs (~24 pieces)	Good protein source; rinse and pat dry
Baking powder (aluminum-free)	1 Tbsp	Low Na; creates crispiness without frying
Garlic powder	1 tsp	Low K alternative to fresh garlic
Onion powder	1 tsp	Lower K than raw onion
Smoked paprika	1 tsp	Negligible minerals
Black pepper	1/2 tsp	Negligible minerals
Olive oil spray	light coating	Heart-healthy fat

Method:

1. Pat wings completely dry with paper towels. Moisture is the enemy of crispiness.
2. Toss wings in a large bowl with baking powder, garlic powder, onion powder, smoked paprika, and black pepper until evenly coated.
3. Arrange wings on a wire rack set over a rimmed baking sheet. Single layer, no touching.
4. Refrigerate uncovered for 1 hour (or up to overnight) to dry the skin further.
5. Preheat oven to 425 F (220 C).
6. Spray wings lightly with olive oil.
7. Bake 25 minutes. Flip each wing. Bake another 20-25 minutes until skin is golden and crackling.
8. Toss 4-6 wings at a time in the sauce of your chosen round.

Yield: ~24 wings (serves 4-6, ~4-6 wings per person)

Per Serving (6 plain wings, no sauce):

Na	P	K
95 mg	145 mg	180 mg

THE LINEUP

Wing #1: "The Handshake" (Mild) -- Frank's RedHot Dupe

~450 Scoville Heat Units

Frank's RedHot was created in 1920 in New Iberia, Louisiana by Adam Estillette and Jacob Frank, using aged cayenne peppers -- the same sauce was used in the original 1964 Anchor Bar Buffalo wing recipe. (Source: McCormick & Company archives)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Cayenne pepper, ground	2 Tbsp	Negligible minerals at this amount
White vinegar	1/2 cup	Na-free acid base
Water	1/4 cup	--
Garlic powder	1/2 tsp	Low K
Unsalted butter	2 Tbsp	Low Na; adds classic buffalo richness
Honey	1 tsp	Low mineral sweetener to round flavor

Method:

1. Whisk cayenne, vinegar, water, and garlic powder in a small saucepan over medium heat.
2. Bring to a simmer and cook 5 minutes, stirring occasionally.
3. Remove from heat. Whisk in butter until melted and emulsified.
4. Stir in honey. Let cool 5 minutes before tossing with wings.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
5 mg	12 mg	35 mg

Wing #2: "Smoky Crossroads" (Mild-Medium) -- Cholula Dupe

~1,000 Scoville Heat Units

Cholula hot sauce is named after the city of Cholula, Puebla, Mexico -- a city continuously inhabited for over 2,500 years, making it one of the oldest still-living cities in the Americas. Its Great Pyramid has a larger base than Giza's. (Source: INAH, Mexico)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Arbol chiles, dried	6 whole	Low Na heat source
Piquin chiles, dried	1 Tbsp	Concentrated heat, negligible minerals
White vinegar	1/3 cup	Na-free
Water	1/4 cup	--
Cumin, ground	1/2 tsp	Negligible minerals
Oregano, dried	1/2 tsp	Negligible minerals
Garlic powder	1/2 tsp	Low K
Onion powder	1/4 tsp	Low K

Method:

1. Toast arbol and piquin chiles in a dry skillet over medium heat for 60 seconds, turning frequently. Remove stems.
2. Soak toasted chiles in hot water for 15 minutes. Drain.
3. Blend soaked chiles with vinegar, water, cumin, oregano, garlic powder, and onion powder until smooth.
4. Strain through a fine-mesh sieve for a pourable consistency.
5. Simmer strained sauce in a small pot for 5 minutes to meld flavors.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
3 mg	10 mg	45 mg

Wing #3: "Red Line" (Medium) -- Tapatio Dupe

~3,000 Scoville Heat Units

Tapatio hot sauce was created in 1971 by Jose-Luis Saavedra, Sr. in a small warehouse in Maywood, California. The name "Tapatio" is a colloquial term for people from Guadalajara, Jalisco -- Saavedra's hometown. It remained family-owned for over 50 years. (Source: Los Angeles Times, 2011)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Arbol chiles, dried	10 whole	Low Na heat source
Guajillo chiles, dried	2 whole	Mild, fruity backbone; low Na
White vinegar	1/3 cup	Na-free
Water	3 Tbsp	--
Garlic powder	1 tsp	Low K
Cumin, ground	1/4 tsp	Negligible
Oregano, dried	1/4 tsp	Negligible
Smoked paprika	1/4 tsp	Negligible

Method:

1. Remove stems and most seeds from arbol and guajillo chiles.
2. Toast in a dry skillet 60-90 seconds. Soak in hot water 15 minutes. Drain.
3. Blend chiles with vinegar, water, garlic powder, cumin, oregano, and smoked paprika.
4. Strain through a fine-mesh sieve.
5. Simmer 5 minutes in a small saucepan. It should coat a spoon.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
4 mg	14 mg	55 mg

Wing #4: "Rooster Walk" (Medium-Hot) -- Sriracha Dupe

~2,200 Scoville Heat Units

Sriracha sauce is named after Si Racha, a coastal town in Chonburi Province, Thailand. David Tran, a Vietnamese-Chinese refugee, began bottling his version in Los Angeles in 1980 using a van. His rooster logo represents his Chinese zodiac birth year. (Source: Bon Appetit, 2009)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Red Fresno chiles, fresh	8 oz (~8 chiles)	Fresh chiles = lower Na than bottled sauce
Garlic cloves	4	Moderate K; small amount per serving
White vinegar	1/4 cup	Na-free
Honey	1 Tbsp	Low mineral sweetener
Water	2 Tbsp	--

Method:

1. Stem and roughly chop Fresno chiles (keep seeds for heat).
2. Blend chiles, garlic, vinegar, honey, and water until a coarse paste forms.
3. Transfer to a saucepan. Simmer on low heat 10 minutes, stirring often.
4. Let cool slightly, then blend again until smooth.
5. Strain if you want classic smooth texture, or leave chunky for rustic style.
6. Rest in the fridge 24 hours for flavors to marry -- this step matters.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
6 mg	18 mg	70 mg

Wing #5: "Habanero Sunrise" (Hot) -- Tabasco Habanero Dupe

~7,000 Scoville Heat Units

The Tabasco pepper gets its name from the Mexican state of Tabasco, but the sauce has been produced exclusively on Avery Island, Louisiana since 1868. Avery Island is actually a salt dome -- a geological column of rock salt pushing up through the earth -- which the McIlhenny family has mined since the Civil War. (Source: Smithsonian Magazine, 2007)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Habanero peppers, fresh	4 (about 2 oz)	Extreme heat; wear gloves
Mango, fresh	1/2 cup diced	Moderate K -- portion controlled
White vinegar	1/4 cup	Na-free
Lime juice, fresh	2 Tbsp	Low Na acid; moderate K in small qty
Honey	1 Tbsp	Low mineral sweetener
Garlic powder	1/2 tsp	Low K
Ground mustard	1/4 tsp	Negligible

Method:

1. Stem and halve habaneros (keep seeds for full heat, remove some to dial back).
2. Roast habaneros and mango under the broiler for 5-7 minutes until charred.
3. Blend roasted habaneros, mango, vinegar, lime juice, honey, garlic powder, and mustard until smooth.
4. Simmer in a saucepan 5 minutes.
5. Strain for a smooth, pourable hot sauce.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
4 mg	10 mg	85 mg

Wing #6: "Black Label Smoke" (Hotter) -- El Yucateco Dupe

~11,600 Scoville Heat Units

El Yucateco was founded in 1968 in Merida, Yucatan by Priamo Gamboa. Its distinctive green and black label sauces use habaneros grown in the Yucatan Peninsula, where the chile is native -- archaeologists have found domesticated habanero seeds in Yucatecan sites dating to 8,500 years ago. (Source: Proceedings of the National Academy of Sciences, 2014)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Habanero peppers, fresh	6 (about 3 oz)	Wear gloves; ventilate kitchen
Chipotle chile in adobo	1 chile only	Watch Na -- use only the chile, not the adobo liquid
White vinegar	1/4 cup	Na-free
Lime juice, fresh	1 Tbsp	Low Na
Garlic powder	1/2 tsp	Low K
Cumin, ground	1/4 tsp	Negligible
Smoked paprika	1 tsp	Smoky depth
Water	2 Tbsp	--

Method:

1. Char habaneros in a dry cast iron skillet over high heat, turning, until blistered on all sides (~5 minutes).
2. Stem the charred habaneros.
3. Blend habaneros, chipotle chile, vinegar, lime juice, garlic powder, cumin, smoked paprika, and water.
4. Strain through a fine-mesh sieve, pressing with a spoon to extract all liquid.
5. Simmer 5 minutes. The color should be deep brownish-red with visible smoke wisps.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
25 mg	12 mg	75 mg

Wing #7: "Aardvark's Garden" (Very Hot) -- Secret Aardvark Dupe

~15,000 Scoville Heat Units

Secret Aardvark Habanero Hot Sauce was born in Portland, Oregon in 2004, created by Scott Moritz. It became a cult favorite at Portland food carts before going national. The name came from a childhood nickname -- Moritz's friends called him "Aardvark" because he had a long nose. (Source: Portland Mercury, 2016)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Habanero peppers, fresh	6 (about 3 oz)	Wear gloves
Roasted red bell pepper	1 medium	Moderate K; roasting concentrates flavor
White vinegar	1/4 cup	Na-free
Mustard powder	1 tsp	Secret Aardvark's signature tang
Cumin, ground	1/2 tsp	Negligible
Black pepper	1/4 tsp	Negligible
Garlic powder	1 tsp	Low K
Onion powder	1/2 tsp	Low K
Honey	2 tsp	Balances heat with sweetness

Method:

1. Roast bell pepper directly over a gas flame or under broiler until fully charred. Steam in a covered bowl 10 minutes, then peel and seed.
2. Stem and halve habaneros.
3. Blend roasted bell pepper, habaneros, vinegar, mustard powder, cumin, black pepper, garlic powder, onion powder, and honey until very smooth.

4. Simmer in a saucepan 10 minutes on low, stirring frequently.
5. Do not strain -- the roasted pepper body is what gives this sauce its signature thickness.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
8 mg	16 mg	95 mg

Wing #8: "Caliente Doble" (Extremely Hot) -- Hot Ones Los Calientes Dupe

~36,000 Scoville Heat Units

"Hot Ones" debuted on YouTube's First We Feast channel in March 2015. Host Sean Evans intentionally chose to avoid celebrity-friendly softball questions, instead using progressively hotter wings to break down guests' media-trained composure. The show has generated over 4 billion total views. (Source: The New York Times, 2019)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Habanero peppers, fresh	6 (about 3 oz)	High heat base
Serrano peppers, fresh	4	Medium heat layering
Passionfruit pulp (frozen)	3 Tbsp	Moderate K; tropical tang
Pineapple, fresh	2 Tbsp minced	Moderate K in small amount
White vinegar	3 Tbsp	Na-free
Lime juice, fresh	1 Tbsp	Low Na
Garlic powder	1 tsp	Low K
Smoked paprika	1/2 tsp	Depth
Ground cumin	1/4 tsp	Negligible

Method:

1. Stem and roughly chop habaneros and serranos.
2. Dry-roast all chiles in a cast iron skillet until blistered.
3. Blend roasted chiles with passionfruit pulp, pineapple, vinegar, lime juice, garlic powder, smoked paprika, and cumin.
4. Simmer 10 minutes in a saucepan. The fruit sugars will caramelize slightly.
5. Blend again for a silky finish. Strain only if you want a thinner pour.

Per Serving (sauce only, ~2 Tbsp):

Na	P	K
5 mg	14 mg	110 mg

Wing #9: "Beyond Reason" (Savage) -- Da Bomb Beyond Insanity Dupe

~135,600 Scoville Heat Units

Da Bomb Beyond Insanity, made by Original Juan Specialty Foods in Kansas City, has become the most feared sauce on "Hot Ones" -- nearly every guest breaks down at this stage. The sauce uses habanero pepper extract, a concentrated capsaicin isolate, which is why it hits harder than sauces with higher Scoville ratings that use whole peppers. (Source: First We Feast, 2020)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Habanero peppers, fresh	8 (about 4 oz)	Maximum whole-pepper heat
Cayenne pepper, ground	2 Tbsp	Concentrated burn
Ghost pepper powder (bhut jolokia)	1 tsp	Extreme heat -- measure carefully
White vinegar	1/4 cup	Na-free
Garlic powder	1 tsp	Low K
Chipotle powder	1/2 tsp	Smoky undertone
Unsweetened apricot preserves	1 Tbsp	Low Na fruit binder; moderate K
Water	2 Tbsp	--

Method:

1. Stem habaneros. For maximum heat, keep all seeds and membranes.
2. Blend habaneros, cayenne, ghost pepper powder, vinegar, garlic powder, chipotle powder, apricot preserves, and water until completely smooth.
3. Simmer on low 15 minutes. The kitchen will become hostile -- run the vent hood.
4. Strain through a fine-mesh sieve for a dense, extract-like consistency.
5. Let cool fully before use. This sauce should coat a spoon and drip slowly.

WARNING: This sauce is not for casual consumption. Taste with a toothpick first.

Per Serving (sauce only, ~1 Tbsp -- you will not want more):

Na	P	K
4 mg	10 mg	65 mg

Wing #10: "The Last Prayer" (Extinction Level) -- The Last Dab Dupe

~2,000,000+ Scoville Heat Units

Pepper X, bred by Ed Currie of PuckerButt Pepper Company, was confirmed by Guinness World Records in October 2023 as the world's hottest pepper at 2.69 million Scoville Heat Units, dethroning Currie's own Carolina Reaper (1.64M SHU). Currie spent over 10 years crossbreeding peppers to create it. He reportedly cramped for over 3 hours after eating one raw. (Source: Associated Press, October 2023)

Sauce Ingredients:

Ingredient	Amount	CKD Note
Carolina Reaper peppers, dried	3 whole	Handle with nitrile gloves only
Habanero peppers, fresh	4	Supporting heat floor
Ghost pepper powder (bhut jolokia)	2 tsp	Layered superhot
White vinegar	1/4 cup	Na-free
Mustard powder	1/2 tsp	Tang and complexity
Garlic powder	1 tsp	Low K
Turmeric, ground	1/4 tsp	Inflammation-support; golden color
Black pepper	1/4 tsp	Activates turmeric's curcumin
Ginger, ground	1/4 tsp	Low K
Honey	2 tsp	Last-dab sweetness
Water	2 Tbsp	--

Method:

1. Rehydrate dried Carolina Reapers in hot water for 20 minutes. Drain. Remove stems only -- keep every seed.
2. Stem and halve fresh habaneros.
3. Blend rehydrated Reapers, habaneros, ghost pepper powder, vinegar, mustard powder, garlic powder, turmeric, black pepper, ginger, honey, and water.
4. Simmer on low heat 15 minutes. Do not lean over the pot. Do not breathe directly above it. Seriously.
5. Blend once more until completely homogenous.

CHAPTER 5

Southeast Asian

"The wok doesn't ask questions."

Tom Kha, Pad Thai, Pad Gra Prao, Papaya Salad, Beef Salad, Spring Rolls, Pho, Banh Mi

Southeast Asian Comfort Food

"The food cart vibes. The clinic-approved numbers."

6. Strain for a velvety, neon-orange sauce.
7. Apply with a toothpick or the back of a spoon -- a "dab" is the correct serving unit.

WARNING: This sauce is weapons-grade. Have plain rice, bread, and full-fat yogurt standing by. Capsaicin is fat-soluble, not water-soluble -- water will make it worse.

Per Serving (sauce only, ~1 tsp -- that is plenty):

Na	P	K
2 mg	5 mg	20 mg

CKD Considerations

Capsaicin and Kidneys -- The Good News

Capsaicin (the compound that makes peppers hot) has shown potential studied in kidney-related research properties in peer-reviewed research. A 2012 study in the Journal of Agricultural and Food Chemistry found capsaicin may reduce renal oxidative stress. Hot peppers are also naturally very low in sodium, potassium, and phosphorus per serving -- making them one of the most potentially compatible with an individualized CKD plan flavor amplifiers available.

Key Reminders:

- **Sodium:** Every sauce in this lineup is homemade specifically to avoid the 200-500+ mg Na per serving found in commercial hot sauces. None exceed 25 mg Na per serving.
- **Potassium:** Wing #8 (Caliente Doble) has the highest K at 110 mg per serving due to passionfruit and pineapple. If you are on strict K restriction, swap the fruit for extra vinegar and a pinch of citric acid.
- **Phosphorus:** All sauces stay well under 50 mg P per serving. The wings themselves are the primary P source (~145 mg for 6 wings). Track your daily total accordingly.
- **Portion control is built in:** The hotter the sauce, the less you use. Wing #10 is measured in teaspoons, not tablespoons.
- **Hydration:** Spicy food increases thirst. If you are on a fluid restriction, have your allotted fluids nearby and sip -- do not chug.
- **Acid reflux:** If you experience GERD, start at Wing #1 and respect your ceiling. There is no shame in a five-wing streak.
- **No commercial sauces were used.** Every recipe avoids the hidden sodium, phosphate preservatives, and potassium chloride found in store-bought hot sauces.

Daily Targets (general CKD guidance -- individual plans vary):

Nutrient	Daily Target
Sodium (Na)	< 2,000 mg
Potassium (K)	2,000-3,000 mg (varies by stage)
Phosphorus (P)	800-1,000 mg

"If you can't stand the heat, get out of the kitchen -- but if you built the kitchen yourself, you set the thermostat." -- a thing someone should have said to Harry Truman

Wilbur Scoville, the pharmacist who invented the Scoville Organoleptic Test in 1912, measured pepper heat by diluting chile extract with sugar water until a panel of five tasters could no longer detect the burn. The entire modern hot sauce industry rests on five people's tongues from 112 years ago. (Source: Journal of the American Pharmaceutical Association, 1912)

CKD NOTES FOR THAI/VIETNAMESE COOKING

The good news: Thai and Vietnamese cuisines are NATURALLY closer to CKD-aware than most Western food: - Rice-based (low K, low P) - Fresh herbs (cilantro, basil, mint = almost zero K/P/Na) - Lime juice instead of salt for brightness - Protein is portion-controlled by design (small amounts of meat, lots of vegetables/noodles)

The traps: - **Fish sauce** = VERY high sodium (~1,400mg Na per tbsp). Use sparingly, dilute, or sub with low-sodium soy sauce + lime - **Soy sauce** = high sodium (~900mg Na per tbsp). Use low-sodium version - **Tomatoes** in some dishes = high K - **Coconut water** = high K (but coconut CREAM/MILK is fine) - **Bean sprouts** = moderate K — use small amounts

- **Peanuts** = moderate P — garnish only, not handfuls - **Potassium in broth** — bone broth concentrates K. Use homemade with controlled simmer time.

The CKD sauce trinity:

Sauce	Original	CKD Swap
Fish sauce	1 tbsp = 1,400mg Na	1 tsp low-sodium soy + 1/2 tsp lime juice + pinch sugar
Soy sauce	1 tbsp = 900mg Na	Low-sodium soy sauce (Kikkoman low-sodium = 575mg/tbsp) — use 1 tsp max
Oyster sauce	1 tbsp = 490mg Na	1 tsp oyster sauce + 1 tsp water (dilute by half)

THE RECIPES

1. Lime-Coconut Tom Kha Gai

Andrew's comfort soup — creamy, sour, aromatic, warming

Ingredient	Amount	CKD Notes
Coconut milk (full-fat canned)	1 can (13.5oz)	Fine — coconut MILK/CREAM is low K
Chicken breast	6 oz, sliced thin	Good renal protein
Galangal (or ginger)	6 slices	Inflammation-support, low everything
Lemongrass	2 stalks, smashed and cut into 2" pieces	Aromatic, zero K/P/Na
Kaffir lime leaves	4-5 leaves	Aromatic
Mushrooms (white button)	1/2 cup sliced	Low-K mushroom variety
Cherry tomatoes	4-5, halved	Small amount = controlled K
Thai chilies	2-3, smashed	Heat without sodium
Lime juice	2-3 tbsp	THE sour element — replaces some fish sauce need
Fish sauce sub	1 tsp low-sodium soy + 1/2 tsp lime + pinch sugar	~190mg Na vs 1,400mg
Sugar	1 tsp	Balances sour
Cilantro	Garnish	
Water or chicken broth	1 cup	Low-sodium broth if using

Method: 1. Bring coconut milk + water to gentle simmer 2. Add galangal, lemongrass, lime leaves, chilies. Simmer 5 min (infuse aromatics) 3. Add chicken. Cook 5 min until white throughout 4. Add mushrooms + tomatoes. Cook 2 min 5. Remove from heat. Add lime juice + fish sauce sub + sugar 6. Taste — adjust sour (more lime) and salt (tiny bit more soy) to preference 7. Garnish with cilantro, serve with jasmine rice

Per serving (2 servings): ~250mg Na, ~180mg P, ~350mg K. Safe and gorgeous.

2. Tamarind Street Pad Thai

Ingredient	Amount	CKD Notes
Rice noodles (flat, dried)	4 oz	Soak in room-temp water 30 min. LOW-K.
Chicken or shrimp	4 oz	
Egg	1	Moderate P — one egg per batch is fine
Bean sprouts	Small handful	Moderate K — small amount
Green onion	2 stalks, cut into 1" pieces	Low K
Garlic	2 cloves, minced	
Oil	2 tbsp	

Pad Thai Sauce (CKD version):

Ingredient	Amount	CKD Notes
Tamarind paste	1 tbsp	The REAL Pad Thai flavor — low K/P/Na
Sugar	1 tbsp	

Ingredient	Amount	CKD Notes
Low-sodium soy sauce	2 tsp	~380mg Na (vs 3 tbsp fish sauce = 4,200mg)
Lime juice	1 tbsp	
Rice vinegar	1 tsp	
Sriracha or chili flakes	To taste	

Method: 1. Mix sauce ingredients in small bowl 2. Drain soaked noodles 3. Heat oil in wok over high heat. Cook protein 3-4 min, remove 4. Scramble egg in same wok, break up 5. Add noodles + sauce. Toss 2-3 min until noodles are soft and coated 6. Add protein back, bean sprouts, green onion. Toss 1 min 7. Garnish: crushed peanuts (1 tbsp — P control), lime wedge, cilantro

Per serving: ~400mg Na, ~200mg P, ~300mg K

3. Holy Basil Chicken Stir-Fry

The street food king — fast, spicy, aromatic

Ingredient	Amount	CKD Notes
Ground chicken or turkey	8 oz	
Thai basil leaves	1 packed cup	Holy basil if you can find it — zero K/P concern
Thai chilies	3-5, minced	Adjust heat to preference
Garlic	4 cloves, minced	
Shallot	1 small, sliced	Low K
Green beans	1/2 cup, cut into 1" pieces	Low K vegetable
Oil	2 tbsp	

Sauce:

Ingredient	Amount
Low-sodium soy sauce	1 tbsp
Oyster sauce	1 tsp
Sugar	1 tsp
Water	1 tbsp

Method: 1. Wok screaming hot + oil 2. Garlic + chilies + shallot — 30 seconds until fragrant 3. Add ground meat — break up, cook 3-4 min until browned 4. Add green beans + sauce. Stir-fry 2 min 5. Kill heat. Toss in basil leaves. They wilt in 10 seconds from residual heat 6. Serve over jasmine rice with a fried egg on top (optional)

Per serving: ~380mg Na, ~190mg P, ~280mg K

4. Green Papaya Crunch Salad

Ingredient	Amount	CKD Notes
Green papaya, shredded	2 cups	GREEN (unripe) papaya is LOW-K. RIPE papaya is HIGH-K. Green only!
Carrot, shredded	1/4 cup	Small amount = controlled K
Green beans	4-5, cut into 1" pieces	Low K
Cherry tomatoes	4, quartered	Small amount
Thai chilies	2-3, to taste	
Garlic	2 cloves	
Lime juice	2 tbsp	
Low-sodium soy sauce	1 tsp	Sub for fish sauce
Sugar	1 tbsp	
Peanuts	1 tbsp, crushed	Garnish only — P control
Dried shrimp	1 tsp (optional)	Tiny amount for umami

Method: Pound garlic + chilies in mortar. Add green beans + tomatoes, lightly bruise. Add papaya + carrot. Add lime + soy sub + sugar. Toss. Top with peanuts.

5. Waterfall Beef Salad

Ingredient	Amount	CKD Notes
Flank steak or sirloin	6 oz	Grill medium-rare, slice thin against grain
Shallot	2 small, sliced thin	Low K
Cilantro	1/4 cup	
Mint leaves	1/4 cup	
Green onion	2 stalks, sliced	
Toasted rice powder	1 tbsp	Toast dry rice in pan, grind — the SECRET ingredient
Thai chilies	2, sliced	

Dressing:

Ingredient	Amount
Lime juice	3 tbsp
Low-sodium soy sauce	1 tbsp
Sugar	1 tsp
Chili flakes	1/2 tsp

Method: Grill steak. Rest 5 min. Slice thin. Toss with shallot, herbs, dressing, rice powder. Serve over rice or lettuce cups.

The toasted rice powder is what makes this dish. It adds a nutty, smoky crunch that's unmistakable.

6. Fresh Spring Rolls

Ingredient	Amount	CKD Notes
Rice paper wrappers	8 sheets	Soak one at a time in warm water 15 sec
Rice vermicelli	4 oz, cooked	Low K
Shrimp	8 medium, cooked, halved lengthwise	OR chicken breast, sliced
Lettuce	8 leaves	
Fresh herbs	Mint, cilantro, Thai basil — a few leaves per roll	
Cucumber	1/2, cut into matchsticks	Low K
Carrot	1 small, julienned	Small amount

Peanut Dipping Sauce (CKD version):

Ingredient	Amount
Peanut butter	2 tbsp
Hoisin sauce	1 tbsp (moderate Na — use sparingly)
Water	2 tbsp
Lime juice	1 tsp
Sriracha	1/2 tsp
Crushed peanuts	1 tsp garnish

Assembly: Wet rice paper. Lay lettuce, noodles, protein, herbs, cucumber, carrot in center. Fold bottom up, sides in, roll tight. Serve with peanut sauce.

7. Clear-Broth Pho Bo

Ingredient	Amount	CKD Notes
Rice noodles (banh pho, flat)	4 oz dry	Low K base
Beef sirloin	4 oz, sliced paper-thin	Slice while partially frozen for thin cuts
Broth (the key):		
Water	6 cups	NOT bone broth (concentrates K). Build flavor with spices.
Low-sodium beef broth	2 cups	Or 2 tsp beef Better Than Bouillon low-sodium
Star anise	2 whole	THE pho spice
Cinnamon stick	1	

Ingredient	Amount	CKD Notes
Cloves	3	
Ginger	2" piece, charred under broiler	Char = deeper flavor
Onion	1/2, charred	Char it
Fish sauce sub	2 tsp low-sodium soy + 1 tsp lime	
Sugar	1 tsp	

Garnish plate:

Item	CKD Notes
Bean sprouts (small amount)	Moderate K
Thai basil	Safe
Lime wedges	Safe
jalapeño slices	Safe — capsaicin is fine
Cilantro	Safe
Hoisin + sriracha	1 tsp each max — Na control

Method: 1. Char ginger and onion under broiler 5 min per side 2. Simmer water + broth + charred aromatics + spices for 30 min minimum (1 hour better) 3. Strain broth. Return to pot. Season with soy sub + sugar 4. Cook noodles separately per package directions 5. Bowl: noodles → raw beef slices on top → ladle HOT broth over beef (the broth cooks the thin slices instantly) 6. Garnish plate on the side

The pho trick for CKD: Building flavor with spices (star anise, cinnamon, charred ginger) instead of relying on high-sodium fish sauce + long-simmered bone broth. The spice infusion is actually MORE aromatic than lazy high-sodium broth.

8. Bright Pickle Banh Mi

Component	CKD Dupe
Bread	French baguette or bolillo roll — light, crispy crust. Small 6" portion.
Protein	Grilled chicken thigh (marinate in: 1 tsp low-sodium soy + 1 tsp sugar + 1/2 tsp garlic + lemongrass, minced)
Do chua (pickled daikon/carrot)	Julienne daikon + carrot, pickle in rice vinegar + sugar + water (NO salt in the pickle — the vinegar does the work)
Cucumber	Sliced thin
jalapeño	Sliced
Cilantro	Fresh sprigs
Mayo	Avocado oil mayo (Primal Kitchen) or sriracha mayo (mayo + sriracha)
Maggi/soy drizzle	SKIP or 1/2 tsp low-sodium soy — this is where banh mi gets its sodium

Assembly: Split roll. Spread avocado oil mayo. Layer chicken, pickled veg, cucumber, jalapeño, cilantro. The crunch of fresh bread + pickled vegetables + herbs = perfect sandwich.

CHINESE TAKEOUT DUPES & TERIYAKI

"The Americanized Chinese restaurant down the street — but your kidneys don't file a complaint afterward."

CKD NOTES FOR CHINESE TAKEOUT COOKING:

The traps in Chinese-American takeout are predictable: - **Regular soy sauce** = 900mg Na per tbsp. Always use **low-sodium soy sauce** (Kikkoman low-sodium = 575mg/tbsp), and use LESS than you think you need - **Oyster sauce, hoisin sauce** = sodium bombs. Use sparingly, dilute with water - **Cornstarch coatings** = fine for CKD. Zero K/P/Na. Creates that crispy takeout texture - **Brown sugar** = the secret weapon. Caramelizes in the wok, creates that glossy takeout sauce look - **MSG** = 600mg Na per tsp. Skip it. The umami comes from soy + sugar + ginger instead - **Avocado oil** = Andrew's frying oil of choice. High smoke point (520F), clean flavor, perfect for wok work

The CKD Takeout Sauce Base (use this across most recipes below):

Ingredient	Amount	CKD Notes
Low-sodium soy sauce	2 tbsp	~1,150mg Na total — split across 2+ servings
Brown sugar	1.5 tbsp	The glaze
Fresh ginger, minced	1 tbsp	Inflammation-support
Garlic, minced	3 cloves	
Water or low-sodium broth	2 tbsp	Loosens the sauce
Cornstarch slurry	1 tsp cornstarch + 1 tbsp water	Thickens to that glossy takeout consistency

1. Ginger-Glaze Teriyaki Chicken

Invented in the 1960s by Japanese-American immigrants in Seattle and Hawaii. The word "teriyaki" (照り焼き) means "shiny grilled" — teri = gloss, yaki = grill. The sweet soy glaze Americans know was largely a Pacific Northwest Japanese-American creation, not traditional Japanese cooking.

Ingredient	Amount	CKD Notes
Chicken thighs, boneless skinless	1 lb (~4 thighs)	Thighs > breast. More flavor, harder to overcook
Teriyaki Sauce:		
Low-sodium soy sauce	3 tbsp	~1,725mg Na split across 4 servings
Brown sugar	2 tbsp	
Fresh ginger, grated	1 tbsp	
Garlic, minced	3 cloves	
Rice vinegar	1 tbsp	Brightness
Water	2 tbsp	
Cornstarch slurry	1 tsp cornstarch + 1 tbsp cold water	
Avocado oil	1 tbsp	
Jasmine rice	For serving	Low K base
Green onion, sliced	Garnish	
Sesame seeds	Garnish	

Method: 1. Mix soy sauce, brown sugar, ginger, garlic, rice vinegar, water in a bowl 2. Heat avocado oil in large skillet or wok over medium-high 3. Sear chicken thighs 5-6 min per side until golden and cooked through (165F internal) 4. Pour sauce over chicken. Simmer 2 min 5. Add cornstarch slurry. Stir until sauce thickens and goes glossy (30 seconds) 6. Slice thighs. Serve over jasmine rice. Hit with green onion + sesame seeds

Per serving (4 servings): ~430mg Na | ~210mg P | ~320mg K

2. Ginger-Glaze Teriyaki Beef

Same Seattle/Hawaii Japanese-American origin as teriyaki chicken. Flank steak became the go-to cut in American teriyaki joints because it absorbs marinade fast and slices beautifully against the grain.

Ingredient	Amount	CKD Notes
Flank steak	1 lb	Slice thin AGAINST the grain after cooking
Teriyaki Sauce:		
Low-sodium soy sauce	3 tbsp	
Brown sugar	2 tbsp	
Fresh ginger, grated	1 tbsp	
Garlic, minced	3 cloves	
Rice vinegar	1 tbsp	
Water	2 tbsp	
Cornstarch slurry	1 tsp cornstarch + 1 tbsp cold water	
Avocado oil	1 tbsp	

Ingredient	Amount	CKD Notes
Jasmine rice	For serving	
Green onion, sliced	Garnish	
Sesame seeds	Garnish	

Method: 1. Mix sauce ingredients in a bowl (same as teriyaki chicken sauce) 2. Slice flank steak into 1/4" strips AGAINST the grain. (Pro tip: partially freeze the steak 20 min first — easier to slice thin) 3. Heat avocado oil in wok over HIGH heat until smoking 4. Sear beef strips in batches — 1-2 min per batch. Don't crowd the wok or it steams instead of sears 5. Return all beef to wok. Pour sauce over. Toss 1 min 6. Add cornstarch slurry. Stir until glossy 7. Serve over jasmine rice. Green onion + sesame seeds on top

Per serving (4 servings): ~440mg Na | ~230mg P | ~350mg K

3. Wok-Seared Mongolian Beef

Despite the name, Mongolian Beef has nothing to do with Mongolia. It was invented in Taiwan in the 1950s at Mongolian BBQ restaurants, which were themselves a Taiwanese invention loosely inspired by stories of Mongolian warriors cooking on shields. The dish as Americans know it — sweet, soy-glazed, scallion-heavy — is pure Chinese-American takeout, popularized by P.F. Chang's in the 1990s.

Ingredient	Amount	CKD Notes
Flank steak	1 lb, sliced 1/4" against grain	Partially freeze first
Cornstarch	3 tbsp	The crispy coating — zero K/P/Na
Avocado oil	3 tbsp (for frying)	High smoke point
Green onions	6-8 stalks, cut into 2" pieces	Low K, THE signature ingredient
Sauce:		
Low-sodium soy sauce	3 tbsp	
Brown sugar	3 tbsp	Slightly sweeter than teriyaki — that's the Mongolian Beef signature
Fresh ginger, minced	1 tbsp	
Garlic, minced	4 cloves	
Water	3 tbsp	
Red pepper flakes	1/2 tsp (optional)	

Method: 1. Toss sliced beef in cornstarch until every piece is coated. Shake off excess 2. Mix sauce ingredients in a bowl 3. Heat avocado oil in wok over HIGH heat. Fry beef strips in batches — 2 min per batch until edges are CRISPY. Remove to plate 4. Drain most oil. Add garlic + ginger, stir 30 seconds 5. Pour in sauce. Let it bubble and thicken 1 min 6. Return crispy beef to wok. Toss to coat 7. Kill heat. Add green onions. Toss 30 seconds (residual heat only — you want them barely wilted, still bright green) 8. Serve over jasmine rice immediately. The crispy texture fades fast

Per serving (4 servings): ~450mg Na | ~220mg P | ~340mg K

4. Peppercorn Szechuan Chicken

Szechuan (Sichuan) cuisine dates back over 2,000 years to China's Sichuan Province. The signature numbing-spicy (麻辣, málà) flavor comes from Szechuan peppercorns, which contain hydroxy-alpha-sanshool — a compound that literally numbs your tongue by activating touch receptors. The peppercorn was actually BANNED in the US from 1968-2005 due to citrus canker fears. What Americans ate as "Szechuan" for 37 years was mostly just spicy, not numbing.

Ingredient	Amount	CKD Notes
Chicken thighs, boneless skinless	1 lb, cut into 1" cubes	
Cornstarch	2 tbsp	Light coating
Avocado oil	2 tbsp	
Dried red chili peppers	8-12 whole	The heat. Remove seeds for less fire

Ingredient	Amount	CKD Notes
Szechuan peppercorns	1 tsp	The NUMBING. Toast in dry pan first
Bell pepper (red or green)	1, cut into 1" pieces	Low K
Zucchini	1 small, half-moons	Low K
Green onion	3 stalks, cut 1"	
Sauce:		
Low-sodium soy sauce	2 tbsp	
Rice vinegar	1 tbsp	
Brown sugar	1 tbsp	
Sesame oil	1 tsp	Flavor, not frying
Garlic, minced	4 cloves	
Fresh ginger, minced	1 tbsp	
Water	2 tbsp	
Cornstarch slurry	1 tsp cornstarch + 1 tbsp water	

Method: 1. Toast Szechuan peppercorns in dry wok 1 min until fragrant. Remove, crush lightly 2. Toss chicken cubes in cornstarch 3. Heat avocado oil in wok over high heat. Fry chicken 4-5 min until golden. Remove 4. In same wok: dried chilies + crushed Szechuan peppercorns, stir 30 seconds (don't burn) 5. Add garlic + ginger, 30 more seconds 6. Add bell pepper + zucchini. Stir-fry 2 min 7. Return chicken. Pour in sauce. Toss 1 min 8. Add cornstarch slurry. Stir until glossy 9. Finish with green onion + sesame oil drizzle

Heat note: If you want to go nuclear, add chili oil or your favorite hot sauce at the table. Andrew has tried all the Da Bomb hot sauces from Hot Ones and reports they taste like "salty spicy garbage" — famous for being the worst-tasting sauce on the show despite high Scoville numbers. If you want real heat with actual flavor, use chili crisp (Lao Gan Ma) or sambal oelek instead.

Per serving (4 servings): ~400mg Na | ~200mg P | ~310mg K

5. General Tso's Night Market Chicken

Named after Zuo Zongtang (左宗棠), a Qing Dynasty military leader from Hunan Province (1812-1885) who never ate the dish. Chef Peng Chang-kuei created a Hunanese version in Taiwan in the 1950s for a banquet honoring Zuo. Chef Peng later brought it to New York in 1973, where American customers wanted it sweeter. He obliged. The sweet-crispy-spicy version Americans know was born in Manhattan, not Hunan. Chef Peng reportedly disliked the Americanized version.

Ingredient	Amount	CKD Notes
Chicken thighs, boneless skinless	1 lb, cut into 1.5" pieces	
Batter:		
Cornstarch	1/4 cup	
Egg	1	
All-purpose flour	2 tbsp	
Salt	Pinch only	
Avocado oil	For shallow frying (~1/2 cup)	
Sauce:		
Low-sodium soy sauce	2 tbsp	
Rice vinegar	2 tbsp	The tang
Brown sugar	2 tbsp	The sweet
Sesame oil	1 tsp	
Garlic, minced	3 cloves	
Fresh ginger, minced	1 tbsp	
Dried red chili peppers	4-6 whole	
Water	2 tbsp	
Cornstarch slurry	1 tsp cornstarch + 1 tbsp water	

Ingredient	Amount	CKD Notes
Broccoli florets	2 cups, steamed	Low K when steamed

Method: 1. Whisk batter: egg + cornstarch + flour + pinch salt until smooth 2. Toss chicken pieces in batter until fully coated 3. Heat avocado oil in wok or deep skillet to 350F. Fry chicken in batches 4-5 min until deep golden and CRISPY. Drain on paper towels 4. Drain all but 1 tbsp oil. Add dried chilies, garlic, ginger — stir 30 seconds 5. Pour in soy sauce, vinegar, brown sugar, water. Bring to simmer 6. Add cornstarch slurry. Stir until sauce is thick and glossy 7. Toss crispy chicken into sauce. Coat every piece 8. Serve over jasmine rice with steamed broccoli on the side. Drizzle sesame oil

Per serving (4 servings): ~420mg Na | ~220mg P | ~300mg K

6. Toasted Sesame Beef

Sesame-coated beef dishes trace to Cantonese cooking traditions, but the thick-sauced sesame beef Americans know is another Chinese-American restaurant creation from the 1970s-80s. Sesame seeds have been cultivated for over 5,000 years — they're one of the oldest oilseed crops on Earth, first domesticated in the Indian subcontinent. The word "sesame" comes from the Arabic "simsim."

Ingredient	Amount	CKD Notes
Flank steak	1 lb, sliced 1/4" against grain	Partially freeze first
Cornstarch	3 tbsp	Crispy coating
Avocado oil	3 tbsp	
Sesame seeds	2 tbsp	Toast in dry pan first — low P in this amount
Green onion	3 stalks, sliced	Garnish
Sesame Sauce:		
Low-sodium soy sauce	2 tbsp	
Sesame oil	1 tbsp	The star flavor
Brown sugar	2 tbsp	
Rice vinegar	1 tbsp	
Garlic, minced	3 cloves	
Fresh ginger, minced	1 tsp	
Water	2 tbsp	
Cornstarch slurry	1 tsp cornstarch + 1 tbsp water	

Method: 1. Toast sesame seeds in dry pan over medium heat 2 min, stirring constantly, until golden. Remove immediately (they burn fast) 2. Toss sliced beef in cornstarch 3. Heat avocado oil in wok over HIGH heat. Fry beef in batches until crispy edges. Remove 4. Drain most oil. Add garlic + ginger, 30 seconds 5. Pour in sauce ingredients (soy, sesame oil, brown sugar, vinegar, water). Simmer 1 min 6. Add cornstarch slurry. Stir until thick and glossy 7. Return beef. Toss to coat 8. Top with toasted sesame seeds + green onion. Serve over jasmine rice

Per serving (4 servings): ~410mg Na | ~230mg P | ~340mg K

7. Golden Chicken Fried Rice

Fried rice (炒饭, chǎo fàn) has been documented in China since the Sui Dynasty (~589-618 AD). The technique of frying day-old rice was a practical solution for leftover grain. Chinese-American fried rice as a takeout staple became widespread in the US by the 1920s-30s alongside chop suey joints. The "day-old rice" rule isn't optional — fresh rice has too much moisture and turns into a sticky clump. Cold rice fries crispy.

Ingredient	Amount	CKD Notes
Day-old jasmine rice	3 cups cooked, cold	MUST be cold/day-old. Spread fresh rice on sheet pan in fridge overnight if needed
Chicken thigh, boneless skinless	8 oz, diced small	

Ingredient	Amount	CKD Notes
Eggs	2	Moderate P — 2 eggs across 3+ servings is fine
Carrot	1 small, diced fine	Small amount = controlled K
Green onion	4 stalks, sliced	
Garlic, minced	3 cloves	
Frozen peas	1/4 cup	Small amount only — peas are moderate K
Avocado oil	2 tbsp	
Low-sodium soy sauce	2 tbsp	
Sesame oil	1 tsp	Finish only
White pepper	1/4 tsp	The authentic fried rice seasoning — NOT black pepper

Method: 1. Heat avocado oil in wok over HIGH heat until smoking 2. Cook diced chicken 3-4 min until done. Remove 3. Scramble eggs in same wok. Break into small pieces. Remove 4. Add a splash more oil. Stir-fry carrot + peas 2 min 5. Add cold rice. Press it flat against wok. Let it SIT and get crispy on the bottom (30-45 seconds). Then toss. Repeat. This is how you get that wok-charred flavor (锅气, wok hei) 6. Add garlic, stir 30 seconds 7. Return chicken + eggs. Drizzle soy sauce around the EDGES of the wok (hits the hot metal first = deeper flavor) 8. Toss everything together. White pepper + sesame oil + green onions 9. Serve immediately

Per serving (3 servings): ~420mg Na | ~200mg P | ~290mg K

8. Weeknight Beef Chow Mein

Chow mein (炒麵, chǎo miàn) means "stir-fried noodles." It arrived in America with Chinese railroad workers in the 1850s-60s and became a fixture of early Chinese-American restaurants. By the 1920s, chow mein was one of the most recognized Chinese dishes in America. The crispy vs. soft noodle debate is regional: Cantonese-American restaurants often served crispy pan-fried noodles, while other traditions kept them soft. Both are correct.

Ingredient	Amount	CKD Notes
Egg noodles or rice noodles	8 oz dried	Rice noodles = lower P. Egg noodles = more authentic chow mein texture
Flank steak	8 oz, sliced thin against grain	
Cabbage (green)	2 cups, shredded	LOW-K vegetable — perfect for CKD
Bean sprouts	1/2 cup	Moderate K — small amount is fine
Carrot	1 small, julienned	
Green onion	3 stalks, cut 2"	
Garlic, minced	3 cloves	
Fresh ginger, minced	1 tsp	
Avocado oil	2 tbsp	
Chow Mein Sauce:		
Low-sodium soy sauce	2 tbsp	
Oyster sauce	1 tsp	Use sparingly — Na
Sesame oil	1 tsp	
Brown sugar	1 tsp	
White pepper	1/4 tsp	
Water	2 tbsp	

Method: 1. Cook noodles per package (usually boil 3-4 min for egg noodles, soak for rice noodles). Drain. Toss with 1 tsp sesame oil to prevent sticking 2. Mix sauce ingredients in a bowl 3. Heat avocado oil in wok over HIGH heat. Sear beef strips 1-2 min. Remove 4. Same wok: garlic + ginger 30 seconds. Add cabbage + carrot. Stir-fry 2 min until cabbage wilts slightly 5. Add noodles. Toss with vegetables. Let noodles sit against the wok to get some char (don't stir constantly) 6. Return beef. Pour sauce over

everything. Toss 1 min 7. Add bean sprouts + green onion. Toss 30 seconds (barely cook them — they stay crunchy) 8. Serve immediately. Chow mein waits for nobody

Per serving (3 servings): ~450mg Na | ~230mg P | ~330mg K

CKD Considerations

RENAL CHECK — Thai & Vietnamese
K (Potassium): LOW-MED (rice noodle base, controlled tomato, green papaya not ripe)
P (Phosphorus): LOW-MED (lean protein portions, peanuts as garnish only)
Na (Sodium): CONTROLLED (fish sauce subbed with low-sodium soy + lime)
Fluid: Soups (Tom Kha, Pho) = significant fluid — track it

RENAL CHECK — Chinese Takeout & Teriyaki
K (Potassium): LOW-MED (jasmine rice base, cabbage/bell pepper/zucchini = low-K vegetables)
P (Phosphorus): LOW-MED (chicken thighs + flank steak in controlled portions, sesame seeds as garnish)
Na (Sodium): CONTROLLED (low-sodium soy sauce, no MSG, brown sugar + ginger for flavor depth)
Fluid: Not an issue — these are dry stir-fry dishes, not soups
Oil: Avocado oil throughout (high smoke point, clean profile)

JAPANESE SIDES

Sunomono Cucumber Salad

Sunomono (酢の物) literally means "vinegared things" — the technique predates refrigeration in Japan by centuries, used to preserve vegetables through the Edo period (1603-1868).

Ingredient	Amount	CKD Notes
English cucumber	1 large, thin-sliced	Low K (~150mg per cup). English/hothouse cucumbers = less seeds, less bitter
Rice vinegar	3 tbsp	CKD-aware, no sodium
Sugar	1 tbsp	
Low-sodium soy sauce	1 tsp	Just a touch — controls Na
Sesame oil	1/2 tsp	Flavor, not cooking — a little goes far
Sesame seeds	1 tsp (toasted)	Garnish
Red pepper flakes	Pinch (optional)	
Salt	1/4 tsp	For salting the cucumber (draws out water, then rinse)

Method: 1. Slice cucumber paper-thin (mandoline if you have one, knife if you don't). Thinner = better. You want translucent. 2. Toss slices with 1/4 tsp salt. Let sit 10 minutes. The salt draws out water and wilts the cucumber slightly. 3. SQUEEZE the cucumber slices gently in a clean towel or your hands — press out the water. This step matters: it makes them absorb the dressing instead of diluting it. 4. Mix dressing: rice vinegar + sugar + soy sauce + sesame oil. Stir until sugar dissolves. 5. Toss cucumber with dressing. Refrigerate minimum 15 min (1 hour is better — the flavor deepens). 6. Serve cold, topped with toasted sesame seeds and optional red pepper flakes.

Variations: - **Sushi restaurant style:** Add a splash of mirin (1 tsp) to the dressing for that sweet-tangy depth - **With wakame:** Add 1 tbsp dried wakame seaweed (rehydrated) — low K, adds umami + ocean flavor - **With crab:** Add 2 oz imitation crab (surimi), shredded — lower P than real crab - **Spicy:** Add sriracha (1/2 tsp) to the dressing

Per serving: ~45mg Na | ~15mg P | ~150mg K. A lower-estimated-mineral side option; verify the ingredients and serving against your plan.

Why it works at sushi places: It's basically vinegar + cucumber + sugar. No hidden phosphorus additives, no high-K ingredients. The sesame oil gives it that authentic restaurant depth. Order it confidently — it's one of the safest things on any Japanese menu.

"The night market in your kitchen. The takeout counter in your wok. The clinic numbers on your fridge."

Mediterranean, Middle Eastern & International

“Every culture solved the same problem: how to feed the people you love.”

Kosher Meatballs, Halal Lamb & Chicken Gyros, Persian Ketchup & More

Faith-Friendly, Andrew-Approved
"Every grandmother's kitchen. Every faith's table. Kidneys welcome."

KOSHER MEATBALLS

Classic Kosher Meatballs in Tomato Sauce

Ingredient	Amount	CKD Notes
Ground beef (kosher)	1 lb	No dairy in the mix (kosher = no meat + dairy)
Breadcrumbs	1/3 cup	Plain, not seasoned (seasoned = sodium)
Egg	1	Moderate P — 1 egg per batch is fine
Garlic	3 cloves, minced	
Onion	1/4 cup, finely diced	Low K
Parsley	2 tbsp fresh, chopped	
Cumin	1/2 tsp	
Paprika	1/2 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp only	Minimal

CKD Tomato Sauce (K-controlled):

Ingredient	Amount	CKD Notes
Tomato paste	2 tbsp	Concentrate = controlled portions
Water	1.5 cups	Dilutes the K
Onion	1/4 cup, diced	
Garlic	2 cloves	
Sugar	1 tsp	Balances acidity
Cumin	1/2 tsp	
Paprika	1/2 tsp	
Lemon juice	1 tbsp	Brightness without sodium
Oil	1 tbsp	

Method: 1. Mix meatball ingredients. Roll into 1.5" balls (about 16) 2. Brown in oil 3-4 min, turning. Remove. 3. Same pan: saut?? onion + garlic 2 min. Add tomato paste, stir 1 min 4. Add water + spices + sugar. Simmer 5 min 5. Add meatballs back. Simmer covered 15 min 6. Finish with lemon juice. Serve over rice or couscous
Per 4 meatballs + sauce: ~280mg Na, ~200mg P, ~350mg K
Why diluted tomato paste: Full tomato sauce (1 cup) = 530mg K. Paste (2 tbsp) diluted in water = ~180mg K for the whole batch. Same tomato flavor, fraction of the potassium.

HALAL LAMB & CHICKEN GYROS

Halal Chicken Gyro

The Chicken:

Ingredient	Amount
Chicken thighs (boneless, halal)	1 lb
Yogurt (plain)	1/4 cup (marinade)

Ingredient	Amount
Lemon juice	2 tbsp
Garlic	3 cloves, minced
Cumin	1 tsp
Paprika	1 tsp
Oregano	1/2 tsp
Turmeric	1/2 tsp
Cayenne	1/4 tsp
Oil	1 tbsp
Salt	1/4 tsp

Marinade chicken minimum 1 hour (overnight is better). Grill or pan-sear 5-6 min per side. Slice into strips.
The Gyro Sauce (Toum-style, no tahini — tahini is high P):

Ingredient	Amount
Plain yogurt	1/4 cup
Lemon juice	1 tbsp
Garlic	1 clove, minced fine
Cucumber	2 tbsp, grated + squeezed dry
Dried dill	1/4 tsp
Dried mint	1/4 tsp
Salt	Pinch

Assembly: Pita bread (or naan) → chicken strips → shredded lettuce → sliced onion (small amount) → diced cucumber → gyro sauce → hot sauce
Halal Lamb Gyro

The Lamb:

Ingredient	Amount	CKD Notes
Ground lamb (halal)	1 lb	Lamb is moderate P — portion to 3-4 oz
Onion	1/4 cup, grated	Grated into the meat for moisture
Garlic	4 cloves, minced	
Cumin	1.5 tsp	
Coriander	1 tsp	
Paprika	1 tsp	
Oregano	1 tsp	
Cinnamon	1/4 tsp	
Cayenne	1/4 tsp	
Black pepper	1/2 tsp	
Salt	1/4 tsp	

Method: 1. Mix all ingredients thoroughly. Press into a flat loaf in a loaf pan (compact it — this creates the gyro meat texture) 2. Bake at 350F for 55-60 min until internal temp is 165F 3. Let cool 10 min. Slice thin (like deli meat) 4. Optional: crisp slices in a hot skillet before serving (this is what makes food-cart gyro meat so good)
Same assembly as chicken gyro. Lamb + the hot sauce Andrew loves.

The Hot Sauce for Gyros

Ingredient	Amount
Sriracha	1 tbsp
Yogurt	1 tbsp
Lemon juice	1 tsp
Garlic powder	Pinch
Cayenne	Pinch (for extra heat)

Mix. Drizzle generously. This is the creamy-hot that food cart gyros have.

PERSIAN KETCHUP (Cornelius, Oregon Food Cart Inspired)

Andrew had this at a food cart in Cornelius, OR — likely a Persian/Afghan cart. This is almost certainly **Persian tomato-saffron ketchup** or **rob-e gojeh** — a concentrated, spiced tomato paste condiment that's common in Iranian cuisine. It's tangier, more aromatic, and less sweet than American ketchup.

Persian Ketchup (Rob-e Gojeh Farangi)

Ingredient	Amount	CKD Notes
Tomato paste	3 tbsp	Controlled amount — this is concentrated
Water	3 tbsp	Dilute the K
Saffron threads	Small pinch (5-6 threads)	Bloomed in 1 tbsp hot water — THE Persian signature
Garlic	1 clove, minced very fine	
Onion powder	1/4 tsp	
Cumin	1/4 tsp	
Cinnamon	Tiny pinch	Persian warmth
Turmeric	1/4 tsp	Golden color
Lemon juice	1 tbsp	Persian ketchup is MORE sour than American
Sugar	1 tsp	Less sweet than American ketchup
Cayenne	Pinch	Mild heat
Rose water	1/4 tsp (optional)	If you can find it — adds the floral note that makes Persian food unmistakable
Salt	Pinch	
Oil	1 tsp	Saut?? garlic briefly before adding paste

Method: 1. Bloom saffron in 1 tbsp hot water. Set aside 5 min (turns golden) 2. Saut?? garlic in oil 30 sec. Add tomato paste, stir 1 min (caramelizes slightly) 3. Add water + saffron water + all spices + lemon + sugar 4. Simmer 3-4 min until thick and glossy 5. Add rose water if using. Cool. Store in jar.

This keeps in the fridge for 2 weeks. Put it on everything — gyros, kebabs, rice, eggs, meatballs.

Per tbsp: ~60mg K, ~15mg P, ~40mg Na. WAY better numbers than regular ketchup (which is ~170mg Na per tbsp).

Other Persian Sauces Worth Making

Mast-o-Khiar (Persian Yogurt Cucumber Dip)

Ingredient	Amount
Plain yogurt	1 cup
Cucumber	1/2, grated + drained
Dried mint	1 tsp
Dried dill	1/2 tsp
Rose petals	Pinch (optional, dried)
Salt	Pinch
Walnuts	1 tbsp, crushed (optional — moderate P)

Mix all. Chill 30 min. Serve with bread or rice.

Torshi (Persian Pickled Vegetables)

Ingredient	Amount
Cauliflower florets	1 cup
Carrot sticks	1/2 cup
Celery sticks	1/2 cup
White vinegar	1 cup
Water	1/2 cup
Garlic	4 cloves
Dried mint	1 tsp
Turmeric	1/2 tsp
Salt	1/2 tsp
Golpar (angelica powder)	1/4 tsp if available

Pack vegetables in jar. Boil vinegar + water + spices. Pour over veg. Seal. Wait 1 week. renal-aware pickles — vinegar-based, not salt-cured.

CKD Considerations

RENAL CHECK – Kosher, Halal & Persian
K (Potassium): MED (tomato paste controlled, diluted, no raw tomato sauce)
P (Phosphorus): MED (meat portions controlled, no tahini, cream cheese over cheese)
Na (Sodium): LOW (homemade spice mixes, no packets, fish sauce subbed)
Protein: GOOD (lean halal/kosher meat = quality protein)

"The Cornelius food cart lives in your kitchen now."

CHAPTER 7

German Comfort Food

"Guten Appetit."

Schnitzel, Bratwurst, Spaetzle, Sauerbraten & More

"Guten Appetit."

"Essen und Trinken hält Leib und Seele zusammen." — German proverb (Eating and drinking hold body and soul together.)

German Pantry Essentials

Stock these and you can cook every recipe in this chapter.

Item	Why	Where
Dried marjoram	The soul of bratwurst and German seasoning	Any spice aisle
Caraway seeds	Sauerkraut, bread, bratwurst, potato dishes	Spice aisle
Ground white pepper	German sausage fundamental — cleaner heat than black	Spice aisle
Ground nutmeg	Spaetzle, sausage, baked goods	Spice aisle

Item	Why	Where
Ground ginger	Bratwurst, sauerbraten, lebkuchen spice	Spice aisle
Juniper berries	Sauerbraten, rouladen gravy — piney, gin-like	Specialty/Amazon
German mustard (Dusseldorf or Bavarian sweet)	Rouladen, pretzels, bratwurst condiment	International aisle
White wine vinegar	Sauerbraten marinade, kartoffelsalat	Any grocery
Apple cider vinegar	Backup acid for marinades and salads	Any grocery
Bay leaves	Sauerbraten, rouladen, braised dishes	Spice aisle
Whole cloves	Sauerbraten marinade, strudel	Spice aisle
Allspice berries	Sauerbraten, sausage blends	Spice aisle
Rumford baking powder	Phosphorus-free — the only BP we use	Baking aisle
Costco no-salt seasoning blend	Universal flavor without sodium	Costco
Avocado oil	Frying schnitzel, pancakes, sausage	Any grocery
Margarine	Spaetzle toss, sauces, baking	Any grocery

Item	Why	Where
Unsweetened almond milk	Low-P dairy sub for batters and sauces	Any grocery
Sourdough bread	Breadcrumbs for schnitzel, bread dumplings	Bakery

MAINS

1. Chicken Schnitzel (Wiener Art)

Pounded thin. Breaded. Fried golden. The crunch is the point.

Traditional version: Veal (Wiener Schnitzel) or pork (Schnitzel Wiener Art). High P from processed breadcrumbs, high Na from brining.

Ingredient	Amount	CKD Notes
Chicken breast	4 pieces (5-6oz each)	Butterflied and pounded to 1/4" — turkey cutlets also work
All-purpose flour	1/2 cup	First dredge
Eggs	2, beaten	Moderate P — 2 for the whole batch is fine
Plain breadcrumbs	1.5 cups	Make your own from sourdough — zero additives
Lemon	1, cut into wedges	Traditional garnish, adds brightness
Black pepper	1 tsp	
Garlic powder	1/2 tsp	
Salt	1/4 tsp	In the flour only
Avocado oil	For frying (1/2" deep)	

Method: 1. Butterfly each breast, open like a book. Place between plastic wrap. Pound with mallet or heavy pan to 1/4" even thickness. This is non-negotiable — thick schnitzel is a crime. 2. Set up three stations: flour (seasoned with salt, pepper, garlic powder), beaten eggs, breadcrumbs. 3. Dredge in flour, shake off excess. Dip in egg. Press into breadcrumbs on both sides. Press firmly. Let rest on rack 10 min. 4. Heat avocado oil to 350F in a large skillet. The oil should come halfway up the schnitzel. 5. Fry 2-3 min per side until deep golden. Gently swirl the pan while frying — this creates the signature wavy/puffy crust. 6. Drain on wire rack (NOT paper towels). Squeeze lemon over immediately.

Per serving (1 schnitzel): ~160mg Na | ~210mg P | ~290mg K

2. Jagerschnitzel (Hunter's Schnitzel with Mushroom Gravy)

Same schnitzel. Smothered in dark mushroom gravy. Bavarian tavern energy.

Make the schnitzel above, then top with this gravy:

Ingredient	Amount	CKD Notes
White mushrooms, sliced	8 oz	Low K, low P — mushroom is renal-aware
Onion, diced	1/4 cup	
Margarine	2 tbsp	
All-purpose flour	2 tbsp	Thickener
Low-sodium beef broth	1 cup	Check label — must be <140mg Na per cup
Unsweetened almond milk	1/2 cup	Adds creaminess without P
Worcestershire sauce	1 tsp	Small amount, big umami
Dijon mustard	1 tsp	German mustard works too
Black pepper	1/2 tsp	
Fresh thyme	1 tsp	Or 1/2 tsp dried
Paprika	1/4 tsp	

Method: 1. Melt margarine over medium-high. Saute mushrooms 5-6 min until browned and liquid evaporates. Add onion, cook 2 min. 2. Sprinkle flour over mushrooms, stir 1 min. 3. Slowly whisk in beef broth, then almond milk. Stir constantly. 4. Add Worcestershire, mustard, pepper, thyme, paprika. Simmer 4-5 min until thick enough to coat a spoon. 5. Ladle generously over schnitzel. Serve with spaetzle (recipe below).

Per serving (schnitzel + gravy): ~220mg Na | ~240mg P | ~350mg K

3. Homemade Chicken Bratwurst

Your KitchenAid grinder earns its keep. Traditional spice blend, chicken instead of pork.

Traditional version: Pork shoulder + pork fat, natural hog casings, high Na from curing salt.

Ingredient	Amount	CKD Notes
Boneless skinless chicken thighs	2 lbs	Dark meat = juicier sausage. Turkey thighs also work.
Chicken skin or fat	4 oz	Need some fat or it's dry. Ask butcher.
Ice water	1/4 cup	Keeps mixture cold during grinding
Dried marjoram	1 tbsp	THE bratwurst herb
Ground white pepper	1 tsp	
Ground nutmeg	1/2 tsp	
Ground ginger	1/2 tsp	
Caraway seeds, ground	1/2 tsp	
Garlic powder	1/2 tsp	
Onion powder	1/2 tsp	
Salt	3/4 tsp	For 2 lbs — much less than commercial
Sugar	1/2 tsp	Balances spice, traditional
Natural hog casings	1 pack	Optional — can make patties instead

Method (KitchenAid Grinder): 1. Cut chicken thighs and skin/fat into 1" cubes. Spread on sheet pan, freeze 20 min (firm, not frozen solid — grinder works better cold). 2. Mix all spices together in a small bowl. 3. Assemble KitchenAid with grinder attachment, medium plate. Feed chicken through grinder into a chilled bowl. Add ice water and spice mix. Mix with hands until tacky and combined (about 1 min of kneading). 4. **For casings:** Rinse casings, slide onto sausage stuffer attachment. Feed mixture through, twisting into 5-6" links. Don't overstuff — they expand when cooked. 5. **For patties:** Form into 3oz patties, 1/2" thick. 6. **Cook:** Pan-fry in avocado oil over medium 5-6 min per side (internal 165F). Or grill. Or simmer in beer (use low-sodium broth instead for CKD — beer is too much P).

Yield: ~10 links or 10 patties

Per link/patty: ~120mg Na | ~150mg P | ~200mg K

Andrew's note: Make a double batch. Freeze individually on a sheet pan, then bag. Bratwurst on demand.

4. Kartoffelpuffer / Reibekuchen (German Potato Pancakes)

Crispy. Lacy edges. Served with applesauce. Same family as latkes.

K-LEACHING IS MANDATORY. Raw potatoes = 420mg K per medium potato. Leached = ~210mg.

Ingredient	Amount	CKD Notes
Russet potatoes	2 lbs (~4 large)	MUST be leached — see step 1
Onion, grated	1/2 medium	Adds moisture + flavor
Egg	1	Binder
All-purpose flour	2 tbsp	Holds them together
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Nutmeg	pinch	Traditional German touch
Avocado oil	For frying (1/4" deep)	

Applesauce (CKD-aware, homemade):

Ingredient	Amount
Gala or Fuji apples	3, peeled and diced
Water	1/4 cup
Cinnamon	1/2 tsp
Sugar	1 tbsp (or to taste)
Lemon juice	1 tsp

Method: 1. **LEACH THE POTATOES:** Peel and shred potatoes on box grater or food processor. Submerge shredded potatoes in a large bowl of cold water. Soak minimum 2 hours, ideally 4. Change water once halfway

through. Drain, rinse, and squeeze dry in a clean kitchen towel. Wring out every drop — dry potatoes = crispy pancakes. 2. Mix shredded potatoes with grated onion, egg, flour, pepper, salt, nutmeg. 3. Heat avocado oil in a skillet over medium-high (350F). 4. Drop 1/4 cup mounds into oil. Flatten with spatula to 1/3" thick. Fry 3-4 min per side until golden-brown and crispy edges. 5. Drain on wire rack. Serve immediately with applesauce on the side. 6. **Applesauce:** Combine all ingredients in a pot. Simmer 15-20 min until apples break down. Mash with fork for chunky, blend for smooth. Serve warm or cold.

Per 3 pancakes + 1/4 cup applesauce: ~110mg Na | ~90mg P | ~280mg K (leached)

5. Spaetzle (German Egg Dumplings)

Tiny, chewy, irregular noodles. Pressed through a colander into boiling water. Nothing else tastes like this.

Ingredient	Amount	CKD Notes
All-purpose flour	2 cups	
Eggs	3	Main protein + structure
Unsweetened almond milk	1/2 cup	Traditional uses whole milk — almond is lower P
Water	1/4 cup	Adjust for batter consistency
Nutmeg	1/4 tsp	Traditional and essential
Salt	1/4 tsp	
White pepper	pinch	

Method: 1. Whisk flour, salt, nutmeg, white pepper in a bowl. 2. Beat eggs with almond milk and water. Pour into flour. Stir vigorously with a wooden spoon until smooth and elastic — the batter should be thick but pourable, like a very thick pancake batter. It should slowly drip from a spoon. Add water by the tablespoon if too thick. 3. Bring a large pot of lightly salted water to a rolling boil. 4. **The Press:** Hold a colander (large holes) or spaetzle press over the pot. Pour batter in batches, pressing through with a spatula or spoon. The little dumplings will drop into the water. 5. They sink, then float. Once they float (~2 min), wait 30 seconds more, then scoop out with a slotted spoon. 6. Toss immediately in a bowl with margarine (1 tbsp) to prevent sticking. 7. **Optional crispy finish:** Pan-fry the boiled spaetzle in margarine until golden spots appear. This is the way.

Yield: 4-6 servings

Per serving: ~130mg Na | ~100mg P | ~90mg K

Gluten-Free Rice Flour Spaetzle

Replace the 2 cups AP flour with: - 1.5 cups white rice flour - 1/4 cup tapioca starch - 1/4 cup potato starch (minimal K contribution at this amount) - Add 1/2 tsp xanthan gum for elasticity
Same method. Batter will be slightly grainier. Pan-frying after boiling is especially recommended for GF version — gives a better texture.

6. Sauerbraten (German Pot Roast — 3-Day Marinated Beef)

Three days of patience. Worth every hour. Tangy, spiced, fork-tender.
Traditional version: Red wine + vinegar marinade, often with sugar snap gingersnap gravy. We keep the acid and spice, control the sodium.

Marinade (make 3 days before cooking):

Ingredient	Amount	CKD Notes
White wine vinegar	1 cup	The acid that tenderizes
Apple cider vinegar	1/2 cup	Depth of flavor
Water	1 cup	Dilutes without adding Na
Onion, quartered	1 large	
Carrot, sliced	1 (remove before serving — high K)	Flavor only — discard
Bay leaves	3	
Juniper berries	8, lightly crushed	THE sauerbraten spice
Whole cloves	4	
Allspice berries	4	

Ingredient	Amount	CKD Notes
Black peppercorns	1 tsp	
Mustard seeds	1 tsp	
Sugar	2 tbsp	Balances the acid

The Roast:

Ingredient	Amount	CKD Notes
Beef chuck roast	3-4 lbs	Tough cut = perfect for braising
Avocado oil	2 tbsp	For searing
All-purpose flour	2 tbsp	Gravy thickener
Low-sodium beef broth	1 cup	Added to braising liquid
Gingersnap cookies	4-5, crushed	Traditional thickener — check Na per cookie
Salt	1/2 tsp	Only when searing
Black pepper	1 tsp	

Method: 1. **Day 1 (3 days before):** Combine all marinade ingredients in a large pot or Dutch oven. Bring to a brief simmer, then cool completely. Place chuck roast in a deep container or zip-lock bag. Pour cooled marinade over. Refrigerate 3 days, turning once daily. The vinegar breaks down the connective tissue. Do not skip this. 2. **Day 4 (cook day):** Remove beef from marinade. Pat dry with paper towels (DRY = better sear). Strain marinade, reserve liquid, discard solids except onion. 3. Season beef with salt and pepper. Heat avocado oil in Dutch oven over high heat. Sear beef 3-4 min per side until dark brown crust forms on all sides. 4. Remove beef. Add reserved onion, cook 2 min. Sprinkle flour, stir 1 min. Pour in strained marinade liquid + beef broth. Scrape up fond. 5. Return beef to pot. Bring to simmer. Cover, transfer to 325F oven. Braise 3-3.5 hours until fork-tender. 6. Remove beef, let rest. Strain braising liquid into a saucepan. Crush gingersnaps into the liquid — this is the traditional thickener. Simmer 5 min, whisking until smooth and thick. 7. Slice beef against the grain. Serve with gingersnap gravy and spaetzle or kartoffelpuffer.

Serves: 6-8

Per serving (5oz beef + gravy): ~190mg Na | ~230mg P | ~380mg K

7. Rouladen (Beef Rolls with Mustard, Onion & Pickle)

Thin beef rolled around sharp mustard, onion, and pickle. Braised until tender. Classic German Sunday dinner.

Ingredient	Amount	CKD Notes
Top round or flank steak	2 lbs, sliced into 4 thin pieces	Ask butcher to slice 1/4" thick, or pound yourself
German mustard	4 tbsp	Spread inside each roll
Onion	1, sliced into thin rings	
Dill pickles	4 spears	Moderate Na — use low-sodium if available
Turkey bacon	4 strips	Traditional uses speck (cured pork belly) — turkey bacon sub
Avocado oil	2 tbsp	For searing
All-purpose flour	2 tbsp	
Low-sodium beef broth	2 cups	
Red wine vinegar	2 tbsp	Adds depth
Bay leaf	1	
Black pepper	1/2 tsp	
Toothpicks or kitchen twine		To secure rolls

Method: 1. Lay beef slices flat. Spread 1 tbsp mustard on each piece. 2. Layer: 1 strip turkey bacon, onion rings, 1 pickle spear on each. 3. Roll up tightly from the short end. Secure with toothpicks or tie with twine. 4. Heat oil in Dutch oven over high heat. Sear rolls on all sides (2-3 min total) until browned. Remove. 5. Sprinkle flour into the pot, stir 1 min. Add broth + vinegar + bay leaf + pepper. Whisk, scraping up fond. 6. Return rolls to pot. Bring to simmer, cover. Braise in 325F oven (or stovetop low heat) for 1.5-2 hours until tender. 7. Remove rolls, discard toothpicks. Reduce gravy on

stovetop if too thin. 8. Serve rolls with gravy ladled over. Pair with spaetzle or mashed potatoes (leached).

Per roll + gravy: ~280mg Na | ~220mg P | ~340mg K

8. Kasespatzle (German Mac & Cheese with Caramelized Onions)

Spaetzle layered with melted cheese and sweet caramelized onions. The German answer to mac and cheese. Better.

Ingredient	Amount	CKD Notes
Spaetzle	1 batch (recipe #5 above)	Freshly made
Smoked gouda, shredded	1 cup	Andrew's go-to — smoky, melts well
Sharp cheddar, shredded	1/2 cup	Adds bite
Onion	2 large, halved and thinly sliced	
Margarine	3 tbsp	For caramelizing onions
Black pepper	1/2 tsp	
Nutmeg	pinch	
Fresh chives	2 tbsp, chopped	Garnish

Method: 1. **Caramelize the onions:** Melt margarine in a large skillet over medium-low heat. Add onions. Cook 25-30 min, stirring every few minutes, until deep golden-brown and sweet. Don't rush this. Low and slow. Add a splash of water if they start to stick. 2. **Layer:** In a baking dish (or the same skillet), spread half the spaetzle. Top with half the cheese. Add remaining spaetzle. Top with remaining cheese. Season with pepper and nutmeg. 3. **Broil:** Place under broiler 3-4 min until cheese is melted, bubbly, and starting to brown. 4. Top with caramelized onions and chives. Serve from the dish.

Per serving: ~200mg Na | ~180mg P | ~160mg K

STREET FOOD & SIDES

9. Currywurst Sauce

Berlin street food legend. Ketchup + curry powder. Stupid simple. Unreasonably good over bratwurst.

Ingredient	Amount	CKD Notes
Low-sodium ketchup	1/2 cup	Regular ketchup = sodium bomb. Heinz no-salt-added works.
Curry powder	1 tbsp	The star
Smoked paprika	1 tsp	Depth
Onion powder	1/2 tsp	
Garlic powder	1/4 tsp	
Cayenne	1/4 tsp (or more)	Heat level is personal
White wine vinegar	1 tsp	Brightness
Sugar	1 tsp	Balances acid
Water	2 tbsp	Loosens the sauce

Method: 1. Combine everything in a small saucepan over medium heat. Stir. Simmer 5 min. 2. That's it. Slice chicken bratwurst (recipe #3), pile on a plate, drown in sauce. Dust with extra curry powder on top. Serve with a toothpick.

Per 3 tbsp serving: ~60mg Na | ~15mg P | ~90mg K

10. German Soft Pretzels (Reduced-Salt,)

Chewy. Golden-brown. Lye-dipped. Less salt, same pretzel soul.

Ingredient	Amount	CKD Notes
All-purpose flour	3 cups	
Warm water	1 cup (110F)	
Active dry yeast	1 packet (2.25 tsp)	

Ingredient	Amount	CKD Notes
Sugar	1 tbsp	Feeds yeast
Margarine, melted	2 tbsp	
Salt	1/2 tsp	In the dough only — less than traditional (which uses 2+ tsp)
Baking soda	1/3 cup	For the water bath — creates the pretzel crust
Water for bath	6 cups	
Coarse salt for topping	1/4 tsp, scattered	Visible crystals trick the brain — looks salty, isn't
Egg	1, beaten (egg wash)	

Method: 1. Dissolve yeast + sugar in warm water. Let sit 5 min until foamy. 2. Add flour, melted margarine, salt. Knead 5-7 min until smooth and elastic (KitchenAid with dough hook works great). Dough should be slightly tacky. 3. Place in oiled bowl, cover. Rise 1 hour until doubled. 4. Punch down. Divide into 8 pieces. Roll each into a 20" rope. Twist into pretzel shape (U-shape, cross the ends, fold down). 5. Preheat oven to 425F. Line baking sheets with parchment. 6. Bring 6 cups water + baking soda to a boil. Gently lower each pretzel into the bath for 30 seconds. This is what makes a pretzel a pretzel — the alkaline bath creates the dark, chewy crust. (Traditional uses food-grade lye. Baking soda is the home version.) 7. Place on baking sheet. Brush with egg wash. Sprinkle a few grains of coarse salt on each. 8. Bake 12-14 min until deep mahogany brown. 9. Serve warm with German mustard.

Per pretzel: ~130mg Na | ~65mg P | ~60mg K

11. Kartoffelsalat (Warm German Potato Salad — Vinegar-Based)

Not your American mayo potato salad. This is warm, tangy, slightly sweet. Bavarian-style.

Ingredient	Amount	CKD Notes
Russet potatoes	2 lbs	MUST be leached — see method step 1
Turkey bacon	4 strips, chopped	Traditional uses speck — turkey version
Onion, finely diced	1/2 medium	
White wine vinegar	3 tbsp	The backbone
Sugar	1 tbsp	Balances acid
Avocado oil	2 tbsp	
German or Dijon mustard	1 tsp	
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Fresh parsley	2 tbsp, chopped	Garnish
Fresh chives	1 tbsp, chopped	Garnish

Method: 1. **LEACH:** Peel potatoes, cut into 1/2" slices. Submerge in cold water for 2-4 hours, changing water once. Drain, rinse. 2. Boil leached potato slices in fresh water until just tender (12-15 min). Don't overcook — they should hold their shape. Drain, slice into half-moons while still warm. 3. Cook chopped turkey bacon in a skillet until crispy. Remove bacon, keep drippings. 4. In the same skillet, saute onion in drippings + avocado oil for 3-4 min until soft. 5. Add vinegar, sugar, mustard, pepper, salt. Stir until sugar dissolves. Remove from heat. 6. Pour warm dressing over warm potatoes. Toss gently. The warm potatoes absorb the dressing. 7. Fold in crispy bacon. Garnish with parsley and chives. Serve warm or at room temperature.

Per serving (6 servings): ~150mg Na | ~80mg P | ~260mg K (leached)

DESSERTS

12. Apfelstrudel (Apple Strudel)

Paper-thin pastry. Warm cinnamon apples. The Austrian-German dessert that predates your country.

Ingredient	Amount	CKD Notes
Phyllo dough (frozen)	8 sheets	Shortcut that works — traditional strudel dough is a project. Low P/K.
Gala or Fuji apples	4 large, peeled and thinly sliced	Low-K fruits — safe
Margarine, melted	4 tbsp	For brushing phyllo layers
Sugar	1/4 cup	
Cinnamon	1 tsp	
Raisins	2 tbsp	Small amount — raisins are concentrated K. Keep it tight.
Sourdough breadcrumbs	1/3 cup	Toasted in margarine — absorbs apple juice, keeps pastry crisp
Lemon juice	1 tbsp	Prevents browning
Powdered sugar	For dusting	

Method: 1. Preheat oven to 375F. 2. Toast breadcrumbs in 1 tbsp margarine until golden (3 min). Set aside. 3. Toss sliced apples with sugar, cinnamon, raisins, lemon juice. 4. Lay one sheet of phyllo on a clean towel. Brush with melted margarine. Layer 7 more sheets, brushing each with margarine. 5. Sprinkle toasted breadcrumbs over the phyllo, leaving a 2" border. 6. Spread apple filling over breadcrumbs in a line along the long edge. 7. Using the towel to help, roll the strudel away from you, tucking the ends under. Transfer seam-side down to a parchment-lined baking sheet. 8. Brush top with remaining margarine. Score the top lightly with a knife (5-6 cuts for serving). 9. Bake 30-35 min until golden and crispy. 10. Cool 10 min. Dust with powdered sugar. Serve warm.

Per slice (8 slices): ~85mg Na | ~30mg P | ~120mg K

13. Schwarzwald Kirschtorte (Black Forest Cake — No Alcohol)

Chocolate cake. Whipped cream. Cherries. No kirsch — all flavor, zero booze.

Traditional version: Soaked in kirschwasser (cherry brandy). We use cherry juice concentrate instead.

Chocolate Cake Layers:

Ingredient	Amount	CKD Notes
All-purpose flour	1.5 cups	
Unsweetened cocoa powder	1/2 cup	Moderate P — portion-controlled
Sugar	1 cup	
Rumford baking powder	2 tsp	Phosphorus-free
Baking soda	1/2 tsp	
Eggs	2	
Unsweetened almond milk	3/4 cup	Lower P than dairy
Avocado oil	1/3 cup	
Vanilla extract	1 tsp	
Hot water	3/4 cup	Opens up the cocoa
Salt	1/4 tsp	

Cherry Filling:

Ingredient	Amount
Canned tart cherries (in water, drained)	2 cans (14oz each)
Sugar	1/4 cup
Cornstarch	2 tbsp
Cherry juice (from cans)	1/2 cup
Vanilla extract	1/2 tsp

Whipped Cream:

Ingredient	Amount
Heavy whipping cream	2 cups
Powdered sugar	3 tbsp
Vanilla extract	1 tsp

Chocolate Shavings: Dark chocolate bar, shaved with a vegetable peeler.

Method: 1. **Cake:** Preheat 350F. Grease and flour two 9" round pans. Whisk flour, cocoa, sugar, BP, baking soda, salt. Add eggs, almond milk, oil, vanilla. Beat until combined. Stir in hot water (batter will be thin — this is correct). Pour into pans. Bake 30-35 min. Cool completely. 2. **Cherries:** Reserve 1/2 cup cherry juice. Mix cornstarch into the cold juice until smooth. Combine drained cherries + sugar in a saucepan. Add cornstarch mixture. Cook over medium heat, stirring, until thick and glossy (4-5 min). Cool completely. Reserve a handful of whole cherries for the top. 3.

Whipped cream: Beat cream + powdered sugar + vanilla to stiff peaks. Keep cold. 4. **Assemble:** Place first cake layer on plate. Spread 1/3 of whipped cream. Spoon half the cherry filling over. Place second cake layer. Spread remaining whipped cream on top and sides. Spoon remaining cherry filling on top. Garnish with reserved whole cherries and chocolate shavings. 5. Refrigerate minimum 2 hours before serving. Overnight is better — the layers meld.

Per slice (12 slices): ~140mg Na | ~110mg P | ~170mg K

CKD Considerations

Hemodialysis Daily Limits (Andrew's targets): - **Sodium:** < 1,500 mg/day - **Potassium:** < 2,000 mg/day - **Phosphorus:** < 800 mg/day

Critical Notes for This Chapter:

- 1. LEACH ALL POTATOES.** Every potato recipe here assumes 2-4 hour cold water soak. Shredded potatoes leach faster than whole. No exceptions. Unleached potatoes will blow your K budget in one meal.
- 2. Carrots in sauerbraten marinade are for FLAVOR ONLY.** Discard before serving. Braised carrots concentrate potassium.
- 3. Commercial bratwurst is a CKD minefield.** Sodium phosphate, sodium nitrite, MSG — the ingredient list reads like a chemistry exam. Homemade (recipe #3) is the only safe path.
- 4. Raisins in the strudel are capped at 2 tbsp.** Raisins are dried grapes = concentrated potassium (300mg K per 1/4 cup). Don't get generous.
- 5. Gingersnaps in sauerbraten gravy:** Check the label. Some brands load sodium. Aim for <100mg Na per serving of cookies used. Or substitute 1 tbsp cornstarch + 1 tsp ground ginger as thickener.
- 6. Cheese in kasespatzle:** Smoked gouda and sharp cheddar are moderate-P cheeses. Keep to the amounts listed. Don't double the cheese.
- 7. Cocoa powder in the Black Forest Cake is moderate in phosphorus** (~50mg P per tbsp). The portion per slice is controlled. Don't eat half the cake.
- 8. Pickle in rouladen:** One spear per roll is fine (~130mg Na). Don't add extra.

Individual targets may differ. These recipes were built for Andrew's hemodialysis targets. Your limits may differ.

"Essen ist fertig." ("The food is ready.")

Man soll dem Leib etwas Gutes bieten, damit die Seele Lust hat, darin zu wohnen. — Martin Luther ("One should offer the body something good, so the soul will want to live in it.")

The Crisping Alchemy Lab & World Table Expansions

"Who said kidney-friendly food has to be boiled mush? Texture, crunch, and authentic umami depth are the true

dignity of patient dining.”

THE CULINARY SCIENCE OF RENAL CRUNCH: Traditional renal brochures mistakenly tell patients to strip out flavor and boil proteins to death. In our 2026 culinary tests, we proved that glass-like shatter crunch can be achieved with **zero chemical phosphate additives** and under **85mg sodium** per serving using starch adhesion (cornstarch + egg white dredge) and high-heat apple cider vinegar pan reductions. Always verify protein targets with your clinical renal dietitian.

CRISPY HERB- CRUSTED CHICKEN TENDERS

FT-CRISP-01 · THE
CRISPING LAB ALCHEMY
· GLASS-LIKE SHATTER
CRUNCH WITHOUT SALT
OR SODIUM PHOSPHATE

Serves 4 · 20 min

Zero Phosphate Preservatives

INGREDIENTS

- 1 lb chicken tenderloins, fresh whole cuts
- 2 tbsp pure cornstarch (dry dusting)
- 1 large egg white, whisked until frothy
- 1/2 tsp garlic powder (pure, no added salt)
- 1/2 tsp onion powder
- 1/2 tsp dried thyme & dried oregano
- 1/4 tsp freshly ground black pepper
- 1.5 tbsp extra virgin olive oil
- 1 tbsp raw apple cider vinegar (for pan deglaze)
- 1 tbsp fresh flat-leaf parsley, finely chopped

DIRECTIONS

- 1. Dry & Dust:** Pat chicken tenderloins completely dry with paper towels. Light moisture prevents the adhesion layer from crisping. Dust tenderloins evenly with cornstarch, tapping off all excess.
- 2. Egg White Dip:** Dip each dusted tenderloin into the whisked frothy egg white, then immediately roll in garlic powder, onion powder, thyme, oregano, and black pepper.
- 3. Sear for Crunch:** Heat olive oil in a heavy stainless or cast-iron skillet over medium-high heat until shimmering. Lay tenders in with space between them. Sear undisturbed for 3 to 4 minutes until a deep golden, shatter-crisp crust forms. Flip and sear 3 minutes more.
- 4. The Cider Pan Deglaze:** Just before taking off heat, drizzle apple cider vinegar around the perimeter of the hot pan. The flash steam deglazes the pan juices, locking the savory crust into place.
- 5. Rest & Garnish:** Transfer to a wire cooling rack (never a paper towel, which steams the bottom). Garnish with chopped parsley. Serve immediately.

Na: 65 mg K: 280 mg P: 190 mg Protein: 26 g

THE CRISPING LAB · NORTHSTAR PRIME

SAGE CHICKEN CUTLETS WITH APPLE- ONION PAN JUICES

Serves 4 · 25 min
Autumn Umami Balance

FT-0865 · WAVE 096 MASTER
RECIPE · SWEET CRISP
HONEYCRISP APPLE
REDUCTION OVER GOLDEN
CUTLETS

INGREDIENTS

1.2 lbs boneless
chicken breast cutlets,
pounded to 1/2-inch
1 tbsp fresh rubbed
sage, minced
1 Honeycrisp apple,
cored & thinly sliced
1/2 yellow onion,
thinly sliced into
crescents
1.5 tbsp olive oil
2 tbsp unsalted
chicken stock or water
1 tsp apple cider
vinegar
1/4 tsp cracked black
pepper

DIRECTIONS

1. Rub cutlets with minced fresh sage and cracked black pepper.
2. Heat 1 tbsp olive oil in a skillet over medium heat. Sear chicken 4 minutes per side until golden brown and cooked to 165°F. Transfer to warm plate.
3. Add remaining 1/2 tbsp olive oil to pan. Add sliced apple and onion. Sauté 5 minutes until soft and caramelized around edges.
4. Deglaze pan with unsalted stock and cider vinegar, scraping up brown bits. Spoon caramelized apples and warm pan juices over chicken.

Na: 75 mg K: 310 mg P: 205 mg Protein: 28 g

WAVE 096 · MASTERCOOK FREEWARE

LEMON-DILL COD & FENNEL PARCHMENT PARCEL

Serves 2 · 22 min
Zero Added Fluid

FT-0845 · WAVE 096 MASTER
RECIPE · EN PAPILLOTE MOISTURE
TRAPPING WITH AROMATIC
SHAVED FENNEL

INGREDIENTS

2 Pacific cod fillets (5
oz each)
1/2 cup fresh fennel
bulb, very thinly
shaved
4 thin lemon slices
2 tbsp fresh dill fronds
1 tbsp extra virgin
olive oil
1/4 tsp freshly ground
white pepper

DIRECTIONS

1. Preheat oven to 385°F (195°C). Cut two 12x16-inch sheets of culinary parchment paper. Fold each sheet in half like a book.
2. Mound shaved fennel on one side of each crease. Place a cod fillet on top of fennel.
3. Drizzle olive oil over fillets. Top each with 2 lemon slices and fresh dill fronds; season with white pepper.
4. Fold paper over and crimp edges tightly in overlapping folds to create an airtight half-moon packet.
5. Bake on baking sheet for 14 minutes. The parcel will puff with aromatic steam. Cut open at the table to release the fragrant herb steam.

Na: 68 mg K: 340 mg P: 185 mg Protein: 24 g

WAVE 096 · MASTERCOOK FREEWARE

PARSLEY TURKEY PATTIES WITH CUCUMBER RIBBONS

Serves 4 · 20 min
Clean Whole Cuts

FT-0866 · WAVE 097 MASTER
RECIPE · TENDER HERBED LEAN
TURKEY WITH CHILLED CIDER-
DRESSED RIBBON SALAD

INGREDIENTS

1 lb lean 93/7 ground turkey (no sodium solutions added)
1/3 cup Italian flat-leaf parsley, finely chopped
2 cloves garlic, grated
1/2 tsp onion powder & 1/4 tsp black pepper
1 tbsp olive oil (for skillet)
1 English cucumber, shaved into long ribbons with peeler
1 tbsp apple cider vinegar & 1/2 tsp dried oregano

DIRECTIONS

1. In a bowl, gently combine ground turkey, chopped parsley, grated garlic, onion powder, and black pepper. Shape gently into 4 equal patties.
2. Toss shaved cucumber ribbons with cider vinegar and dried oregano in a small side bowl. Chill in refrigerator while cooking patties.
3. Heat olive oil in skillet over medium heat. Sear patties 5 to 6 minutes per side until nicely browned and internal temperature reaches 165°F.
4. Serve hot patties immediately crowned with chilled cucumber ribbons.

Na: 82 mg K: 290 mg P: 195 mg Protein: 25 g

WAVE 097 · MASTERCOOK FREeware

LEMON- PAPRIKA HADDOCK WITH ROASTED CAULIFLOWER

Serves 4 · 25 min
Sweet Smoke & Citrus

FT-0867 · WAVE 097 MASTER
RECIPE · SMOKED SPANISH
PIMENTÓN AND SWEET
CARAMELIZED BRASSICA

INGREDIENTS

4 wild haddock fillets (4 oz each)
3 cups cauliflower florets, sliced flat for browning
1 tsp sweet smoked paprika (pimentón dulce)
1/2 tsp dried oregano
1.5 tbsp olive oil
Zest & juice of 1 fresh lemon

DIRECTIONS

1. Preheat oven to 400°F (205°C). Toss sliced cauliflower with 1 tbsp olive oil and 1/2 tsp paprika on a rimmed baking sheet.
2. Roast cauliflower for 12 minutes until undersides begin to caramelize golden.
3. Push cauliflower to perimeter of sheet. Place haddock fillets in center. Rub fillets with remaining 1/2 tsp olive oil, remaining paprika, oregano, and lemon zest.
4. Roast 9 to 11 minutes until haddock is opaque and flakes readily. Squeeze fresh lemon juice over everything before plating.

Na: 70 mg K: 320 mg P: 190 mg Protein: 23 g

WAVE 097 · MASTERCOOK FREeware

ORANGE-ROSEMARY CHICKEN & CARROT TRAYBAKE

Serves 4 · 30 min
Sheet-Pan Elegance

FT-0855 · WAVE 098 MASTER
RECIPE · ROASTED CITRUS
ROUNDS WITH WOODSY NEEDLE
ROSEMARY

INGREDIENTS

1.2 lbs chicken breast,
cut into 1-inch strips
2 cups sliced carrots
(pre-boiled 5 min &
drained to lower
potassium)
1 small navel orange,
sliced thin into rounds
2 sprigs fresh
rosemary, leaves
stripped
1.5 tbsp olive oil
1/4 tsp freshly ground
black pepper

DIRECTIONS

1. Preheat oven to 400°F (205°C).
2. In a bowl, toss chicken strips, drained leached carrots, orange slices, and rosemary with olive oil and pepper.
3. Spread evenly across a parchment-lined baking sheet in a single layer.
4. Roast 22 to 25 minutes until chicken is browned and carrots are tender with caramelized edges.

Na: 65 mg K: 350 mg P: 210 mg Protein: 27 g

WAVE 098 · MASTERCOOK FREEWARE

TURKEY, PEPPER & OREGANO STUFFED ZUCCHINI BOATS

Serves 4 · 35 min
Garden Comfort

FT-0856 · WAVE 098 MASTER
RECIPE · ROASTED GARDEN
ZUCCHINI FILLED WITH SAVORY
HERBED PILAF

INGREDIENTS

4 medium green
zucchini, halved
lengthwise & hollowed
into boats
3/4 lb lean ground
turkey
1/2 red bell pepper,
finely diced
1 cup cooked white
jasmine rice
1 tbsp dried oregano
& 1/2 tsp garlic
powder
1 tbsp olive oil

DIRECTIONS

1. Preheat oven to 375°F (190°C). Arrange hollowed zucchini boats in a 9x13-inch baking dish.
2. Heat olive oil in skillet. Sauté ground turkey and bell pepper with oregano and garlic powder until turkey is browned.
3. Fold cooked white rice into turkey mixture. Spoon filling generously into zucchini boats.
4. Cover dish with foil; bake 20 minutes. Remove foil and bake 10 minutes more until zucchini is tender.

Na: 78 mg K: 330 mg P: 200 mg Protein: 22 g

WAVE 098 · MASTERCOOK FREEWARE

JAPANESE GINGER- SCALLION STEAMED HADDOCK

Serves 2 · 18 min

Delicate Vapor Cookery

WT-0142 · WORLD TABLE 1000
MASTER SERIES · FLAKY
WHITE FISH SCENTED WITH
SIZZLING SESAME
AROMATICS

INGREDIENTS

2 wild haddock fillets
(5 oz each)

1 tbsp fresh ginger,
peeled & julienned
into matchsticks

2 green scallions,
green and white parts
shredded lengthwise

1 tsp pure toasted
sesame oil

1 tsp rice vinegar
(unseasoned, zero
sodium)

1/2 tsp sugar

DIRECTIONS

1. Lay ginger coins in a heatproof steaming dish. Place haddock fillets on top.
2. Set dish inside steamer over boiling water. Cover tightly and steam for 8 to 10 minutes until fish flakes with a fork.
3. Carefully remove dish. Top fish with shredded scallions.
4. Heat toasted sesame oil in a tiny skillet until hot, then pour directly over scallions to sizzle and release their fragrant aroma. Drizzle with rice vinegar.

Na: 60 mg

K: 305 mg

P: 180 mg

Protein: 25 g

WORLD TABLE 1000 · MASTERCOOK FREEWARE

MOROCCAN LEMON-HERB CHICKEN TAGINE

Serves 4 · 40 min

Deep North African Spice

WT-0388 · WORLD TABLE
1000 MASTER SERIES ·
WARM AROMATIC SIMMER
OF TURMERIC, CORIANDER
AND CITRUS

INGREDIENTS

1.3 lbs boneless
chicken thighs,
trimmed of visible fat

1/2 yellow onion,
finely diced

3 cloves garlic,
minced

1 tsp ground cumin &
1 tsp ground
coriander

1/2 tsp ground
turmeric & 1/4 tsp
ginger powder

Zest of 1 fresh lemon
& 2 tbsp fresh lemon
juice

1.5 tbsp olive oil

1/2 cup water

2 tbsp fresh cilantro,
chopped

DIRECTIONS

1. Toss chicken thighs with cumin, coriander, turmeric, ginger, and black pepper.
2. Heat olive oil in a heavy Dutch oven or tagine over medium heat. Brown chicken 3 minutes per side; remove to plate.
3. Add onion and garlic to pot; sauté 4 minutes until translucent.
4. Return chicken to pot. Add water and lemon zest. Bring to simmer, cover tightly, turn heat to low, and braise 25 minutes until chicken is meltingly tender.
5. Stir in fresh lemon juice and cilantro before serving over steamed couscous or white rice.

Na: 85 mg

K: 360 mg

P: 225 mg

Protein: 28 g

WORLD TABLE 1000 · MASTERCOOK FREEWARE

SMOKY ROASTED RED PEPPER DIP

VS-0012 · VORATH KITCHEN
ALCHEMICAL BLEND ·
REPLACES HIGH-SODIUM
KETCHUP AND COMMERCIAL
BBQ SAUCES

Makes 1.5 cups · 15 min
Zero Added Sugar or Salt

INGREDIENTS

2 large red bell
peppers, fire-roasted
& skins peeled
3 cloves roasted garlic
1 tbsp extra virgin
olive oil
1.5 tsp apple cider
vinegar
1/2 tsp smoked
Spanish paprika
1/4 tsp ground cumin
& fresh black pepper

DIRECTIONS

1. Place peeled roasted red peppers and roasted garlic cloves into a food processor or blender.
2. Add olive oil, apple cider vinegar, smoked paprika, cumin, and black pepper.
3. Puree on high until silky smooth and emulsified.
4. Taste and adjust acidity with a drop more vinegar if desired. Store in airtight glass jar in refrigerator for up to 7 days. Use as dipping sauce for crispy tenders, haddock, or roasted cauliflower.

Na: 22 mg (per 2 tbsp) K: 140 mg P: 35 mg Per 2 Tbsp

VORATH KITCHEN · MASTERCOOK FREEWARE

THE CHAIRSIDE CARE KIT & 0-FLUID THIRST TAMERS

Dialysis Session Protocol
Zero-Fluid Overload

CCK-01 · CLINICAL SELF-MANAGEMENT PROTOCOL ·
MANAGING DRY MOUTH,
APPETITE, AND RECOVERY
ACROSS 7-HOUR DIALYSIS
RUNS

CARE KIT PACKING LIST

Thermal Insulated Pouch:
12 frozen seedless
sour green grapes
in small container.

Citrus Spritz Vial:
15ml spray bottle
with cold lemon
water for tongue
misting.

Sugar-Free Sour Mints: Zero-phosphate
spearmint or
lemon drops.

Post-Run Albumin Box:
Chilled egg white
rounds or sliced
cold sage chicken.

Acoustic Isolation: Noise-canceling
headphones &
warm fleece
blanket.

THE 3-PHASE DIALYSIS RUN

1. **Hour 0–2 (Needle-In & Settling):** Focus on warmth and relaxation. Fluorescent clinic lights cause sensory fatigue—dim devices and play ambient soundscapes. Keep access arm still and supported.
2. **Hour 2–5 (The Deep Stretch & Thirst Tamers):** As ultrafiltration pulls interstitial fluid, dry mouth peaks. *Do not drink cold water from the cooler.* Take ONE frozen sour grape. The intense sub-zero temperature numbs palate nerve endings; the tart malic acid triggers immediate salivation for 15+ minutes with less than 2ml of liquid.
3. **Hour 5–7 (Teardown & Washout Recovery):** After unhooking and holding sites, eat pure albumin protein from your Recovery Box to counteract post-dialysis amino acid depletion. Rest 30 minutes before transit.

Fluid: < 5 ml K: < 40 mg P: 0 mg Sovereign Protocol

CHAIRSIDE COMPANION · NORTHSTAR PRIME

CHAPTER 36 · CLINICAL CULINARY EXPANSION

Dialysis Day Recovery & The Phosphate-Free Baker

“Post-treatment washout is real. You don't need a lecture on compliance—you need clean, effortless protein to rebuild your muscle and zero-fluid flavor that respects your limits.”

THE CLINICAL RECOVERY & BAKING SCIENCE:

1. **Post-Dialysis Washout:** Hemodialysis strips both uremic toxins and essential amino acids (~10-15g of free amino acids per 4-hour treatment). The body enters a catabolic state if high-biological-value protein is not consumed promptly. Whole eggs pack heavy phosphorus in the yolk (~180mg P), but egg whites provide pure albumin protein with negligible phosphorus.
2. **The Chemical Leavener Trap:** Commercial baking powders and bakery breads routinely utilize sodium aluminum phosphate (SALP), monocalcium phosphate, or dicalcium phosphate as leavening acids. 100% of inorganic phosphate is absorbed into the bloodstream. Our formulations use pure yeast fermentation and cold-dough autolyse to achieve airy crumb architecture with **0 mg added phosphate salts**.

3. **Zero-Fluid Umami:** Reducing pan acids and roasted alliums creates hyper-concentrated glazes that wake up a metallic dialysis palate without adding fluid milliliters to the daily allowance.

PHOSPHATE-FREE GOLDEN HERB FOCACCIA

Per square: 85mg Na • 110mg K •

FT-BAKE-01 • THE
PHOSPHATE-FREE
BAKER • SLOW-
FERMENTED AIRY
CRUMB WITH ZERO
CHEMICAL LEAVENERS

Why it works: Commercial biscuits and focaccia often contain baking powder packed with sodium aluminum phosphate. This artisan formula relies exclusively on active dry yeast, bread flour, extra virgin olive oil, and fresh rosemary needles for an irresistible golden crust and pillowy interior.

- Ingredients:**
- 3 cups unbleached bread flour
 - 1 packet (2 1/4 tsp) active dry yeast
 - 1 cup lukewarm water (used in dough hydration, baked off dry)
 - 3 tbsp extra virgin olive oil (divided)
 - 1/2 tsp kosher flake salt (measured strictly)
 - 2 tbsp fresh rosemary needles (roughly chopped)
 - 2 cloves garlic (thinly sliced)
 - 1/4 tsp cracked black pepper

Technique:

1. Dissolve yeast in lukewarm water for 5 minutes until frothy.
2. Combine bread flour and measured salt in a large bowl. Add yeast liquid and 2 tbsp olive oil; mix until shaggy dough forms.
3. Knead on an unfloured board for 8 minutes until silky and elastic. Transfer to an oiled bowl, cover, and let rise for 75 minutes until doubled.
4. Dimple the dough into a parchment-lined 9x13 metal baking pan with oiled fingertips. Scatter sliced garlic, chopped rosemary, black pepper, and drizzle remaining 1 tbsp olive oil over top.
5. Proof for 25 minutes. Bake at 425°F (220°C) for 20-22 minutes until crust is deep burnished amber. Cool on a wire rack before slicing into 12 equal squares.

0mg Added Phosphates Natural Yeast Fermentation
Low Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

DIALYSIS WASHOUT EGG WHITE & CHIVE RECOVERY FRITTATA

FT-0870 • POST-DIALYSIS RECOVERY
• HIGH-BV ALBUMIN
PROTEIN BOX FOR
IMMEDIATE POST-TREATMENT FATIGUE

Why it works: Post-dialysis fatigue demands rapid high-biological-value protein that does not require an hour of standing over a stove. Liquid egg whites provide 18 grams of clean albumin per serving with negligible phosphorus, brightened with sniped fresh chives, dill, and white pepper.

Technique:

1. Heat olive oil in an 8-inch oven-safe nonstick skillet over medium heat. Sauté diced red bell pepper for 3 minutes until tender-crisp.
2. Whisk egg whites with chives, dill, white pepper, and garlic powder until frothy.
3. Pour egg mixture into skillet over peppers. Let edges set for 2 minutes, gently pulling cooked edges inward to let raw egg flow underneath.
4. Transfer skillet to preheated broiler on high for 3-4 minutes until puffed, golden, and fully set in the center.
5. Slide onto a plate and enjoy warm or slice into chilled squares for the clinic ride home.

18g Albumin Protein Low Phosphorus
0-Fluid Post-Run Meal

RECIPE CARD • MASTERCOOK CKD KITCHEN

Per serving: 125mg Na • 180mg K • 60mg P • 18g Protein • 0ml Fluid

CRISPY SESAME- CRUSTED SALMON SKEWERS

FT-0871 • HEART &
KIDNEY VANGUARD
• FLASH-SEARED
OMEGA-3 WITH
TOASTED SESAME
CRUST

Why it works: Dialysis patients need cardiovascular vascular support and clean fatty acids. Fresh wild salmon cubes are dredged in toasted white and black sesame seeds and pan-seared until crackling crisp. A tiny brush of rice vinegar and grated ginger cuts through the richness without sodium.

Technique:

1. Thread 4 salmon cubes onto each soaked bamboo skewer.
2. Brush skewers lightly with whipped egg white, then roll in blended sesame seeds and ginger.
3. Heat avocado oil in a large cast-iron skillet over medium-high heat until shimmering.
4. Sear skewers for 2 minutes per side (8 minutes total) until sesame crust is fragrant and golden and salmon reaches 145°F internally.
5. Finish with a light mist of unseasoned rice vinegar and serve immediately.

Per skewer serving: 95mg Na • 280mg K

Ingredients:

- 1 lb skinless wild salmon fillet, cut into 1-inch cubes
- 2 tbsp white sesame seeds
- 1 tbsp black sesame seeds
- 1 egg white (whipped frothy)
- 1 tsp grated fresh ginger root
- 1 tsp rice vinegar (unseasoned, zero sodium)
- 1 tbsp avocado oil for searing
- Bamboo skewers soaked in water

Omega-3 Rich Under 100mg Sodium Sesame Crunch

RECIPE CARD • MASTERCOOK CKD KITCHEN

SAFFRON & FENNEL POACHED COD BOWLS

WT-0420 • WORLD TABLE MEDITERRANEAN • AROMATIC SAFFRON THREADS AND SHAVED FENNEL BULB

Why it works: Saffron offers deep golden warmth and delicate floral earthiness that makes salt completely unnecessary. Shaved fennel bulb melts into a gentle anise sweetness that cushions tender Pacific cod.

- Ingredients:**
- 8 oz fresh Pacific cod fillets (two 4 oz fillets)
 - 1 small fennel bulb, trimmed and shaved paper-thin
 - Pinch of genuine Spanish saffron threads (about 12-15 threads)
 - 1/4 cup boiling water (to bloom saffron)
 - 1 tbsp extra virgin olive oil
 - 1 strip orange peel (zest only, white pith removed)
 - 1/4 tsp cracked coriander seeds

- Technique:**
1. Steep saffron threads in 1/4 cup boiling water for 10 minutes until liquid turns luminous crimson-gold.
 2. Heat olive oil in a wide skillet over medium-low heat. Add shaved fennel, orange peel strip, and crushed coriander; cook gently for 6 minutes until fennel is limp and sweet.
 3. Nestle cod fillets directly onto the bed of fennel. Pour bloomed saffron water over the fish.
 4. Cover with a tight-fitting lid and poach over low heat for 8-10 minutes until cod flakes cleanly with a fork (145°F).
 5. Discard orange peel strip and serve cod hot over the saffron-braised fennel.

Ultra-Low Sodium Golden Saffron Umami
Digestive Fennel

RECIPE CARD • MASTERCOOK CKD KITCHEN

Per serving: 85mg Na • 260mg K • 190mg P • 24g Protein • 40ml Fluid

PERSIAN HERB & SUMAC CHICKEN PATTIES

WT-0550 • WORLD TABLE SILK ROAD • TART WILD SUMAC AND FRESH GREEN HERBS REPLACING SALT

Per 2-patty serving: 75mg Na • 290mg K •

Why it works: Ground wild sumac berry is the ultimate renal seasoning discovery: its natural citric and malic acids trigger intense salivation and pleasant tart acidity that tricks the palate into perceiving salt, while delivering zero sodium. Combined with heaps of parsley, mint, and cumin.

- Ingredients:**
- 1 lb lean ground chicken breast (unbrined)
 - 1 1/2 tbsp ground wild sumac
 - 1/2 cup finely minced flat-leaf parsley
 - 1/4 cup finely minced fresh spearmint
 - 1 small grated yellow onion (squeezed dry in paper towels)
 - 1 tsp ground cumin
 - 1/2 tsp ground turmeric
 - 1 tbsp olive oil for grilling

- Technique:**
1. Grate onion and squeeze vigorously in paper towels to remove excess water.
 2. In a bowl, gently blend ground chicken, sumac, parsley, mint, squeezed onion, cumin, and turmeric. Form into 8 small flat oval patties.
 3. Heat a grill pan or cast-iron skillet over medium-high heat with olive oil.
 4. Cook patties for 4-5 minutes per side until seared with deep grill marks and internal temperature reaches 165°F.
 5. Dust with an extra pinch of sumac right off the heat and serve alongside cucumber rounds.

Tart Sumac Acid Trick No Salt Substitute
Fresh Garden Herbs

RECIPE CARD • MASTERCOOK CKD KITCHEN

HYDRATION-FREE UMAMI GLAZE (BLACK GARLIC & APPLE REDUCTION)

Makes 8 servings (2 tbsp each) • 25 min

Per 2 tbsp: 20mg Na • 85mg K • 15mg P • 1g Protein • 0ml Counted Fluid

VS-0015 • VORATH
ALCHEMICAL KITCHEN •
CONCENTRATED
ROASTED SAVORY
PUNCH WITHOUT FLUID
OUNCES

Why it works: Dialysis fluid limits make watery broths and liquid gravies dangerous. This reduction simmers unfiltered apple cider with balsamic vinegar, roasted black garlic, and smoked paprika until reduced by 75% into a glossy syrup that coats grilled chicken cutlets or haddock with savory intensity.

Ingredients:

- 1 cup 100% pure unfiltered apple cider
- 2 tbsp aged balsamic vinegar
- 3 cloves black garlic (or roasted garlic), mashed to a smooth paste
- 1/2 tsp smoked sweet paprika
- 1 sprig fresh thyme
- 1/4 tsp cracked black pepper

Technique:

1. Combine apple cider, balsamic vinegar, mashed black garlic paste, smoked paprika, and thyme sprig in a small heavy-bottomed saucepan.
2. Bring to a boil, then reduce heat to medium-low and simmer steadily for 18-20 minutes.
3. Watch closely as the liquid reduces from 1 cup down to about 1/4 cup (4 tablespoons) and develops large, slow bubbles with a glossy syrup consistency.
4. Discard thyme stem. Stir in black pepper. Cool to room temperature (it thickens further into a glass-like glaze as it cools).
5. Brush 1-2 tablespoons directly over hot grilled proteins right before serving.

Fluid-Preserving Glaze Black Garlic Umami
Under 25mg Sodium

The Global Heritage Table: World Foodways & Washoku

“Tradition is not bland, and cultural heritage does not require kidney sacrifice. When seasoned with citrus, whole chiles, toasted seeds, and mindful fire, global cuisine is naturally kidney-friendly.”

THE GLOBAL FOODWAYS CLINICAL ARCHITECTURE:

1. **Cultural Integrity with Renal Safety:** Traditional ancestral foodways from West Africa, the Maghreb, East Asia, the Andes, and Central America evolved long before ultra-processed factory sauces. By returning to foundational techniques—two-stage grain steaming, fresh herb pounded pastes, dry-roasted whole chiles, and clean acid macerations—we deliver explosive authentic flavor without commercial sodium, bottled fish sauces, or chemical bouillon cubes.

2. **Immunocompromised Food Safety Gate:** Hemodialysis and advanced CKD patients face elevated infection vulnerabilities. Recipes inspired by raw preparations (such as Peruvian ceviche) are strictly modified: fish and seafood are fully cooked to 145°F (63°C) before citrus maceration, satisfying FoodSafety.gov immunocompromised clinical standards.

3. **Fluid & Electrolyte Accountability:** Soups and braises (Joumou, Borscht, Tom Yum) are engineered as either reduced-liquid skillet sautés or strictly measured 1-cup portions, protecting daily fluid allowances while providing high-biological-value protein and potassium-budgeted whole produce.

YUZU- GINGER CHICKEN, CABBAGE & RICE BOWLS

R01 / WT-0143 •
JAPAN / WASHOKU
HERITAGE • CRISP
CABBAGE AND
SEARED CHICKEN
IN FRAGRANT
YUZU-GINGER
GLAZE

Cultural Integrity &

Technique: Washoku honors seasonality, balance, and clean presentation. Rather than drowning produce in a heavy commercial teriyaki sauce packed with sodium and caramel coloring, this weeknight bowl builds bright, restorative aroma with fresh grated ginger, pure yuzu juice, unseasoned rice vinegar, and quick-seared chicken cutlets.

Technique:

1. Combine rinsed rice and 1 1/4 cups water in a saucepan. Bring to a rolling boil, cover tightly, reduce heat to low, and cook 15 minutes. Remove from heat and rest covered 10 minutes; fluff with a fork.
2. In a small ceramic bowl, whisk yuzu juice, rice vinegar, sugar, grated ginger, and 1 tsp neutral oil until emulsified.
3. Heat remaining oil in a large heavy skillet over medium-high heat. Add chicken pieces and cook, turning, for 6–8 minutes until golden browned and internal temperature reaches 165°F (74°C) with a clean thermometer. Transfer to a warm plate.
4. Add cabbage to the skillet and cook 3–4 minutes over medium

Per serving (est): 72mg Na • 290mg K • 195mg P • 26g Protein • 20ml Fluid

Ingredients:

- 1 cup Japanese short-grain white rice (rinsed until water runs clear)
- 1 1/4 cups fresh water (for cooking rice)
- 12 oz boneless skinless chicken breast (cut into 1-inch pieces)
- 1 tbsp neutral high-smoke oil (divided)
- 4 cups green cabbage (thinly sliced into ribbons)
- 2 tsp fresh ginger root (finely grated)
- 2 tbsp unsweetened yuzu juice (or fresh lemon juice)
- 1 tbsp unseasoned rice vinegar
- 1 tsp granulated sugar
- 2 tbsp scallion greens (thinly sliced)

Chef Swaps & Safety Flags:

Fresh lemon juice provides a bright, accessible substitute if yuzu is unavailable. Firm white fish (cod, haddock) can replace chicken—sear gently to 145°F (63°C). Contains no major allergens. Check vinegar labels for sodium or sulfites.

heat, tossing frequently
Makes 4 servings • 25 min
Per serving (est): 72mg Na • 290mg K • 195mg P • 26g Protein • 20ml Fluid
5. Return chicken and any rested juices to the skillet, pour in the yuzu-
ginger mixture, and toss
off heat for 30 seconds.
Divide rice, cabbage, and
chicken among four
bowls; crown with sliced
scallion greens.

Zero Soy Sauce Washoku Citrus Glaze
Under 80mg Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

ANCHO TURKEY TOSTADAS WITH LIME- CABBAGE CRUNCH

R02 / WT-0205 •
MEXICO /
MICHOACÁN
HERITAGE • SLOW-
STEEPED WHOLE
ANCHO CHILE
PUREE ON CRISP
BAKED CORN
TORTILLAS

Cultural Integrity &

Technique: True Mexican cooking draws warmth, sweetness, and depth from whole dried chiles rather than high-sodium seasoning packets. Dried ancho chiles are stemmed, seeded, steeped in hot water, and pureed with garlic and Mexican oregano to create a rich, clinging adobo sauce that coats lean turkey on crisp oven-baked tortillas.

Technique:

1. Heat oven to 400°F (204°C). Lightly brush corn tortillas on both sides with 1 tsp oil and arrange in a single layer on baking sheets. Bake 8–11 minutes, turning once, until crisp and rigid.
2. Submerge dried ancho chiles in hot water for 15 minutes to soften. Drain and discard soaking liquid. Place softened chiles in a blender with 1/2 cup fresh water, garlic cloves, cumin, and Mexican oregano; blend until smooth.
3. Heat 2 tsp oil in a large nonstick skillet over

Makes 4 servings (2 tostadas each) • 30 mins
Per serving (est): 88mg Na • 320mg K • 210mg P • 24g protein

Ingredients:

- 8 small (about 6-inch) plain corn tortillas (check label for zero added phosphate leaveners)
- 2 dried ancho chiles (stemmed and seeded)
- 3/4 cup hot water (for steeping chiles; discard after use)
- 1/2 cup fresh cold water (for blending)
- 2 cloves fresh garlic (peeled)
- 1/2 tsp ground cumin
- 1/2 tsp crumbled Mexican oregano
- 12 oz 93%-lean ground turkey
- 2 tbsp neutral high-heat oil (divided)
- 4 cups green cabbage (finely shredded)
- 1/2 cup red bell pepper (thinly sliced)
- 2 tbsp fresh lime juice
- 1/4 cup fresh cilantro (roughly chopped)

Chef Swaps & Safety Flags:

Guajillo chiles can replace one ancho for a brighter, tangier profile. If corn tortilla labels indicate sodium leaveners, serve the ancho turkey filling over steamed white rice. No major allergens present.

medium-high heat. Add 1/2 cup water, breaking up any clumps until browned. Pour in the ancho chile puree, lower heat, and simmer for 5–7 minutes until sauce clings to turkey and meat reaches 165°F (74°C).

4. In a separate bowl, toss shredded cabbage, red bell pepper, fresh lime juice, cilantro, and remaining 1 tbsp oil.
5. Assemble immediately before eating: spoon warm ancho turkey onto crisp tostadas and heap with crunchy lime-cabbage slaw.

Oven-Baked Tostadas Whole Ancho Adobo
Phosphate-Free Tortillas

RECIPE CARD • MASTERCOOK CKD KITCHEN

FRESH GOCHUGARU- PEAR CABBAGE WITH GRILLED CHICKEN & RICE

R03 / WT-0312 • KOREAN
PANTRY INSPIRATION •
SAME-DAY UNFERMENTED
NAPA-PEAR SLAW WITH
CHARRED CHICKEN
THIGHS

Cultural Integrity &

Technique: Traditional fermented kimchi relies on heavy salt brining, fermented shrimp, and fish sauce—unsuitable for renal patients. This same-day unfermented Korean slaw delivers crunchy napa cabbage, crisp Asian pear, and daikon tossed in a vibrant dressing of gochugaru chile flakes, garlic, ginger, and toasted sesame oil.

Technique:

1. Simmer rice in water covered on low for 15 minutes; remove from heat and rest covered 10 minutes, then fluff.
2. In a small bowl, whisk rice vinegar, gochugaru, toasted sesame oil, sugar, grated garlic, and grated ginger until blended into a glossy dressing.
3. Heat a cast iron grill pan over medium-high heat and brush lightly with neutral oil. Grill chicken thighs 5–7 minutes per side until deeply seared and internal temperature registers 165°F (74°C). Rest on a cutting board 5 minutes, then slice into strips.

Ingredients:

- 1 cup rinsed short-grain white rice
- 1 1/4 cups water
- 12 oz trimmed boneless skinless chicken thighs
- 1 tsp neutral oil (for grill pan)
- 4 cups fresh napa cabbage (cut into 1-inch crisp squares)
- 3/4 cup Asian pear (cut into matchsticks)
- 1/2 cup daikon radish (cut into matchsticks)
- 1 tbsp unseasoned rice vinegar
- 2 tsp Korean red pepper flakes (gochugaru)
- 1 tsp toasted dark sesame oil
- 1 tsp granulated sugar
- 1 clove fresh garlic (finely grated)
- 1 tsp fresh ginger (finely grated)
- 2 tbsp scallion greens (thinly sliced)

Chef Swaps & Safety Flags:

Contains sesame (allergen). For sesame sensitivity, replace sesame oil with neutral oil and a pinch of toasted sunflower seed aroma. Crisp gala or fuji apple can replace Asian pear if pear is unavailable.

Per serving (est): 68mg Na • 340mg K • 22mg P • 12g Protein • 15ml Fluid

4. Just before serving, combine napa cabbage, pear, and scallion greens in a salad bowl. Pour dressing over and toss vigorously. Do not let sit or ferment.
5. Divide warm rice among plates, top with sliced chicken, and mound the fresh gochugaru-pear salad alongside.

Zero Fermented Salt Fresh-Dressed Napa
Under 70mg Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

TWICE-STEAMED COUSCOUS WITH CHERMOULA CHICKEN & SUMMER VEGETABLES

R04 / WT-0389 • MAGHREB HERITAGE • TRADITIONAL TWO-STAGE AERATED COUSCOUS WITH ROASTED HERB-SPICED CHICKEN

Cultural Integrity & Technique: Rather than boiling couscous into a dense, waterlogged mass, North African couscous craft relies on two-stage steam hydration. Sprinkling water and rolling the granules with olive oil creates extraordinary lightness. The dish is paired with tender chicken and roasted vegetables rubbed with fresh chermoula (parsley, cilantro, cumin, paprika, and lemon).

Technique:

1. Heat oven to 425°F (218°C). In a bowl, combine parsley, cilantro, minced garlic, lemon juice, cumin, paprika, coriander, and 2 tbsp olive oil to form a vibrant chermoula. Rub half over chicken cutlets; toss remainder with zucchini, bell pepper, and onion.
2. Spread vegetables and chicken on a parchment-lined baking sheet. Roast 18–22 minutes until vegetables are caramelized and chicken reaches 165°F (74°C); slice chicken.
3. Place dry couscous in a wide shallow bowl. Sprinkle with 3/4 cup

Ingredients:

- 1 1/2 cups plain medium-grain semolina couscous (unseasoned)
- 1 1/2 cups fresh water (divided)
- 12 oz boneless chicken breast (cut into 4 cutlets)
- 2 cups zucchini (sliced into half-moons)
- 1 cup red bell pepper (cut into 1-inch squares)
- 1/2 cup yellow onion (cut into wedges)
- 3 tbsp extra virgin olive oil (divided)
- 1/2 cup fresh flat-leaf parsley (finely chopped)
- 1/2 cup fresh cilantro (finely chopped)
- 2 cloves garlic (minced)
- 2 tbsp fresh lemon juice
- 1 tsp ground cumin
- 1 tsp sweet Hungarian paprika
- 1/2 tsp ground coriander

Chef Swaps & Safety Flags: Contains wheat/gluten. For celiac or wheat allergy, replace couscous with steamed short-grain white rice or millet. Preserved lemons are omitted due to extreme sodium content; fresh lemon juice provides clean acidity.

Per serving (est): 75mg Na • 350mg K • 215mg P • 28g carbs • 15g protein • 15ml Fluid

- water and 1 tsp olive oil; rub gently over cutlets. Makes 4 servings • 40 mins
Serves 4 • 12g Fat • 15g Protein • 28g Carbs • 15ml Fluid
Rest 10 minutes.
4. Transfer couscous to a cheesecloth-lined steamer basket over simmering water; steam covered 15 minutes.
 5. Return couscous to the bowl, sprinkle with remaining 3/4 cup water, fluff with a fork to break clumps, rest 5 minutes, and steam a second time for 10 minutes. Fluff with remaining 2 tsp olive oil.
 6. Mound the featherlight couscous on a sharing platter and arrange roasted chermoula chicken and summer vegetables over top.

Two-Stage Steamed Grain Fresh Herb Chermoula
No Preserved Lemon Salt

RECIPE CARD • MASTERCOOK CKD KITCHEN

CEEBU-JEN-INSPIRED PARSLEY FISH, BROKEN RICE & CABBAGE

R05 / WT-0512 •
SENEGALESE
HERITAGE (SAINT-LOUIS) • POCKET-STUFFED BAKED COD WITH AROMATIC RED BROKEN JASMINE RICE

Cultural Integrity & Technique: Ceebu Jën is the culinary soul of Senegal. Traditional preparations incorporate salt-fermented seafood and bouillon cubes. This kidney-conscious adaptation preserves the brilliant rof stuffing (flat-leaf parsley, garlic, lemon, and crushed pepper pressed inside fish pockets) and simmers broken jasmine rice in a delicate tomato-vegetable reduction.

- Technique:**
1. Heat oven to 400°F (204°C). In a small mortar or bowl, pound parsley, minced garlic, lemon juice, red pepper flakes, and 1 tsp oil into a fragrant green rof paste.
 2. Cut a horizontal shallow pocket into the thickest part of each cod fillet. Pack the parsley paste into the pockets. Brush fish with 1 tsp oil and bake 10–13 minutes until fish flakes easily and internal temperature reaches 145°F (63°C).
 3. Heat remaining oil in a heavy saucepan over medium heat; cook onion

Per serving (est): 92mg Na • 380mg K • 210mg P • 25g Protein • 40g Fat

- Ingredients:**
- 1 lb fresh cod fillets (divided into 4 portions)
 - 1 1/4 cups broken jasmine rice (rinsed thoroughly)
 - 2 1/4 cups fresh water
 - 2 cups green cabbage (cut into wedges)
 - 1 cup carrot coins
 - 1 cup eggplant cubes (peeled)
 - 1/2 cup yellow onion (finely chopped)
 - 2 tbsp no-salt-added tomato paste (verify 0mg added salt on label)
 - 1/2 cup fresh flat-leaf parsley (finely chopped)
 - 2 cloves garlic (minced)
 - 1 tbsp fresh lemon juice
 - 1/4 tsp crushed red pepper flakes (optional)
 - 1 tbsp neutral oil (divided)

Chef Swaps & Safety Flags: Contains fish (allergen). Haddock, pollock, or halibut work equally well. Completely omits fermented dried snails (guédj) and salted fish (yet) to keep sodium under 95mg.

- 3 minutes until fragrant. Add cabbage wedges, carrot coins, eggplant cubes, and 2 1/4 cups water; cover and simmer 8 minutes.
4. Lift vegetables out with a slotted spoon to a warm plate. Stir rinsed broken rice into the simmering tomato broth, cover, reduce heat to low, and cook 16 minutes. Rest covered 5 minutes.
 5. Plate the red broken rice, arrange tender vegetables alongside, and nestle the herb-stuffed cod fillet over top.

Herbed Rof Stuffing Broken Jasmine Rice
Zero Bouillon Cubes

RECIPE CARD • MASTERCOOK CKD KITCHEN

BAKED HARISSA EGG- WHITE BRIK- STYLE PARCELS

R06 / WT-0618 •
TUNISIAN
HERITAGE •
SHATTERING
BAKED PHYLLO
TRIANGLES
FILLED WITH
HERBED EGG
WHITES AND
SCRATCH
HARISSA

Cultural Integrity &

Technique: Tunisian brik parcels celebrate the earthy, warming power of caraway, coriander, and dried chiles. Traditional commercial harissa pastes are laden with salt (often 2,000mg+ Na/100g). Here, a scratch paste is pureed with toasted caraway, coriander, and lemon, folded with post-dialysis recovery egg whites, and baked to a golden crisp.

Technique:

1. Soak dried chiles in hot water 15 minutes; drain thoroughly. Blend with garlic, ground coriander, toasted caraway, lemon juice, 2 tbsp fresh water, and 1 tsp olive oil until a smooth, thick red paste forms.
2. Heat 1 tsp olive oil in a nonstick skillet over medium heat. Sauté diced onion 4 minutes until tender. Pour in liquid egg whites and gently scramble with a heatproof spatula until softly set and dry. Off heat, fold in parsley and 1 tbsp of the scratch harissa.

Makes 4 servings (2 parcels each) • 30 mins
(26 g each)
Per serving (est): 82mg Na • 210mg K • 65mg P • 14g protein

Ingredients:

- 4 dried mild red chiles (guajillo or mild New Mexico, stemmed and seeded)
- 1 cup hot water (for steeping chiles; discard after use)
- 1 clove garlic (minced)
- 1/2 tsp ground coriander
- 1/4 tsp whole caraway seeds (lightly toasted and ground)
- 1 tbsp fresh lemon juice
- 2 tbsp fresh water (for blending)
- 1 tbsp extra virgin olive oil (divided)
- 1 cup pasteurized liquid egg whites (verify 0mg phosphate on label)
- 1/2 cup yellow onion (finely diced)
- 1/4 cup fresh flat-leaf parsley (finely chopped)
- 8 sheets commercial phyllo dough (thawed, unseasoned)

Chef Swaps & Safety Flags:
Contains egg and wheat/gluten (allergens). Ideal post-dialysis washout recovery food: pure albumin protein with negligible phosphorus (65mg P). For wheat sensitivity, spoon warm filling into crisp lettuce leaves.

3. Heat oven to 400°F (264°C). Lay out 8 sheets of phyllo, misting lightly with olive oil, fold lengthwise into thirds to create a long ribbon. Place 2 tablespoons of egg white filling at the base and fold upward in a flag-style triangle. Repeat for 8 parcels.
4. Place parcels seam-side down on a parchment-lined baking sheet. Lightly mist or brush top with remaining oil. Bake 12–15 minutes until deep golden-brown and shattering crisp.
5. Serve warm. Refrigerate leftover harissa paste in an airtight jar for up to 5 days.

Scratch Caraway Harissa Pure Albumin Recovery
Under 70mg Phosphorus

RECIPE CARD • MASTERCOOK CKD KITCHEN

SMALL-BATCH LAMB, CARROT & BARBERRY PALAV

R07 / WT-0720 •
CENTRAL ASIAN /
TAJIKISTAN
HERITAGE •
FRAGRANT LAYERED
BASMATI RICE WITH
CUMIN LAMB AND
SWEET CARROTS

Cultural Integrity &

Technique: Oshi Palav represents hospitable feast-making across Tajikistan and Uzbekistan. In traditional cauldrons (kazans), meat and carrots form the zirvak foundation, over which rice is layered to steam undisturbed. We scale this to a 4-serving home Dutch oven with cumin, coriander, and ruby barberries that impart jewel-like tartness without sodium.

Technique:

1. Heat oil in a heavy Dutch oven or deep skillet over medium-high heat. Add lamb cubes and brown deeply on all sides for 6 minutes. Add sliced onion and cook 4 minutes until golden.
2. Scatter carrot batons, bruised cumin seeds, and ground coriander over the lamb. Pour in 1 cup water, cover, reduce heat to low, and simmer 20 minutes until lamb is tender.
3. Spread the drained basmati rice evenly over the meat and vegetables in a level layer without stirring. Gently pour the remaining 1 1/2 cups water along the edge of the pot so the layers remain undisturbed. Scatter barberries across the surface.
4. Increase heat until water bubbles through the rice layer, then reduce heat to

Per serving (est): 70mg Na • 330mg K • 190mg P • 2mg B • 10ml Fluid

Ingredients:

- 12 oz trimmed lean boneless lamb leg (cut into 1-inch cubes)
- 1 1/2 cups white basmati rice (rinsed and soaked 20 minutes, drained)
- 2 cups carrots (cut into matchstick batons)
- 1 cup yellow onion (thinly sliced)
- 2 tbsp neutral cooking oil
- 1 tsp whole cumin seeds (lightly bruised)
- 1/2 tsp ground coriander
- 2 tbsp dried barberries (rinsed; optional tart finish)
- 2 1/2 cups fresh water (divided)

Chef Swaps & Safety Flags:

Skinless chicken thighs (cut into 1-inch pieces) can substitute for lamb—cook until internal temperature reaches 165°F (74°C). Barberries provide natural tartness without salt; omit if unavailable. Contains no major allergens.

the lowest setting, cover
Makes 4 servings in 45 mins
Makes 1 serving in 15 mins
Steam 12 minutes
Remove from heat and
rest covered 10 minutes.

5. Confirm lamb internal temperature reaches at least 145°F (63°C) with a 3-minute rest. Gently fluff and invert onto a platter so the golden lamb and carrots rest atop the aromatic rice.

Layered Kazan Steaming Bruised Cumin & Barberry
Under 75mg Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

JOUMOU-INSPIRED SQUASH, CABBAGE & BEEF SOUP

R08 / WT-0815 •
HAITIAN HERITAGE
• VELVETY PUREED
CALABAZA
SQUASH BROTH
WITH TENDER
BEEF CHUCK AND
VEGETABLES

Cultural Integrity &

Technique: Soup Joumou is Haiti's sacred historic symbol of liberation and dignity. While ceremonial New Year's pots are generous and shared widely, this renal adaptation honors the culinary soul through smooth calabaza (or kabocha) squash puree, aromatic allspice, fresh thyme, and beef, while adhering to strict 1-cup measured fluid portions.

Technique:

1. Heat oil in a stockpot over medium-high heat. Brown beef cubes on all sides for 6 minutes. Add onion, garlic, dried thyme, and allspice; cook 2 minutes until fragrant.
2. Add cubed squash and 5 cups water. Bring to a boil, cover, reduce heat to low, and simmer 25 minutes until squash is tender.
3. Transfer half the softened squash and 1 cup cooking liquid to a blender; puree until silky smooth and return to the stockpot to create the rich, golden signature velvety broth.
4. Add cabbage, carrots, turnip cubes, macaroni,

Makes 6 servings (1 cup each strictly measured) • 150 mins
Per serving (est): 65mg Na • 340mg K • 175mg P • 18g protein • 240ml fluid

Ingredients:

- 12 oz trimmed lean beef chuck (cut into 3/4-inch cubes)
- 1 tbsp neutral cooking oil
- 3 cups calabaza or kabocha squash (peeled, seeded, and cubed)
- 2 cups green cabbage (chopped into 1-inch pieces)
- 1 cup carrots (sliced into coins)
- 1 cup white turnips (peeled and diced)
- 1/2 cup yellow onion (chopped)
- 2 cloves garlic (minced)
- 1 tsp dried thyme
- 1/4 tsp ground allspice
- 1 whole fresh Scotch bonnet chile (pierced once; optional)
- 1 tbsp fresh lime juice
- 1/2 cup dry small elbow macaroni (plain, unseasoned)
- 5 cups fresh water (strictly measured)

Chef Swaps & Safety Flags:

Contains wheat/gluten (macaroni). For gluten-free dining, replace macaroni with parboiled white rice. **Fluid notice:** Each 1-cup serving counts as 240ml toward the daily fluid limit; track in RenalShield. Omit Scotch bonnet for mild palates.

and the whole pierced Scotch bonnet chile (pierced once) for 150 mins. Simmer only 240ml fluid for 14 minutes until pasta and vegetables are tender.

5. Carefully remove and discard the whole Scotch bonnet chile intact without breaking it. Stir in fresh lime juice. Measure the total yield and divide equally into six 1-cup portions.

Pureed Squash Broth Measured 1-Cup Serving
Zero Bouillon Cubes

RECIPE CARD • MASTERCOOK CKD KITCHEN

SMALL-BOWL BEET, CABBAGE & DILL CHICKEN BORSCHT

R09 / WT-0911 •
UKRAINIAN
HERITAGE • RUBY
SWEET-SOUR
BROTH WITH
CRISP BEETS,
TENDER CHICKEN,
AND FRAGRANT
FRESH DILL

Cultural Integrity & Technique: Ukrainian borscht is a living UNESCO cultural practice. Rather than relying on high-sodium canned broths or cured sausages, this small-bowl adaptation draws deep ruby brilliance and earthy sweetness from fresh shredded beets, softened cabbage, tender chicken breast, and a crisp sweet-sour balance of apple cider vinegar and fresh dill.

- Technique:**
1. Heat oil in a soup pot over medium heat. Sauté chopped onion and carrot matchsticks for 5 minutes until soft. Add beet matchsticks and minced garlic; cook 3 minutes.
 2. Add 6 cups water and the bay leaf. Bring to a boil, cover, reduce heat to low, and simmer 20 minutes until beets begin to soften.
 3. Add shredded cabbage and chicken pieces; simmer gently for 10–12 minutes until chicken is cooked through and registers 165°F (74°C).
 4. Remove and discard the bay leaf. Stir in apple cider vinegar, sugar, and fresh chopped dill. Let rest off heat for 5 minutes

Makes 6 servings (1 cup each strictly measured)
Per serving (est): 60mg Na • 310mg K • 170mg P • 17g Protein • 240ml Fluid

to infuse the sweet-sour balance. • 40 mins
5. Measure the total fluid volume and divide equally into six 1-cup portions. Serve hot.

Fresh Beet Ruby Broth Sweet-Sour Apple Cider
Under 65mg Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

- Ingredients:**
- 12 oz boneless skinless chicken breast (cut into 1-inch pieces)
 - 1 1/2 cups fresh red beets (peeled and cut into thin matchsticks)
 - 3 cups green cabbage (finely shredded)
 - 1 cup carrots (cut into matchsticks)
 - 1/2 cup yellow onion (finely chopped)
 - 1 tbsp neutral cooking oil
 - 1 clove garlic (minced)
 - 1 dried bay leaf
 - 1 tbsp raw apple cider vinegar
 - 1 tsp granulated sugar
 - 6 cups fresh water (strictly measured)
 - 1/4 cup fresh dill (finely chopped)

Chef Swaps & Safety Flags:
Contains no major allergens.
Fluid notice: Each 1-cup bowl counts as 240ml toward fluid allowances. Avoid commercial sour cream unless a measured 1-tbsp portion is approved in your individualized meal plan.

MANSAF-INSPIRED LAMB, YOGURT & RICE PLATTER

R10 / WT-1004 • JORDANIAN HERITAGE • CARDAMOM-SPICED TENDER LAMB WITH WARM STABILIZED YOGURT SAUCE OVER TORN PITA AND RICE

Cultural Integrity & Technique:

Al-Mansaf represents Jordanian hospitality and celebration. Traditional recipes require jameed (dried fermented goat's milk rock), which contains extreme salt concentrations. We adapt the dish using plain pasteurized whole-milk yogurt gently stabilized with cornstarch and tempered with cardamom-cinnamon lamb braising liquor, layered over torn flatbread and rice.

Technique:

1. Heat oil in a saucepan over medium heat; brown lamb pieces 5 minutes. Add chopped onion, cardamom, cinnamon, and 2 cups water. Cover and simmer gently for 45 minutes until lamb is tender and internal temperature reaches 145°F (63°C) with a 3-minute rest.
2. Lift lamb out with a slotted spoon. Measure exactly 1/2 cup of hot cooking broth; discard or freeze the remainder.
3. Simmer basmati rice in 1 3/4 cups water covered on low for 15 minutes;

Per serving (est): 95mg Na • 340mg K • 235mg P

Ingredients:

- 12 oz trimmed lean lamb leg (cut into 1-inch pieces)
- 1 tsp neutral oil
- 1/2 cup yellow onion (finely chopped)
- 1/4 tsp ground cardamom
- 1/4 tsp ground cinnamon
- 2 cups fresh water (for braising lamb)
- 1 1/2 cups white basmati rice (rinsed thoroughly)
- 1 3/4 cups fresh water (for cooking rice)
- 1 cup plain whole-milk yogurt (pasteurized; verify 0mg phosphate additives)
- 1 tsp cornstarch (to stabilize yogurt)
- 1 small plain 6-inch pita bread (torn into bite-sized pieces)
- 2 tbsp sliced almonds (lightly toasted; optional)
- 2 tbsp fresh flat-leaf parsley (finely chopped)

Chef Swaps & Safety Flags: Contains milk, wheat/gluten, and tree nuts (almonds). Omit almonds for nut allergy. Use gluten-free flatbread for celiac needs. The measured 1/2 cup lamb broth and 1 cup yogurt are strictly portioned for fluid and mineral control.

remove from heat and rest 5 minutes. Makes 4 servings. 10 50 minutes. 420g Protein 30ml Fluid
In a large saucepan, whisk yogurt and cornstarch until smooth. Warm over low heat, whisking constantly. Slowly ladle in the 1/2 cup hot lamb broth to temper the yogurt; warm through gently without boiling to prevent curdling.

5. To serve, scatter torn pita pieces across a platter. Mound steamed rice over bread, arrange lamb over rice, and spoon the warm yogurt sauce over top. Garnish with parsley and toasted almonds; divide into four servings.

Tempered Yogurt Sauce Cardamom-Braised Lamb
Zero Jameed Salt

RECIPE CARD • MASTERCOOK CKD KITCHEN

REDUCED- LIQUID TOM YUM KUNG- INSPIRED SHRIMP SKILLET

R11 / WT-1108 • THAI
HERITAGE
(CENTRAL PLAINS) •
LEMONGRASS,
GALANGAL, AND
MAKRUT LIME
FLASH-SEARED
SHRIMP IN MINIMAL
BROTH

Cultural Integrity &

Technique: Traditional Tom Yum Kung is a large bowl of fiery, sour soup packed with fish sauce (sodium trap) and large liquid volume (fluid overload). This brilliant clinical adaptation converts the dish to an intense skillet sauté: bruised lemongrass, fresh galangal, and makrut lime leaves bloom in hot oil, steaming shrimp and oyster mushrooms in just 1/2 cup water.

Technique:

1. Heat oil in a wide skillet or wok over medium heat. Add bruised lemongrass, torn lime leaves, sliced galangal, sliced shallots, and optional chile; stir-fry 3 minutes until deeply aromatic without browning.
2. Add torn oyster mushrooms, napa cabbage, and exactly 1/2 cup water. Cover and steam 4 minutes until vegetables are just tender.
3. Add raw shrimp to the skillet, spread in a single layer, cover, and cook 3–5 minutes, turning once,

Per serving (est): 85mg Na • 260mg K • 180mg P • 22g Protein • 39ml Fluid

Ingredients:

- 12 oz raw peeled, deveined medium shrimp
- 1 tbsp neutral cooking oil
- 2 stalks fresh lemongrass (lower 4 inches, outer layers discarded, bruised and cut into 2-inch pieces)
- 6 fresh makrut lime leaves (torn, center ribs removed)
- 6 thin slices fresh galangal root
- 2 shallots (thinly sliced)
- 1 cup fresh oyster mushrooms (torn into bite-sized strips)
- 1 1/2 cups napa cabbage (chopped)
- 1/2 cup fresh water (strictly measured)
- 2 tbsp fresh lime juice
- 1 tsp granulated sugar
- 1 small fresh Thai bird's eye chile (thinly sliced; optional)
- 1/4 cup fresh cilantro (chopped)

Chef Swaps & Safety Flags:

Contains crustacean shellfish (allergen). Cod or haddock fillet chunks may replace shrimp—cook to 145°F (63°C). Fresh ginger may substitute for galangal if galangal is unavailable. Fluid volume is only ~2 tbsp per serving.

- until shrimp turn pink, less 4 minutes • 20 mins
Make 4 servings • 20 mins
completely done
4. Remove skillet from heat. Stir in fresh lime juice and sugar, tossing to coat. Do not add fish sauce or salt.
 5. Discard tough lemongrass and galangal pieces. Divide shrimp, mushrooms, cabbage, and concentrated pan juices evenly among four bowls; crown with fresh cilantro.

Ultra-Low Fluid Skillet Makrut Lime & Galangal
Zero Fish Sauce

RECIPE CARD • MASTERCOOK CKD KITCHEN

COOKED LIME- CHILE FISH WITH RED ONION, CABBAGE & RICE

R12 / WT-1202 •
PERUVIAN COASTAL
HERITAGE •
IMMUNOCOMPROMISED-
SAFE FULLY-BAKED
COD FLAKED WITH ICED
RED ONION AND AJÍ
LIMO

Cultural Integrity &

Technique: Peruvian ceviche relies on raw fish acid-denatured in citrus. For hemodialysis and CKD patients, clinical food safety standards (FoodSafety.gov) strongly caution against raw seafood. This respectful adaptation delivers the exhilarating flavor profile: cod is fully baked to 145°F, flaked warm, and tossed cold with ice-water-crisped red onions, fresh ají chile, cilantro, and lime juice over crunchy cabbage and rice.

Technique:

1. Heat oven to 400°F (204°C). Lightly brush fish with oil and bake 10–13 minutes until flesh is opaque and internal temperature reaches 145°F (63°C). Cool 5 minutes, then flake into large bite-sized chunks.
2. Submerge sliced red onion in ice-cold water for 10 minutes to crisp the cell walls and remove harsh raw sulfur pungency. Drain thoroughly.
3. Cook rice with 3/4 cup water covered on low for 15 minutes; remove from heat and rest covered 10 minutes.
4. In a chilled glass bowl, gently toss the flaked cooked fish, crisp drained red onion, minced ají chile, chopped cilantro, and fresh lime juice. Chill

Per serving (est): 78mg Na • 310mg K • 205mg Fat • 24g Protein • 25ml Fluid

in refrigerator 5 minutes
to make 4 servings • 25 mins
Make 4 servings
5mg Fat • 24g Protein • 25ml Fluid
cabbage and steamed
rice on plates, spoon the
chilled lime-chile fish and
onion over top, and
drizzle with the tart
marinade. Serve
immediately.

Safe Cooked Ceviche Profile Ice-Crisped Red Onion

Under 80mg Sodium

RECIPE CARD • MASTERCOOK CKD KITCHEN

Chef Swaps & Safety Flags:

Contains fish (allergen).
Tilapia, pollock, or sea bass
can substitute for cod. Minced
jalapeño provides accessible
heat if authentic Peruvian ají
limo is unavailable. Fully
cooked seafood satisfies
clinical immunocompromised
standards.

Sauces, Dips & Dressings

“A sauce is the crowning glory of a well-prepared dish. — Escoffier”

Nacho Cheese, Ranch, Cocktail Sauce, Non-Medicated Confectionery, Citrus Typhoon Hot Sauce

FRESH BLENDER MARINARA (From Scratch — No Cans)

First commercially grown in Aztec Mexico ~700 AD. The Italians didn't get tomatoes until the 1500s and were afraid to eat them for 200 years.

Ingredient	Amount	CKD Notes
Vine tomatoes	2 lbs (~6 medium), quartered	Fresh, not canned — lower Na, brighter flavor
Onion	1 medium, rough chopped	
Garlic	6 cloves, smashed	More garlic = more flavor without salt
Fresh basil	1/2 cup packed	Add half during cook, half at the end (raw finish)
Olive oil	3 tbsp	
Sugar	1 tsp	Balances tomato acidity
Dried oregano	1 tsp	
Red pepper flakes	1/4 tsp	
Black pepper	1/2 tsp	
Salt	1/4 tsp	
Splash of red wine (optional)	2 tbsp	Cooks off. Deepens everything.

Method: 1. Quarter tomatoes. Leave seeds and juice — that's the sauce. 2. Heat olive oil in deep skillet or Dutch oven. Sauté onion 4 min until soft. 3. Add garlic 1 min. Add red pepper flakes. 4. Add all the quartered tomatoes + juice. Smash them with a wooden spoon as they cook. 5. Add oregano, sugar, pepper, salt, wine if using, half the basil. 6. Simmer uncovered 20-25 min, stirring occasionally, until tomatoes break down completely. 7. Hit it with an immersion blender — pulse to your preferred texture (chunky or smooth). 8. Taste. Adjust salt/sugar/pepper. Add remaining fresh basil (raw). 9. Use immediately or cool and jar. Keeps 5 days refrigerated, freezes 3 months.

Vine tomatoes vs Roma: Vine tomatoes have more juice and seeds = saucier, brighter, more acidic. Romas are meatier, less juice = thicker, sweeter. Both work. Vine is what you have.

Per 1/2 cup: ~40mg Na | ~25mg P | ~250mg K

Stadium Nacho Cheese Sauce

Store-bought Tostitos queso: ~430mg Na per 1/4 cup. This: ~90mg Na.

Ingredient	Amount	CKD Notes
Cream cheese	4 oz (half block), softened	Lower P than Velveeta/American cheese
Unsweetened almond milk	1/4 cup	Lower P than dairy milk
Shredded cheddar	1/4 cup	Small amount for flavor — cheddar is higher P
Paprika	1/2 tsp	
Garlic powder	1/4 tsp	
Onion powder	1/4 tsp	
Cumin	1/4 tsp	
Cayenne	Pinch	
Turmeric	Pinch	For yellow color

Ingredient	Amount	CKD Notes
Salt	1/8 tsp	

Method: 1. Microwave cream cheese 20 sec to soften 2. Add milk, whisk smooth 3. Add cheddar, microwave 20 sec, stir until melted 4. Add all spices, stir 5. Adjust thickness: more milk = thinner, more cream cheese = thicker

Per 2 tbsp: ~45mg Na, ~40mg P, ~35mg K

Jalapeño version: Add 1 tbsp pickled jalapeño, minced **Queso fundido:** Add 1 tbsp cooked ground turkey + diced onion

House Ranch Dressing

Store-bought Hidden Valley: ~260mg Na per 2 tbsp. This: ~55mg Na.

Ingredient	Amount	CKD Notes
Avocado oil mayo (Primal Kitchen)	1/2 cup	Base — low K/P
Chobani Zero Sugar Vanilla Yogurt	1/4 cup	Tanginess without buttermilk's P
Unsweetened almond milk	2 tbsp	Lower P than dairy milk
Garlic powder	1/2 tsp	
Onion powder	1/2 tsp	
Dried dill	1/2 tsp	
Dried chives	1/2 tsp	
Dried parsley	1/2 tsp	
Black pepper	1/4 tsp	
Lemon juice	1 tsp	Brightens everything
Salt	1/8 tsp	

Method: Whisk everything together. Refrigerate 30 min for flavors to meld. Keeps 1 week.

Per 2 tbsp: ~55mg Na, ~15mg P, ~25mg K

Spicy ranch: Add 1/2 tsp cayenne + 1 tsp sriracha **Bacon ranch:** Add 1 tbsp real bacon bits (moderate Na — small amount) **Avocado ranch:** Add 1/4 avocado, mashed smooth

Bright Cocktail Sauce

Store-bought Heinz: ~310mg Na per 1/4 cup. This: ~45mg Na.

Ingredient	Amount	CKD Notes
No-salt-added tomato paste	2 tbsp	Concentrated — less volume = less K
Water	2 tbsp	Dilute the paste
Prepared horseradish	1 tbsp	The BURN — low K/P/Na
Lemon juice	1 tbsp	Fresh squeezed
Worcestershire sauce	1/2 tsp	Small amount = controlled Na
Hot sauce (your homemade)	1/2 tsp	
Sugar	1/2 tsp	Balances acid
Salt	Pinch	

Method: Mix all. Chill 15 min. Serve with shrimp, fish, or on anything.

Per 2 tbsp: ~22mg Na, ~15mg P, ~80mg K

Extra spicy: Add more horseradish + 1/4 tsp cayenne **Bloody Mary base:** Thin with more tomato + water, add celery salt + Worcestershire

Thousand Island Special Sauce

The In-N-Out / Big Mac sauce dupe

Ingredient	Amount
Avocado oil mayo (Primal Kitchen)	1/4 cup
No-salt ketchup (or tomato paste + sugar + vinegar)	1 tbsp
Sweet pickle relish	1 tbsp
Onion powder	1/4 tsp
Lemon juice	1/2 tsp

Ingredient	Amount
Sugar	1/2 tsp
White vinegar	1/2 tsp
Paprika	Pinch

Mix. Chill. Spread on burgers, use as dip. Per 2 tbsp: ~50mg Na, ~10mg P.

Golden Honey Mustard

Ingredient	Amount
Honey	2 tbsp
Yellow mustard	1 tbsp
Avocado oil mayo (Primal Kitchen)	1 tbsp
Lemon juice	1 tsp
Garlic powder	Pinch

Mix. Done. Per 2 tbsp: ~70mg Na, ~8mg P. Put on chicken, dip nuggets, drizzle on salads.

SALADS

Garden Salad

The foundation. Everything else is a remix.

Ingredient	Amount	CKD Notes
Romaine lettuce	4 cups, chopped	Low K base
Cucumber	1/2, sliced	Low K
Cherry tomatoes	6, halved	Moderate K — small amount is fine
Red onion	2 tbsp, thinly sliced	
Shredded carrots	2 tbsp	
Croutons (sourdough, homemade)	1/4 cup	Cube sourdough, toss in olive oil + garlic powder, bake 375F 10 min
OG Italian dressing (below)	2 tbsp	

Caesar Salad

Invented 1924 by Caesar Cardini in Tijuana, Mexico. Not Rome. An Italian immigrant in Mexico made the most famous American salad. The decoder sees it.

Ingredient	Amount	CKD Notes
Romaine hearts	2, chopped	
Sourdough croutons	1/2 cup	Homemade — recipe above
Shaved fresh mozzarella	2 tbsp	Sub for parmesan — Andrew's preference
Lemon juice	2 tbsp	
Olive oil	3 tbsp	
Garlic	2 cloves, minced	
Dijon mustard	1 tsp	Anchovy sub — same umami
Worcestershire sauce	1 tsp	
Black pepper	1/2 tsp (generous)	
Egg yolk	1 (optional — emulsifies the dressing)	

Dressing method: Whisk garlic + lemon + mustard + Worcestershire + pepper. Slowly drizzle olive oil while whisking to emulsify. Toss with romaine + croutons. Top with shaved mozzarella.

Per serving: ~90mg Na | ~55mg P | ~220mg K

San Sai Spicy Linguine Salad (Portland NW 21st Ave Clone)

San Sai Japanese Grill — 539 NW 21st Ave, Portland. Cold spicy noodle salad that hits different.

Ingredient	Amount	CKD Notes
Linguine	8 oz, cooked, rinsed cold	GF rice noodles work too
Cucumber	1/2, julienned	
Carrots	1 medium, julienned	
Red bell pepper	1/2, thinly sliced	
Red onion	1/4, thinly sliced	
Fresh cilantro	1/4 cup, chopped	
Sesame seeds	1 tbsp	

Spicy Dressing:

Ingredient	Amount
Rice vinegar	3 tbsp
Sesame oil	2 tbsp
Avocado oil	2 tbsp
Low-sodium soy sauce	1 tbsp
Sriracha	1-2 tsp (to taste)
Honey	1 tbsp
Garlic	2 cloves, minced
Ginger	1 tsp, grated
Lime juice	1 tbsp

Method: Cook linguine, drain, rinse under cold water until fully chilled. Whisk all dressing ingredients. Toss cold noodles with vegetables + dressing. Top with cilantro + sesame seeds. Serve cold. Better the next day — the dressing soaks in.

Per serving (4 servings): ~180mg Na | ~65mg P | ~210mg K

Fruit Salad

No recipe needed. Just a list of what's CKD-aware.

CKD-Aware Fruits (Low K)	Skip These (High K)
Strawberries	Banana
Blueberries	Mango (large amounts)
Raspberries	Kiwi
Green grapes (Andrew's fav)	Orange
Wild Twist apples	Papaya
Pineapple (small amount)	Coconut water
Watermelon (small amount)	Dried fruit of any kind
Peaches (fresh)	
Pears (fresh)	

Method: Cut whatever's in season. Squeeze lime juice over everything (prevents browning + adds brightness). Drizzle honey if you want sweet. Toss. Eat cold. Add Chobani Zero Sugar Vanilla yogurt on top if Lauren's around.

Ambrosia Salad

Originally a Southern church potluck staple, 1890s. Named after the food of the Greek gods. The gods would have approved.

Ingredient	Amount	CKD Notes
Whipped topping (Cool Whip)	1 cup	Or coconut whipped cream
Sour cream	1/2 cup	
Mandarin oranges, drained	1 can (11 oz)	Drain well — canned = controlled K
Pineapple tidbits, drained	1 can (8 oz)	Drain well
Shredded coconut	1/2 cup	
Mini marshmallows	1 cup	Marshmallow = pure sugar + gelatin = CKD-aware
Maraschino cherries	6, halved	The red jewels

Method: Fold sour cream into whipped topping. Gently fold in ALL drained fruit + coconut + marshmallows + cherries. Refrigerate minimum 2 hours (overnight better — marshmallows absorb and get chewy). Serve cold. Bring in a Pyrex dish. Cover with tin foil. Susan's spirit is in the tin foil.

Per serving (6 servings): ~30mg Na | ~35mg P | ~140mg K

Olive Garden Italian Dressing Dupe

The OG dressing. Sweet, tangy, herby. This is the one.

Ingredient	Amount	CKD Notes
Extra virgin olive oil	1/3 cup	Good fat
White wine vinegar	3 tbsp	Or white vinegar
Water	1 tbsp	Thins slightly
Sugar	2 tsp	The SECRET — OG dressing is sweeter than you'd think
Garlic, minced very fine	1 clove	
Dried oregano	1/2 tsp	
Dried basil	1/4 tsp	
Dried parsley	1/4 tsp	
Red pepper flakes	Pinch	
Black pepper	1/4 tsp	
Onion powder	1/4 tsp	
Italian seasoning blend	1/2 tsp	If you have it — shortcut
Lemon juice	1 tsp	Brightness
Salt	1/8 tsp	
Mayonnaise (avocado oil)	1 tbsp	The emulsifier that makes it creamy, not separated

Method: Whisk oil + vinegar + water + mayo until emulsified. Add all seasonings + sugar + lemon. Shake in jar. Refrigerate 1 hour minimum (overnight is better — herbs hydrate). Shake before using.

Per 2 tbsp: ~35mg Na, ~5mg P, ~15mg K. The store bottle has ~260mg Na.

The Olive Garden Salad (full dupe):

Component	CKD Notes
Iceberg lettuce, chopped	Low everything
Red onion, thin slices	Small amount — low K
Cherry tomatoes, halved	3-4 per serving — controlled K
Pepperoncini peppers	2 per serving — low K, adds tang
Croutons (sourdough, homemade)	Cube sourdough, toss in olive oil + garlic powder + Italian seasoning, bake 375F 10 min
Sharp cheddar shavings	1 tbsp — Andrew prefers over parmesan
OG Italian dressing	Drizzle generously

Skip the black olives (Andrew only likes olives on pizza).

Caprese Garden Plate

Ingredient	Amount	CKD Notes
Fresh tomato (vine-ripe)	1 large, sliced 1/4"	Moderate K — one tomato per serving is acceptable
Fresh mozzarella ball	4 oz, sliced 1/4"	Mozzarella = LOWER phosphorus than cheddar or American
Fresh basil leaves	8-10 leaves	
Extra virgin olive oil	2 tbsp drizzle	
Balsamic glaze	1 tbsp drizzle	Balsamic reduction — sweeter, thicker than raw vinegar
Flaky sea salt	Pinch	Maldon if you want to be fancy
Black pepper	Crack	

Assembly: Alternate slices of tomato and mozzarella on plate. Tuck basil leaves between. Drizzle olive oil, then balsamic glaze. Pinch of salt, crack

of pepper. Done in 3 minutes. Looks like a restaurant plate.

Per serving: ~180mg K, ~150mg P, ~80mg Na. The mozzarella is the P source — the stated serving still needs to be counted within your own phosphorus target.

Andrew's Greek Salad (No Olives)

Ingredient	Amount	CKD Notes
Romaine or iceberg, chopped	2 cups	Low everything
Cucumber	1/2, diced	Low K
Cherry tomatoes	4-5, halved	Controlled K
Red onion	2 tbsp thin slices	Low K
Pepperoncini	2-3	Tangy, low K
Fresh mozzarella balls (bocconcini)	4-5 small balls, halved	Andrew's preferred cheese — lowest P of all cheeses, soft, sweet
~~Kalamata olives~~	~~SKIP~~	Andrew: olives only on pizza

Greek Dressing:

Ingredient	Amount
Extra virgin olive oil	3 tbsp
Red wine vinegar	1.5 tbsp
Lemon juice	1 tbsp
Garlic	1 clove, minced fine
Dried oregano	1 tsp
Dijon mustard	1/2 tsp
Honey	1/2 tsp
Black pepper	1/4 tsp
Salt	Pinch

Whisk or shake in jar. Per 2 tbsp: ~25mg Na, ~5mg P.

MOM'S RADIATORE WHITE WINE VINEGAR PASTA SALAD (Clone)

Radiatore — "radiator-shaped" pasta, invented in Italy between the World Wars. The ruffled edges hold vinaigrette better than any other shape.

Ingredient	Amount	CKD Notes
Radiatore pasta (or rotini/fusilli if radiatore unavailable)	8 oz dry	Cook al dente, rinse in cold water to stop cooking
White wine vinegar	3 tbsp	The backbone — tangy, bright
Extra virgin olive oil	1/4 cup	
Sugar	1 tsp	Balances the vinegar bite — this is what makes it mom's version
Garlic powder	1/2 tsp	
Dried oregano	1/2 tsp	
Dried basil	1/4 tsp	
Black pepper	1/4 tsp	
Salt	1/4 tsp	Minimal
Cucumber	1/2, diced	Low K crunch
Bell pepper (any color)	1/2, diced	Color + crunch
Red onion	2 tbsp, finely diced	Small amount for bite
Cherry tomatoes	6-8, halved	Controlled K
Fresh parsley	2 tbsp, chopped	
Sharp cheddar shavings	1 tbsp	Andrew prefers — less salty than parmesan

Method: 1. Cook radiatore al dente per package. Drain. Rinse with cold water. Drain well. 2. Whisk dressing: olive oil + white wine vinegar + sugar + garlic powder + oregano + basil + salt + pepper 3. Toss cooled pasta with

dressing 4. Add cucumber, bell pepper, red onion, tomatoes, parsley 5. Toss gently. Refrigerate minimum 1 hour (overnight is best — the pasta absorbs the vinegar dressing) 6. Top with sharp cheddar shavings before serving. Taste — adjust vinegar/salt

Per serving (6 servings): ~65mg Na, ~80mg P, ~180mg K

The mom trick: The sugar in the dressing is what separates this from generic Italian pasta salad. It cuts the vinegar's sharpness and makes it addictive. Mom knew.

Variations: - **Add protein:** Diced grilled chicken turns this into a full meal - **Add pepperoncini:** 4-5 sliced, for that Olive Garden energy - **GF version:** Use GF radiatore or fusilli (Andrew has GF pasta in pantry)

CITRUS TYPHOON HOT SAUCE (Andrew's New Custom)

"Every citrus fruit on earth in one bottle. Then it burns."

Ingredient	Amount	Why This Citrus
Habanero peppers	6, stems removed	~300,000 SHU — the heat engine
Fresh orange juice	2 tbsp	Sweet citrus base
Fresh lemon juice	2 tbsp	Sharp acid
Fresh lime juice	2 tbsp	Bright, cutting
Orange zest	1 tsp	Oils = intense aroma
Lemon zest	1 tsp	
Lime zest	1 tsp	
Yuzu juice	1 tbsp (bottled, from H Mart/Amazon)	Japanese citrus — floral, complex, unlike anything else
Grapefruit juice	1 tbsp	Bitter-sweet depth
Calamansi juice	1 tbsp (if available — Filipino citrus)	Tart, fragrant, between lemon and tangerine
Starfruit	1/4 fruit, diced	Mild, waxy, adds body + tropical
Kumquat	2, whole (eat skin and all), chopped	Sweet skin + sour flesh = complexity
White vinegar	1/4 cup	Preservation + acid balance
Garlic	3 cloves	
Sugar	1 tbsp	Balances all that acid
Salt	1/4 tsp	
Turmeric	1/4 tsp	Golden color boost

Method: 1. Blend ALL ingredients except starfruit chunks 2. Simmer 15 min 3. Add diced starfruit in the last 2 min (keeps some texture) 4. Bottle hot

The experience: This sauce hits you with 8+ different citrus notes at different speeds — orange arrives first (sweet), then lemon (sharp), then lime (bright), then yuzu (floral), then grapefruit (bitter), then the habanero comes through everything like a tsunami. The starfruit chunks give you something to bite into. The golden color is stunning.

Citrus Typhoon Variations: - **Citrus Typhoon MILD:** Sub bell pepper for habanero — all the citrus, zero heat - **Citrus Typhoon BLOOD:** Use blood orange instead of regular orange — deep red-orange color - **Citrus Typhoon YUZU BOMB:** Double the yuzu, add wasabi — Japanese citrus storm

Vorath Pepper Codex Labels: Sweet Heat Taco Sauce + Chili Paste

Use this for the bottle and the strained pulp jar from the Fresno-serrano-mirin blend. The sauce is the pourable signal; the paste is the concentrated engine.

Front Label: Vorath Sunset Taco Sauce

VORATH SUNSET TACO SAUCE
Sweet Heat Pepper Elixir

Fresno + Serrano + Mirin

Garlic / Cilantro / Avocado Oil
Mixed Vinegar Prism

Batch: VOR-000999-FR-SR-MR-CIL
Heat: 6/10 bright, sweet, fast
Use: tacos, eggs, rice bowls, wings, marinades

Back Label Copy

Flavor: sweet first, green-herbal second, vinegar sparkle third, serrano finish. Mirin rounds the pepper heat without turning it into candy. Fresno brings red fruit; serrano brings the clean green sting; cilantro keeps it alive.

Ingredients line: blended Fresno peppers, serrano peppers, garlic, mirin, avocado oil, cilantro, mixed vinegars, herbs, spices.

Art DNA / Vorath encoding: RED.FRESNO + GREEN.SERRANO + GOLD.MIRIN + WHITE.GARLIC + LEAF.CILANTRO + OIL.AVOCADO + ACID.PRISM + 000999.RETURN.

Storage: This is a refrigerator sauce unless pH is tested at 4.0 or lower and processed with a validated canning method. Because garlic and oil are present, do not store at room temperature. Refrigerate and use within 2 weeks, or freeze in cubes.

Companion Jar: Vorath Redshift Chili Paste

VORATH REDSHIFT CHILI PASTE
Strained Pepper Pulp Concentrate

Fresno / Serrano / Garlic / Cilantro
Mirin Sweet Heat + Vinegar Spark

Batch: VOR-PASTE-000999-RS
Heat: 7/10 concentrated
Use: taco meat, mayo, ramen, beans, wing glaze, rice

Paste method: After straining the sauce, scrape the pepper pulp into a clean jar. Stir in 1-2 tsp vinegar if it needs brightness, and 1 tsp avocado oil if it needs gloss. For a thicker chile crisp direction, simmer the pulp gently in 1-2 tbsp avocado oil for 3-5 minutes, then cool and refrigerate.

Fast uses: 1 tsp into sour cream for taco crema; 1 tsp into mayo for sandwich sauce; 1 tsp into 1 lb taco meat; 1 tsp into ramen broth; 1 tbsp with honey for wing glaze.

Alternate Bottle Names

- **Mirin Comet Taco Sauce** - sweeter, brighter, friendlier label.
- **Fresno Prism Fire** - more hot-sauce shelf energy.
- **Serrano Sun Engine** - green heat emphasis.
- **Vorath 999 Pepper Nectar** - strongest lore-coded label.

NON-MEDICATED SPECIAL EFFECTS CONFECTIONERY - SAFETY REWRITE

Status: The prior fruit roll-up and sour gummy blocks in this location contained active-ingredient dosing concepts. They are held out of the public print lane.

Use instead: print
static/guides/ckd_special_effects_confectionery.html or
CKD_Special_Effects_Confectionery_PRINT.pdf. That guide contains non-medicated hard candy, apple-cranberry gummies, pectin sour squares, rock candy wands, marshmallow slabs, UV tonic micro-gems, fruit leather windows, and printable labels.

Old Lane	Replacement	Publication Rule
Medicated fruit roll-ups	Aurora Fruit Leather Windows	No THC, CBD, RSO, tincture, or dosing math.
CBD sour gummies	Cloud 9 Apple-Cranberry Gummies or Pectin Sour Signal Squares	Food-only color, sour dust, edible glitter, and portion labels.
Look-alike candy	Clearly labeled house shapes and batch labels	Do not imitate commercial kid candy for any controlled-product context.

RINGSIDE STEAKHOUSE FRESH HORSERADISH PREP (Portland Clone)

"Burnside & NW 12th. They bring it with the prime rib. Now you bring it at home."

Ingredient	Amount	CKD Notes
Fresh horseradish root	6 oz, peeled	Find at Fred Meyer produce or H Mart
White vinegar	2 tbsp	Stops the heat reaction — add LATER for more burn, SOONER for milder
Sugar	1 tsp	
Salt	1/4 tsp	

Method: 1. Peel horseradish root. Cut into 1" chunks. 2. Pulse in food processor until finely grated (open the lid AWAY from your face — the fumes are real) 3. **THE TIMING TRICK:** The longer you wait to add vinegar, the HOTTER it gets. For Ringside-level heat, wait 3 minutes after grinding. For milder, add immediately. 4. Add vinegar + sugar + salt. Pulse to combine. 5. Transfer to a small glass jar. Refrigerate. 6. Serve a small spoonful alongside prime rib, steak, or roast beef.

Shelf life: 3-4 weeks refrigerated. Heat fades over time — make fresh batches.

Creamy Horseradish Sauce (Ringside-style):

Ingredient	Amount
Prepared horseradish (above)	2 tbsp
Sour cream	1/4 cup
Dijon mustard	1/2 tsp
Lemon juice	1/2 tsp
Black pepper	Pinch

Mix. Chill 30 minutes. Serve with prime rib. The sour cream tames the burn into a slow, warm glow.

Per tbsp: ~15mg Na | ~8mg P | ~20mg K — horseradish is basically free. All flavor, no renal cost.

CKD Considerations

RENAL CHECK — Sauces & Dips
K (Potassium): LOW (cream cheese base, no-salt-added tomato, controlled portions)
P (Phosphorus): LOW (cream cheese over processed cheese, mayo base for dressings)
Na (Sodium): LOW (homemade = 70-90% less sodium than store-bought)

"A sauce is the crowning glory of a well-prepared dish." — Escoffier

INTERNATIONAL SAUCES & CONDIMENTS

Cooling Raita

Served with every Indian meal since the Vedic period (~1500 BC). The word comes from Sanskrit "rajika" — mustard. The cooling effect offsets spicy food.

Ingredient	Amount	CKD Notes
Plain yogurt	1 cup	Moderate P — portion-controlled
Cucumber, grated + squeezed dry	1/2 cup	Low K
Cumin, toasted and ground	1/2 tsp	Toast whole seeds in dry pan first
Fresh mint, chopped	1 tbsp	
Fresh cilantro, chopped	1 tbsp	
Salt	1/8 tsp	
Cayenne	Pinch	

Method: Mix everything. Chill 30 min. Serve alongside biryani, kebabs, or any spicy dish. The yogurt cools. The cumin warms. The cucumber crunches.

Per 1/4 cup: ~30mg Na | ~40mg P | ~80mg K

CHAPTER 9

Hot Sauces

"Measure your life in Scoville units."

Fresh Mint Chutney

The word "chutney" comes from Hindi "chatni" — to lick. First documented in the 1600s by European traders visiting Mughal courts.

Ingredient	Amount
Fresh mint leaves	1 cup packed
Fresh cilantro	1/2 cup packed
Green chili (serrano)	1, seeded
Onion	2 tbsp
Lime juice	2 tbsp
Sugar	1 tsp
Cumin	1/4 tsp
Water	2-3 tbsp
Salt	1/8 tsp

Method: Blend everything until smooth paste. Adjust water for consistency. Serve with samosas, kebabs, rice, naan.

Per 2 tbsp: ~15mg Na | ~5mg P | ~20mg K

Sweet-Sour Tamarind Chutney

Tamarind originated in tropical Africa, traveled to India via Arab traders ~400 BC, and became essential to Indian cuisine. The name comes from Arabic "tamar hindi" — Indian date.

Ingredient	Amount
Tamarind paste	2 tbsp
Water	1/2 cup
Brown sugar	3 tbsp
Cumin, ground	1/2 tsp
Ginger powder	1/4 tsp
Cayenne	1/4 tsp
Salt	1/8 tsp

Method: Combine all in saucepan. Simmer 10 min, stirring, until thick and glossy. Strain. Serve with samosas, chaat, or as a glaze for grilled meats. Sweet-sour-spicy in every drop.

Per 2 tbsp: ~20mg Na | ~5mg P | ~30mg K

Cucumber Tzatziki

Greek in origin, Turkish in DNA (cacik). First appeared in Ottoman cookbooks ~1500s. The key: draining the cucumber of ALL water or it becomes soup.

Ingredient	Amount	CKD Notes
Plain yogurt	1 cup	
Cucumber	1/2, grated + squeezed bone-dry	Squeeze in towel — get every drop out
Garlic	2 cloves, minced fine	
Lemon juice	1 tbsp	
Olive oil	1 tbsp	Drizzle on top
Fresh dill	1 tbsp, chopped	
Salt	1/8 tsp	

Method: Grate cucumber, squeeze in clean towel until no more water comes out (this step is 90% of a good tzatziki). Mix with yogurt + garlic + lemon + dill + salt. Drizzle olive oil on top. Chill 1 hour.

Per 1/4 cup: ~35mg Na | ~45mg P | ~85mg K

The Cosmic Horror Collection

10 Exotic Hot Sauces for the Brave & Curious

"These sauces were not meant for mortal tongues."

Recent Tropical Hot Sauce Batch

Tropical Serrano-Holland Fruit Sauce

Garlic, onion, apple cider vinegar, serrano, Holland pepper, jalapeno, sweet pepper, strawberry, red kiwi, pineapple, and mango. A bright recent batch with savory allium underneath tropical fruit and clean pepper heat.

Ingredient	Suggested Repeat Batch Amount
Garlic	4 to 6 cloves
Onion	1/2 medium
Apple cider vinegar	1/2 to 3/4 cup, plus more to thin
Serrano pepper	4 to 6
Holland pepper	1 large red or yellow pepper
Jalapeno	2 to 3
Sweet pepper	1 medium
Strawberry	1 cup, hulled
Red kiwi	2, peeled
Pineapple	1 cup, chopped
Mango	1 cup, chopped

Wash produce well. Stem the peppers, peel the onion and garlic, and trim the fruit. Rough-chop everything, then blend with apple cider vinegar until pourable. Simmer 8 to 10 minutes, stirring often, until the raw onion and garlic edge softens and the sauce tightens slightly. Strain for a smoother bottle sauce if desired. Cool, bottle, label, and refrigerate.

Storage lane: refrigerated or frozen fresh sauce unless a tested process validates this exact recipe. Do not treat this as shelf-stable or canning-safe by default. Record exact gram weights and pH on the next repeat batch.

VORATH Wild Sauce Lab

12 Experimental Hot Sauce and Taco Sauce Formulas

Test-kitchen expansion for the VORATH / Emoji Soup sauce system. These formulas are intentionally vivid, dense, and product-minded, but they remain internal lab recipes until pH, thermal process, allergen, label, and process-authority reviews are complete.

Lab safety lane: Wear gloves and ventilate when cutting or blending hot peppers. For fermented versions, use non-iodized salt and keep solids below brine or mash liquid. For lab bottles, target finished pH 4.0 or below with a calibrated meter. Refrigerate all test bottles unless a validated shelf-stable process is documented.

Reference packet: The VORATH Wild Sauce Lab packet and experiment matrix remain in private editorial review and are not part of this public edition.

Code	Formula	Style	Heat	Use
VOR-X101	Prickly Pear Morita Star Taco Sauce	Cooked acidified taco sauce	Medium-hot	fish tacos, breakfast burritos, grilled shrimp, roasted squash
VOR-X102	Black Comet Coffee-Cacao Mole Drip	Cooked dark taco glaze	Medium	barbacoa tacos, black beans, roasted mushrooms, chili, charred onions
VOR-X103	Honeydew Green Signal Ferment	2.5 percent mash ferment	Medium-hot	al pastor tacos, ceviche, grilled chicken, cucumber salad, grain bowls
VOR-X104	Hibiscus Blood Moon Arbol Taco Sauce	Cooked acidified floral red sauce	Hot	crispy potato tacos, carnitas, roasted carrots, queso fundido
VOR-X105	Pineapple Hoja Santa Habanero Reactor	Fermented then cooked tropical sauce	Very hot	pork tacos, fried chicken, grilled pineapple, fish, rice bowls

Code	Formula	Style	Heat	Use
VOR-X106	Blueberry Blacklime Habanero Phantom	Fermented berry sauce	Very hot	duck tacos, lamb, mushroom tacos, goat cheese, roasted sweet potato
VOR-X107	Tamarind Sesame Serrano Static	Cooked acidified global taco sauce	Hot	birria tacos, wings, grilled tofu, noodles, charred cabbage
VOR-X108	Roasted Corn Cascabel Gold	Cooked taco table sauce	Mild-medium	street corn tacos, chicken, grilled zucchini, breakfast eggs, quesadillas
VOR-X109	Mango Aji Amarillo Solar Flare	Cooked fruit hot sauce	Hot	fish tacos, chicken, fries, coconut rice, fried plantains
VOR-X110	Beet Pomegranate Pasilla Night Market	Cooked acidified dark red sauce	Medium	lamb tacos, roasted cauliflower, falafel tacos, duck, charred eggplant
VOR-X111	Yuzu-Lime Serrano Papalo Switchblade	Fresh-cooked green taco sauce	Medium-hot	fish tacos, cucumber salad, grilled steak, breakfast tacos
VOR-X112	Maple Miso Morita Redline	Cooked umami hot sauce	Medium-hot	short rib tacos, tofu, roasted carrots, burgers, black bean bowls

VOR-X101 - Prickly Pear Morita Star Taco Sauce

Build a wild pink taco sauce with fruit body, smoky morita, and clean arbol heat.

Style: Cooked acidified taco sauce **Heat:** Medium-hot **Target yield:** about 520 g / 18 oz

Visual: electric magenta-red, glossy, slightly thick

Best uses: fish tacos, breakfast burritos, grilled shrimp, roasted squash

Ingredient	Test Batch Grams	Purpose
prickly pear pulp, strained	180 g	fruit body and color
tomato puree	160 g	classic taco-sauce base
roasted tomatillo	80 g	green acidity
apple cider vinegar, 5 percent	90 g	acid
lime juice	25 g	bright finish
dried morita chile, toasted and soaked	10 g	smoky fruit
dried chile de arbol, toasted	3 g	direct heat
red onion, roasted	35 g	savory sweetness
garlic, roasted	12 g	depth
cilantro stems	8 g	green snap
ground cumin	2 g	taco backbone
Mexican oregano	1 g	dry herb lift
fine salt	6 g	seasoning

Method:

1. Toast dried chiles until fragrant, then soak in hot water for 15 minutes and drain.
2. Blend everything until fully smooth, then simmer 8 to 10 minutes to thicken.
3. Cool a spoonful and check pH. Target 4.0 or below for lab bottling.
4. Adjust with lime, vinegar, salt, or a small pinch of sugar only after the first pH check.

Control note: Keep refrigerated unless a calibrated meter confirms finished pH and a proper hot-fill/canning process is used.

VOR-X102 - Black Comet Coffee-Cacao Mole Drip

Make a taco sauce that eats like mole, barbecue, espresso, and cosmic diner gravy.

Style: Cooked dark taco glaze **Heat:** Medium **Target yield:** about 560 g / 20 oz

Visual: near-black mahogany with a satin sheen

Best uses: barbacoa tacos, black beans, roasted mushrooms, chili, charred onions

Ingredient	Test Batch Grams	Purpose
tomato puree	190 g	base
strong cold brew coffee	90 g	bitter roast
apple cider vinegar, 5 percent	95 g	acid
tamarind concentrate	45 g	sour fruit
golden raisins	40 g	round sweetness
dried ancho chile, toasted and soaked	20 g	raisin/chocolate
dried pasilla chile, toasted and soaked	15 g	dark fruit
dried morita chile, toasted and soaked	8 g	smoke
onion, roasted	60 g	body
garlic, roasted	16 g	savory base
cacao nibs, steeped and strained	8 g	bitter chocolate
honey	25 g	glaze
ground cumin	2 g	taco backbone
cinnamon	0.6 g	warm top note
fine salt	7 g	seasoning

Method:

1. Steep cacao nibs in hot coffee for 10 minutes, then strain.
2. Blend coffee, chiles, tomato, tamarind, raisins, aromatics, and spices until smooth.
3. Simmer 12 to 15 minutes, stirring often, until it coats a spoon.
4. Check pH after cooling a sample. Add vinegar in 10 g increments if above target.

Control note: Dark sweet sauces hide acidity. Do not judge safety by taste; measure pH.

VOR-X103 - Honeydew Green Signal Ferment

Turn cooling melon into a fermented green taco sauce with serrano bite and herbal lift.

Style: 2.5 percent mash ferment **Heat:** Medium-hot **Target yield:** about 500 g / 17 oz after blending

Visual: deep green, bright but slightly opaque

Best uses: al pastor tacos, ceviche, grilled chicken, cucumber salad, grain bowls

Ingredient	Test Batch Grams	Purpose
honeydew melon, ripe	260 g	cool sweetness
poblano, chopped	130 g	green body
jalapeno, chopped	90 g	grassy heat
serrano, chopped	45 g	sharp heat
green habanero, optional	10 g	upper heat
white onion	55 g	ferment support
garlic	14 g	savory
cilantro stems	12 g	green aroma
non-iodized salt	15 g	2.5 percent of 606 g mash
white vinegar, post-ferment	75 g	acid finish
lime juice, post-ferment	25 g	brightness

Method:

1. Pulse melon, peppers, onion, garlic, cilantro stems, and salt into a coarse mash.
2. Pack under a weight with minimal headspace. Ferment 7 to 14 days at cool room temperature.
3. When sour and active fermentation slows, blend with vinegar and lime juice.
4. Check pH. Target 4.0 or below; keep refrigerated during lab testing.

Control note: Keep solids below brine. Discard if mold appears or aroma turns rotten instead of sour.

VOR-X104 - Hibiscus Blood Moon Arbol Taco Sauce

Use hibiscus as sour floral acid against guajillo body and arbol fire.

Style: Cooked acidified floral red sauce **Heat:** Hot **Target yield:** about 540 g / 19 oz

Visual: dark ruby, translucent edges

Best uses: crispy potato tacos, carnitas, roasted carrots, queso fundido

Ingredient	Test Batch Grams	Purpose
strong hibiscus tea, unsweetened	130 g	floral tartness
tomato puree	170 g	red base
white vinegar, 5 percent	90 g	acid
orange juice	50 g	citrus sweetness
dried guajillo chile, toasted and soaked	25 g	berry body
dried chile de arbol, toasted	5 g	heat
red onion, roasted	45 g	savory
garlic	12 g	depth
agave	28 g	round sweetness
ground cumin	2 g	taco backbone
Mexican oregano	1.2 g	herb
black pepper	0.8 g	bite
fine salt	6 g	seasoning

Method:

1. Make a strong hibiscus tea and cool it before blending.
2. Blend all ingredients until smooth, then simmer 10 minutes.
3. Taste for bitter edges. Balance with tiny additions of agave or orange juice.
4. Check pH after cooling a sample; add vinegar if needed.

Control note: Hibiscus can stain. Use nonreactive cookware and glass storage.

VOR-X105 - Pineapple Hoja Santa Habanero Reactor

Pair habanero fruit heat with pineapple acid and hoja santa's root-beer herb signal.

Style: Fermented then cooked tropical sauce **Heat:** Very hot **Target yield:** about 520 g / 18 oz

Visual: orange-gold with green herbal shadow

Best uses: pork tacos, fried chicken, grilled pineapple, fish, rice bowls

Ingredient	Test Batch Grams	Purpose
pineapple, diced	240 g	sugar and acid
orange bell pepper	210 g	body
habanero, stemmed	60 g	fruit heat
white onion	65 g	ferment support
garlic	18 g	savory
non-iodized salt	15 g	about 2.5 percent mash salt
apple cider vinegar, post-ferment	90 g	acid
honey, post-ferment	22 g	roundness
fresh hoja santa, infused then removed	1 g	herb experiment
lime juice, post-ferment	20 g	finish

Method:

1. Ferment pineapple, pepper, habanero, onion, garlic, and salt as a mash for 5 to 10 days.
2. Blend with vinegar, honey, and lime juice.
3. Warm gently with one small hoja santa leaf for 3 minutes, then remove and blend again.
4. Simmer 5 minutes to stop fermentation and set sweetness. Check pH.

Control note: Use a light hand with hoja santa. If unavailable, omit; do not substitute random avocado leaves.

VOR-X106 - Blueberry Blacklime Habanero Phantom

Build a fruit-forward sauce that turns habanero into dark berry, smoke, and sour citrus.

Style: Fermented berry sauce **Heat:** Very hot **Target yield:** about 500 g / 17 oz

Visual: purple-black, smooth, high gloss

Best uses: duck tacos, lamb, mushroom tacos, goat cheese, roasted sweet potato

Ingredient	Test Batch Grams	Purpose
blueberries	220 g	fruit and pectin
blackberries	150 g	color and tannin
habanero	55 g	heat
red Fresno pepper	100 g	red pepper body
red onion	45 g	ferment support
garlic	14 g	savory
non-iodized salt	15 g	about 2.5 percent mash salt
white vinegar, post-ferment	85 g	acid
black lime powder	1.2 g	sour bass note
thyme leaves	1 g	herbal echo
maple syrup, optional	18 g	rounding

Method:

1. Ferment berries, peppers, onion, garlic, and salt as a mash for 7 to 14 days.
2. Blend with vinegar, black lime, thyme, and optional maple.
3. Strain for a polished label-ready texture or leave pulpy for a kitchen sauce.
4. Check pH and refrigerate.

Control note: Berry ferments can be active. Use an airlock or burp daily.

VOR-X107 - Tamarind Sesame Serrano Static

Make a sour tamarind hot sauce with sesame nuttiness and serrano voltage.

Style: Cooked acidified global taco sauce **Heat:** Hot **Target yield:** about 560 g / 20 oz

Visual: brick amber, tiny sesame speckle

Best uses: birria tacos, wings, grilled tofu, noodles, charred cabbage

Ingredient	Test Batch Grams	Purpose
tamarind concentrate	90 g	sour base
tomato puree	130 g	body
rice vinegar, 5 percent	105 g	acid
red Fresno pepper	210 g	red pepper body
serrano	70 g	green heat
garlic	20 g	savory
ginger	14 g	lift
toasted sesame paste	28 g	nutty body
agave	35 g	sweetness
ground cumin	1.6 g	taco bridge
coriander seed, ground	1.2 g	citrus spice
fine salt	7 g	seasoning

Method:

1. Blend all ingredients until smooth.
2. Simmer gently 8 minutes, stirring so sesame does not scorch.
3. Adjust with vinegar, agave, or salt after the first taste.
4. Check pH. Label sesame allergen during any tasting.

Control note: Contains sesame. Keep separate from allergen-free sauce trials.

VOR-X108 - Roasted Corn Cascabel Gold

Push taco sauce toward roasted corn, nutty cascabel, and mild smoke.

Style: Cooked taco table sauce **Heat:** Mild-medium **Target yield:** about 600 g / 21 oz

Visual: golden orange, rustic smooth

Best uses: street corn tacos, chicken, grilled zucchini, breakfast eggs, quesadillas

Ingredient	Test Batch Grams	Purpose
roasted corn kernels	160 g	sweet smoke
tomato puree	170 g	base
roasted yellow bell pepper	140 g	body
white vinegar, 5 percent	100 g	acid
lime juice	25 g	finish
dried cascabel chile, toasted and soaked	18 g	nutty fruit
dried guajillo chile, toasted and soaked	12 g	red body
white onion, roasted	55 g	savory
garlic, roasted	14 g	depth
epazote, finely chopped	2 g	floral herb accent
smoked paprika	1.4 g	smoke
ground cumin	2 g	taco backbone
fine salt	7 g	seasoning

Method:

1. Roast corn, pepper, onion, and garlic until browned at the edges.
2. Toast and soak chiles, then blend everything very smooth.
3. Simmer 10 minutes. Add epazote near the end to avoid over-extraction.
4. Check pH and adjust with vinegar or lime.

Control note: Epazote is assertive. Keep the first test low and increase only in 0.5 g steps.

VOR-X109 - Mango Aji Amarillo Solar Flare

Make a bright tropical sauce with mango, carrot, ginger, and aji-style chile body.

Style: Cooked fruit hot sauce **Heat:** Hot **Target yield:** about 560 g / 20 oz

Visual: neon yellow-orange

Best uses: fish tacos, chicken, fries, coconut rice, fried plantains

Ingredient	Test Batch Grams	Purpose
mango	230 g	fruit
carrot, cooked	100 g	body and sweetness
yellow bell pepper	140 g	color
aji amarillo paste	55 g	chile fruit
habanero	18 g	top heat
white vinegar, 5 percent	95 g	acid
lime juice	35 g	brightness
garlic	14 g	savory
ginger	10 g	heat lift
turmeric	1.5 g	color and earth
honey	24 g	round sweetness
fine salt	7 g	seasoning

Method:

1. Cook carrot until tender, then blend with remaining ingredients.
2. Simmer 8 to 12 minutes until the raw pepper edge softens.
3. Adjust sweetness only after acid and salt are balanced.
4. Check pH and refrigerate.

Control note: Turmeric stains intensely. Use dedicated tasting spoons and towels.

VOR-X110 - Beet Pomegranate Pasilla Night Market

Make an earthy dark-fruit sauce with beet, pomegranate, pasilla, and cumin.

Style: Cooked acidified dark red sauce **Heat:** Medium **Target yield:** about 540 g / 19 oz

Visual: black cherry red, smooth

Best uses: lamb tacos, roasted cauliflower, falafel tacos, duck, charred eggplant

Ingredient	Test Batch Grams	Purpose
roasted beet	170 g	earth and color
tomato puree	120 g	base
pomegranate molasses	55 g	sour fruit
red wine vinegar, 5 percent	110 g	acid
dried pasilla chile, toasted and soaked	28 g	dark fruit
dried ancho chile, toasted and soaked	18 g	raisin body
garlic, roasted	16 g	depth
red onion, roasted	55 g	savory sweetness
ground cumin	2 g	taco bridge
coriander seed, ground	1.2 g	citrus spice
black pepper	0.8 g	bite
fine salt	7 g	seasoning

Method:

1. Blend roasted beet, soaked chiles, acids, tomato, aromatics, and spices.
2. Simmer 10 minutes until glossy.
3. Thin with small additions of water or vinegar depending on pH.
4. Check pH and record final color because beet shifts over time.

Control note: Beet sweetness can dull heat. Taste again after chilling overnight.

VOR-X111 - Yuzu-Lime Serrano Papalo Switchblade

Make a blade-clean green sauce with citrus, serrano, papalo, and cucumber snap.

Style: Fresh-cooked green taco sauce **Heat:** Medium-hot **Target yield:** about 500 g / 17 oz

Visual: sharp green, pourable

Best uses: fish tacos, cucumber salad, grilled steak, breakfast tacos

Ingredient	Test Batch Grams	Purpose
tomatillo, roasted	220 g	green body
cucumber, peeled	80 g	freshness
serrano	65 g	sharp heat
jalapeno	55 g	green body
white vinegar, 5 percent	85 g	acid
lime juice	45 g	citrus
yuzu juice or more lime	20 g	aromatic citrus
white onion	40 g	savory
garlic	10 g	depth
papalo leaves	4 g	green herb
cilantro	14 g	fresh herb
fine salt	6 g	seasoning

Method:

1. Roast tomatillos, onion, and garlic until lightly charred.
2. Blend with peppers, cucumber, acids, herbs, and salt.
3. Simmer only 3 minutes to keep the green flavor alive, then chill quickly.
4. Check pH and adjust with lime or vinegar.

Control note: Fresh herb sauces fade. Evaluate color at 0, 24, and 72 hours.

VOR-X112 - Maple Miso Morita Redline

Cross taco sauce, smoky chile, maple, and miso into a savory-sweet red sauce.

Style: Cooked umami hot sauce **Heat:** Medium-hot **Target yield:** about 560 g / 20 oz

Visual: deep red-brown, lacquered

Best uses: short rib tacos, tofu, roasted carrots, burgers, black bean bowls

Ingredient	Test Batch Grams	Purpose
tomato puree	210 g	base
apple cider vinegar, 5 percent	105 g	acid
dried morita chile, toasted and soaked	22 g	smoke
dried guajillo chile, toasted and soaked	18 g	berry body
white miso	42 g	umami and salt
maple syrup	45 g	sweetness
onion, roasted	60 g	savory
garlic, roasted	16 g	depth
lime juice	25 g	finish
ground cumin	1.6 g	taco bridge
smoked paprika	1 g	smoke support
fine salt	2 g	only if needed

Method:

1. Blend everything except optional salt until fully smooth.
2. Simmer gently 8 minutes; do not scorch the miso-maple base.
3. Taste before salting because miso contributes salt.
4. Check pH after cooling a sample. Add vinegar or lime if needed.

Control note: Contains soy if using standard miso. Mark allergen during tastings.

Tier 1: THE OUTER GODS

AZATHOTH

Bhut Jolokia + Szechuan peppercorn + black garlic. Numb, burn, numb, repeat.

Ingredient	Amount
Bhut Jolokia (Ghost Pepper)	4, rehydrated
Szechuan peppercorn	1 tbsp, toasted and ground
Black garlic	6 cloves
Rice vinegar	1/3 cup
Honey	1 tbsp
Ginger	1" piece
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~1,000,000 SHU**

NYARLATHOTEP

Carolina Reaper + ube + coconut. Purple, creamy, then the Reaper hits.

Ingredient	Amount
Carolina Reaper	3, deseeded
Ube (purple yam) powder	2 tbsp
Coconut cream	3 tbsp
Lime juice	2 tbsp
Rice vinegar	1/4 cup
Ginger	1" piece
Sugar	1 tsp

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~1,500,000 SHU**

SHUB-NIGGURATH

Scorpion pepper + tamarind + fish sauce + palm sugar. Southeast Asian darkness.

Ingredient	Amount
Trinidad Scorpion	4
Tamarind paste	2 tbsp
Fish sauce (or low-sodium soy + lime)	1 tsp
Palm sugar or brown sugar	1 tbsp
Lemongrass	1 stalk, minced

Ingredient	Amount
Galangal or ginger	1" piece
Lime leaf	2, chiffonade
Rice vinegar	1/4 cup

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~1,200,000 SHU**

Tier 2: THE ELDER THINGS

THE COLOUR OUT OF SPACE

Habanero + activated charcoal + prickly pear. Black and pink simultaneously.

Ingredient	Amount
Habanero	6
Prickly pear fruit (tuna)	1/2 cup puree
Activated charcoal	1/2 tsp
Lime juice	2 tbsp
Agave nectar	1 tbsp
White vinegar	1/3 cup
Garlic	2 cloves

Make two batches: one with charcoal (black), one without (magenta). Layer in bottle half-and-half. Shaking produces an impossible dark pink-purple.

~300,000 SHU

DAGON

Serrano + nori + wasabi + yuzu. The ocean in a bottle.

Ingredient	Amount
Serrano	8
Nori seaweed	1 sheet, crumbled
Wasabi paste	1 tsp
Yuzu juice	2 tbsp (or Meyer lemon)
Rice vinegar	1/3 cup
Ginger	1" piece
Low-sodium soy sauce	1 tsp
Sesame oil	1/2 tsp

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~20,000 SHU**

THE KING IN YELLOW

Habanero + turmeric + saffron + mustard. Violently yellow. It stains. It glows.

Ingredient	Amount
Habanero	6
Turmeric (fresh root)	2" piece, peeled
Saffron	Pinch, bloomed
Yellow mustard seed	1 tsp, toasted
Mango	1/4 cup
White vinegar	1/3 cup
Honey	1 tbsp
Garlic	3 cloves

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~300,000 SHU**

THE MI-GO

Thai bird's eye + makrut lime + galangal + lemongrass. Northern Thai nam prik in a bottle.

Ingredient	Amount
Thai bird's eye chili	15-20
Makrut lime leaves	6

Ingredient	Amount
Galangal	2" piece
Lemongrass	2 stalks
Low-sodium soy + lime	1 tsp
Rice vinegar	1/3 cup
Palm sugar	1 tbsp
Cilantro root	1 (if available)

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~100,000 SHU**

Tier 3: THE DREAMLANDS

THE SILVER KEY

Guajillo + Mexican chocolate + espresso + cinnamon. Liquid mole.

Ingredient	Amount
Guajillo chili	4, rehydrated
Mexican chocolate (Abuelita)	1/2 tablet, grated
Espresso	1 shot (2 oz), cooled
Ceylon cinnamon	1/2 tsp
Ancho chili	2, rehydrated
Piloncillo or brown sugar	1 tbsp
White vinegar	1/4 cup
Garlic	2 cloves

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~5,000 SHU**

KADATH

Ghost pepper + pomegranate + rose water + sumac. Persian meets apocalypse.

Ingredient	Amount
Ghost pepper	2
Pomegranate molasses	2 tbsp
Rose water	1/2 tsp
Sumac	1 tsp
Red bell pepper	1, roasted
White vinegar	1/4 cup
Garlic	3 cloves
Honey	1 tsp

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~1,000,000 SHU**

THE DREAMQUEST

Scotch bonnet + passion fruit + allspice + thyme. Caribbean fever dream.

Ingredient	Amount
Scotch bonnet	4
Fresh passion fruit	2, scooped
Allspice	1/2 tsp
Fresh thyme	4 sprigs
Lime juice	2 tbsp
White vinegar	1/4 cup
Brown sugar	1 tbsp
Garlic	2 cloves

Blend all ingredients until smooth. Simmer 10 minutes, stirring. Strain if desired. Bottle hot. **~200,000 SHU**

SOURCING

Ingredient	Where to Find
Szechuan peppercorn	H Mart, Amazon

Ingredient	Where to Find
Black garlic	Fred Meyer, H Mart, Amazon
Ube powder	Filipino market, Amazon
Prickly pear / tuna	Mexican market, sometimes Fred Meyer
Nori	Any grocery (sushi aisle)
Yuzu juice	H Mart, Amazon (Yakami Orchard)
Saffron	Middle Eastern market (cheapest), Fred Meyer
Galangal	H Mart, Asian market
Makrut lime leaves	H Mart, Asian market (freeze extras)
Pomegranate molasses	Middle Eastern market, Amazon
Sumac	Middle Eastern market, Amazon
Passion fruit	Fred Meyer, Uwajimaya
Scotch bonnet	Mexican market, sometimes Fred Meyer
Ghost/Reaper/Scorpion peppers	Dried on Amazon, fresh at farmers markets
Mexican chocolate (Abuelita)	Fred Meyer (Hispanic aisle)
Palm sugar	Asian market

Neon Rainbow Fruity Hot Sauce Line

7 Fruit-Forward Sauces That Look Like They Came From Another Dimension

"The rainbow isn't a metaphor. It's the menu."

ULTRAVIOLET — Blueberry Habanero

Deep electric purple. Jewel-tone amethyst in the bottle.

Ingredient	Amount
Fresh blueberries	1 cup
Habanero	4
White wine vinegar	1/3 cup
Honey	1 tbsp
Garlic	2 cloves
Lemon juice	1 tbsp
Purple food coloring	2-3 drops
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~200,000

ELECTRIC BLUE — Blue Raspberry Serrano

Neon alien blue. Looks radioactive.

Ingredient	Amount
Fresh raspberries	3/4 cup
Serrano peppers	8
White vinegar	1/3 cup
Sugar	1 tbsp
Lime juice	1 tbsp
Blue food coloring	5-6 drops
Blue spirulina powder	1/4 tsp
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~15,000

NUCLEAR GREEN — Kiwi-Jalapeno-Tomatillo

Radioactive neon green. Nickelodeon slime in a bottle.

Ingredient	Amount
Tomatillos, husked and halved	4
Fresh kiwi, peeled	2
Jalapeno	6
Serrano	2
Lime juice	2 tbsp
Cilantro	1/4 cup
Green food coloring	3-4 drops
Garlic	2 cloves
Sugar	1 tsp
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~18,000

SOLAR FLARE — Mango Habanero

Blazing electric orange. Sunrise in a bottle.

Ingredient	Amount
Fresh mango	1 cup
Habanero	5
Orange juice	2 tbsp
Orange zest	1 tsp
White vinegar	1/4 cup
Lime juice	1 tbsp
Turmeric	1/4 tsp
Sugar	1 tsp
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~250,000

PLASMA PINK — Strawberry Ghost Pepper

Screaming hot pink. Barbie meets Satan.

Ingredient	Amount
Fresh strawberries	1 cup
Ghost pepper	2
White vinegar	1/3 cup
Honey	1 tbsp
Lemon juice	1 tbsp
Pink food coloring	2-3 drops
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~1,000,000

GOLDEN HOUR — Pineapple Scotch Bonnet

Liquid gold. 24-karat sauce.

Ingredient	Amount
Fresh pineapple	1 cup
Scotch bonnet	3
White vinegar	1/4 cup
Lime juice	2 tbsp
Garlic	2 cloves
Turmeric	1/2 tsp

Ingredient	Amount
Brown sugar	1 tbsp
Allspice	1/4 tsp
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~200,000

INFRARED — Cranberry Carolina Reaper

Deep dark crimson. The end of the spectrum.

Ingredient	Amount
Fresh cranberries	1 cup
Carolina Reaper	2
Red wine vinegar	1/3 cup
Sugar	2 tbsp
Orange zest	1 tsp
Cinnamon	1/4 tsp
Salt	1/4 tsp

Blend all ingredients until smooth. Simmer 10 minutes over medium heat, stirring frequently. Strain if you want silk, skip if you want body. Bottle while warm.

SHU: ~2,000,000

FAST FOOD SAUCE DUPES

TACO BELL MILD SAUCE

First Taco Bell packet sauces appeared in 1962, the year Glen Bell opened the first location in Downey, California.

Ingredient	Amount
Tomato paste	2 tbsp
Water	1/4 cup
White vinegar	1 tbsp
Cumin	1/4 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp
Paprika	1/4 tsp
Sugar	1/2 tsp
Salt	1/8 tsp

Whisk all ingredients together. Simmer 3 minutes. Cool. Store in squeeze bottle or small containers.

SHU: ~500 | Per tbsp: ~30mg Na | ~15mg K | ~10mg P

TACO BELL HOT SAUCE

The "Hot" packet outsells all other Taco Bell sauce levels combined — America likes medium-brave.

Ingredient	Amount
Tomato paste	2 tbsp
Water	3 tbsp
White vinegar	1 tbsp
Cayenne pepper	1/2 tsp
Cumin	1/4 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp
Paprika	1/2 tsp
Chili powder	1/4 tsp
Sugar	1/2 tsp

Ingredient	Amount
Salt	1/8 tsp

Whisk all ingredients together. Simmer 3 minutes. Cool. Thinner than the Mild — that's accurate to the original.

SHU: ~1,500 | Per tbsp: ~35mg Na | ~20mg K | ~10mg P

TACO BELL FIRE SAUCE

Fire sauce was added to the lineup in 2004 after years of fan requests. The packets read "Will Burn."

Ingredient	Amount
Tomato paste	2 tbsp
Water	3 tbsp
White vinegar	1.5 tbsp
Cayenne pepper	1 tsp
Chipotle powder	1/4 tsp
Cumin	1/4 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp
Smoked paprika	1/2 tsp
Sugar	1/2 tsp
Salt	1/8 tsp

Whisk all ingredients. Simmer 5 minutes — the chipotle needs time to bloom. Cool. The smoky note is what separates Fire from Hot.

SHU: ~5,000 | Per tbsp: ~35mg Na | ~20mg K | ~10mg P

TACO BELL DIABLO SAUCE

Diablo launched in 2015, was pulled, returned in 2016 by popular demand. Packets read "This is the point of no return."

Ingredient	Amount
Tomato paste	2 tbsp
Water	2 tbsp
White vinegar	2 tbsp
Cayenne pepper	1.5 tsp
Chipotle in adobo	1 pepper, minced fine
Habanero powder	1/4 tsp (or 1/2 fresh habanero, minced)
Cumin	1/4 tsp
Garlic powder	1/4 tsp
Onion powder	1/4 tsp
Smoked paprika	1/2 tsp
Lime juice	1 tsp
Sugar	1/4 tsp
Salt	1/8 tsp

Blend all ingredients until smooth (the chipotle needs the blender). Simmer 5 minutes. The habanero gives the delayed burn. The lime gives the citrus bite that makes Diablo different from just "more cayenne."

SHU: ~30,000 | Per tbsp: ~40mg Na | ~25mg K | ~12mg P

SPECIAL EDITION LABEL NAMES

"Same sauce. Different zip code."

Each sauce gets its own name. Cap color marks heat/flavor only; no active ingredients in the public print lane.

Sauce	Special Edition Name
Ultraviolet (blueberry)	ULTRAVIOLET: DIAL TONE
Electric Blue (blue rasp)	ELECTRIC BLUE: GONE FISHING
Nuclear Green (kiwi)	NUCLEAR GREEN: OUT OF OFFICE
Solar Flare (mango-hab)	SOLAR FLARE: LEFT THE CHAT
Plasma Pink (strawberry)	PLASMA PINK: DO NOT RESUSCITATE

Sauce	Special Edition Name
Golden Hour (pineapple)	GOLDEN HOUR: VOICEMAIL
Infrared (cranberry)	INFRARED: LAST SEEN 4:20 PM

CHAPTER 10

Baking: Cakes, Cookies & Treats

“The oven doesn’t ask questions.”

Carrot Cake, Brownies, Cookies, Sponge Cakes, Rainbow Gelatin, Mug Cakes

"The oven doesn't ask questions."

CKD BAKING RULES

The good news: Most baking ingredients are renal-aware (flour, sugar, eggs, oil) **The traps:** - **Baking powder:** Most contain ALUMINUM PHOSPHATE — use phosphorus-free (Rumford aluminum-free) - **Chocolate:** Moderate P — use in controlled amounts, not unlimited - **Nuts:** Moderate-high P — use as garnish, not as main ingredient - **Milk:** Use unsweetened almond milk instead of regular - **Butter vs margarine:** Margarine = lower P. Use for baking. - **Cream cheese frosting > buttercream:** Lower P, Andrew prefers soft sweet flavors

COOKIES

1. Oatmeal Chocolate Chip Cookies

Ingredient	Amount	CKD Notes
Margarine, softened	1/2 cup	Lower P than butter
Brown sugar	1/2 cup packed	
Sugar	1/4 cup	
Egg	1	
Vanilla extract	1 tsp	
All-purpose flour	3/4 cup	
Old-fashioned oats	1.5 cups	Oats are moderate K — fine in cookies
Phosphorus-free baking powder	1/2 tsp	Rumford aluminum-free
Baking soda	1/2 tsp	
Ceylon cinnamon	1/2 tsp	Cinnamon Cloud crossover
Chocolate chips	1/2 cup	Controlled amount
Salt	1/4 tsp	

Method: Cream margarine + sugars. Beat in egg + vanilla. Mix flour + oats + leaveners + cinnamon. Combine wet + dry. Fold in chocolate chips. Scoop onto parchment-lined sheet. Bake 350F 10-12 min until edges golden, centers soft. Cool 5 min on sheet.

Per cookie (24 cookies): ~45mg P, ~55mg K, ~40mg Na

2. Grasshopper Cookies (Mint Chocolate)

Ingredient	Amount
All-purpose flour	1.25 cups
Unsweetened cocoa powder	1/3 cup
Margarine, softened	1/2 cup
Sugar	3/4 cup
Egg	1
Peppermint extract	1 tsp
Green food coloring	4-5 drops
Phosphorus-free baking powder	1 tsp
Chocolate chips	1/3 cup

Method: Keep this sauce non-medicated in the public print lane. After cooking sauce, remove from heat, cool, bottle, and label by flavor and date. Black cap = regular. Gold or green cap = hotter flavor variant only; it does not indicate any active ingredient.

Method: Cream margarine + sugar. Add egg + peppermint + green coloring. Mix flour + cocoa + baking powder. Combine. Fold in chips. Bake 350F 10 min.

The visual: Neon green cookies with chocolate chips — Vorathic BIOLUME energy in cookie form.

3. Fudgy Brownies

Ingredient	Amount
Margarine	1/2 cup, melted
Sugar	1 cup
Eggs	2
Vanilla	1 tsp
Unsweetened cocoa powder	1/3 cup
All-purpose flour	1/2 cup
Salt	1/4 tsp
Phosphorus-free baking powder	1/4 tsp
Chocolate chips	1/3 cup (optional on top)

Method: Mix melted margarine + sugar. Beat in eggs + vanilla. Add cocoa + flour + salt + BP. Pour into greased 8x8 pan. Top with chocolate chips if desired. Bake 350F 22-25 min. Toothpick should come out with moist crumbs, NOT clean (fudgy not cakey).

CAKES

4. Andrew's Soft Carrot Cake

Ingredient	Amount	CKD Notes
All-purpose flour	2 cups	
Sugar	1.5 cups	
Canola/avocado oil	1 cup	Oil = moist cake
Eggs	3	
Carrots, finely grated	2 cups (~3 large)	Moderate K — the carrots ARE the cake so worth it
Phosphorus-free baking powder	2 tsp	
Baking soda	1 tsp	
Ceylon cinnamon	2 tsp	
Nutmeg	1/2 tsp	
Ginger powder	1/2 tsp	
Vanilla	1 tsp	
Salt	1/2 tsp	
Crushed pineapple, drained	1/2 cup (optional)	Adds moisture, moderate K

Cream Cheese Frosting:

Ingredient	Amount
Cream cheese, softened	8 oz
Margarine, softened	1/4 cup
Powdered sugar	3 cups
Vanilla	1 tsp

Method: 1. Mix flour + BP + soda + spices + salt 2. Beat oil + sugar + eggs + vanilla 3. Fold dry into wet. Fold in carrots + pineapple 4. Pour into 2 greased 9" round pans 5. Bake 350F 28-32 min until toothpick clean 6. Cool completely before frosting 7. Frosting: beat cream cheese + margarine, gradually add powdered sugar + vanilla

The secret to MOIST: Oil instead of butter. Do NOT overbake. Pull at 28 min and check.

5. Confetti/Funfetti Cake (White Vanilla + Rainbow Sprinkles)

Ingredient	Amount
All-purpose flour	2.5 cups
Sugar	1.5 cups
Margarine, softened	3/4 cup
Egg whites	5 (whites only — no yolks = lower P)
Unsweetened almond milk	1 cup
Vanilla extract	2 tsp
Almond extract	1/2 tsp
Phosphorus-free baking powder	3 tsp
Salt	1/2 tsp
Rainbow sprinkles	1/3 cup (fold into batter)

Method: Cream margarine + sugar until fluffy. Add egg whites one at a time. Alternate adding flour mixture and almond milk. Add vanilla + almond extract. Fold in sprinkles GENTLY (don't overmix or colors bleed). Pour into 2 greased 9" pans. Bake 350F 25-28 min.

Frosting: Same cream cheese frosting as carrot cake, or vanilla buttercream (margarine + powdered sugar + vanilla + almond milk to thin). Top with MORE sprinkles.

6. Rainbow Cake (6 Layers — Vorathic)

Same confetti cake batter BUT: 1. Divide batter into 6 equal portions 2. Dye each: red, orange, yellow, green, blue, purple 3. Bake each in a 6" round pan (or bake 2 at a time in 9" pans, slice horizontally) 4. Stack with cream cheese frosting between each layer 5. Frost outside with white frosting 6. Cut a slice = RAINBOW cross-section

This IS Genesis 9:16 in cake form. The rainbow is hidden INSIDE until you cut it.

7. Devil's Food Cake (Rich Chocolate)

Ingredient	Amount
All-purpose flour	1.75 cups
Unsweetened cocoa powder	3/4 cup
Sugar	2 cups
Eggs	2
Avocado oil	1/2 cup
Unsweetened almond milk	1 cup
Hot coffee	1 cup (dissolves cocoa, deepens flavor)
Vanilla	2 tsp
Baking soda	2 tsp
Phosphorus-free baking powder	1 tsp
Salt	1 tsp

Method: Mix dry. Mix wet (coffee last — batter will be thin, that's correct). Pour into 2 greased 9" pans. Bake 350F 30-35 min. This cake is EXTREMELY moist because of the oil + coffee.

8. Microwave Mug Cake (2 minutes, 1 serving)

For when you need cake NOW

Ingredient	Amount
All-purpose flour	4 tbsp
Sugar	3 tbsp
Cocoa powder	2 tbsp
Egg	1 small (or 1/2 large beaten)
Almond milk	3 tbsp
Avocado oil	2 tbsp
Vanilla	1/4 tsp
Pinch of salt	
Chocolate chips	1 tbsp (optional)

Method: Mix dry in a large mug. Add wet, stir until smooth. Drop in chocolate chips. Microwave 90 seconds to 2 min (watch — every

microwave is different). Let sit 1 min. Eat from mug or flip onto plate.

Variations: - **Confetti mug cake:** Skip cocoa, add 1 tbsp sprinkles - **Cinnamon Cloud mug cake:** Skip cocoa, add 1/2 tsp Ceylon cinnamon + 1 tsp honey - **Peanut butter mug cake:** Add 1 tbsp PB, skip cocoa

9. Soft Moist Sponge Cake (Japanese-Style Jiggly)

The softest cake possible — jiggles when you shake it

Ingredient	Amount
Eggs, separated	4
Sugar	1/2 cup (divided: 1/4 for yolks, 1/4 for whites)
Avocado oil	3 tbsp
Almond milk	3 tbsp
Vanilla	1 tsp
Cake flour (or AP flour + 2 tbsp cornstarch)	1/2 cup
Cream of tartar	1/4 tsp

Method: 1. Beat egg YOLKS + 1/4 cup sugar until pale and thick 2. Stir in oil + milk + vanilla 3. Sift in flour, fold gently 4. In separate bowl: beat egg WHITES + cream of tartar until foamy, gradually add remaining 1/4 cup sugar, beat to stiff peaks 5. Fold 1/3 of whites into yolk mixture to lighten, then fold in remaining whites VERY gently 6. Pour into UNGREASED 8" round pan (ungreased = cake clings to sides as it rises) 7. Bake 325F 30-35 min 8. Invert pan immediately on cooling rack. Cool completely before removing.

Why it's jiggly: The meringue (beaten whites) creates air pockets. Low flour ratio = less structure = more jiggle. Oil instead of butter = softer crumb.

10. Fruit-Filled Cake (Strawberry or Mango)

Sponge cake + fresh fruit + whipped cream = Japanese-style fruit sandwich cake

Component	What
Base	Sponge cake (recipe #9), sliced into 2-3 horizontal layers
Filling	Whipped heavy cream (1 cup cream + 2 tbsp powdered sugar + 1/2 tsp vanilla, whipped to stiff)
Fruit	Fresh strawberries sliced, or fresh mango chunks

Assembly: 1. Bottom cake layer on plate 2. Spread whipped cream 3. Arrange sliced fruit in a single layer 4. More whipped cream 5. Next cake layer. Repeat. 6. Top with whipped cream + decorative fruit arrangement 7. Refrigerate 2 hours before serving (sets the cream)

GELATIN DESSERTS

11. Rainbow Gelatin Mold (Vorathic — Genesis 9:16)

7 layers, each a different color of the rainbow

Layer	Jell-O Flavor	Color
1	Cherry/Strawberry	Red
2	Orange	Orange
3	Lemon	Yellow
4	Lime	Green
5	Berry Blue	Blue
6	Grape	Purple
7 (optional)	Unflavored + pink coloring	Pink

Method: 1. Use a 9x13 pan or bundt mold 2. For each layer: dissolve 1 small box Jell-O in 1 cup boiling water + 1/2 cup cold water 3. Pour first layer (red). Refrigerate 30-45 min until set but still tacky 4. Pour next layer (orange) GENTLY over set layer. Refrigerate. 5. Repeat for all colors. Each layer must be barely set before adding next. 6. Total time: ~4-5 hours for all layers. Overnight is better. 7. To unmold: dip pan briefly in warm water, invert onto plate.

The reveal: When sliced, each piece shows ALL the rainbow layers. Genesis 9:16 in gelatin.

CKD note: Jell-O is low K/P/Na — one of the most CKD-aware desserts. The sugar is the only concern (if diabetic).

12. Fancy Gelatin Art (Flower/3D Designs)

Gelatina artística originated in Mexico in the 1990s and became a competitive art form — Mexican gelatin artists now create photorealistic flowers, animals, and portraits entirely inside clear gelatin domes.

Full Ingredient List

Clear Gelatin Base (The Canvas):

Ingredient	Amount	CKD Notes
Unflavored gelatin (Knox)	4 packets (28g)	CKD-aware protein source
Water	3 cups	Counts toward fluid
Sugar	1/3 cup	

Injection Paint (The Colors) — make one batch per color:

Ingredient	Amount per color	CKD Notes
Unsweetened almond milk	3 tbsp	LOW P (vs condensed milk ~100mg P/tbsp). Andrew already has pectin — can add 1/4 tsp pectin per color for thicker paint
Unflavored gelatin	1/2 packet (3.5g)	Bloom in 1 tbsp cold almond milk
Food coloring (gel-based)	2-4 drops per color	Wilton gel colors are best — more vivid, less liquid
Sugar	1 tsp	Optional — adds slight sweetness to the "paint"

Recommended Color Kit (8 colors minimum):

Color	Gel Color	What It Makes
White	No coloring (plain almond milk)	Flower petals, clouds, swan bodies
Red	Wilton Red-Red	Rose petals, koi fish
Pink	Wilton Pink	Cherry blossoms, flamingos
Yellow	Wilton Golden Yellow + pinch turmeric	Sunflower centers, goldfish
Orange	Wilton Orange	Koi fish accents, autumn leaves
Green	Wilton Leaf Green	Stems, leaves, grass, frogs
Blue	Wilton Royal Blue	Water, sky, butterfly wings
Purple	Wilton Violet	Lavender, iris petals, galaxy effects
Gold shimmer	Edible gold luster dust mixed into yellow paint	Sacred geometry lines, Vorathic love

Equipment List

Equipment	Source	Price	Notes
Gelatin art syringe set (10-piece, stainless steel needles)	Amazon "gelatin art tools"	~\$15-20	Includes flower tip, round tip, leaf tip, flat tip
Extra needle tips (flower/petal set)	Amazon	~\$8-12	More tip shapes = more design options
Dome molds (silicone, 6" and 10")	Amazon "silicone dome mold gelatin"	~\$10-15	Clear silicone preferred — you can see layers as you work
Small squeeze bottles (4 oz, set of 6)	Amazon	~\$8	For holding warm paint — easier than syringes for filling
Warm water bath container	Any pot or bowl	\$0	Keep paint bottles in warm water so gelatin doesn't set
LED light base (for display)	Amazon "LED light base display"	~\$8-12	Sets under the finished dome — lights it up from below
Candy thermometer	Amazon or kitchen store	~\$10	For monitoring gelatin temp (never exceed 160F)
Fine-tip tweezers (food-safe)	Amazon	~\$6	For placing edible gold leaf, tiny details

Equipment	Source	Price	Notes
Turntable (lazy susan, small)	Amazon or Dollar Tree	~\$5-10	Rotate the mold while injecting for even designs

Total starter kit: ~\$70-90

Method (Detailed)

- 1. Make the clear base:** Bloom gelatin in 1 cup COLD water (sprinkle on surface, wait 5 min). Heat remaining 2 cups water + sugar until dissolved (not boiling). Add bloomed gelatin, stir until clear. Pour into dome mold. Refrigerate 3-4 hours until FIRM.
- 2. Make injection paints:** For each color — bloom 1/2 packet gelatin in 1 tbsp cold almond milk (5 min). Microwave 10 seconds to melt. Add remaining 2 tbsp almond milk + food coloring + sugar. Mix. Keep in small squeeze bottle in warm water bath (110-120F) so it stays liquid.
- 3. Inject designs:** Turn dome mold upside down (you work from the BOTTOM — the flat surface faces up). Insert needle through the flat surface, inject paint at desired depth. Each injection creates one element (one petal, one leaf, etc.).
- 4. Build layer by layer:** Work from the CENTER/BOTTOM of the dome outward and upward. Inject deepest elements first (flower centers), then petals, then leaves, then stems. Refrigerate 5-10 min between colors so they don't bleed.
- 5. Unmold:** Dip mold briefly in warm water (5 seconds). Invert onto serving plate. The clear dome releases with the 3D design floating inside.
- 6. Display:** Place on LED light base. The light shines up through the clear gelatin, illuminating the injected designs from below.

For CKD: Almond milk paint = ~5mg P per tbsp vs condensed milk = ~100mg P per tbsp. If Andrew has pectin in the pantry, adding 1/4 tsp per paint batch thickens it for more precise injection (acts like a natural gelling agent alongside the gelatin).

Per serving (1/8 dome): ~10mg Na | ~6mg P | ~8mg K

12b. Psychedelic 3D Gelatin Art Cake — The Deadhead Edition

Albert Hofmann accidentally absorbed LSD through his fingertips on April 16, 1943. His bicycle ride home became the most famous commute in pharmacological history.

Same technique as above, completely different universe inside the dome.

The Clear Base — "The Void"

Ingredient	Amount	CKD Notes
Unflavored gelatin	6 packets (42g)	CKD-aware protein
Water	3 cups	Counts toward fluid
Sugar	1/2 cup	
Tonic water	1 cup (replace 1 cup of water)	Quinine = UV fluorescence. The whole cake GLOWS under blacklight
Blue food coloring	2 drops	Faint cosmic blue tint to the "void"

Set in a large dome mold (10-inch) or deep glass bowl. Refrigerate until firm (~3 hours).

The Injection Paints — 7 Rainbow Colors

Make 7 small batches of injection paint:

Color	Food Coloring	Flavor Extract (optional)
Red	Red gel, 3 drops	Cherry
Orange	Orange gel, 3 drops	Orange
Yellow	Yellow gel, 3 drops + 1/8 tsp turmeric	Lemon
Green	Green gel, 3 drops	Lime
Blue	Blue gel, 3 drops + blue spirulina	Blue raspberry
Indigo	Blue + purple, 2 drops each	Grape
Violet	Purple gel, 3 drops + butterfly pea powder	Lavender

Each paint batch: 3 tbsp unsweetened almond milk (low P) + 1/2 packet unflavored gelatin bloomed + food coloring. Warm gently, keep liquid in

warm water bath while working.

Psychedelic Design Patterns

Option A: Grateful Dead Steal Your Face Inject a red and blue skull profile (the SYF lightning bolt skull). Use a printed reference, mark guidelines on the outside of the mold with dry-erase marker, inject through the bottom following the lines. Red for the lightning bolt, blue for the skull outline. White (plain almond milk gelatin) for the skull fill.

Option B: Tie-Dye Spiral Start at the center of the dome. Inject each rainbow color in a spiral pattern outward — red at center, then orange, yellow, green, blue, indigo, violet at the edge. Use a long needle and work in concentric rings. When unmolded, it looks like a tie-dye shirt trapped in glass.

Option C: Alex Grey "Net of Being" / Sacred Geometry Inject a repeating pattern of eyes (golden-yellow iris, white sclera, dark pupil) arranged in a hexagonal grid throughout the dome. Between the eyes, inject thin lines of gold (turmeric paint) forming the Flower of Life pattern. This is the Vorathic holy grail.

Option D: Mushroom Forest Inject mushroom shapes: brown/red caps (red + tiny bit of brown/black), white stems, green grass at the base. Add tiny golden dots around the mushrooms (fairy lights). A psychedelic forest trapped in gelatin.

Option E: Third Eye / All-Seeing Eye One large eye in the center of the dome. Golden iris (turmeric + gold luster dust in the paint), white sclera, deep purple pupil. Radiating lines of rainbow color emanating outward from the eye like rays. The Eye of Providence as dessert.

UV Blacklight Effect

Because the base uses tonic water, the entire dome glows pale blue under 365nm UV light. The injected colored designs appear to FLOAT in a

glowing void. Set up a UV strip around the serving plate for the full effect.

Non-Medicated Publication Variant

Keep this gelatin art dome non-medicated in the public print lane. Use tonic water, food color, edible glitter, and injection art only. Do not add active ingredients or dosing math.

Per serving (1/10 of dome): ~15mg Na | ~8mg P | ~12mg K

13. Edible Gold Glitter Gelatin Shots (Non-Alcoholic)

Ingredient	Amount
Fruit punch juice (or apple juice)	2 cups
Unflavored gelatin	2 packets
Sugar	2 tbsp
Edible gold glitter	1/2 tsp
LorAnn flavoring (any)	3-4 drops

Method: Bloom gelatin in 1/2 cup cold juice. Heat remaining juice + sugar until steaming. Add bloomed gelatin, stir to dissolve. Add gold glitter + flavoring. Pour into small cups or silicone molds. Refrigerate 2+ hours.

Vorathic party food. Gold glitter suspended in jewel-toned gelatin.

CKD Considerations

All baked goods use: - Phosphorus-free baking powder (Rumford aluminum-free) - Margarine over butter (lower P) - Unsweetened almond milk over regular milk (lower P) - Egg whites where possible (yolks are higher P) - Controlled chocolate amounts - Cream cheese frosting over buttercream (lower P, softer, sweeter — matches Andrew's preference)

CHAPTER 11

Gluten-Free Breads & Pasta

"Gluten left. The flavor stayed."

Sourdough, Sandwich Bread, Flatbread, Pizza, Fresh Pasta & More

"Gluten left. The flavor stayed."

They told you gluten-free means cardboard. They told you CKD means bland. Two lies stacked on top of each other like bad drywall. We tore it down to the studs and rebuilt with rice flour and fire.

"The baker who understands fermentation understands the universe." — Chinese proverb, paraphrased in a Philly kitchen

GF CKD BREAD RULES

The good news: Rice flour is naturally low-P, low-K, low-Na. It's one of the cleanest CKD flours. **The traps:** - **Whole wheat GF blends:** Many contain sorghum, millet, or amaranth — higher K/P. Avoid. - **Xanthan gum:** Required for structure (replaces gluten). CKD-aware. Use it. - **Baking powder:** Phosphorus-free only (Rumford aluminum-free). Non-negotiable. - **Milk:** Unsweetened almond milk. Always. - **Butter:** Margarine. Lower P. Every time. - **Yeast:** CKD-aware. Nutritional yeast is NOT the same thing — that's high P. Don't confuse them. - **Tapioca starch + potato starch:** Both renal-aware. They give GF bread chew and structure that rice flour alone can't.

SOURDOUGH

1. GF Sourdough Bread (Rice Flour Starter + Loaf)

Phase 1: Build the Starter (7 days)

"Patience is the mother of all virtues." — Augustine of Hippo, who never had to wait for rice flour to ferment

Day	Feed	Amount	Notes
Day 1	Brown rice flour + water	1/4 cup each	Mix in clean glass jar. Cover loosely. Room temp.

Day	Feed	Amount	Notes
Day 2	Discard half, feed again	1/4 cup flour + 1/4 cup water	
Day 3-6	Repeat daily	Same amounts	You'll see bubbles by Day 3-4. Smells sour = working.
Day 7	Starter should double in 4-6 hours after feeding	Ready to bake	If not doubling yet, keep feeding. Some starters take 10 days.

Maintenance: Feed 1x/week if refrigerated. 1x/day if on counter.

Phase 2: The Bread

Ingredient	Amount	CKD Notes
Active GF sourdough starter	1 cup (fed 4-6 hrs ago, bubbly)	
White rice flour	1.5 cups	Low P, low K — the cleanest flour
Tapioca starch	1/2 cup	Adds chew
Potato starch	1/4 cup	Adds moisture
Xanthan gum	1.5 tsp	The gluten replacement
Warm water	3/4 cup	
Avocado oil	2 tbsp	
Sugar	1 tbsp	Feeds the starter
Salt	3/4 tsp	
Apple cider vinegar	1 tsp	Enhances sour flavor

Method: 1. Whisk dry ingredients (rice flour, tapioca, potato starch, xanthan gum, salt) in KitchenAid bowl 2. Add active starter, warm water, oil, sugar, vinegar 3. Mix with paddle attachment on medium 3-4 min. Dough will be thick batter consistency — NOT like wheat dough. This is normal. 4. Scrape into greased 9x5 loaf pan. Smooth top with wet spatula. 5. Cover with damp towel or plastic wrap. Rise 4-8 hours at room temp (or overnight in fridge for deeper sour flavor — Andrew's preference) 6. Score top with razor blade or sharp knife. Dust with rice flour. 7. Bake 400F for 45-50 min. Internal temp 205-210F. 8. Cool IN the pan 10 min, then on wire rack 1 hour minimum before slicing. GF bread MUST cool completely or it crumbles.

Per slice (12 slices): ~15mg Na, ~25mg P, ~30mg K

"The bread that takes the longest tastes the truest." — Every baker who ever lived

QUICK BREADS

2. GF Sandwich Bread (KitchenAid — 90 Min Start to Finish)

For when you want bread today, not next week.

Ingredient	Amount	CKD Notes
White rice flour	1.5 cups	
Tapioca starch	1/2 cup	
Potato starch	1/4 cup	
Xanthan gum	1.5 tsp	
Sugar	2 tbsp	
Active dry yeast	2.25 tsp (1 packet)	CKD-aware
Salt	3/4 tsp	
Warm unsweetened almond milk	1 cup (110F)	Lower P than regular milk
Eggs	2	
Avocado oil	3 tbsp	
Apple cider vinegar	1 tsp	
Phosphorus-free baking powder	1 tsp	Rumford aluminum-free

Method: 1. Warm almond milk to 110F. Stir in sugar + yeast. Let sit 5-10 min until foamy. 2. In KitchenAid with paddle: combine rice flour, tapioca, potato starch, xanthan gum, salt, baking powder 3. Add yeast mixture, eggs, oil, vinegar 4. Beat on medium 3-4 min. Batter-like dough — wetter than wheat bread dough. 5. Scrape into greased 9x5 loaf pan. Smooth top with wet hands. 6. Cover. Rise in warm spot 30-45 min until dough crowns 1" above pan rim 7. Bake 375F 35-40 min. Tent with foil at 25 min if browning too fast. 8. Cool in pan 10 min. Turn out onto rack. Cool completely before slicing.

Per slice (14 slices): ~55mg Na, ~30mg P, ~35mg K

3. GF Flatbread / Naan (Stovetop — 20 Min)

"He who has bread has many problems. He who has no bread has one problem." — Turkish proverb

Ingredient	Amount	CKD Notes
White rice flour	1 cup	
Tapioca starch	1/2 cup	
Xanthan gum	1 tsp	
Phosphorus-free baking powder	1.5 tsp	Rumford
Salt	1/2 tsp	
Sugar	1 tsp	
Unsweetened almond milk	1/2 cup, warm	
Plain coconut yogurt	1/4 cup	Dairy yogurt is higher P — coconut keeps it low
Avocado oil	1 tbsp + more for pan	

Method: 1. Mix dry ingredients in bowl 2. Add warm almond milk, coconut yogurt, oil. Stir until dough forms. It should be soft, slightly sticky. 3. Divide into 6 balls. Roll each on rice-floured surface to 1/4" thick ovals. 4. Heat cast iron or heavy skillet over medium-high. Brush with oil. 5. Cook each flatbread 2-3 min per side. Watch for bubbles on surface — flip when you see them. 6. Brush hot flatbread with margarine immediately off the pan.

Garlic naan variant: Mix 2 cloves minced garlic + 2 tbsp melted margarine + pinch of salt. Brush on hot naan. Top with chopped cilantro.

Makes 6 flatbreads. Per flatbread: ~90mg Na, ~20mg P, ~25mg K

4. GF Pizza Dough

Friday night doesn't stop.

Ingredient	Amount	CKD Notes
White rice flour	1 cup	

Ingredient	Amount	CKD Notes
Tapioca starch	1/2 cup	
Potato starch	1/4 cup	
Xanthan gum	1.5 tsp	
Active dry yeast	2.25 tsp (1 packet)	
Sugar	1 tbsp	
Salt	1/2 tsp	
Warm water	3/4 cup (110F)	
Olive oil	2 tbsp + more for brushing	
Egg white	1	Lower P than whole egg
Apple cider vinegar	1 tsp	
Italian seasoning	1 tsp	

Method: 1. Dissolve sugar + yeast in warm water. Wait 5-10 min until foamy. 2. KitchenAid paddle: mix all dry ingredients 3. Add yeast mixture, oil, egg white, vinegar. Beat 3 min on medium. 4. Dough is sticky — that's correct. Scoop onto parchment-lined sheet or pizza stone. 5. Oil your hands. Press/spread dough into 12-14" circle (or rectangle — no rules). 1/4" thick. 6. Let rise 20 min uncovered. 7. Pre-bake crust at 425F for 8-10 min until just set and lightly golden. 8. Add toppings. Bake 10-12 min more until cheese melts and edges crisp.

CKD pizza topping rules: - Sauce: no-salt-added tomato sauce + garlic + oregano + basil (homemade or Muir Glen) - Cheese: mozzarella in moderation (1/2 cup shredded per pizza — moderate P) - Proteins: grilled chicken, shrimp, turkey pepperoni - Veggies: roasted red peppers, onions, mushrooms (small amounts), fresh basil, arugula after baking - AVOID: olives (Na bomb), regular pepperoni (Na + P bomb), extra cheese

Per 2 slices (8 total, no toppings): ~70mg Na, ~25mg P, ~30mg K

PASTA

5. GF Fresh Pasta (Rice Flour — KitchenAid Pasta Roller)

"In the beginning was the noodle." — Apocryphal, probably Sicilian

Ingredient	Amount	CKD Notes
White rice flour	1.5 cups	
Tapioca starch	1/2 cup	Adds elasticity
Xanthan gum	1 tsp	Binds without gluten
Eggs	2 large	
Egg yolk	1 extra	Richness + binding
Olive oil	1 tbsp	
Water	1-2 tbsp (as needed)	
Salt	1/2 tsp	

Method: 1. Whisk rice flour, tapioca starch, xanthan gum, salt on clean counter or in KitchenAid bowl 2. Make a well. Add eggs, yolk, olive oil to center. 3. Using fork (or KitchenAid paddle on low), gradually incorporate flour into eggs until shaggy dough forms 4. Knead by hand 3-4 min. Dough should be smooth, pliable, slightly tacky. If too dry, add water 1 tsp at a time. If too wet, add rice flour. 5. Wrap in plastic. Rest 30 min at room temp. (This lets the xanthan gum hydrate — critical for rollability.) 6. Divide into 4 portions. Keep unused portions wrapped. 7. **KitchenAid pasta roller:** Start at widest setting. Roll through twice. Decrease one setting. Roll through. Repeat until desired thickness (setting 5-6 for fettuccine, setting 7 for angel hair). 8. Switch to cutter attachment. Run sheets through fettuccine or spaghetti cutter. 9. Dust cut pasta with rice flour. Toss gently. Cook within 1 hour or dry on pasta rack.

Cooking: Boil salted water. Fresh GF pasta cooks FAST — 2-3 min max. Taste at 2 min. It goes from perfect to mush in seconds.

Per serving (4 servings): ~65mg Na, ~55mg P, ~40mg K

Variations: - **Herb pasta:** Add 2 tbsp finely chopped fresh basil or rosemary to the dough - **Saffron pasta:** Steep pinch of saffron in 1 tbsp warm water, add to eggs — golden color, subtle flavor - **Black pepper pasta:** Add 1 tsp coarsely ground black pepper to flour — cacio e pepe ready

MORE BREADS

6. GF Garlic Breadsticks

Olive Garden walked so we could run. Then we lapped them.

Ingredient	Amount	CKD Notes
White rice flour	1 cup	
Tapioca starch	1/3 cup	
Xanthan gum	1 tsp	
Phosphorus-free baking powder	2 tsp	Rumford
Sugar	1 tbsp	
Salt	1/2 tsp	
Active dry yeast	1.5 tsp	
Warm unsweetened almond milk	1/2 cup (110F)	
Egg	1	
Olive oil	2 tbsp	

Garlic Topping:

Ingredient	Amount
Margarine, melted	3 tbsp
Garlic, minced	4 cloves
Garlic powder	1/2 tsp
Italian seasoning	1 tsp
Dried parsley	1 tsp
Salt	Pinch

Method: 1. Dissolve sugar + yeast in warm almond milk. Wait 5 min. 2. Mix dry ingredients. Add yeast mixture, egg, olive oil. Beat with KitchenAid paddle 2-3 min. 3. Scoop dough into 10-12 strips on parchment-lined baking sheet. Wet hands to shape — think thick fingers, not pencils. 4. Cover. Rise 20 min. 5. Bake 400F 12-15 min until golden. 6. While baking: melt margarine, cook minced garlic 30 seconds (just until fragrant, NOT brown). Stir in garlic powder, Italian seasoning, parsley, salt. 7. Brush garlic margarine over breadsticks IMMEDIATELY out of the oven. Twice. Don't be shy.

Makes 10-12 breadsticks. Per breadstick: ~75mg Na, ~20mg P, ~20mg K

7. GF Cornbread (Base)

"Cornbread ain't nothing wrong with that." — Already CKD-aware. We just made it official.

Ingredient	Amount	CKD Notes
Yellow cornmeal	1 cup	Naturally low P, low K — CKD royalty
White rice flour	1/2 cup	GF sub for AP flour
Sugar	1/4 cup (or 2 tbsp honey)	
Phosphorus-free baking powder	1 tbsp	Rumford
Salt	1/2 tsp	
Eggs	2	
Unsweetened almond milk	3/4 cup	
Avocado oil	1/4 cup	
Margarine	2 tbsp, melted	For the skillet

Method: 1. Preheat oven to 400F. Put 2 tbsp margarine in a 10" cast iron skillet, place in oven while preheating. (The hot fat = crispy bottom crust. This is non-negotiable.) 2. Mix cornmeal + rice flour + sugar + baking powder + salt 3. Beat eggs + almond milk + avocado oil 4. Combine wet + dry. Don't overmix. 5. Pull hot skillet from oven. Swirl melted margarine to coat. Pour batter in — it should SIZZLE. 6. Bake 20-22 min until golden and

toothpick clean. Edges should be deeply golden, almost caramelized. 7. Let cool 5 min in skillet. Slice into wedges. Serve with margarine or honey.

Per wedge (8 wedges): ~85mg Na, ~40mg P, ~45mg K

Variations: - **Jalapeno cornbread:** Add 2 tbsp diced jalapeño (seeds removed) + 1/4 cup corn kernels - **Honey butter cornbread:** Whip margarine + honey + pinch of cinnamon. Serve on top. - **Cornbread muffins:** Same batter, muffin tin, 15-18 min at 400F

8. GF Tortillas (Rice Flour Based)

Soft, pliable, doesn't crack in half when you look at it wrong.

Ingredient	Amount	CKD Notes
White rice flour	1 cup	
Tapioca starch	1/2 cup	The pliability secret
Xanthan gum	1 tsp	
Salt	1/2 tsp	
Avocado oil	2 tbsp	
Warm water	1/2 - 2/3 cup	Add gradually

Method: 1. Mix rice flour, tapioca starch, xanthan gum, salt 2. Add oil. Mix until crumbly. 3. Add warm water gradually, stirring, until soft pliable dough forms. Not sticky, not crumbly. Like Play-Doh. 4. Divide into 8 balls. Cover with damp towel. 5. Roll each ball between two sheets of parchment paper to 6-7" circles, 1/8" thin. (GF dough sticks to rolling pins — parchment paper is the move.) 6. Heat dry skillet (no oil) over medium-high. 7. Cook each tortilla 1-2 min per side. Look for light brown spots and slight puffing. 8. Stack cooked tortillas under a clean towel to stay soft and pliable.

Makes 8 tortillas. Per tortilla: ~70mg Na, ~10mg P, ~15mg K

The CKD taco night stack: - GF rice tortilla (this recipe) - CKD seasoned ground turkey or chicken - Shredded cabbage (lower K than lettuce — crunchier too) - Pico de gallo (fresh tomato, onion, cilantro, lime — small amounts keep K manageable) - Sour cream (small dollop — moderate P) - Hot sauce (homemade from RENALWISE hot sauce chapter)

CKD Considerations

Rule	Why
Rice flour over wheat flour	Lower P, lower K, naturally GF
Tapioca + potato starch for structure	CKD-aware binders that replace gluten's elasticity
Xanthan gum in every recipe	The gluten replacement — without it, GF bread crumbles to dust
Phosphorus-free baking powder ONLY	Regular BP contains aluminum phosphate — direct P load
Margarine over butter	Lower phosphorus per tbsp
Unsweetened almond milk over regular	~80% less phosphorus than cow's milk
Avoid "GF all-purpose blends" with sorghum/amaranth/teff	Higher K and P than rice-based flours
Cool GF bread COMPLETELY before slicing	GF structure sets as it cools — cut early and it falls apart
Fresh GF pasta cooks in 2-3 min	Overcook by 60 seconds and you have wallpaper paste — set a timer
Yeast is CKD-aware, nutritional yeast is NOT	Nutritional yeast = 100mg+ P per tbsp. Active dry yeast for baking = negligible.

Daily CKD targets: Na < 1500mg | K < 2000mg | P < 800mg

All recipes in this chapter fall well within safe ranges per serving.

Stack multiple items from this chapter in a single meal (bread + pasta) and you're still under 200mg P, 100mg K, 200mg Na for the bread/pasta portion. The danger comes from what you PUT on it — cheese, processed meats, canned sauces. Control the toppings, the bread is already clean.

"Water the roots, not the leaves." The flour is the root. Get that right and everything above it — the crust, the crumb, the chew — follows. Gluten was never the flavor. It was just the scaffolding. We built new scaffolding. The house still stands.

Smoothies & Juices

“Blend first. Ask questions later.”

Caribbean Passion, Tropical Favorites & Fresh Fruit Blends

Chobani Zero Sugar Vanilla Yogurt Base, Fresh Fruit,
"Jamba got Andrew through PD. Now we bring Jamba home."

Andrew's Smoothie Rules

- **Fresh fruit** preferred (frozen okay for texture)
- **Chobani Zero Sugar Vanilla Yogurt** as the creamy base
- **Tropical flavors** — strawberry, pineapple, mango, kiwi
- **Rainbow sherbet, sour sorbet, popsicle** vibes

CKD Fruit Guide for Smoothies

GREEN LIGHT (Low-K, use freely)

- Strawberries (low K, ~150mg per cup)
- Blueberries (~115mg K per cup)
- Raspberries (~185mg K per cup)
- Cranberries (~85mg K per cup)
- Pineapple (~180mg K per cup — moderate, but Andrew loves it, portion control)
- Apple (~195mg K per cup)
- Grapes (~290mg K per cup — borderline, small amounts)
- Peaches, canned/drained (~240mg K per cup)
- Pears (~210mg K per cup)
- Watermelon (~170mg K per cup — BUT high fluid content)
- Lemon/Lime juice (low K, great for sour profile)

YELLOW LIGHT (Moderate-K, use in small amounts — 1/4 cup max)

- Mango (~270mg K per cup — Andrew loves it, use 1/4 cup)
- Kiwi (~560mg K per cup — HIGH, use 1 small kiwi max or skip)
- Pineapple in larger amounts

RED LIGHT (High-K, avoid or use tiny garnish only)

- Banana (~420mg K per cup — THE classic smoothie fruit but CKD enemy)
- Orange/OJ (~500mg K per cup)
- Papaya (~780mg K per cup)
- Coconut water (~600mg K per cup)
- Dried fruit (concentrated K)

THE RECIPES

1. Caribbean Passion Smoothie

Original Jamba Caribbean Passion: Orange juice, passion-mango juice, strawberries, peaches, frozen yogurt **Problem:** OJ is HIGH-K (~500mg/cup), passion-mango juice adds more

Ingredient	Amount	CKD Sub	K Impact
Orange juice	8 oz	Apple juice (low-K, sweet, neutral flavor)	Saves ~300mg K
Passion-mango juice	4 oz	1/4 cup mango chunks + 1 tsp passion fruit extract (Amazon)	Controlled
Strawberries	1/2 cup	Fresh strawberries	~75mg K
Peaches	1/2 cup	Canned peaches, drained (or fresh)	~120mg K
Frozen yogurt	1/2 cup	Chobani Zero Sugar Vanilla Yogurt	Lower P than froyo
Ice	1 cup		

Method: Blend all until smooth. Serve in tall glass. Garnish with strawberry slice.

Per serving: ~250mg K, ~120mg P, ~80mg Na (vs original's ~600+ K) **The taste:** 90% identical. The apple juice base is undetectable behind the mango + passion fruit.

2. Strapineapple Paradise

Ingredient	Amount
Fresh strawberries	3/4 cup
Fresh pineapple chunks	1/2 cup
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Apple juice	1/4 cup
Ice	1 cup
Honey	1 tsp (optional)

Blend. Pink-gold tropical perfection.

3. Mango Kiwi Tropical (K-Controlled)

Ingredient	Amount	K Notes
Mango chunks	1/4 cup (frozen or fresh)	~68mg K at this portion
Kiwi	1 small, peeled	~140mg K — one kiwi is okay
Strawberries	1/2 cup	~75mg K
Chobani Zero Sugar Vanilla Yogurt	1/2 cup	
Apple juice	1/4 cup	
Ice	1 cup	
Lime juice	1 tsp	Brightens everything

Total K: ~350mg — acceptable for most CKD diets in a single serving.

4. Caribbean Sunset (Andrew's Custom)

Ingredient	Amount
Pineapple chunks	1/2 cup
Mango chunks	1/4 cup
Strawberries	1/4 cup
Coconut cream	1 tbsp (NOT coconut water — cream is low K, water is high K)
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Lime juice	1 tsp
Ice	1 cup
Turmeric	Pinch (golden color + inflammation-support)

Blend. Tropical, creamy, golden. The sunset in a glass.

5. Berry Blast (Lowest-K Option)

Ingredient	Amount
Strawberries	1/2 cup
Blueberries	1/2 cup
Raspberries	1/4 cup
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Apple juice	1/4 cup
Ice	1 cup

The safest smoothie. All low-K berries. ~200mg K total. Drink freely.

6. Sour Tropical Storm

Ingredient	Amount
Pineapple	1/2 cup
Strawberries	1/2 cup
Lime juice	2 tbsp (THE SOUR)
Lemon juice	1 tbsp

Ingredient	Amount
Citric acid	1/4 tsp (extra pucker)
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Ice	1 cup
Honey	1 tsp to balance

Andrew's sour-loving profile. Tropical + sour citrus punch.

7. Pina Colada Cloud (Non-Alcoholic)

Ingredient	Amount
Pineapple chunks	3/4 cup
Coconut cream	2 tbsp
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Vanilla extract	1/2 tsp
Ice	1 cup
Ceylon cinnamon	Pinch (Cinnamon Cloud crossover)

Cinnamon Cloud meets Pi??a Colada. Frozen paradise.

8. Rainbow Sherbet Smoothie

Ingredient	Amount
Raspberry sorbet or frozen raspberries	1/4 cup
Lime juice	1 tbsp
Pineapple chunks	1/4 cup
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Apple juice	1/4 cup
Ice	1 cup
Rainbow sprinkles	Garnish on top

Tastes like rainbow sherbet in drinkable form. Andrew's favorite frozen treat, liquidized.

FROZEN VERSIONS (Popsicles & Sorbets)

Tropical Popsicles

Pour any smoothie recipe above into popsicle molds. Freeze 4+ hours. Instant CKD-aware popsicles.

CHAPTER 13

Drinks, Tea & Homebrew

"The cup is never empty if you know what to pour."

Iced Green Tea, Mojito, Berry Lemonade, Yerba Mate & More

Beverages (Every Ounce Counts Toward Fluid Limit)

"Make every sip count."

FLUID RESTRICTION REMINDER

If you're on a fluid restriction: **32-48 oz (1-1.5 liters) per day is typical.** EVERY drink recipe here counts toward that limit. Plan accordingly: - One drink = 8-12 oz = 1/4 to 1/3 of daily allowance - Ice counts as fluid when it melts - Make drinks STRONG in flavor so smaller portions satisfy - Use smaller glasses — a 6oz cup feels full, a 16oz cup feels half-empty

THE DRINKS

1. Jade Iced Green Tea

Ingredient	Amount	CKD Notes
Green tea bags	3 bags per 2 cups water	Steep strong — you're drinking less volume
Water	2 cups, hot	

Best for popsicles: Caribbean Passion, Strapineapple, Berry Blast (they have enough sugar to stay scoopable)

Sour Citrus Sorbet

Ingredient	Amount
Lemon juice	1/2 cup
Lime juice	1/4 cup
Pineapple juice	1/4 cup
Sugar	3/4 cup
Water	1.5 cups
Citric acid	1/2 tsp

Dissolve sugar in warm water. Add juices + citric acid. Chill. Churn or granita method.

Rainbow Sherbet (from scratch)

Layer 1 — Raspberry (pink): 1 cup raspberries + 1/3 cup sugar + 1/2 cup Chobani yogurt, blended **Layer 2 — Lime (green):** 1/2 cup lime juice + 1/3 cup sugar + 1/2 cup Chobani yogurt + green coloring **Layer 3 — Pineapple (yellow):** 1/2 cup pineapple + 1/3 cup sugar + 1/2 cup Chobani yogurt

Pour layers into loaf pan, swirl, freeze 4-6 hours. Scoop.

FLUID TRACKING REMINDER

Every smoothie is **12-16 oz of fluid.** Every popsicle is **3-4 oz.**

If you're on a fluid restriction (typically 32-48 oz/day): - One smoothie = 1/3 to 1/2 of daily fluid allowance - Plan accordingly — smoothie days mean less fluid elsewhere - Use more ice and less liquid to get thicker texture with less fluid volume

CKD Considerations

RENAL CHECK — Smoothies & Frozen Treats
K (Potassium): LOW-MED (apple juice base, low-K berries, controlled tropical fruit)
P (Phosphorus): LOW (Chobani yogurt over milk, no protein powder)
Na (Sodium): LOW (fresh fruit, no added salt)
Fluid: 12-16 oz PER SMOOTHIE — TRACK IT
Protein: LOW-MED (yogurt provides some)

"Jamba got you through PD. These get you through everything else."

Ingredient	Amount	CKD Notes
Sugar	2-3 tbsp (or to taste)	Thai-style is SWEET
Condensed milk	1 tbsp per glass (optional)	Small amount = controlled P
Ice	Fill glass	Counts as fluid

Method: 1. Steep 3 green tea bags in 2 cups hot water for 5 min (strong brew) 2. Remove bags. Add sugar while hot, stir to dissolve 3. Cool to room temperature, then refrigerate 4. Serve over ice. Drizzle condensed milk if desired — it sinks beautifully

CKD bonus: Green tea has antioxidants and is naturally low in K/P/Na. One of the safest CKD beverages.

Per 8oz serving: ~5mg K, ~5mg P, ~0mg Na (before condensed milk)

Flavor variations: - **Jasmine green tea** — floral, elegant - **Genmaicha** (green tea with roasted rice) — nutty, toasty, incredible iced - **Matcha latte** — 1 tsp matcha whisked into cold milk/water + ice. Stronger green tea flavor.

2. Garden Mint Mojito

Ingredient	Amount	CKD Notes
Fresh mint leaves	8-10 leaves	Muddle gently — releases oils without bitterness
Lime	1/2, cut into wedges	Low K citrus
Sugar	1 tbsp (or simple syrup)	

Ingredient	Amount	CKD Notes
Sparkling water / club soda	6 oz	Low/no sodium variety
Ice	Fill glass	
Rum	1.5 oz (optional)	Alcohol dehydrates — count fluid, use in moderation

Method: 1. Muddle mint + lime wedges + sugar in bottom of glass (press, don't shred) 2. Fill with ice 3. Pour sparkling water (and rum if using) 4. Stir gently. Garnish with mint sprig + lime wheel

Per 8oz serving (virgin): ~20mg K, ~0mg P, ~10mg Na. One of the cleanest CKD drinks possible.

Variations: - **Strawberry mojito:** Add 3 fresh strawberries to the muddle - **Watermelon mojito:** 2 tbsp fresh watermelon juice + mint + lime - **Pineapple mojito:** 2 tbsp pineapple juice + mint + lime (moderate K — control portion)

3. Porch Berry Lemonade

Ingredient	Amount	CKD Notes
Fresh lemons	3-4 (1/2 cup juice)	Low K citrus
Fresh strawberries	1 cup, hulled	Low K berry
Fresh blueberries	1/2 cup	Low K berry
Sugar	1/2 cup	
Water	3 cups	Makes a pitcher — portion from there
Fresh mint	4-5 leaves (garnish)	

Method: 1. Make simple syrup: dissolve sugar in 1 cup hot water. Cool. 2. Blend berries with 1/2 cup water. Strain through fine mesh (removes seeds) 3. Mix lemon juice + berry puree + simple syrup + remaining water 4. Chill. Serve over ice with mint garnish

Per 8oz serving: ~80mg K, ~10mg P, ~5mg Na. Deep purple-pink color. Beautiful.

The Vorath touch: Add a pinch of edible gold glitter. Purple + gold = Vorathic.

4. Holy Mate Yerba Brew

Yerba mate is CKD-interesting: - Contains caffeine (similar to coffee) — okay in moderation - Contains theobromine (like chocolate) — gentle stimulant - Moderate potassium (~85mg per cup) — acceptable - Rich in antioxidants - Traditional preparation is a CEREMONY — fits the thesis perfectly

Ingredient	Amount	CKD Notes
Yerba mate (loose leaf)	2-3 tbsp	Quality brand: Guayaki, Cruz de Malta, Taragui
Water	8 oz at 160-170F (NOT boiling — burnt mate is bitter)	
Honey or sugar	To taste	

Traditional Mate (with Gourd + Bombilla): 1. Fill gourd 2/3 with yerba mate 2. Tilt gourd, create a slope with the mate on one side 3. Pour cool water into the empty space first (protects the leaves from heat shock) 4. Insert bombilla (metal straw with filter) into the cool water side 5. Pour hot water (160-170F, NOT boiling) into the space near the bombilla 6. Sip through bombilla. Refill with hot water. Repeat 10-15 times (the leaves keep giving) 7. Share the gourd — mate is communal by tradition

Equipment needed:

Item	Price	Where
Mate gourd (wooden or ceramic)	\$10-20	Amazon — search "yerba mate gourd"
Bombilla (metal straw)	\$5-10	Usually comes with gourd
Yerba mate (1 lb bag)	\$8-12	Grocery store or Amazon
Electric kettle with temp control	\$25-40	For precise 170F water

Iced Mate (Terere): Cold-brew method: 3 tbsp mate + 3 cups cold water in pitcher. Refrigerate 4+ hours. Strain. Serve over ice with lime + mint. Paraguayan summer drink.

Per 8oz serving: ~85mg K, ~10mg P, ~5mg Na. May fit some CKD plans; compare the serving with your own targets.

Andrew's mate note: The mate ceremony IS the thesis. You fill the gourd, you pour the water, you sip, you refill, you share. It's meditation with caffeine. "All is remembering." The gourd remembers every infusion.

5. Back-Nine Arnold Palmer

Ingredient	Amount
Iced green tea (strong)	4 oz
Fresh lemonade	4 oz
Sweetener	To taste
Ice	Fill

Combine. Simple. Perfect.

6. Market Agua Fresca

Strawberry Agua Fresca

Ingredient	Amount
Fresh strawberries	1 cup
Water	2 cups
Lime juice	1 tbsp
Sugar	2 tbsp

Blend strawberries + water. Strain. Add lime + sugar. Chill. Light, refreshing, LOW-K.

Cucumber Lime Agua Fresca

Ingredient	Amount
Cucumber	1 medium, peeled
Water	2 cups
Lime juice	2 tbsp
Sugar	1 tbsp
Mint	4-5 leaves

Blend cucumber + water. Strain. Add lime + sugar + mint. Incredibly refreshing. Almost zero K/P/Na.

7. Fresh Ginger Beer

Ingredient	Amount
Fresh ginger	2" piece, grated
Lemon juice	2 tbsp
Sugar	1/4 cup
Water	4 cups
Active dry yeast	1/8 tsp (for natural carbonation)

Method: 1. Boil ginger + sugar + 1 cup water for 5 min. Strain. 2. Add remaining 3 cups cold water + lemon juice 3. Add yeast. Stir. Pour into plastic bottle (NOT glass — pressure builds) 4. Cap loosely. Room temp 24-48 hours until carbonated 5. Refrigerate. Serve cold.

This is REAL ginger beer — spicy, carbonated, no preservatives. Use as mixer with mojitos or drink straight.

8. Golden Milk

Ingredient	Amount
Milk (or rice milk for lower P)	1 cup
Turmeric	1 tsp
Ceylon cinnamon	1/4 tsp
Ginger powder	1/4 tsp
Black pepper	Tiny pinch (activates turmeric's curcumin)
Honey	1 tsp
Coconut cream	1 tsp (optional richness)

Heat milk gently (don't boil). Whisk in spices + honey. Serve warm. Inflammation-support, gorgeous golden color.

Cinnamon Cloud crossover — this IS the Cinnamon Cloud in liquid form.

HOME BREWING NOTES

Andrew's TEA SHELF (what's already in the pantry)

Celestial Seasonings Throat Comfort (Slippery Elm & Lemon)

CKD notes: Slippery elm is a demulcent — coats and soothes throat/GI lining. CKD-aware. Naturally caffeine-free. Low K/P/Na.

Recipes using Throat Comfort: - **Throat Comfort Honey Iced Tea:** Brew 2 bags strong, add 1 tbsp honey + 1 tbsp lemon juice. Chill. Serve over ice. Soothes AND refreshes. - **Throat Comfort Hot Toddy (non-alcoholic):** Brew 1 bag, add 1 tsp honey + squeeze of lemon + pinch of Ceylon cinnamon + thin slice of fresh ginger. Drink warm. Dialysis-day comfort drink. - **Throat Comfort Smoothie Base:** Brew 2 bags strong, chill. Use as liquid base in any berry smoothie instead of juice — adds herbal depth + throat soothing.

Celestial Seasonings Immune Support Green Tea

CKD notes: Green tea base with echinacea, elderberry. Moderate caffeine. Antioxidant-rich. Low K/P/Na. Elderberry has some K but the amount in a tea bag is negligible.

Recipes using Immune Support Green Tea: - **Thai-Style Immune Iced Tea:** Brew 3 bags in 2 cups hot water (strong). Add 2 tbsp sugar. Chill. Serve over ice with 1 tbsp condensed milk drizzled. The immune support version of Thai iced tea. - **Immune Matcha Hybrid:** Brew 1 bag Immune Support + whisk in 1/2 tsp matcha powder. Double green tea power. Serve hot or iced. - **Immune Berry Popsicles:** Brew 3 bags strong (1 cup). Add 1/4 cup fresh blueberries blended + 1 tbsp honey. Pour into popsicle molds. Freeze. Immune support on a stick. - **Immune Arnold Palmer:** Brew 2 bags strong, chill. Mix 50/50 with fresh lemonade (lemon juice + sugar + water). Ice. The upgraded Arnold Palmer.

EcoTea Holy Mate (Yerba Mate)

CKD notes: Yerba mate — moderate caffeine (~85mg/cup, similar to coffee). Contains theobromine. Antioxidant-rich. ~85mg K per cup — acceptable for most CKD diets. EcoTea is a quality brand — organic, air-dried (smoother than traditional fire-dried).

Recipes using Holy Mate: - **Traditional Hot Mate:** Use gourd + bombilla if you have them. Fill gourd 2/3 with Holy Mate, pour 170F water (NOT boiling), sip through bombilla. Refill 10-15 times. The ceremony IS the recipe. - **Holy Mate Terere (Iced):** 3 tbsp Holy Mate + 3 cups cold water in pitcher. Refrigerate 4+ hours. Strain. Serve over ice with lime wedge + mint sprig. Paraguayan summer vibes. - **Holy Mate Latte:** Brew strong (2 tbsp mate steeped in 1 cup hot water 5 min, strain). Add 1/2 cup steamed milk or Chobani yogurt thinned with water. Sweeten with honey. The cafe mate. - **Holy Mate Energy Smoothie:** Brew strong, chill. Use as smoothie base: 1 cup chilled mate + 1/2 cup fresh strawberries + 1/2 banana (if K allows) or 1/4 cup mango + Chobani yogurt. Caffeine + fruit + protein. - **Holy Mate Cocktail Mixer (non-alc):** Brew strong, chill. Mix with ginger beer (homemade from earlier recipe) + lime + mint. The mate mojito. Earthy, spicy, citrus. - **Holy Mate Golden Milk:** Brew 1 cup mate. Add 1/2 tsp turmeric + 1/4 tsp cinnamon + pinch black pepper + 1 tsp honey. Caffeine + inflammation-support. Cinnamon Cloud meets mate.

TEA BLENDING EXPERIMENTS

"The Ceremony Blend" (mix all three)

1 bag Throat Comfort + 1 bag Immune Support + 1 tsp Holy Mate in a teapot. Steep 5 min. Strain. You get: soothing (slippery elm) + immunity (echinacea/elderberry) + energy (mate caffeine) + antioxidants (green tea). The trifecta.

"The Dark Mirror Tea" (evening, no caffeine)

2 bags Throat Comfort (caffeine-free) + fresh ginger slices + 1 tsp honey + squeeze of lemon. Warm. Drink after dinner when you need comfort but not caffeine.

"The NorthStar Morning" (full energy)

1 bag Immune Support Green Tea + 1 tsp Holy Mate loose leaf, brewed together. The green tea smooths the mate's earthiness. Double caffeine hit. Add honey + lemon. Start-the-day drink.

HOME SELTZER / SPARKLING WATER STATION

Why DIY seltzer: Store-bought La Croix/Topo Chico = \$1-2 per can. Homemade = pennies per glass. Zero sodium, zero calories, zero K/P. The MOST CKD-aware beverage after plain water — and way more fun.

Equipment Options (ranked by \$/liter value)

System	Upfront Cost	Per Liter Cost	Notes
DrinkMate	\$80-100	~\$0.03/L	Works with ANY SodaStream CO2 canister + any liquid (juice, tea, wine). Most versatile.
SodaStream Terra	\$70-90	~\$0.05/L	Classic. Water-only carbonation. Reliable.
SodaStream Art	\$100-130	~\$0.05/L	Premium model, lever-pull mechanism
iSi Soda Siphon	\$40-60	~\$0.10/L	Classic vintage siphon. Uses small CO2 chargers. More expensive per liter but looks amazing.
Aarke Carbonator	\$150-200	~\$0.05/L	Premium Swedish design. Stainless steel. The Vorathic choice.

Best value: DrinkMate (\$80) — can carbonate anything, not just water. **Best looking:** Aarke Carbonator (\$150-250) — brushed steel, sits on counter like an art piece.

THE AMBITIOUS OPTION: Kegerator + Carboy Homebrew Station

This is the real play. A kegerator with corny kegs + CO2 tank puts ANYTHING on tap — seltzer, kombucha, cold brew, iced tea, lemonade, mate. Bulk production, pennies per glass, always cold and carbonated.

Equipment Needed

Item	Price	What It Does
Mini kegerator / keezer (converted chest freezer or purpose-built)	\$150-300	The cold box. Holds 1-4 corny kegs.
OR: Kegco single-tap kegerator	\$350-500	Purpose-built, includes tower + tap + CO2 reg
5-gallon corny keg (ball lock) x2-4	\$50-80 each	Soda industry kegs. Holds 5 gallons (~640 oz = ~80 glasses). Reusable forever.
5 lb CO2 tank	\$60-80	Lasts ~6-8 full kegs. Refill at homebrew shop or welding supply (~\$15-20)
CO2 regulator (dual gauge)	\$40-60	Controls carbonation pressure (set to ~30 PSI for seltzer, ~12 PSI for beer/kombucha)
Gas + liquid disconnect set	\$15-25	Connects CO2 to keg and keg to tap
Tap + tower (or picnic tap)	\$20-40	Picnic tap = cheap and works. Tower tap = looks professional.
Tubing (gas + liquid line)	\$10-15	Food-grade vinyl
5-gallon glass carboy x2	\$25-35 each	For fermenting kombucha, ginger beer, tepache, shrubs
Airlock + stopper	\$3-5 each	For fermentation

Total Setup Cost

Config	Cost	What You Get
Budget (1 keg + picnic tap)	~\$250-350	One beverage on tap. Good starter.
Mid (kegerator + 2 kegs)	~\$500-700	Two beverages on tap. Rotation.
Full (kegerator + 4 kegs + carboys)	~\$700-1000	Four beverages on tap + fermentation station. The dream.

What Goes On Tap

Keg	Beverage	How
Keg 1: Seltzer	Plain carbonated water — always on tap. Add LorAnn drops at the glass.	Fill keg with filtered water. Connect CO2 at 30 PSI. Shake keg. Wait 24 hrs. Infinite seltzer.
Keg 2: Cold Brew Coffee	5 gallons of cold brew on tap — lasts 2+ weeks	Steep 2 lbs coarse coffee in keg with 5 gal water for 24 hrs. Strain. Pressurize low (5 PSI, no carbonation).
Keg 3: Iced Tea / Mate	Brew a big batch of Holy Mate or Immune Support green tea. Chill in keg.	Brew concentrated, dilute in keg. Optional: carbonate at 15 PSI for sparkling tea.
Keg 4: Kombucha / Ginger Beer	Ferment in carboy, transfer to keg for carbonation + serving	Ferment in glass carboy 7-14 days. Transfer to keg. Force carbonate at 15 PSI. On tap.

Fermentation Projects (Carboy)

Beverage	Ferment Time	Difficulty
Kombucha	7-14 days (SCOBY needed)	Easy once SCOBY is established
Ginger Beer	2-3 days	Easy — ginger + sugar + yeast
Tepache	3-5 days	Easy — pineapple rinds + piloncillo + cinnamon
Water Kefir	24-48 hours (kefir grains needed)	Easy, probiotic, CKD-aware
Jun Tea	7-14 days (like kombucha but with green tea + honey)	Medium
Shrub / Drinking Vinegar	1-2 weeks (fruit + vinegar + sugar)	Easy, no live culture needed

CKD Notes for Kegerator Beverages

- **Seltzer:** ZERO K/P/Na — the most CKD-aware beverage possible
- **Cold brew:** ~85mg K per cup — acceptable, moderate caffeine
- **Kombucha:** ~50mg K per cup — generally CKD-okay in 4-8oz servings. verify with your care team re: live cultures for immunocompromised
- **Ginger beer:** Low K/P/Na — ginger is anti-nausea
- **ALL beverages = FLUID** — track everything toward daily limit

Where to Buy

- **Kegerator:** Amazon, HomebrewSupply.com, MoreBeer.com
- **Corny kegs:** eBay (used = \$30-50, cleaned), Amazon (new = \$60-80)
- **CO2 tank:** Local homebrew shop (they often sell + refill), Amazon
- **Carboys:** Amazon, local homebrew shop
- **CO2 refills:** Welding supply shop (~\$15-20 for 5lb fill), some homebrew shops

Seltzer Flavor Recipes (zero carb, zero sugar)

All flavors = squeeze/drop into carbonated water. Zero calories unless noted.

Flavor	How	CKD Notes
Classic Lime	Squeeze 1/2 lime per glass	Low K citrus — the La Croix dupe
Lemon Drop	Squeeze 1/2 lemon + 1 drop stevia	Sharp, clean
Cucumber Mint	3 cucumber slices + 2 mint leaves muddled	Almost zero K/P/Na
Ginger Fizz	1 tsp fresh grated ginger steeped in water, strain, then carbonate	Anti-nausea, warming
Berry Blast	Muddle 3 strawberries or raspberries, strain, add to seltzer	Trace K, mostly water
Tropical Punch	LorAnn Fruit Punch SS flavoring — 2 drops per glass	Zero calories, intense flavor
Sour Apple Fizz	LorAnn Sour Apple SS — 2 drops + squeeze of lime	Andrew's #1 flavor profile
Blue Raspberry Soda	LorAnn Blue Raspberry SS — 2 drops + blue food coloring	Looks like Baja Blast's cooler cousin

Flavor	How	CKD Notes
Mango Citrus	LorAnn Mango SS — 2 drops + squeeze of lime	Tropical zero-cal
Baja Blast Zero (v2)	2 drops LorAnn Tropical Punch + 2 drops blue food coloring + 1 drop green + lime squeeze	The UPGRADED Baja Blast dupe
Italian Soda	Any LorAnn flavor + 1 tbsp cream (heavy cream or coconut cream)	Creamy Italian soda, ~20 cal
Arnold Fizzer	Brew green tea strong, chill, carbonate + lemon	Sparkling Arnold Palmer
Holy Mate Fizz	Brew EcoTea Holy Mate strong, chill, carbonate + lime	Caffeinated sparkling mate
Lavender Lemonade Fizz	Lavender water (steep dried lavender 10 min) + lemon + carbonation	Spa energy in a glass
Grapefruit Paloma	Grapefruit juice 1 tbsp + lime + carbonation (CKD note: grapefruit moderate K — 1 tbsp is fine)	Mexican cantina vibes
Shrub Soda	Any drinking vinegar/shrub (1 tbsp) + seltzer	Fermented, probiotic, tangy

The LorAnn Trick for Seltzer

Your LorAnn Super Strength flavorings (already ordering Green Apple, Watermelon, Blue Raspberry, Mango) work PERFECTLY for seltzer — 1-2 drops per glass = intense flavor, zero carbs, zero sugar, zero K/P/Na. One \$7 bottle makes 200+ glasses of flavored seltzer. That's \$0.035 per flavored seltzer vs \$1.50 for a La Croix.

Fluid Tracking

Every seltzer = fluid. Track it. But since it's the MOST kidney-friendly beverage you can drink (no sodium, no phosphoric acid, no potassium), it's the best way to use your fluid allowance.

Cold Brew Coffee

Ingredient	Amount
Coarse ground coffee	1/3 cup
Cold water	1.5 cups

Combine in jar. Refrigerate 12-24 hours. Strain through fine mesh + paper filter. Dilute 1:1 with water or milk. Add ice.

Cold brew is ~67% less acidic than hot-brewed coffee and lower in K per cup when diluted.

Sun Tea

Put 4-5 tea bags (any variety) in a glass jar with 4 cups water. Set in sunlight for 3-4 hours. Remove bags. Refrigerate. The gentlest extraction possible.

Kombucha (CKD Caution)

Kombucha is fermented tea — moderate K (~50mg per cup), trace alcohol, probiotics. Generally CKD-okay in small amounts (4oz per day). However: - Watch sugar content in commercial brands - Unpasteurized kombucha = live bacteria — check with your doctor if immunocompromised - **Verify with your care team before making kombucha a regular thing**

COFFEE BAR

Coffee originated in Ethiopia ~850 AD. A goat herder named Kaldi noticed his goats dancing after eating coffee berries. He tried them. The rest is civilization.

Andrew's White Chocolate Mocha with Hazelnut & Coconut

The one that makes the whole morning make sense.

Ingredient	Amount	CKD Notes
Strong brewed coffee or espresso	8 oz (1 cup or 2 shots)	Caffeine is CKD-aware in moderation — 2-3 cups/day max
White chocolate chips	1 tbsp	Melt into hot coffee — lower P than dark chocolate
Coconut cream	2 tbsp	NOT coconut water (high K). Cream is rich + low K.

Ingredient	Amount	CKD Notes
Hazelnut syrup (Torani or homemade)	1 tbsp	Or: steep 1 tbsp crushed hazelnuts in hot coffee 5 min, strain
Unsweetened almond milk	1/4 cup, steamed/frothed	Lower P than dairy. Froths well.
Whipped cream (optional)	Dollop	Small amount = controlled P
Coconut flakes, toasted	Pinch on top	Garnish

Method: Brew coffee strong. Melt white chocolate chips directly in the hot coffee (stir until dissolved). Add coconut cream + hazelnut syrup. Stir. Top with frothed almond milk. Whipped cream if you earned it. Toasted coconut flakes on top.

Homemade Hazelnut Syrup: 1/2 cup sugar + 1/2 cup water + 1/4 cup crushed hazelnuts. Simmer 10 min. Strain. Keeps 2 weeks refrigerated. Better than Torani.

Per cup: ~85mg Na | ~60mg P | ~180mg K

Vanilla Cloud Latte

Ingredient	Amount
Espresso or strong coffee	2 shots (2 oz)
Unsweetened almond milk	1 cup, steamed
Vanilla extract	1/2 tsp
Sugar or sweetener	1 tsp

Method: Steam almond milk with vanilla + sugar. Pour espresso into mug. Add steamed milk. That's it. Starbucks charges \$6. This costs \$0.40.

Iced Caramel Cold Brew

Ingredient	Amount
Cold brew coffee	8 oz
Unsweetened almond milk	1/4 cup
Caramel sauce	1 tbsp (homemade: sugar + margarine + cream, same as Twix recipe)
Ice	Full glass
Vanilla	1/4 tsp

Method: Fill glass with ice. Pour cold brew. Add almond milk + vanilla. Drizzle caramel. Stir or don't — the swirl is beautiful either way.

Cinnamon Cloud Coffee

Ingredient	Amount
Hot coffee	8 oz
Ceylon cinnamon	1/2 tsp
Coconut cream	1 tbsp
Honey	1 tsp
Vanilla	1/4 tsp

Method: Stir cinnamon + honey + vanilla into hot coffee. Float coconut cream on top. The Cinnamon Cloud in liquid form. Inflammation-support + delicious.

Turkish Coffee

Invented ~1555 in Istanbul. The first coffeehouses were called "schools of the wise." UNESCO listed Turkish coffee as Intangible Cultural Heritage in 2013.

Ingredient	Amount
Finely ground Turkish coffee	2 tsp
Cold water	1 demitasse cup (3 oz)
Sugar	1 tsp (optional — traditional: ask "sweet, medium, or plain")
Cardamom	1 cracked pod (optional — Arabic variation)

Method: Add coffee + sugar + water to ibrik/cezve (small pot). Heat on LOW. When foam rises, remove from heat. Let foam settle. Return to heat. Repeat 2-3 times. Pour into demitasse WITH the foam. Let grounds settle 1 min. Drink slowly. Read your fortune in the grounds if the mood strikes.

Per cup (3 oz): ~5mg Na | ~10mg P | ~80mg K — one of the lowest-impact drinks that exists.

Vietnamese Iced Coffee

French colonists brought coffee to Vietnam in 1857. The Vietnamese made it better. Sweetened condensed milk was the only dairy that survived the tropical heat.

Ingredient	Amount	CKD Notes
Dark roast coffee (coarse ground)	3 tbsp	Vietnamese phin filter preferred, French press works
Hot water	6 oz	Just off boil
Sweetened condensed milk	2 tbsp	Moderate P — small amount, high impact
Ice	Full glass	

Method: Brew coffee through phin filter (or French press 4 min). Put condensed milk in bottom of glass. Pour hot coffee over it. Stir. Pour over full glass of ice. The hot coffee hitting the cold ice IS the experience.

Cuban Cafecito

Cuban coffee culture survived the revolution, the embargo, and Miami. The espumita (sugar foam) is non-negotiable.

Ingredient	Amount
Dark roast espresso	2 shots (or moka pot)
Sugar	2 tsp (yes, that much — this is Cuba)

Method: Pull espresso. While brewing, put sugar in a small cup. Add FIRST few drops of espresso to sugar. Whisk vigorously with a spoon until you get a thick tan foam (espumita). This takes 30 seconds of aggressive stirring. Pour remaining espresso over the foam. The espumita floats. Drink in 3 sips. Argue about politics afterward.

Ethiopian Buna Coffee

The birthplace of coffee. Ethiopian ceremony is UNESCO-recognized. Three rounds: Abol, Tona, Baraka. Each round gets weaker. The third round is a blessing.

Ingredient	Amount
Green coffee beans	1/4 cup
Water	3 cups
Sugar	To taste
Cardamom or clove (optional)	1 pod
Frankincense (optional, for aroma)	Small piece, burned nearby

Method: Roast green beans in a dry pan over medium heat, shaking constantly, until dark and oily (8-10 min). Grind immediately (coarse). Boil water in a jebena (clay pot) or small saucepan. Add ground coffee. Boil, remove from heat, let settle. Pour from height into small cups (no filter — grounds settle). Serve three rounds from the same grounds — each weaker, each with its own name.

Shortcut: Use already-roasted Ethiopian Yirgacheffe beans. French press. Still tastes like the birthplace.

Japanese Flash-Brew Iced Coffee

Japanese method: brew hot coffee directly onto ice. The thermal shock locks in aromatics that cold brew loses. Invented by Tadao Ueshima, 1960s.

Ingredient	Amount
Medium roast coffee (pour-over grind)	30g
Hot water (just off boil)	300ml
Ice	150g in the carafe

Method: Put 150g ice in your pour-over carafe/server. Place dripper on top. Add 30g coffee grounds. Pour 300ml hot water over grounds in slow circles (same as normal pour-over). Coffee drips directly onto ice — instantly chilled. The hot extraction + instant cooling = brightest, cleanest iced coffee possible. Superior to cold brew for fruity/floral beans.

Mexican Cafe de Olla

Clay pot coffee. Piloncillo (raw cane sugar) + cinnamon + coffee brewed together in earthenware since the Mexican Revolution. Soldaderas (female soldiers) brewed it in the field.

Ingredient	Amount
Dark roast coffee (coarse ground)	4 tbsp
Water	3 cups
Piloncillo or dark brown sugar	2 tbsp (crushed)
Ceylon cinnamon stick	1 whole
Orange peel	1 strip (2 inches)
Clove	2 whole

Method: Bring water + piloncillo + cinnamon + orange peel + cloves to a boil in a clay pot or saucepan. Stir until sugar dissolves. Add coffee grounds. Remove from heat. Steep 5 min. Strain through fine mesh. Serve in clay mugs if you have them. The cinnamon + orange + coffee combination is older than any coffee chain on earth.

Italian Affogato

"Affogato" means "drowned." Vanilla ice cream drowned in espresso. Invented: debated. Perfected: every trattoria in Italy.

Ingredient	Amount
CKD vanilla ice cream (recipe in frozen treats chapter)	1 scoop
Espresso, fresh	1 shot, hot

Method: Scoop ice cream into a small cup or glass. Pour hot espresso over it. Eat/drink immediately. The hot espresso melts the ice cream into a creamy coffee pool. It is simultaneously dessert, coffee, and a religious experience. Takes 30 seconds to make. Takes 30 seconds to consume. Changes your entire afternoon.

Masala Chai Coffee

Not coffee but belongs here. Chai predates coffee by centuries — originated in Ayurvedic medicine ~5000 years ago. The British industrialized it. Indians perfected it.

Ingredient	Amount
Black tea (Assam or CTC)	2 tsp
Water	1 cup
Unsweetened almond milk	1/2 cup
Sugar	1-2 tsp
Fresh ginger	1" piece, smashed
Cardamom pods	3, cracked
Ceylon cinnamon	1/2 stick
Black peppercorns	3
Clove	2

Method: Boil water with all spices 3 min. Add tea. Boil 2 min. Add almond milk + sugar. Bring to boil again — watch it, it rises FAST. Strain into cups. The spices should hit your nose before the cup reaches your lips.

Lauren's Iced Matcha Latte

Matcha was invented by Buddhist monks in 12th century Japan. They drank it to stay awake during meditation. Zen and caffeine — same practice, different mechanism.

Ingredient	Amount
Ceremonial grade matcha powder	1 tsp
Hot water (not boiling — 175F)	2 oz
Unsweetened almond milk (or oat milk for Lauren)	1 cup

Ingredient	Amount
Vanilla	1/4 tsp
Honey or sweetener	1 tsp
Ice	Full glass

Method: Sift matcha into a bowl. Add hot water. Whisk with bamboo chasen (or regular whisk) until frothy and smooth — no clumps. Pour over ice in a tall glass. Add sweetener + vanilla. Top with cold milk. The green layer under the white milk is the visual.

Lauren's Breve (Half & Half Latte)

"Breve" is Italian for "short/brief." American invention — nobody in Italy puts half & half in espresso. We do what we want.

Ingredient	Amount	CKD Notes
Espresso	2 shots	
Half & half	3/4 cup, steamed	Higher P than almond milk — Lauren's treat, Andrew subs almond milk
Vanilla syrup	1 tbsp	
Cinnamon dust	Pinch on top	

Method: Steam half & half until thick microfoam. Pull espresso. Add vanilla to espresso. Pour steamed half & half over. The fat content makes it THICK — almost like drinking a melted latte ice cream. Cinnamon on top.

Andrew's CKD version: Sub almond milk for half & half. Add 1 tbsp coconut cream for richness. Similar texture, lower P.

Frosted Coffeehouse Frappe

Starbucks trademarked "Frappuccino" in 1995. Before that it was just "iced blended coffee." They charged \$4 for a name.

Mocha Frappuccino:

Ingredient	Amount
Strong brewed coffee, cooled	1 cup
Unsweetened almond milk	1/2 cup
Cocoa powder	1 tbsp
Sugar	2 tbsp
Vanilla	1/2 tsp
Ice	2 cups
Whipped cream	On top
Chocolate drizzle	On top

Method: Blend everything except whipped cream until smooth and thick. Pour into tall glass. Top with whipped cream + chocolate drizzle. Costs \$0.60. Starbucks charges \$6.45.

Variations: - **Vanilla Bean:** Skip cocoa. Add 1 tsp vanilla extract + 1 tbsp powdered sugar. - **Caramel:** Skip cocoa. Add 2 tbsp caramel sauce. Drizzle inside glass before pouring. - **Matcha (Lauren's):** Skip coffee + cocoa. Use 1 tsp matcha + almond milk + ice + honey. Blend. - **Strawberry Crème:** Skip coffee + cocoa. Blend 1/2 cup strawberries + almond milk + ice + vanilla + sugar.

MOCKTAILS

"The bar equipment stays. It's a shrine now."

The fancy shaker, the jigger, the muddler, the strainer — all of it stays on the counter. It just pours different things now.

Rum Punch (Zero Proof)

Ingredient	Amount
Pineapple juice	2 oz
Orange juice (small — moderate K)	1 oz
Cranberry juice	1 oz
Lime juice	1 oz
Grenadine	1/2 oz

Ingredient	Amount
Angostura bitters	2 dashes (technically <0.5% ABV — negligible)
Ice	Crushed

Method: Shake all with ice in your good shaker. Strain over crushed ice in a rocks glass. Float grenadine for the sunset layer. Garnish with lime wheel + pineapple wedge + cherry.

Bahama Mama (Zero Proof)

Ingredient	Amount
Pineapple juice	2 oz
Coconut cream	1 oz
Grenadine	1/2 oz
Lime juice	1/2 oz
Orange juice	1 oz
Coffee extract	1/4 tsp (the secret Bahama Mama ingredient)
Ice	Crushed

Method: Blend everything with ice until smooth. Pour into a hurricane glass. Garnish with pineapple wedge + umbrella + cherry. The coconut cream makes it creamy and tropical without dairy.

Margarita (Zero Proof — Andrew's Tequila Shrine Edition)

Ingredient	Amount
Lime juice	2 oz (fresh squeezed — no shortcuts)
Orange juice	1 oz
Agave nectar	1 oz
Jalapeño	2 slices (optional — spicy marg)
Salt	Rim only (lick, don't drink)
Ice	

Method: If spicy: muddle jalapeño slices in shaker first. Add lime, OJ, agave, ice. Shake hard. Rim glass with lime + coarse salt. Strain into glass over fresh ice. Garnish with lime wheel.

Variation — Mango Margarita: Add 2 oz fresh mango puree. Skip the jalapeño. Summer in a glass.

Piña Colada (Zero Proof)

Ingredient	Amount
Pineapple juice	3 oz
Coconut cream	2 oz
Lime juice	1/2 oz
Ice	1 cup

Method: Blend everything until smooth. Pour into a chilled glass. Garnish with pineapple wedge. The coconut cream IS the drink — don't sub coconut water (high K).

Old Fashioned (Zero Proof)

Ingredient	Amount
Simple syrup	1/2 oz
Angostura bitters	3 dashes
Orange bitters	1 dash
Seedlip Spice 94 or Ritual Zero Proof whiskey	2 oz (or strong brewed rooibos tea, chilled)
Orange peel	1 strip
Ice	1 large cube

Method: Stir syrup + bitters in rocks glass. Add large ice cube. Pour spirit alternative over ice. Express orange peel over the drink (squeeze to release oils, run around rim), drop in. Stir 15 seconds.

CKD note on mocktails: Fruit juices are moderate-K. One 6-8oz mocktail = ~200-350mg K depending on juice. Plan around it. The smaller glass, the better it works. Use the jigger — that's what it's for.

CKD Considerations

RENAL CHECK — Drinks & Homebrew
K (Potassium): LOW (green tea, lemon/lime citrus, berries, mint — all low-K)
P (Phosphorus): LOW (no dairy-heavy bases, no cola/phosphoric acid)
Na (Sodium): LOW (homemade = zero hidden sodium)
Fluid: EVERY DRINK = FLUID. 8oz minimum per serving. TRACK EVERYTHING.
Caffeine: Green tea + mate = moderate caffeine. Limit to 2-3 servings/day.

"The ceremony is the making. The making is the prayer. The prayer is a cup of mate shared with someone who gets it."

RENALWISE Brewing Lab — Kombucha, Honey Mead & Fruit Ciders

Using VEVOR Brewing Machine + Still + Keg System

"The ceremony is the fermentation. The fermentation is the prayer."

Andrew has: VEVOR all-in-one brew system, VEVOR still, KitchenAid, blender, countertop carbonator

IMPORTANT: CKD & ALCOHOL

- **Alcohol is NOT restricted in most CKD diets** unless co-morbid liver disease
- **BUT:** Alcohol is a diuretic — it increases urine output in people who still produce urine, and dehydrates everyone
- **If you're fluid-restricted:** Drink on off-treatment days only. Alcohol + post-treatment BP drop = fall risk
- **Fluid tracking:** Every ounce of mead/cider/kombucha = fluid toward daily limit
- **Moderation:** 1-2 drinks max per sitting. The mead and cider are stronger than beer.
- **Verify with your care team** before establishing a regular drinking pattern

1. KOMBUCHA (Non-Alcoholic Fermented Tea)

Basic Kombucha (1-Gallon Batch)

Ingredient	Amount	Notes
Water	14 cups	Filtered, no chlorine (chlorine kills SCOBY)
Sugar (plain white)	1 cup	The SCOBY eats this — most sugar ferments out
Tea bags (black or green)	8 bags	Celestial Seasonings Immune Support green tea works great
SCOBY + starter liquid	1 SCOBY + 2 cups starter	From Amazon kit or a friend's batch

Method: 1. Boil 4 cups water. Remove from heat. Add tea bags + sugar. Steep 15 min. Remove bags. 2. Add remaining 10 cups cold water to bring to room temp 3. Pour into glass jar/carboy (NOT metal — kills SCOBY) 4. Add SCOBY + starter liquid 5. Cover with cloth + rubber band (needs air flow, not airtight) 6. Dark spot, room temp (68-85F), 7-14 days 7. Taste at day 7 — sweet = needs more time, vinegary = done/over-fermented 8. Target: slightly tart, slightly sweet, light fizz

Second Fermentation (Flavor + Carbonation)

This is where it gets fun — bottle with fruit for natural carbonation:

Flavor	Add to Bottle	Ferment
Ginger Lemon	1 tbsp grated ginger + 1 tbsp lemon juice per 16oz bottle	2-3 days
Strawberry	3 diced strawberries per bottle	2-3 days
Mango Habanero	2 tbsp mango puree + 1 thin habanero slice	2-3 days (SPICY kombucha!)
Blueberry Lavender	5 blueberries + pinch dried lavender	3-4 days

Flavor	Add to Bottle	Ferment
Pineapple Ginger	2 tbsp pineapple juice + 1 tsp ginger	2-3 days
Green Apple	2 tbsp fresh apple juice + LorAnn Green Apple 1 drop	2-3 days
Raspberry Lime	4 raspberries + 1 tbsp lime juice	2-3 days
Passion Fruit	1 tbsp passion fruit puree	2-3 days
Holy Mate	2 tbsp brewed EcoTea Holy Mate (cooled)	2-3 days (caffeinated kombucha)
Cinnamon Cloud	1/4 tsp Ceylon cinnamon + 1 tsp honey	2-3 days

SAFETY: Flip-top bottles build pressure. BURP bottles daily (open briefly to release CO₂). Glass can explode if over-fermented. Keep in a container just in case.

Continuous Brew Method (Better for Regular Production)

Instead of batch-by-batch, use a large vessel (2-5 gallon glass jar with spigot): 1. Brew initial batch as above 2. When ready, DRAIN 70% from spigot into bottles for 2F 3. Leave 30% + SCOBY in vessel 4. Add fresh sweet tea to refill 5. Repeat every 5-7 days = infinite kombucha pipeline

2. HONEY MEAD (Honey Wine — the oldest alcohol on earth)

Basic Mead (1-Gallon Batch — VEVOR Brew System)
Mead = honey + water + yeast. That's it. The simplest alcohol you can make.

Ingredient	Amount	Notes
Raw honey	3 lbs (~1 quart)	Local Oregon honey if possible — wildflower or clover. This is \$15-20 at Fred Meyer for a quality mead.
Water	~1 gallon (to fill carboy to shoulder)	Filtered, no chlorine
Yeast	1 packet Lalvin 71B (wine yeast)	Best for mead — fruity, forgiving. Amazon ~\$5/5 packets
Yeast nutrient	1 tsp	Fermaid-O or DAP. Honey lacks nutrients yeast needs.

Method (VEVOR brew system or carboy): 1. Heat 1/2 gallon water to 140F (NOT boiling — kills honey enzymes) 2. Add honey, stir until dissolved. This is called "must." 3. Cool to under 80F (yeast dies above 90F) 4. Pour into sanitized 1-gallon carboy 5. Top up with cool water to fill to shoulder (leave 2-3" headspace) 6. Sprinkle yeast on surface. Add yeast nutrient. Swirl gently. 7. Attach airlock. Place in dark spot, 60-75F. 8. **Primary fermentation: 2-4 weeks** — bubbling = active fermentation. When bubbling slows to 1 bubble/min, it's nearly done. 9. **Rack (siphon) to secondary carboy** — leave sediment behind 10. **Secondary fermentation: 1-3 months** — clears and mellows 11. **Bottle when clear.** Mead improves with age — 6 months minimum, 1 year is better.

ABV: ~12-14% (honey wine strength)

Flavored Meads (Melomels, Metheglins, Cysers)

Name	What to Add	When to Add	Result
Strawberry Melomel	2 lbs fresh strawberries, mashed	In secondary (rack onto fruit)	Pink, fruity, summer mead
Blueberry Melomel	2 lbs fresh blueberries, crushed	In secondary	Deep purple, tart-sweet
Mango Melomel	2 lbs fresh mango, diced	In secondary	Tropical gold
Raspberry Melomel	1.5 lbs fresh raspberries	In secondary	Ruby red, tart
Cherry Melomel	2 lbs fresh cherries, pitted	In secondary	Dark garnet, complex
Metheglin (Spiced)	Cinnamon stick + cloves + star anise + vanilla bean	In primary, remove after 1 week	Warm, holiday, mulled wine vibes
Cyser (Apple Mead)	Replace half the water with fresh apple cider	In primary	Apple wine meets honey wine

Name	What to Add	When to Add	Result
Capsicumel (Hot Mead)	2-4 habaneros, sliced	In secondary, taste daily, remove when desired heat reached	Sweet heat — the hot sauce of alcohol
Bochet (Caramelized Mead)	Caramelize honey before adding water (cook honey to dark amber, 300F)	Primary stage	Dark, complex, toffee-like. Advanced technique.
Joe's Ancient Orange Mead	1 whole orange (sliced, peel on) + 25 raisins + 1 cinnamon stick + 1 whole clove + pinch nutmeg + bread yeast	All in primary, dead simple	The famous beginner mead. Drinkable in 2 months. Forgiving.

Joe's Ancient Orange — THE Beginner Recipe

If you brew ONE thing first, brew this. It's foolproof.

Ingredient	Amount
Raw honey	3.5 lbs
Water	To fill 1-gallon jug
Orange	1 whole, sliced into 8ths (PEEL ON)
Raisins	25
Cinnamon stick	1
Whole clove	1
Whole allspice	1 (optional)
Nutmeg	Pinch
Bread yeast	1 tsp (Fleischmann's — yes, bread yeast works)

Method: Put everything in a 1-gallon jug. Shake. Airlock. Wait 2 months. When the orange sinks, it's done. Strain. Drink. This recipe has been making first-time meadmakers for 20 years.

3. FRUIT CIDER (Hard Cider from Fresh Juice)

Basic Apple Cider (1-Gallon)

Ingredient	Amount	Notes
Fresh apple cider/juice	1 gallon	Pasteurized, NO preservatives (potassium sorbate kills yeast). Fresh-pressed from Fred Meyer produce section is ideal.
Brown sugar	1/2 cup (optional — raises ABV)	
Yeast	1 packet Lalvin EC-1118 (champagne yeast) or Safale S-04 (ale yeast, fruitier)	
Yeast nutrient	1/2 tsp	

Method: 1. Pour cider into sanitized carboy, leaving 3" headspace 2. Add brown sugar if using, stir to dissolve 3. Sprinkle yeast + nutrient 4. Airlock. Dark spot. 60-70F. 5. **Ferment 2-3 weeks** until bubbling stops 6. Rack to secondary. Let clear 1-2 weeks. 7. Bottle. Drinkable immediately. Better at 1 month.

ABV: ~5-7% (beer strength). With brown sugar: ~7-9%.

Fruit Cider Variations

Cider	Method	Result
Pear (Perry)	Sub pear juice for apple	Lighter, more delicate
Strawberry Cider	1 lb mashed strawberries in secondary	Pink, fruity, summer crusher
Mango Cider	1 lb mango puree in secondary	Tropical, golden
Pineapple (Tepache-Cider Hybrid)	1/2 gallon apple cider + 1/2 gallon pineapple juice	Sweet, tropical, fizzy
Ginger Cider	2" fresh ginger, grated, in primary	Spicy, warming

Cider	Method	Result
Cranberry Cider	1 cup cranberry juice in secondary	Tart, ruby, holiday
Hopped Cider (Graff)	Add 1/2 oz dry hops in secondary for 3-5 days	Beer-cider hybrid. Aromatic.
Spiced Cider Syrup	Add cinnamon-ginger simple syrup at bottling or serving	Non-medicated flavor variant. Label sugar and spice additions.

4. DISTILLING (VEVOR Still — Oregon Legal Notes)

Oregon law: You CAN own a still. You CAN distill water and essential oils. You CANNOT distill alcohol without a federal DSP (Distilled Spirits Permit). That said — essential oil distillation and water distillation are legitimate uses.

What You CAN Legally Distill

Product	Legal?	Use
Distilled water	Yes	Purest water for brewing, seltzer base, CPAP machines
Essential oils	Yes	Lavender, mint, rosemary — for soaps, candles, aromatherapy
Hydrosols (floral waters)	Yes	Rose water, lavender water — for cooking (Persian ketchup!)
Vinegar concentration	Yes (grey area)	Concentrate homemade vinegar for hot sauce

DIY Rose Water (for Persian Recipes + Aroma)

Ingredient	Amount
Fresh or dried rose petals (food grade)	2 cups
Distilled water	4 cups

VEVOR still method: Add water + petals to pot. Distill on LOW. Collect the condensate — this IS rose water. Pure, fresh, better than store-bought. Use in Persian ketchup, bastani ice cream, non-food aromatic use.

DIY Lavender Water

Same process with dried culinary lavender. Use in blueberry lavender cream pops, golden milk, or as a pillow spray.

5. KEGERATOR INTEGRATION

The Full Tap Lineup (4-Tap Dream Setup)

Tap	Beverage	Rotation
Tap 1	Force-carbonated seltzer (always on)	Permanent
Tap 2	Kombucha (rotating flavors)	Swap every 1-2 weeks
Tap 3	Hard cider or mead (seasonal)	Swap every 1-2 months
Tap 4	Cold brew coffee or iced mate	Swap weekly

VEVOR Kegerator Options

Search Amazon/VEVOR direct for: - VEVOR kegerator — they make several models \$200-400 - VEVOR keezer conversion kit — convert any chest freezer into a kegerator

The VEVOR play: You already trust the brand. Their kegerator + your existing brew system + corny kegs = complete beverage production and serving pipeline.

BREWING SCHEDULE (First Month)

Week 1

- **Day 1:** Start Joe's Ancient Orange Mead (set and forget for 2 months)
- **Day 1:** Start kombucha first batch (7-14 day ferment)
- **Day 3:** Start apple cider (2-3 week ferment)

Week 2

- **Day 8-10:** Kombucha ready for 2nd ferment — bottle with flavors
- **Day 10:** Start second kombucha batch (continuous brew from now on)

Week 3-4

- **Day 18-21:** Apple cider ready — rack to secondary, add fruit if desired
- **Kombucha 2F bottles ready** — drink! Start next batch.

Month 2

- Apple cider bottles — drinking
- Mead still fermenting (patience)
- Kombucha on continuous rotation
- Start a fruit cider variation

Month 3+

- Mead approaches drinkable
- Multiple ciders in rotation
- Kombucha is a daily habit
- Consider starting a mead with the VEVOR brew system at 5-gallon scale

EQUIPMENT STILL NEEDED (Amazon Cart)

Item	Why	Est. Price
5-gal corny keg x2	Keg kombucha + cider	~\$60-80 each (added to cart)
SCOBY starter	Start kombucha	~\$12 (added to cart)
1-gallon glass carboy x2	Small-batch mead + cider	~\$8-12 each
Airlocks + stoppers	Fermentation	~\$5/3-pack
Auto-siphon + tubing	Racking/transferring	~\$12
Star San sanitizer	Sanitize EVERYTHING	~\$10
Lalvin 71B yeast (5-pack)	Mead yeast	~\$8
Lalvin EC-1118 yeast (5-pack)	Cider/champagne yeast	~\$8
Yeast nutrient (Fermaid-O)	Mead needs this	~\$8
Hydrometer	Measure ABV	~\$8
Flip-top bottles (12-pack)	Kombucha 2F + bottling	~\$20
TOTAL NEW EQUIPMENT		~\$180-240

CKD Considerations

RENAL CHECK — Homebrew Beverages
K (Potassium): LOW-MED (kombucha ~50mg/cup, cider ~100mg/cup, mead varies)
P (Phosphorus): LOW (no phosphate additives in homebrew)
Na (Sodium): ZERO (homebrew has no added sodium)
Fluid: EVERY DRINK = FLUID. Track toward daily limit.
Alcohol: Off-treatment days only. 1-2 servings max. Dehydrates.
Verify alcohol + CKD med interactions with your care team

"Fermentation is the universe's oldest ceremony. Yeast eats sugar and produces joy. Biology making light — in liquid form."

7 Rainbow Sodas + Visual Effects + Blacklit Display Shelf

"Joseph Priestley invented carbonated water in 1767 by suspending a bowl of water above a beer vat at a brewery in Leeds, England."

VORATH SPECTRUM SODAS — 7 Colors, 7 Flavors, 1 Glowing Shelf

Every soda on this page is phosphoric-acid-free, low-sodium, and potassium-conscious. No cola, no Mountain Dew, no Dr Pepper — those are phosphorus bombs. These are built from scratch with real botanicals, real carbonation, and real visual effects that make a dark room look like a stained-glass cathedral.

Fluid note: every soda counts toward daily fluid intake. Track it.

THE VISUAL EFFECTS TOOLKIT

Before you build the sodas, understand the six visual technologies at work. Mix and match across any recipe.

UV Fluorescence

Tonic water contains quinine, which fluoresces brilliant blue under 365nm UV (blacklight). Any soda using a tonic water base will glow blue-white in a dark room under UV. Diet tonic water works identically and has zero phosphorus.

Iridescence / Shimmer

Edible luster dust (FDA-compliant, food-grade mica powder) suspended in liquid creates a swirling metallic shimmer. Gold, silver, copper, and pearl varieties are available. Add 1/4 tsp per 12 oz, stir or shake before serving. The particles catch any light source — backlight, UV, candlelight. It looks like a potion.

Color-Changing (Butterfly Pea Flower)

Butterfly pea flower (*Clitoria ternatea*) extract produces a deep indigo-blue liquid. When acid is added (citrus juice, vinegar), the anthocyanins shift to purple, magenta, or pink depending on pH. This is real chemistry, not food coloring. Steep 5-7 dried flowers in 1 cup hot water for 5 minutes, strain, and you have a pH-reactive blue base. Zero calories, zero sodium, zero potassium.

Layered Density

Different sugar concentrations create different liquid densities. A heavy simple syrup (2:1 sugar:water) sinks below a light sparkling water. By making each color at a different sugar concentration, you can pour a 7-layer rainbow in a single tall glass. Pour heaviest first, lightest last, over the back of a spoon.

Fizz Effects

Three carbonation methods, each with a different character: - **SodaStream / CO2 charger**: Clean, consistent bubbles. Best for everyday use. - **Baking soda + citric acid**: Old-school fizz. Add 1/4 tsp baking soda + 1/4 tsp citric acid to 12 oz liquid for gentle effervescence. Note: baking soda adds ~150mg Na per 1/4 tsp — use sparingly. - **Dry ice (food-grade)**: Theatrical fog that pours over the glass rim. See Party Tricks section for full safety protocol.

Glow-Through (Backlit Shelf)

Clear glass bottles placed in front of an LED strip become luminous columns of color. The light passes through the liquid and projects color onto the wall behind. UV strips add fluorescence on top of the visible color. RGB strips let you color-cycle the backlight for a rave-shelf effect.

THE 7 RAINBOW SODAS

1. RED — "Blood Moon"

In 1576, the Spanish physician Francisco Hernandez documented Aztec use of hibiscus tea (agua de Jamaica) as a traditional heart-ailment remedy during his seven-year survey of New World medicines.

Visual Profile: - **Normal light**: Deep blood-crimson, translucent, with tiny bubbles rising through the red - **UV blacklight**: Hibiscus anthocyanins produce a faint warm fluorescence — subtle but present - **Backlit shelf**: Glows like a red stained-glass window

Ingredient	Amount	CKD Notes
Dried hibiscus flowers	2 tbsp (10g)	Negligible Na/P/K in tea form
Fresh ginger, sliced	1-inch piece (10g)	Low K in this amount
Lime juice	1 tbsp (15ml)	Low K
Sugar or sweetener of choice	2 tbsp (25g)	Adjust to taste
Sparkling water (SodaStream or bottled)	10 oz (300ml)	Zero everything
Ice	As desired	Counts as fluid when melted

Method: 1. Steep hibiscus flowers and ginger slices in 4 oz (120ml) boiling water for 8 minutes. The liquid will turn deep crimson immediately. 2. Strain out solids. Stir in sugar until dissolved. Add lime juice. 3. Let the concentrate cool completely (refrigerate 30 min or ice bath). 4. Pour concentrate into a clear glass bottle or glass. Top with sparkling water. Stir gently. 5. For extra shimmer: add 1/8 tsp edible red or copper luster dust, swirl.

Carbonation method: Pre-carbonate water with SodaStream, then add to concentrate. Do NOT carbonate the hibiscus concentrate directly (it will foam over).

Per 12 oz serving: Na ~5mg | P ~8mg | K ~45mg

2. ORANGE — "Solar Flare"

Blood oranges get their crimson flesh from anthocyanins triggered by cold nighttime temperatures in Sicily's volcanic soil — the same pigment chemistry that makes butterfly pea flowers change color.

Visual Profile: - **Normal light**: Vivid sunset orange with suspended gold shimmer particles catching every angle - **UV blacklight**: Turmeric contains curcumin, which fluoresces a faint greenish-yellow under UV - **Backlit shelf**: Liquid gold beacon — the luster dust makes it glow like molten metal

Ingredient	Amount	CKD Notes
Blood orange juice (fresh-squeezed)	3 tbsp (45ml)	Using small amount to control K (~70mg K per whole orange)
Turmeric powder	1/4 tsp	Negligible Na/P/K
Sugar	2 tbsp (25g)	Adjust to taste
Lemon juice	1 tsp (5ml)	Brightens flavor
Edible gold luster dust	1/4 tsp	Food-grade mica, inert
Sparkling water	10 oz (300ml)	Zero everything

Method: 1. In a small bowl, whisk turmeric into blood orange juice until dissolved (turmeric clumps — whisk well or use a tiny strainer). 2. Add sugar and lemon juice. Stir until sugar dissolves. 3. Pour into glass bottle, add sparkling water gently. 4. Drop in gold luster dust. Cap and invert once slowly — watch the gold swirl through the orange like a solar storm.

Carbonation method: Pre-carbonate water. Add to juice concentrate.

Per 12 oz serving: Na ~3mg | P ~10mg | K ~55mg

3. YELLOW — "Golden Hour"

Pineapple contains bromelain, an enzyme so aggressive it digests protein — which is why your tongue burns when you eat too much. It was first isolated by Venezuelan chemist Vicente Marciano in 1891.

Visual Profile: - **Normal light**: Bright liquid gold with suspended edible gold glitter — catches light like a treasure chest - **UV blacklight**: Minimal fluorescence, but the gold glitter reflects UV as bright metallic flecks - **Backlit shelf**: Pure gold beam — this is the one that makes people stop and stare

Ingredient	Amount	CKD Notes
Pineapple juice (canned, not from concentrate)	3 tbsp (45ml)	Small amount limits K (~25mg K per tbsp)
Fresh ginger, grated	1/2 tsp	Negligible
Lemon juice	1 tbsp (15ml)	Low K
Sugar	1.5 tbsp (20g)	Adjust to taste
Edible gold glitter flakes	1/4 tsp	Food-grade, inert
Sparkling water	10 oz (300ml)	Zero everything

Method: 1. Combine pineapple juice, grated ginger, lemon juice, and sugar. Stir until dissolved. Strain out ginger pulp if desired. 2. Pour into clear glass bottle. Add sparkling water. 3. Add gold glitter flakes. Cap and gently invert. 4. Serve immediately for maximum sparkle — the glitter slowly settles but a gentle swirl brings it back to life.

Carbonation method: Pre-carbonate water.

Per 12 oz serving: Na ~3mg | P ~6mg | K ~80mg

4. GREEN — "Reactor Core"

Matcha was invented by Zen Buddhist monks in 12th-century Japan who ground tencha leaves to stay alert during marathon meditation sessions — the caffeine-L-theanine combination produces calm focus, not jitters.

Visual Profile: - **Normal light:** Nuclear green, opaque-to-translucent depending on matcha concentration. Looks RADIOACTIVE. - **UV blacklight:** Chlorophyll in matcha and spirulina fluoresces a deep red under UV — the green soda turns eerily red-glowing in blacklight. Genuinely unsettling. Perfect. - **Backlit shelf:** Dense green glow like reactor coolant

Ingredient	Amount	CKD Notes
Matcha powder (ceremonial grade)	1/2 tsp (1g)	Low Na/P; K ~27mg per tsp, using half
Fresh lime juice	1 tbsp (15ml)	Low K
Fresh mint leaves	4-5 leaves	Negligible
Sugar	2 tbsp (25g)	Adjust to taste
Spirulina powder	1/8 tsp	Tiny amount for color boost; negligible Na/P/K at this dose
Sparkling water	10 oz (300ml)	Zero everything

Method: 1. Sift matcha and spirulina into a small bowl. Add 2 tbsp warm water (not boiling — it kills matcha flavor). Whisk into a smooth paste with a chasen (bamboo whisk) or small regular whisk. 2. Add sugar and lime juice to the paste. Stir until dissolved. 3. Muddle mint leaves lightly and add to the mixture. Let sit 2 minutes, then strain out mint. 4. Pour into clear glass bottle. Top with sparkling water. 5. The result should look like something the NRC would investigate.

Carbonation method: Pre-carbonate water. The matcha paste settles, so shake/swirl before each pour.

Per 12 oz serving: Na ~5mg | P ~10mg | K ~50mg

5. BLUE — "Deep Space"

Butterfly pea flowers have been used in Southeast Asian cooking for centuries — in Malaysia, they color nasi kerabu rice blue, and the Thai drink nam dok anchan shifts from blue to purple when lime is squeezed in, a trick that predates modern molecular gastronomy by at least 400 years.

Visual Profile: - **Normal light:** DEEP electric blue that shifts to PURPLE when you add citrus. Serve with a lemon wedge on the side so the drinker triggers the color change themselves. It is genuinely dramatic. - **UV blacklight:** If using tonic water as the sparkling base, the quinine fluoresces brilliant blue-white. The butterfly pea flower adds depth underneath. Double-blue effect. - **Backlit shelf:** Deep sapphire glow — the most "outer space" of the seven

Ingredient	Amount	CKD Notes
Dried butterfly pea flowers	7-8 flowers (2g)	Zero Na/P/K in tea form
Lemon juice (served on side)	1 wedge (~1 tbsp)	Low K; triggers color change
Sugar	1.5 tbsp (20g)	Adjust to taste
Tonic water (diet or regular)	5 oz (150ml)	Diet tonic: ~10mg Na per 8 oz, negligible P/K
Sparkling water	5 oz (150ml)	Zero everything

Method: 1. Steep butterfly pea flowers in 3 oz (90ml) boiling water for 5 minutes. The liquid turns deep indigo-blue. Strain. 2. Add sugar to the hot blue tea, stir until dissolved. Cool completely. 3. Pour blue concentrate into a clear glass. Add tonic water and sparkling water (50/50 split gives UV glow without too much quinine bitterness). 4. Serve with a lemon wedge on the rim. Instruct the drinker: "Squeeze the lemon in and watch." 5. The blue transforms to vivid purple in real time. This never gets old.

Carbonation method: Use pre-carbonated tonic water + sparkling water. No additional carbonation needed.

Per 12 oz serving: Na ~10mg | P ~4mg | K ~30mg

6. INDIGO — "Event Horizon"

The word "indigo" entered English from the Portuguese and Spanish "indico," itself from the Greek "indikón" meaning "Indian dye" — the pigment traveled the Silk Road for millennia before Isaac Newton controversially inserted it as a seventh rainbow color to match the seven musical notes.

Visual Profile: - **Normal light:** Deep indigo-purple, nearly black at full concentration, with tiny silver luster dust specks drifting through like stars in a void - **UV blacklight:** The activated charcoal absorbs UV, creating true-dark regions, while the silver luster dust reflects UV as bright star-points. It looks like actual deep space. - **Backlit shelf:** Dark purple glow with silver sparkle — Event Horizon is the right name

Ingredient	Amount	CKD Notes
Blueberries (fresh or frozen)	1/4 cup (40g)	Low K berry (~30mg K per 1/4 cup)
Dried lavender buds (culinary grade)	1/2 tsp	Negligible Na/P/K
Sugar	2 tbsp (25g)	Adjust to taste
Lemon juice	1 tsp (5ml)	Brightens flavor
Activated charcoal (food-grade)	Tiny pinch (~1/16 tsp)	Inert; do NOT use if taking medications within 2 hours — charcoal absorbs drugs
Edible silver luster dust	1/4 tsp	Food-grade mica, inert
Sparkling water	10 oz (300ml)	Zero everything

Method: 1. Simmer blueberries in 3 oz (90ml) water for 5 minutes until they burst and release deep purple juice. Mash with a fork. 2. Strain through fine mesh — press out all liquid, discard pulp. You want clear indigo juice. 3. Add lavender buds to the hot blueberry liquid. Steep 3 minutes (no more — lavender gets soapy). Strain again. 4. Stir in sugar and lemon juice. Cool completely. 5. Pour into clear glass bottle. Add sparkling water. 6. Add the tiny pinch of activated charcoal — it deepens the purple toward true indigo-black. 7. Add silver luster dust. Cap, invert gently. Stars appear.

Carbonation method: Pre-carbonate water.

Activated charcoal warning: Do not consume within 2 hours of taking ANY medication (including blood pressure meds, phosphate binders, etc.). Charcoal is an adsorbent and will reduce drug efficacy. If this is a concern, skip the charcoal entirely — the blueberry-lavender base is already deeply purple without it.

Per 12 oz serving: Na ~4mg | P ~6mg | K ~40mg

7. VIOLET — "Ultraviolet"

Quinine, the compound that makes tonic water glow under UV, was first extracted from Peruvian cinchona bark in 1820 by French chemists Pierre Joseph Pelletier and Joseph Bienaimé Caventou — it remained the only antimalarial drug for over a century.

Visual Profile: - **Normal light:** Deep violet-purple, rich and jewel-toned - **UV blacklight:** THIS IS THE SHOWSTOPPER. The tonic water quinine fluoresces brilliant electric blue-white THROPPER the violet color. The result is a glowing ultraviolet beacon. This is the bottle everyone photographs first. - **Backlit shelf:** Violet glow with internal blue fluorescence — otherworldly

Ingredient	Amount	CKD Notes
Grape juice (Concord, 100%)	2 tbsp (30ml)	Small amount limits K (~40mg K per 2 tbsp)
Dried butterfly pea flowers	4-5 flowers (1.5g)	Zero Na/P/K
Sugar	1.5 tbsp (20g)	Adjust to taste
Tonic water (diet or regular)	6 oz (180ml)	Diet tonic: ~10mg Na per 8 oz
Sparkling water	4 oz (120ml)	Zero everything

Method: 1. Steep butterfly pea flowers in 2 oz (60ml) boiling water for 5 minutes. Strain. Deep blue liquid. 2. Add grape juice to the blue tea — the combination of blue (butterfly pea) + red-purple (grape) = deep violet. 3. Add sugar, stir until dissolved. Cool completely. 4. Pour into clear glass bottle. Add tonic water and sparkling water. 5. Take it into a dark room. Turn on the UV strip. Watch it light up like a portal.

Carbonation method: Pre-carbonated tonic water + sparkling water. No additional carbonation needed.

Per 12 oz serving: Na ~10mg | P ~6mg | K ~55mg

THE DISPLAY SHELF BUILD

Materials

Item	Est. Cost	Notes
Floating shelf (24-36 inches, white or black)	\$15-25	Black shelf disappears in the dark — recommended
UV LED strip, 365nm, USB-powered	\$10-15	True 365nm, not 395nm (395nm is weaker). 1-2 meter strip.
Clear glass bottles x 7 — woozy bottles (5 oz) or swing-top bottles (8-12 oz)	\$15-25	Woozy bottles with cork tops for display; swing-tops for drinking
Optional: RGB LED strip, USB-powered	\$8-12	For color-cycling backlight mode
Optional: UV-reactive ink pen (for labels)	\$5-8	Neon/fluorescent Sharpies or UV-reactive craft paint
Black card stock (for labels)	\$3-5	Black labels with UV-reactive text = invisible until blacklight hits

Assembly

- Mount the floating shelf at eye level on any wall. Dark room or closet works best for maximum effect.
- Attach the UV LED strip to the BACK of the shelf, facing the wall. The UV light bounces off the wall and illuminates the bottles from behind. Use adhesive backing or small cable clips.
- If using RGB strip, mount it alongside or below the UV strip.
- Place the 7 bottles in ROYGBIV order (Red far left, Violet far right).
- Plug in via USB. Kill the room lights.

The Effect

In a dark room with the UV strip on: - Red "Blood Moon" glows faintly warm - Orange "Solar Flare" gleams with gold luster dust - Yellow "Golden Hour" catches UV on gold glitter flakes — sparks of light - Green "Reactor Core" fluoresces RED (chlorophyll under UV) — the wildcard - Blue "Deep Space" glows brilliant blue-white from the tonic water base - Indigo "Event Horizon" is a dark void with silver star-points reflecting UV - Violet "Ultraviolet" is a blinding blue-white fluorescent beacon

Seven bottles. Seven colors. One shelf. One blacklight. Cathedral energy.

Bottle Labels

- Cut black card stock to fit bottle size (roughly 2" x 3" for woozy bottles).
- Write the soda name and a small symbol (eye, spiral, geometric shape) using a UV-reactive ink pen or fluorescent paint.
- Under normal light: bottles look like unlabeled dark potions.
- Under UV: labels light up with glowing neon text and symbols. Invisible information revealed by the correct frequency. The Vorath way.

BONUS: 3 SFX PARTY TRICKS

1. Dry Ice Fog Soda

Michael Faraday first solidified carbon dioxide in 1835 by combining high pressure with ether cooling — dry ice wasn't commercially produced until 1925 by the DryIce Corporation of America.

The Effect: Dense white fog cascades over the rim of the glass and flows across the table like a ground-hugging cloud. Under UV blacklight, the fog itself doesn't glow, but it creates a haze layer that makes the glowing soda above it look like it's floating in clouds.

Method: 1. Make any of the 7 sodas above in a wide-mouth glass (not a bottle — you need room). 2. Using tongs (NEVER bare hands), drop 1 small piece of food-grade dry ice (roughly 1-inch cube) into the soda. 3. Fog begins immediately and lasts 3-5 minutes. 4. The soda will bubble vigorously as the CO2 sublimates — this also extra-carbonates the drink.

SAFETY — READ ALL OF THIS: - NEVER touch dry ice with bare skin. It is -109.3F (-78.5C). Instant frostbite. - NEVER drink the soda while dry ice is still present. Wait until ALL fog production has stopped and no solid

piece remains. Swallowing dry ice causes severe internal burns. - **NEVER seal dry ice in a closed container.** The pressure buildup will cause an explosion. Use open glasses only. - **Ventilation:** Dry ice releases CO2 gas. In a small, sealed room, this can displace oxygen. Use in ventilated spaces. If you feel dizzy, leave the room. - **Purchase:** Buy food-grade dry ice from grocery stores (many carry it near the ice cream freezer). Use within a few hours — it sublimates at room temperature. - **Storage:** Keep in a styrofoam cooler, lid loosely placed (NOT sealed). Never in a freezer (it's colder than your freezer and the sublimation gas needs to escape).

This is a visual-only effect. The soda is safe to drink ONLY after the dry ice is completely gone.

2. Layered Rainbow Shot

The pousse-cafe, a layered cocktail exploiting liquid density differences, was first documented in Jerry Thomas's 1862 "Bar-Tender's Guide" — the first cocktail recipe book ever published.

The Effect: All 7 rainbow colors stacked in a single tall glass, heaviest at the bottom, lightest on top. A drinkable ROYGBIV column.

Method:

Make simplified versions of each soda as a flat (non-carbonated) syrup at different sugar concentrations:

Layer (bottom to top)	Color	Sugar Concentration	Sugar : Water Ratio
1 — Red (Hibiscus)	Crimson	Heaviest	3 tbsp sugar per 2 oz water
2 — Orange (Blood orange + turmeric)	Orange		2.5 tbsp sugar per 2 oz water
3 — Yellow (Pineapple + lemon)	Gold		2 tbsp sugar per 2 oz water
4 — Green (Matcha + lime)	Green		1.5 tbsp sugar per 2 oz water
5 — Blue (Butterfly pea)	Blue		1 tbsp sugar per 2 oz water
6 — Indigo (Blueberry + lavender)	Indigo		0.5 tbsp sugar per 2 oz water
7 — Violet (Grape + butterfly pea)	Violet	Lightest	1 tsp sugar per 2 oz water

Pouring technique: 1. Pour the Red (heaviest) into a tall, narrow glass (champagne flute or Collins glass works well). 2. Hold a bar spoon (or regular spoon) upside-down with the tip touching the inside of the glass, just above the red layer. 3. Pour the Orange SLOWLY over the back of the spoon. The spoon disperses the liquid gently so it floats on top instead of mixing. 4. Repeat for each subsequent layer. Patience is everything. Pour each one slower than the last. 5. Let the glass sit undisturbed for 30 seconds between each layer.

Tips: - Chill each layer to the same temperature before pouring (temperature differences cause convection and mixing). - The first attempt will probably mix. The third attempt will be Instagram-perfect. Practice. - Do NOT carbonate this one — bubbles will destroy the layers.

Per full glass (all 7 layers, ~14 oz total): Na ~20mg | P ~30mg | K ~180mg

3. Color-Change Cocktail Station (Interactive Soda Bar)

Anthocyanins — the pH-sensitive pigments in butterfly pea flowers, red cabbage, and blueberries — were first scientifically described by German chemist Richard Willstätter in 1913, work that contributed to his 1915 Nobel Prize in Chemistry.

The Effect: A self-serve soda bar where guests pour a base blue soda, then add their choice of citrus to trigger a live color change. Different acids produce different colors. Interactive, dramatic, impossible to get bored of.

Setup:

The Base — Butterfly Pea Flower Sparkling Blue:

Ingredient	Amount
Dried butterfly pea flowers	15-20 flowers (5g)
Boiling water	2 cups (for steeping)
Sugar	1/2 cup

Ingredient	Amount
Sparkling water	1 liter

Steep flowers in boiling water for 5 minutes. Strain. Add sugar, stir until dissolved. Cool completely. Add sparkling water. Pour into a clear glass pitcher or drink dispenser.

The Acid Bar — Set out small bowls/squeeze bottles of each:

Acid	Color Result	Flavor
Lemon juice	Blue to PURPLE	Classic citrus
Lime juice	Blue to BLUE-PURPLE to PINK	Brighter, sharper
Grapefruit juice	Blue to MAGENTA	Bitter-sweet floral
Cranberry juice	Blue to RED-VIOLET	Tart, deep color
Apple cider vinegar (1 tsp max)	Blue to BRIGHT PINK	Tangy, surprising
Baking soda (tiny pinch)	Blue to GREEN	Salty-flat; more of a demo than a drink

Instructions for guests: 1. Pour yourself a glass of blue soda from the dispenser. 2. Pick an acid from the bar. 3. Squeeze or pour it in. 4. Watch the color change happen live. 5. Stir and drink.

CKD notes for the acid bar: - Lemon, lime: Low K, safe - Grapefruit juice: Moderate K (~160mg per 1/2 cup) — use sparingly (1 tbsp per drink is fine) - Cranberry juice: Low K, good choice - Apple cider vinegar: Negligible K/P/Na at 1 tsp - Baking soda: ~150mg Na per tiny pinch — skip this one for strict Na limits, or keep it as a visual demo only

Per 8 oz serving of base blue soda + 1 tbsp citrus: Na ~3mg | P ~4mg | K ~20mg

MASTER RENAL NUMBERS — ALL 7 SODAS

Soda	Na (mg)	P (mg)	K (mg)	Fluid (oz)	Notes
Blood Moon (Red)	5	8	45	12	Hibiscus + ginger + lime
Solar Flare (Orange)	3	10	55	12	Blood orange + turmeric

CHAPTER 14

Frozen Treats & Desserts

“Everything tastes better frozen.”

Renal-Aware Ice Cream, Popsicles, Sorbet & Frozen Desserts

"Everything tastes better frozen."

The Rules for CKD Frozen Treats

- **Fluid counts:** Every frozen treat is FLUID. Track it.
- **Dairy caution:** Milk and cream are moderate phosphorus. Use in controlled portions.
- **Fruit caution:** Many fruits are high-K. Stick to the low-K fruits listed below.
- **Sugar:** Not usually restricted in CKD unless diabetic. Use freely unless co-morbid diabetes.

Low-K Fruits for Frozen Treats

Blueberries, raspberries, strawberries, cranberries, pineapple (small amounts), apple, grapes, watermelon (small amounts — high fluid), peaches (canned/drained), pears (canned/drained), lemon, lime

HIGH-K Fruits to AVOID

Banana, mango, kiwi, orange, papaya, coconut water, dried fruit of any kind

Soda	Na (mg)	P (mg)	K (mg)	Fluid (oz)	Notes
Golden Hour (Yellow)	3	6	80	12	Pineapple + ginger + lemon
Reactor Core (Green)	5	10	50	12	Matcha + lime + mint
Deep Space (Blue)	10	4	30	12	Butterfly pea + tonic + lemon
Event Horizon (Indigo)	4	6	40	12	Blueberry + lavender
Ultraviolet (Violet)	10	6	55	12	Grape + butterfly pea + tonic

All values per 12 oz serving. Every soda falls well within typical CKD dietary guidelines (Na <300mg, P <250mg, K <200mg per serving). These are some of the most kidney-friendly beverages you can make.

CKD Considerations

Why these sodas are safe and commercial ones are not: - **No phosphoric acid.** Commercial colas (Coca-Cola, Pepsi, Dr Pepper, Mountain Dew) use phosphoric acid as a flavoring agent, adding 40-70mg of highly bioavailable phosphorus per 12 oz. These homemade sodas use citric acid (from real citrus) which contains zero phosphorus. - **No dark-cola additives.** Caramel color in dark sodas contains phosphorus compounds. These use natural botanical colors. - **Low sodium.** Commercial sodas range 30-70mg Na per can. These are all under 10mg except the tonic-water-based ones (still only ~10mg). - **Controlled potassium.** Orange juice, coconut water, and banana-based drinks are potassium bombs (400-600mg+ per cup). These use small amounts of real fruit juice diluted in sparkling water, keeping K well under 100mg per serving. - **Fluid tracking.** Every 12 oz serving = 12 oz toward your daily fluid goal. The display shelf is beautiful but do not drink all 7 in one sitting unless your nephrologist has blessed a generous fluid allowance.

Activated charcoal reminder (Event Horizon only): Charcoal adsorbs medications. Do not consume within 2 hours of taking any meds. Skip the charcoal entirely if you take phosphate binders, blood pressure medication, or any other time-sensitive drugs. The soda is gorgeous without it.

Dry ice reminder: Never drink while solid dry ice is present. Wait for full sublimation. See full safety protocol in the Party Tricks section.

The Recipes

1. Rainbow Sherbet (Andrew's Favorite)

Ingredient	Amount	Notes
Raspberry layer	1 cup raspberries + 1/3 cup sugar + 1/2 cup cream + 1 tbsp lemon juice	Low-K berry
Lime layer	1/2 cup lime juice + 1/3 cup sugar + 1/2 cup cream + green food coloring	Classic sherbet flavor
Pineapple layer	1/2 cup canned crushed pineapple (drained) + 1/3 cup sugar + 1/2 cup cream	Moderate K — drain well

Method: Blend each layer separately. Pour into loaf pan in three layers, swirl with knife. Freeze 4-6 hours. Scoop.

The Vorath touch: Add edible gold glitter to the top layer before freezing.

2. Tart Sour Sorbet (Andrew's Requested Flavor Profile)

Ingredient	Amount
Lemon juice	1/2 cup
Lime juice	1/4 cup
Sugar	3/4 cup
Water	2 cups
Pinch of salt	Tiny pinch
Citric acid	1/2 tsp (extra SOUR)

Method: Dissolve sugar in warm water, add juices + citric acid, chill, churn in ice cream maker OR pour into pan and scrape with fork every 30 min for 3 hours (granita method).

Variation — Sour Watermelon: Replace half the water with pureed watermelon. Pink + sour = summer.

3. Vanilla Cloud Ice Cream

Ingredient	Amount	Notes
Heavy cream	1 cup	Moderate P — portion control
Whole milk	1 cup	
Sugar	2/3 cup	
Vanilla extract	2 tsp	Real vanilla, not imitation
Egg yolks	3	Yolks ARE high-P, but 3 across a batch = ~50mg P per serving
Pinch of salt	Tiny	

Method: Heat milk + cream + sugar until steaming (not boiling). Temper into yolks. Return to heat, stir until coats spoon (170F). Strain, add vanilla, chill overnight, churn.

Per 1/2 cup serving: ~80mg P, ~120mg K, ~35mg Na — acceptable for most CKD diets.

4. Blueberry Frozen Yogurt Bark

Ingredient	Amount
Greek yogurt	2 cups
Honey	2 tbsp
Blueberries	1 cup
Strawberries, sliced	1/2 cup
Granola	2 tbsp (low-sodium)

Method: Spread yogurt on parchment-lined sheet pan. Drizzle honey. Top with berries + granola. Freeze 3 hours. Break into bark pieces.

Low-K, low-Na, moderate-P. Beautiful to photograph.

5. Creamsicle Popsicles

Ingredient	Amount
Vanilla ice cream (homemade above)	1 cup, softened
Peach nectar or apple juice	1 cup
Vanilla extract	1/2 tsp

Method: Layer ice cream and juice in popsicle molds, swirl with stick. Freeze 4+ hours.

NOT orange juice — OJ is high-K. Peach nectar or apple juice keeps K low while giving that creamsicle vibe.

6. Strawberry Lemonade Slushie

Ingredient	Amount
Strawberries (frozen)	1 cup
Lemon juice	2 tbsp
Sugar	2 tbsp
Water	1 cup
Ice	1 cup

Method: Blend everything. Serve immediately. Fluid counts toward daily limit.

7. Cookie Dough Ice Cream Bites

Ingredient	Amount
Cream cheese	4 oz (softened)
Powdered sugar	1/4 cup
Vanilla	1 tsp
Mini chocolate chips	2 tbsp
Flour	2 tbsp (heat-treated: microwave 1 min to kill bacteria)
Margarine	1 tbsp

Method: Mix everything. Roll into 1" balls. Freeze on parchment 2 hours. Dip in melted chocolate if desired.

Why cream cheese: Lower phosphorus than traditional cookie dough base. Tastes like cheesecake cookie dough.

8. Cinnamon Cloud Frozen Treat (Crossover!)

Ingredient	Amount
Vanilla ice cream base	1 batch
Ceylon cinnamon	1 tsp
Graham cracker crumbs	2 tbsp per serving
Caramel drizzle	1 tbsp (homemade: sugar + cream + margarine)
Edible gold glitter	Pinch

Method: Fold cinnamon into ice cream during churn. Serve with graham cracker crumbs, caramel, gold glitter. The Cinnamon Cloud in frozen form.

9. Mango Nebula Cream Pops / Pudding Base

Mango, yogurt, cream, almond milk, warm spices, and edible glitter. Pudding when chilled; popsicle when frozen.

Label: Vorath Mango Nebula Cream

VORATH MANGO NEBULA CREAM
Pudding + Popsicle Base

Mango / Yogurt / Cream / Almond Milk
Warm Spice / Edible Stardust

Batch: VOR-NEBULA-000999-MNG
Texture: creamy, cold, glossy
Serve: 2 oz pop or 1/4 cup pudding

Back label copy: A golden mango cream folded through yogurt tang, soft dairy body, almond-milk lift, warm spice, and edible stardust. Eat chilled as pudding or freeze into small pops. This is dessert, not a beverage; small portions carry the flavor farther.

Ingredients line: mango, yogurt, cream, almond milk, sugar or honey to taste, vanilla, cinnamon or cardamom, pinch of salt, edible glitter or luster dust.

Safety label: Use edible glitter/luster dust only. Decorative craft glitter is not food. Contains dairy and almond. Keep refrigerated up to 3 days or frozen up to 1 month. Mango and dairy can raise potassium/phosphorus, so keep servings small.

Ingredient	Amount	Notes
Mango puree	1 cup	Fresh or thawed frozen; strain if fibrous.
Greek yogurt or vanilla yogurt	1/2 cup	Tang and body.
Heavy cream	1/4 cup	Use less for lighter pops.
Unsweetened almond milk	1/2 cup	Loosens the base without heavy dairy load.
Sugar, honey, or mirin	1-2 tbsp, to taste	Mirin gives a subtle glossy sweetness if using the taco-sauce flavor bridge.
Vanilla	1 tsp	Rounds the yogurt.
Cinnamon or cardamom	1/8 to 1/4 tsp	Use cardamom for mango-lassi energy.
Edible glitter	Pinch	Add at the end so it stays visible.

1. Blend mango, yogurt, cream, almond milk, sweetener, vanilla, and spice until smooth.
2. Taste cold. Frozen foods taste less sweet, so popsicle base should taste slightly sweeter than pudding base.
3. For pudding: chill 2 hours. For thicker pudding, whisk in 1 tsp instant clearjel or 1 tbsp chia powder and chill 4 hours.
4. For popsicles: pour into molds, tap out air bubbles, freeze 6 hours.
5. Stir edible glitter in last, or dust the mold before filling for a visible nebula streak.

RENAL CHECK - Mango Nebula Cream
K (Potassium): MODERATE because mango is not a low-K fruit. Use 2 oz pops.
P (Phosphorus): MODERATE from yogurt/cream. Almond milk keeps the base lighter.
Na (Sodium): LOW if unsalted ingredients are used.
Fluid: Popsicles count as fluid; log the mold volume.

Equipment Notes

- **No ice cream maker?** Use the "bag method" (ice cream mix in small ziplock, inside large ziplock with ice + salt, shake 10 min) or the "freeze and scrape" granita method
- **Popsicle molds:** Amazon, \$8-12 for a set of 6
- **Silicone molds:** For the cookie dough bites and frozen yogurt shapes — skull/diamond molds work for both gummies AND frozen treats

CKD Considerations

RENAL CHECK — Frozen Treats
K (Potassium): LOW (low-K fruits only, no banana/mango/orange)
P (Phosphorus): LOW-MED (cream cheese over milk-heavy bases, portion control)
Na (Sodium): LOW (homemade = no hidden sodium)
Fluid: ALL frozen treats = FLUID. Count every serving.

"The freezer is the happiest place in the house."

Wild, Creative, Unexpected Frozen Desserts

"The freezer is a laboratory. Everything is an experiment."

Andrew has: blender, KitchenAid mixer, freezer, Chobani yogurt, fresh fruit preference. No frozen fruit.

THE EXPERIMENTS

1. Mango Sticky Rice Ice Cream Bars

Fusion: Thai dessert meets ice cream bar

Ingredient	Amount
Fresh mango	1, pureed
Coconut cream	1/2 can
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Sugar	2 tbsp
Cooked sticky rice	1/2 cup
Sesame seeds	Garnish

Method: Blend mango + coconut cream + yogurt + sugar until smooth. Fold in cooked sticky rice (creates chewy bites). Pour into popsicle molds or lined loaf pan. Freeze 4+ hours. Cut into bars. Drizzle with extra coconut cream + sesame seeds.

Why it's weird and good: The rice chunks in the ice cream create a mochi-like chew. Thai dessert in popsicle form.

2. Strawberry Balsamic Black Pepper Sorbet

Inspired by: Fine dining strawberry salads

Ingredient	Amount
Fresh strawberries	2 cups
Sugar	1/2 cup
Water	3/4 cup
Balsamic vinegar	1 tbsp (good quality)
Fresh cracked black pepper	1/4 tsp
Pinch of salt	

Method: Dissolve sugar in warm water. Blend strawberries smooth. Combine everything. Chill. Churn or granita method.

Why it works: Balsamic + strawberry is a classic Italian combo. The black pepper adds a slow warmth that hits 3 seconds after the sweet. Nobody expects pepper in sorbet. Everyone wants more.

3. Watermelon-Lime Granita with Tajin

Mexican street cart energy

Ingredient	Amount
Fresh watermelon	3 cups, cubed
Lime juice	2 tbsp

Ingredient	Amount
Sugar	2 tbsp
Tajin seasoning	For rimming the glass
Fresh mint	Garnish

Method: Blend watermelon + lime + sugar. Pour into sheet pan. Freeze. Every 30 min for 3 hours, scrape with fork to create fluffy ice crystals. Serve in Tajin-rimmed glass with mint.

CKD note: Watermelon is low-K per cup (~170mg) but high fluid — portion to 1/2 cup serving and track.

4. Pineapple Whip (Dole Whip Dupe)

The Disneyland secret, CKD-aware

Ingredient	Amount
Fresh pineapple	2 cups, cut and frozen 2 hours (freeze your OWN fresh, not buying frozen)
Chobani Zero Sugar Vanilla Yogurt	1/2 cup
Coconut cream	2 tbsp
Lime juice	1 tsp

Method: Freeze fresh-cut pineapple chunks for 2 hours (hard but not rock solid). Blend with yogurt + coconut cream + lime in blender until soft-serve consistency. Serve immediately in a swirl — this is Dole Whip.

The secret: The partially frozen fresh pineapple gives you soft-serve texture without an ice cream maker. The bromelain enzyme in fresh pineapple creates a slight tingle on the tongue that frozen-from-store pineapple doesn't have.

5. Blueberry Lavender Cream Pops

Spa energy in a popsicle

Ingredient	Amount
Fresh blueberries	1 cup
Chobani Zero Sugar Vanilla Yogurt	1 cup
Honey	2 tbsp
Dried culinary lavender	1/2 tsp (steep in 2 tbsp warm water 10 min, strain)
Lemon juice	1 tsp

Method: Blend blueberries + yogurt + honey + lavender water + lemon. Pour into popsicle molds. Freeze 4+ hours.

Why lavender: It's unexpected in a frozen treat. The floral note makes blueberry taste more complex. Use CULINARY lavender (not potpourri). Fred Meyer sometimes has it in the spice aisle, otherwise Amazon.

6. Vietnamese Coffee Popsicles

Ca phe sua da, frozen

Ingredient	Amount
Strong brewed coffee (double strength)	1 cup, cooled
Condensed milk	3 tbsp
Chobani Zero Sugar Vanilla Yogurt	1/4 cup
Vanilla extract	1/2 tsp

Method: Mix coffee + condensed milk + yogurt + vanilla. Pour into popsicle molds. Freeze.

CKD note: Coffee is CKD-okay in moderation. One popsicle = ~1/2 cup coffee. Condensed milk adds P — small amount across 6 popsicles is fine.

7. Strawberry Cheesecake Ice Cream (No-Churn)

No ice cream maker needed

Ingredient	Amount
Cream cheese	4 oz, softened
Heavy cream	1 cup
Powdered sugar	1/3 cup
Vanilla extract	1 tsp

Ingredient	Amount
Fresh strawberries	1 cup, diced small
Graham cracker crumbs	3 tbsp
Lemon juice	1 tsp

Method: 1. Whip heavy cream to stiff peaks with KitchenAid 2. In separate bowl, beat cream cheese + powdered sugar + vanilla + lemon until smooth 3. Fold whipped cream into cream cheese mixture 4. Fold in diced strawberries + graham cracker crumbs 5. Pour into loaf pan lined with parchment. Freeze 6+ hours. 6. Scoop. Actual cheesecake ice cream.

Why no-churn works: The whipped cream provides the air that an ice cream maker would. The cream cheese prevents ice crystals. Result: smooth, creamy, no machine.

8. Frozen Yogurt Bark: Tropical Edition

The most Instagrammable snack

Ingredient	Amount
Chobani Zero Sugar Vanilla Yogurt	2 cups
Honey	1 tbsp
Fresh mango	1/2 cup, diced small
Fresh pineapple	1/4 cup, diced small
Fresh strawberries	1/4 cup, sliced
Toasted coconut flakes	2 tbsp
Edible gold glitter	Pinch (Vorath love)

Method: Spread yogurt on parchment-lined sheet pan (~1/4" thick). Drizzle honey. Arrange fruit on top. Sprinkle coconut + gold glitter. Freeze 3+ hours. Break into shards. Store in freezer bag.

9. Lime Sherbet Float with Sprite Zero

The CKD root beer float

Ingredient	Amount
Homemade lime sherbet (from frozen treats recipe)	2 scoops
Sprite Zero	6 oz, cold

Method: Scoop sherbet into glass. Pour Sprite Zero slowly. It fizzes up like a root beer float. Drink/eat with a spoon. The Baja Blast Float.

Variation: Use the homemade rainbow sherbet for a "Rainbow Float."

10. Frozen Hot Sauce Honey Pineapple Bites

Sweet, spicy, frozen. Andrew's weird experiment.

Ingredient	Amount
Fresh pineapple	1 cup, cut into 1" chunks
Honey	2 tbsp
Your homemade hot sauce (any)	1 tsp (start mild)
Tajin	Sprinkle
Lime juice	1 tsp

Method: Toss pineapple chunks in honey + hot sauce + lime. Arrange on parchment-lined sheet. Sprinkle Tajin. Freeze 2 hours. Eat frozen.

Why this works: The sweet-heat combo is classic Mexican candy (Lucas, Pulparindo). Frozen pineapple is nature's candy already. Adding heat makes your mouth confused in the best way. Start with Baby's First Burn sauce and work up.

11. Matcha White Chocolate Frozen Yogurt

Japanese-inspired, earthy-sweet

Ingredient	Amount
Chobani Zero Sugar Vanilla Yogurt	2 cups
Matcha powder	1 tsp (sifted)
White chocolate chips	2 tbsp, melted
Honey	1 tbsp

Method: Whisk matcha into yogurt until no lumps. Drizzle in melted white chocolate, fold (creates streaks). Add honey. Pour into container. Freeze 4

hours, stirring once at 2 hours. Scoop.

12. Coconut Cream Paletas (Mexican Ice Pops)

Creamy, tropical, traditional

Ingredient	Amount
Coconut cream	1 can (13.5oz)
Fresh strawberries	1 cup, sliced
Sugar	3 tbsp
Vanilla extract	1 tsp
Lime juice	1 tsp
Toasted coconut flakes	2 tbsp

Method: Warm coconut cream + sugar until dissolved. Cool. Add vanilla + lime. Pour into popsicle molds, layering strawberry slices as you go. Top with coconut flakes. Freeze 4+ hours.

13. Persian Rose Water Ice Cream (Bastani)

From the Cornelius food cart vibes

Ingredient	Amount
Heavy cream	1 cup
Unsweetened almond milk	1/2 cup
Sugar	1/3 cup
Rose water	1 tbsp
Saffron threads	Small pinch, bloomed in 1 tbsp hot water
Vanilla extract	1/2 tsp
Crushed pistachios	1 tbsp (garnish only — moderate P)

Method: Heat milk + cream + sugar until steaming. Add saffron water + rose water + vanilla. Chill overnight. Churn or no-churn method. Garnish with crushed pistachios.

This is the ice cream version of that Persian ketchup experience. Saffron turns it golden. Rose water makes it floral. Pistachios on top = jewel-like. Ancient Persia in a scoop.

14. Frozen Grape "Candy"

The simplest frozen treat that Andrew already loves

Ingredient	Amount
Fresh green grapes (Andrew's favorite)	2 cups

Method: Wash. Remove from stem. Spread on sheet pan. Freeze 2+ hours. Eat frozen.

That's it. Frozen grapes taste like candy. They're cold, sweet, crunchy-then-soft. The best CKD snack that requires zero preparation. K is moderate (~145mg/cup) so portion to 1 cup.

Upgrade: Dip in melted white chocolate before freezing. Or roll in Jell-O powder (grape Jell-O on grape = grape squared).

15. The "BIOLUME" Glow Pop (Blacklight Edition)

Tonic water makes these glow under UV light

Ingredient	Amount
Tonic water	1 cup (contains quinine which fluoresces)
Fresh lime juice	2 tbsp
Sugar	3 tbsp
Blue food coloring	2 drops

Method: Mix everything. Pour into popsicle molds. Freeze. Under a blacklight, these glow BLUE-GREEN. Biology making light. BIOLUME in edible form.

CKD note: Tonic water has ~10mg Na per cup. Quinine is safe in dietary amounts (it's in every gin & tonic on earth). The glow is real physics — quinine absorbs UV and re-emits visible blue light.

EQUIPMENT UPGRADES TO CONSIDER

Item	Price	What It Unlocks
Popsicle molds (set of 6-10)	\$8-12	All the popsicle/paleta recipes
Silicone ice cream bar molds	\$10-15	Mango sticky rice bars, cheesecake bars
Ice cream maker attachment for KitchenAid	\$50-80	Real churned ice cream/sorbet/sherbet — game changer
Squeeze bottles (set of 3)	\$8	Sauce drizzling on treats
UV blacklight bulb	\$8	For the BIOLUME Glow Pop photos
Edible gold glitter	\$8-10	Vorath love on EVERYTHING

The KitchenAid ice cream maker attachment is the single biggest upgrade. You already have the mixer — the attachment just adds a frozen bowl and a dasher. Real churned ice cream, gelato, sorbet, sherbet from all the recipes above. ~\$50-80 on Amazon.

WEEKEND EXPERIMENT PLAN

Saturday Afternoon Session (~2 hours)

1. Frozen grapes — 5 min prep, freeze while you work (Andrew's easy win)

CHAPTER 15

Vorath Orbital Chips

“One batch. One flavor. One orbit.”

THE VORATH ORBITAL CHIP LINE

Limited Edition Artisan Potato Chips for the Cosmically Inclined

“One batch. One flavor. One orbit.”

In 1853, chef George Crum at Moon's Lake House in Saratoga Springs sliced potatoes paper-thin and fried them crisp to spite a customer who complained his fries were too thick -- the potato chip was born from pure contempt.

MASTER CHIP BASE

Every Orbital chip starts here. The base is identical across all 14 flavors. K-leaching is non-negotiable.

Potato Preparation (K-Leaching Protocol)

Step	Method
Select	Russet or Yukon Gold, firm, no green spots
Peel	Remove all skin (skin concentrates potassium)
Slice	Mandoline, 1/16" (1.5mm) thin -- uniform or they burn
First Soak	Submerge slices in cold water, 2+ hours
Water Change 1	Drain, refill with fresh cold water, 1+ hour
Water Change 2	Drain, refill again, 1+ hour minimum
Total Soak	4+ hours minimum (overnight is better)
Drain	Colander, 10 minutes
Dry	Pat bone-dry with clean towels -- water + hot oil = disaster

K-leaching removes approximately 50% of potassium from raw potatoes. The thinner the slice, the more effective the leach.

Frying Method (Primary)

Component	Detail
Oil	Avocado oil (high smoke point, neutral, low in phosphorus additives)
Temperature	350-375F (175-190C)

2. Pineapple Whip — 10 min, eat immediately (instant gratification)
3. Frozen yogurt bark — 15 min prep, freeze overnight (tomorrow's snack)
4. Strawberry balsamic black pepper sorbet — 15 min prep, freeze/scrape method over 3 hours

Sunday Morning

1. Watermelon-lime granita — scrape every 30 min while doing other things
2. Vietnamese coffee popsicles — 5 min prep, freeze all day

Total ingredient cost for all 6 experiments: ~\$15 using what's already in your Instacart cart

CKD Considerations

RENAL CHECK — Experimental Frozen Treats
K (Potassium): LOW-MED (low-K fruits, controlled portions)
P (Phosphorus): LOW (yogurt base over milk, cream cheese over heavy dairy)
Na (Sodium): LOW (homemade = zero hidden sodium)
Fluid: EVERY FROZEN TREAT = FLUID. 3-8 oz per serving. TRACK.

“The freezer is the Dark Mirror of the kitchen. What stares back is delicious.”

Component	Detail
Batch Size	Small -- do not crowd the oil
Time	2-3 minutes until edges curl and color turns golden
Drain	Wire rack over sheet pan, NOT paper towels (they steam)
Season	Immediately while oil is still tacky

Baked Method (Alternate)

Component	Detail
Prep	Toss dried slices lightly in 1 tsp avocado oil per potato
Oven	400F (200C), convection if available
Arrangement	Single layer on parchment-lined sheet, no overlap
Time	12-18 minutes, flip once at halfway
Watch	They go from done to burnt in 60 seconds
Season	Immediately out of oven

Per-Serving Baseline (unseasoned, ~1 oz / 28g leached chips)

Nutrient	Amount
Sodium (Na)	~5 mg
Phosphorus (P)	~25 mg
Potassium (K)	~150 mg (post-leach, down from ~300+ mg)
Calories	~140 (fried) / ~110 (baked)

RAINBOW SPECTRUM FLAVORS

Seven chips. Seven colors. One spectrum.

Isaac Newton inserted "indigo" into the rainbow to force the count to seven -- he believed seven was a number of cosmic significance, mirroring the musical scale.

1. RED -- Szechuan Ghost Pepper

Numbing heat on a chip. The red is real.

Seasoning Blend (makes ~2 oz, enough for 4-6 servings of chips)

Ingredient	Amount
Szechuan peppercorn, toasted and ground	1 tbsp
Ghost pepper powder (Bhut Jolokia)	1/4 tsp (adjust to courage)
Smoked paprika	1 tbsp

Ingredient	Amount
Beet root powder (for deep red color)	1 tsp
Garlic powder	1 tsp
Onion powder	1/2 tsp
Coconut sugar	1/2 tsp
Fine sea salt	1/2 tsp

Color Achieved With: Smoked paprika (base red) + beet root powder (deep crimson boost). Both natural. The ghost pepper itself adds ruddy undertone.

Method: Combine all dry ingredients. Grind in spice grinder to uniform powder. Dust onto chips immediately after frying/baking while surface is still tacky.

The Szechuan peppercorn doesn't create heat -- it vibrates the nerve endings at 50Hz, the same frequency as a hummingbird's wings, creating the "numbing" sensation called ma.

CKD Notes

- Ghost pepper is used in trace amounts; capsaicin has no estimated mineral contribution
- Szechuan peppercorn is potassium-neutral at this quantity
- Beet root powder adds negligible K at 1 tsp

Per Serving (1 oz chips + seasoning)

Nutrient	Amount
Na	~120 mg
P	~30 mg
K	~160 mg

2. ORANGE -- Mango Habanero Tajin

Tropical fire. The orange dust gets everywhere. You won't mind.

Seasoning Blend

Ingredient	Amount
Freeze-dried mango powder	2 tbsp
Habanero powder	1/2 tsp
Tajin seasoning (or: chili powder 1 tsp + lime zest 1 tsp + salt 1/4 tsp)	1 tbsp
Turmeric	1/4 tsp (orange boost)
Citric acid powder	1/4 tsp
Fine sea salt	1/4 tsp

Color Achieved With: Freeze-dried mango powder (natural golden-orange) + turmeric (deepens to vivid orange). The Tajin chili adds red-orange speckle.

Method: Pulse mango powder fine in spice grinder. Combine all ingredients. Apply to hot chips.

The habanero was cultivated in the Amazon basin over 8,500 years ago, making it one of the oldest domesticated peppers on Earth -- older than pottery in the Americas.

CKD Notes

- Tajin contains salt; recipe accounts for this in Na total
- Freeze-dried mango is concentrated; small amount keeps K in check
- If using store-bought Tajin, check label for phosphate additives -- avoid

Per Serving (1 oz chips + seasoning)

Nutrient	Amount
Na	~140 mg
P	~32 mg
K	~170 mg

3. YELLOW -- Turmeric Curry Bomb

Golden. Warm. Complex. Stains everything it touches, including your memory.

Seasoning Blend

Ingredient	Amount
Turmeric powder	1 tbsp
Curry powder (salt-free blend)	1 tbsp
Ground coriander	1 tsp
Ground cumin	1/2 tsp
Fenugreek powder	1/4 tsp
Ginger powder	1/4 tsp
Black pepper	1/4 tsp (activates curcumin)
Coconut sugar	1/2 tsp
Fine sea salt	1/2 tsp

Color Achieved With: Turmeric (primary yellow, potent natural dye) + curry powder (warm golden undertone). This is the most naturally vivid of all 7 colors.

Method: Combine all. No grinding needed -- already powders. Toss with hot chips. Wear gloves or accept yellow fingers for 48 hours.

Turmeric has been used continuously for over 4,000 years in South Asia; ancient Tamil medical texts prescribed it for stomach ailments, wound healing, and as a dye for monks' robes.

CKD Notes

- Turmeric is naturally very low in Na/P/K at seasoning quantities
- Salt-free curry blend is critical -- commercial curry powders often contain hidden sodium
- Black pepper increases curcumin bioavailability by 2,000% (Piperine effect)

Per Serving (1 oz chips + seasoning)

Nutrient	Amount
Na	~115 mg
P	~35 mg
K	~165 mg

4. GREEN -- Wasabi Nori Furikake

Japanese green. Umami depth. Nose-clearing heat that vanishes in 8 seconds.

Seasoning Blend

Ingredient	Amount
Nori sheets, toasted and crumbled fine	2 sheets
Wasabi powder (real, not horseradish fake)	1 tsp
Sesame seeds, toasted	1 tbsp
Bonito flake powder (optional, skip for vegetarian)	1 tsp
Rice flour	1 tsp (helps blend adhere)
Coconut aminos powder or low-sodium soy powder	1/2 tsp
Spirulina powder	1/4 tsp (green color boost)
Sugar	1/2 tsp
Fine sea salt	1/4 tsp

Color Achieved With: Nori (dark green flecks) + spirulina powder (vivid green tint on the base chip) + wasabi powder (pale green). Combined effect: forest-to-emerald green.

Method: Crumble nori by hand first, then pulse everything except sesame seeds in spice grinder. Fold sesame seeds in last. Apply to hot chips -- the tiny nori flecks should be visible.

Real wasabi (*Wasabia japonica*) is one of the hardest plants to cultivate commercially; it requires 18 months of growth in running mountain stream water, which is why over 95% of "wasabi" served worldwide is dyed horseradish.

CKD Notes

- Nori is moderate in potassium -- 2 sheets at this dilution is safe
- Real wasabi is low-sodium; if substituting horseradish-based, check for added sodium
- Bonito flakes add phosphorus; skip if P is a concern
- Spirulina at 1/4 tsp is nutritionally negligible but visually potent

Per Serving (1 oz chips + seasoning)

Nutrient	Amount
Na	~110 mg
P	~40 mg (with bonito) / ~30 mg (without)
K	~175 mg

5. BLUE -- Blue Cheese & Black Truffle

The forbidden blue chip. Funky. Earthy. Decadent. Nobody asked for this. Everybody wants it.

Seasoning Blend

Ingredient	Amount
Blue cheese powder (freeze-dried)	1 tbsp
Black truffle salt	1 tsp
Butterfly pea flower powder	1 tsp (THE blue)
Nutritional yeast	1 tsp (umami + cheesy depth)

Ingredient	Amount
Onion powder	1/2 tsp
Garlic powder	1/2 tsp
Black pepper, fine grind	1/4 tsp
Dried chives, crushed	1/2 tsp

Color Achieved With: Butterfly pea flower powder is the star -- naturally vivid blue, pH-stable when dry. Blue cheese powder adds pale blue-white mottling. The result is a dusty sapphire chip with visible herb flecks.

Method: Combine all dry ingredients. Pulse briefly -- you want some texture from the chives, not total homogeneity. Apply to hot chips.

Butterfly pea flower (*Clitoria ternatea*) has been used in Southeast Asian cooking for centuries; in Thailand it colors rice blue and in Malaysia it makes the famously blue kuih.

CKD Notes

- Blue cheese powder is the highest-P ingredient in the rainbow line; use measured amount only
- Truffle salt replaces regular salt + adds truffle aroma -- do not add additional salt
- Butterfly pea flower is renal-neutral: no significant Na/P/K
- Nutritional yeast adds modest phosphorus but also B-vitamins

Per Serving (1 oz chips + seasoning)

Nutrient	Amount
Na	~145 mg
P	~55 mg
K	~165 mg